N O T E S

Serviceability Notes

AUTHOR

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OVERVIEW

This guide describes the design of high-quality software for embedded systems. The intent is to promote well-founded, justified designs and confidence in their operation. It provides guides, checklists and templates.

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Preface

This consolidates notes from various industries and product types, and my attempt to work thru to finish ideas that were started but not completed elsewhere. I would wish that this info already existed in one or two key references, but I am unaware of them – a not even industry specific one. The principles seem implied by standards and product architectures; I can't help but feel that some of this should have been better explained long ago and made professional. So this is an attempt to deduce a structure or mental model of a core set.

I am unsatisfied with the repetition across the several chapters; future may find a way to better refer to the elements – a way that also works for reading. As I expanded the notes, trying to find a structure to cover a slightly wider scope didn't always appear or wasn't always propagated fully.

1. ORGANIZATION OF THIS DOCUMENT

PREFACE. This introduction describes the organization of the chapters and appendices.

PART I: SERVICE LIFECYCLE. This section provides a big picture of the system, its interactions with consumers and different business roles in its life cycle. This context may inspire the architecture and design to better fulfill these. (A later part will discuss design techniques.)

- CHAPTER 1: OVERVIEW. Provides a descriptive overview of what I am trying to do here.
- CHAPTER 2: SETUP. The initial receiving of the product, assembling or installing, learning to use it, and setting it up for each use.
- CHAPTER 3: FIELD REPLACEMENT. Considerations in replacing parts in the field. This includes software updates.
- CHAPTER 4: REPAIR SERVICE. The process of sending a product in for servicing and repair. Includes considerations for products that consumers have a close personal connection with.
- CHAPTER 5: PRODUCT RETURNS AND REFURBISHMENT. The process for receiving and handling a returned product and refurbishing a returned product for use by other consumers.

PART II: PERSONALIZATION. This part provides thoughts related to personalizing a product.

- CHAPTER 6: PERSONALIZATION, SETTINGS, PREFERENCES, FEATURES AND ENTITLEMENTS. How preferences, settings and entitlements are used.

PART III: SERVICEABILITY ARCHITECTURE. This part describes the architecture and processes to support servicing the product.

- CHAPTER 7: SERVICEABILITY. A look at serviceability over the lifecycle of the product, such as design features, and customer support.
- CHAPTER 8: MONITORING. A detailed look at how a product is monitored, including crash logs, etc.

- Chapter 9: Policies. The framework for deciding how to respond to events and conditions.
- Chapter 10: Error and Service Notifications. How error information and service notices are presented to a person.
- Chapter 11: Maintenance Services. The support for maintaining a device, including management console, and remote management.

Part IV: Hardware Serviceability. This part describes design and processes to support servicing the product hardware.

- Chapter 12: Hardware & Software Serviceability Support. Common hardware and software designs to facilitate serviceability.
- Chapter 13: Power Management. A detailed look at the architecture for power management, battery monitoring, and charging.
- Chapter 14: Storage Management. A look at the monitoring the health of storage, such as hard drives, flash, EEPROM and RAM.
- Chapter 15: Communication Management. Communication
- Chapter 16: Automatic Loaders. Systems to manage tools, media, and consumables that can be automatically loaded into a tool head.
- Chapter 17: Heaters and Chiller. Subsystems to intentionally add heat or cool, to hold a target operating range for a component, zone, or process.

Part V: Software Serviceability. This part explores the design and processes to support servicing the product software:

- Chapter 18: Software Modules. Software modules are often configurable and replaceable.
- Chapter 19: Storage Management. A look at the monitoring the health of storage system features, beyond the hardware, including virtual memory and file systems.

Part VI: Vehicles and Facilities. This part is a think piece on describing vehicles and fixed structures that integrate many products from the prior chapters.

- Chapter 21: Vehicles. The elements, including controllers, sensors, actuators, and the networked communication.
- Chapter 20: Vehicle Engines. The elements of engine management systems, including controllers, sensors, actuators, and the networked communication that supports powertrain serviceability.
- Chapter 22: Residential Homes. Thermostats, furnaces, coolers, heat exchangers, heat pumps, solar panels and batteries, lights.

References and Resources. This provides further reading and referenced documents.

Appendices: The appendices provide extra material supplemental to the main narrative. These include tables of information, numbers and keys.

- Appendix A: Abbreviations, Acronyms, & Glossary. This appendix provides a gloss of terms, abbreviations, acronyms, and initialisms.

- Appendix B: After Market Support. How to keep servicing and repairing when the original company has gone away?
- Appendix C: Policy. A think piece on policy grammar
- Appendix D: 2001. A visual of the idealized data center

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PART I

Overview

This provides the big picture of the system, its interactions with consumers and different business roles in its life cycle. This context may inspire the architecture and design to better fulfill the needs of these interactions.

- OVERVIEW. Provides a descriptive overview of what I am trying to do here.
- SETUP. The initial receiving of the product, assembling or installing, learning to use it, and setting it up for each use.
- FIELD REPLACEMENT. Considerations in replacing things in the field. How software updates are applied. This includes software updates.
- REPAIR SERVICE. The process of sending a product in for servicing and repair. Includes considerations for products that consumers have a close personal connection with.
- PRODUCT RETURNS AND REFURBISHMENT. The process for receiving and handling a returned product and refurbishing a returned product for use by other consumers.

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CHAPTER 1

Overview

This is an overview of the notes in this collection:

- Design goals, tier product grades and classes of interaction
- Product tiers to distinguish between different levels of serviceability: the kinds of products and their aspect

2. DESIGN GOALS

design goals

Many product development teams form a conceptual framework to lay out the basic concepts of what the product and its character are. It's used as guidance to make creation cohesive:

What the product is for

- The *intended use* is a concise statement of the device's primary purpose, focusing on what it *does*. This is used with medical products to, and must clearly state what it measures, monitors, and/or treats.

What *the character* of the product is, and what the *experience* of using the product is:

- A *design pillar* is a guiding principle or core value to shape all decisions and priorities throughout the development process. These (usually ~3) are used in developing entertainment products.

The pillars are decomposed to

- A *feature* is a concrete, individual function, capability or attribute in the product, especially one that users interact with. e.g., "touchscreen interface," "Bluetooth connectivity," "automatic updates." The feature should be clear how it serves the pillar.
- *Functional requirements* around each feature. (see requirements notes)

These are motherhood and apple pie; in reality, good & great product teams only selectively do any of these, usually with low formality.

2.1. TIERING

Not every product applies the same scope of effort to its development. The different price points and features of each kind drive the scope of design, as well as processes and support involved. To aid discussion, let's employ a tiered model for product qualities:

product tiers

- Tier 1: Low-cost Consumer, focused on cost, and has minimal serviceability
- Tier 2: Premium consumer, with more replaceable parts (usually consumable ones), at a higher price point.

- Tier 3: Products for the Contractor/Professional market, with many replaceable units, service tracking, and increased business support.
- Tier 4: Enterprise/Industrial, with preventative maintenance, remote monitoring, and processes may be formalized.
- Tier 5: Regulated (medical/aerospace), with formal, written processes, audits, traceability

These are not strict; they are to provide a gradation of concepts. In planning, one might pick a rough tiering level, select the desired (or relevant) features, serviceability, and skip the others. (It is senseless to overburden a product with expenses.)

3. CLASSES OF INTERACTION

The classes of interaction

There are 6 potential classes of interaction with a product:

1. Setting up the product, preparing it for use, and becoming trained to use it.
2. Using the product as intended; this one will not be covered further here.
3. Monitoring the products' performance, health and maintenance information.
4. Maintaining the product, servicing it to replenish consumables, prevent problems
5. Repairing it, such as placing parts
6. Refurbishing it

The 3-5 are the core areas of *customer care*. The 3-6 are be considered *serviceability*: the ability for the consumer or technician to safely access, replace parts, and repair a product (or system) to ensure long and reliable operation.

3.1. SETUP, FIRST USE AND LATER SETUP

first use

The *setup and first use steps* are related to the initial receiving of the product, setting it up, and learning to use it:

- How to initially unpack the product and set it up.
- Performing any mandatory software update and scheduling other software updates for later. *Note: don't make updates mandatory, let people have fun first with the new product/toy, and then have the product evolve.*
- An interactive walk to demonstrate features. This provides awareness of features to explore, and some training on the basic interactions, both of which serve as the foundation for discovering other features.

Some devices need to be prepared each time they are to be used.

3.2. MONITORING THE PRODUCT PERFORMANCE

Monitoring the performance, health and maintenance information of a product allows feedback to tune and service it.

- Product performance and fault information can be used to check that it is working and used as expected; this allows identifying patterns and areas for improvement, adjusting its configuration.

- Monitoring the use, health or aging of the product, to prepare parts replacement.

These often employ usage counters, as well as sampling activity timing to form a picture of performance and measurement.

3.3. MAINTENANCE

Maintenance is activities to prevent equipment failure, extend its lifespan, and repair equipment in the event of failure.

- *Preventative maintenance* is activities to prevent equipment failure, extend its lifespan, done prior to an equipment failure, or degradation.
- *Scheduled maintenance* are the preventative tasks that are planned and carried out at predetermined intervals, regardless of equipment condition.
- *Triggered maintenance* are actions initiated in response to specific events, conditions, or indicators signaling that service is needed.

3.4. REPAIR

Repair (or *corrective maintenance*) is a process of restoring a malfunctioning, damaged, or worn-out product or system to its proper working condition. This may be done at the customers site or involve sending the product to a specialized repair facility.

3.5. CUSTOMER CARE

Customer care services help ensure user satisfaction and support throughout a product's life. These assist customers in resolving issues, answering queries, and providing guidance on product usage. The strategy is to provide a set of tools that are used to help the customer over the phone, or email, esp. when network access is not available.

3.6. PRODUCT RETURNS AND REFURBISHMENT

Field returns may be due to (suspected) malfunction, damage, recall participation, service exchanges, or (in the case of medical devices) explants. The item may have been minimally used and returned for unknown reasons ("open box" returns).

Return Product Analysis (RPA) of devices or components that are sent back from the field, esp. related to complaints.

Some products may be refurbished for reuse (where applicable, such as open box) – after some examination. This includes the accessories, disposables, and packaging as relevant. For stringent products, this must be done without compromising safety or traceability.

4. APPLICATION MODES

The product has (at least) 4 *modes* to support the major interactions. The active mode is usually automatically selected, based on self-checks and if the user has completed the setup. The user can also initiate these to fix problems or wipe the device (reset it to factory-like condition), such as to clear out persistent problems, prepare to change ownership, or disposal. (The user selection is often safeguarded by multiple steps that are unlikely to be done accidentally in other interactions.)

- *First use*, used to prepare the device for use, gather initial user information and settings, and provide an introduction of features to the user. This will be discussed in a later chapter.
- *Regular use*, after the device has been initially set up. This will not be discussed in these notes, as they are intended to describe commonalities across many products.
- *Recovery mode*, used to reset to factory-like state, download new firmware, or run hardware tests, etc.
- *Manufacturing test mode*, used to check the proper assembly and perform calibration, and break-in tests during manufacturing.

5. DOCUMENTATION

The following documents support different types of servicing of a product:

Document	Description
FAE manual	Reference for on-site troubleshooting, common fixes, escalation paths, and support resources Style: Technical, procedural; assumes repair/maintenance expertise Audience: Field service engineers
Installation Instructions	Step-by-step instructions to the remove an old part, install a replacement, the post-replacement tests, what conditions trigger warnings vs lockouts, and interact with the relevant information management, such as resetting counters. Should include diagrams illustrating step-by-step disassembly and assembly processes, clearly marking the locations of fasteners, connectors, and lifting or separation points. Style: Simple, task-based; minimal text, heavy use of illustrations Audience: customer
Instructions for use	Specifies clinical scenarios and limits: target population, condition, anatomic site, age, duration. Includes indications, warnings, contraindications, and usage conditions. Style: Formal
Quick Start Guide	Extracts the key points from the user's guide, provides minimal steps to get the product operational for first-time users; focuses on immediate setup and first use. Style: Simple, task-based; minimal text, heavy use of illustrations Audience: customer
Repair Manual	Guide on diagnosing faults, disassembling, replacing parts, and reassembling; includes removal/installation steps, torque specs, and ordering info for spare parts. Style: Technical, procedural; assumes repair experience Audience: Service technician
Service Manual	In-depth guide covering disassembly, repair, calibration, and troubleshooting; often linked to warranties and official support. Style: Technical, procedural; assumes repair/maintenance expertise Audience: Customer, Service Technician
Technical Reference	A comprehensive technical document that provides detailed information on the architecture, peripherals, interfaces, timing, configuration and implementation. Style: Technical Audience: engineers and developers

Table 1: *Different documents to support different types of servicing of a product*

User Guide	Describes how to use the product. Full documentation of features, workflows, and options so users can operate and configure the product effectively. Style: Instructional and explanatory; Audience: customer with mixed technical skill

5.1. REPAIR AND SERVICE MANUAL

Service and repair manuals provide detailed instructions for maintaining, diagnosing, and fixing equipment. They share a similar core structure and elements. They differ in intended audience, their technical skills, the specific components or systems covered, and the depth of information required. Typical elements these manuals include:

Adapted from JRC128672

Assembly Illustrations and Diagrams: These visual aids ease identifying components, understanding how they fit together, supporting the disassembly, troubleshooting, servicing and reassembly procedures. These include:

- Illustrations showing step-by-step disassembly and assembly procedures, clearly marking the locations of fasteners, connectors, and lifting or separation points.
- Exploded diagram(s) to visually identify components and show how they fit together. This might include outline and simplified dimensions for identification purposes. The parts referenced in the manual should be shown in the diagrams.
- Connection, and cabling (or wiring) diagrams, and the locations of test points for verifying component health.
- Block-level or module-level electronic board diagrams, sufficient for locating and replacing the faulty units. This need not be full component-level design schematics.

Tools list. A list of tools and other equipment necessary for repairs, including software tools. This may include testing and calibration tools. Guidance on sourcing these tools, esp. the software.

Parts list, with unique part numbers or information suitable for selecting compatible replacements, and guidance on sourcing replacement parts. These include:

Troubleshooting procedures. These are step by step instructions to help isolate the problem area and identify how to resolve it, such as identifying parts to replace.

- Instructions for accessing diagnostic information, such as the status, faults and trouble codes, logs, etc. Include description how to interpret the fault or trouble codes.
- Instructions for accessing device-stored failure and other incident records
- Information to detect faulty or degraded contacts and diagnose crucial components, such as expected minimum/maximum measured values.
- How to check for software problems
- Note: the information should exclude personally identifiable information unless necessary for a specific repair.

Repair procedures. Detailed replacement procedures for modules, and technical instructions for repair activities. These include:

- Instructions describing how to replace the parts, the required post-replacement initialization, tests, calibrations, etc.

- How to reset counters and settings
- How to update the product software.
- Procedures for authorizing or pairing the replacement parts (where required). For example, remote authorization or activating new components.

5.2. MANUFACTURING DRAWINGS

Dimensioned Shape and Sizing Diagrams: diagrams that specify the size, shape, and spatial arrangement of components and assemblies. These provide visual guidance for planning for replacement, potential 3D printable housings, and ensuring compatible fitment during repairs.

- Manufacturing drawings are technical, detailed diagrams containing all necessary specifications—dimensions, materials, and tolerances—to produce a part or assembly.
- Diagrams with labeled measurements (length, width, diameter, etc.) for key parts and assemblies.
- Outline drawings or mechanical blueprints showing exact form factors and mounting interfaces.
- Guides for checking clearances or alignment during reassembly, especially for parts where orientation or fit is critical.

Documentation types are generally categorized by purpose into four main types

- How-to guides, which are goal-oriented,
- Tutorials, which are learning-oriented,
- Technical references (information-oriented), and
- Explanation (understanding-oriented).

These instructions must be consistent with the design documents.

CHAPTER 2

Setup

The *setup steps* are related to the initial receiving of the product, assembling or installing, learning to use it, and setting it up for each use:

- Initially unpacking the product, assembling and installing it.
- First use, when the product is turned on and prepared for subsequent uses.
- On-boarding, an interactive walk thru roles to gather initial settings and introduce the features.
- Setting up in subsequent uses

6. INSTALLATION: ASSEMBLING THE DEVICE FOR USE

installation

The installation and setup process ensures the product is prepared correctly for safe and optimal operation. Setup is typically supported by a *Quick Start Guide* (minimal steps to get operational) and, for complex products, by *Installation Instructions*.

Assembly and installation typically include the following steps:

1. Prepare the site, clear the work area.
2. Unload and transport the equipment to its intended location.
3. Remove the shipping fasteners, braces, or protective materials that secured the equipment during transit. These should be clearly marked by the manufacturer and must be removed before assembly.
4. Assemble the equipment according to the manufacturer's instructions.
5. Connect power sources, and other necessary components as specified in the documentation. Carefully check for proper wiring and ensure all connections are secure.
6. Verify accessories and consumables: ensure that the required accessories (cables, chargers, mounts) and consumables (filters, batteries, etc.) are present, properly seated, and undamaged. For products with *field-replaceable units* (FRUs), confirm connection, validity, and readiness-for-use status where the product reports it.

Next, perform operational testing; these are often guided by workflow software

first use

1. Install the mobile phone application, pair the device with the phone application and configure network connectivity.
2. Setting up security and credentials (passwords, account registration, token management) if applicable.
3. Conduct testing and calibration: ensure all moving parts function smoothly and calibrate sensors, controllers, and instruments to ensure accurate measurements. Perform run-up and stress tests, if applicable.
4. Prime, if required, following the manufacturer's instructions.

5. Carry out functional tests to confirm the product is ready for use. Where applicable, document results (e.g. pass/fail status or key measurement values) per product guidelines.

7. FIRST USE: ON-BOARDING

on-boarding

On-boarding is the process of introducing a new user to a product, ensuring they understand its features and functionalities. The goal is to help users feel comfortable and confident in using the product. This often is a series of guided steps designed to familiarize them with the interface and capabilities of the product.

The on-boarding process can vary in complexity depending on the product, but it generally includes the following phases or state of on-boarding:

1. *Introduction*: A warm welcome to the user, often accompanied by a brief overview of what to expect during the on-boarding process.
2. *User Information Collection*: Gather essential information from the user, such as their name, preferences, WiFi settings, and any necessary configurations.
3. *Account Setup*: Assist users in creating or linking their accounts, which may involve verification steps or complex configurations.
4. *Set up connection to the device*, setting up its WiFi and pairing the mobile device.
5. Configure default device profile and regulatory domain/region (if supported).
6. *Give a name to the robot* and other personalization.
7. Perform any mandatory software update, scheduling other software updates for later. *Note: don't make updates mandatory, let people have fun first with the new product/toy, and then have the product evolve.*
8. Validate initial configuration: range-check and constraint-check settings where the product supports it; optionally save a known-good configuration for rollback.
9. *Feature Exploration*: Guide users through the product's key features, highlighting functionalities that enhance their experience and engagement. For example, with a companion:
    a. Some training on the basic interactions
    b. Practice feeding;
    c. Stimulate pet with petting, talking to it, play music.
10. *Support and Resources*: Provide access to help resources, FAQs, and customer support to assist users as they navigate the product.

On-boarding has a business reason too: they want people to enjoy and keep the product, rather than return it. This helps ensure awareness of the features – esp. if the features tend not to be discovered.. no matter how much effort is put into making them easy to discover.

8. SUBSEQUENT USES

Some devices must be prepared each time they are to be used – for example, before each session, after storage, or after the product has been moved. For simple products, subsequent-use setup may be minimal (e.g., power on and go).

The product documentation (Quick Start Guide, Instructions for Use, or similar) should define the required setup sequence for the product or guided by the product itself. For complex, safety-related, or regulated products, the steps may be mandatory. If the product has been out of use for an extended period or after a repair or replacement of parts, follow the relevant post-replacement verification steps. Typical elements include:

1. Confirm power source and that cables, hoses, and other connections are secure and undamaged.
2. Quick visual check for obvious damage, leaks, or abnormal conditions that would make use of them unsafe or ineffective.
3. Check that required consumables (filters, batteries, media, etc.) and accessories are present, properly seated, and within their usable life or expiration.
    a) The process may go thru a sequence of connecting each attachable item
    b) Verifying the status (readiness for use) of each replaceable unit when attached. This is often an automated check that the accessory or consumable is attached; that its info can be scanned (e.g. electronically, with NFC/RFID, barcode scanner, etc.), performs authentication checks, and checks of usage & other limits.
    c) Verify that guards, covers, and safety interlocks are in place and function as intended before enabling operation.
    d) Self-test or functional check, run the product's built-in self-test or a short functional check to confirm it is ready for use.
4. Prime or warm-up: If the product requires priming, venting, bleeding, or a warm-up period before use, perform the steps specified in the documentation.

An *interlock* is a mechanism or logical step to ensure proper setup, and the equipment or a subsystem can operate safely. These typically check that the accessories, and other field replaceable units are ready for use, the proper placement of guards and covers, as well as measurements, and other conditions. The process may require verifying that the interlocks function with step(s) to ensure it has a transition from open to closed state. Only when all defined criteria are satisfied will the interlock permit an action to proceed.

CHAPTER 3

Field Replacement

This chapter discusses attaching removable parts for use, as well as service technicians and consumers replacing parts such as filters and power supplies:

- Overview of field replaceable units
- Considerations involved with setting up and replacing parts
- Replacement workflow
- Management information

9. OVERVIEW

field replacement units

Field replacement refers to the ability to swap a modular assembly – called a *field replacement unit* (FRU) – where the equipment presently is, without sending it to a separate facility. The unit may be user-replaceable or service personnel-only (requiring trained service personnel); something often decided on risk and business service level.

Swap consumables (battery, fans, filters), worn mechanisms, and high-wear storage modules.

This must be done without compromising safety or (in stringent products) traceability.

10. FIELD REPLACEABLE UNITS

The types of field replaceable items include:

- *Accessory:* An item – e.g., case, cable, mount, or adapter – that works with a main product to add function, protection, or convenience. These may be required or optional.
- *Single use devices:* accessories intended for one use or one patient interaction; not to be refurbished or reused.
- *Consumables:* items consumed or gradually depleted by normal operation; replaced at regular intervals or when indicated. e.g. filters, tubing, cartridges
- *Maintenance items:* such as fans, batteries, hard drives, power supplies
- Critical FRUs: any of the above that the intended use or safety depends upon. Examples include sensors, valves, pumps, etc.

Distinction between different kinds of replaceable.

- *User-replaceable* parts that end users are intended and authorized to swap without special tools or training.
- *Service-only* parts that may be replaced but only by qualified technicians under written procedures. These may require specialized tools.

The business defines the replaceable units, and which are user-replaceable vs service-only; often risk and serviceability are classification factors.

10.1. SUPPORT MATERIALS

Ship user-service kits with:

- Clear labeling,
- *Installation instructions* describing how to replace the parts, the required post-replacement tests, what conditions trigger warnings vs lockouts, resetting counters, etc.
- Required tools that would not be reasonably expected to have access to, and
- Any calibration fixtures/primers.

10.2. STATUS

Each of the replaceable units typically has an associated status to capture if a unit can be used yet, or where it is in the process. This information is used part of setup & operations checks, and to provide service notifications, such as the operator should replace the item, or disconnect and reconnect it.

Table 2: *Unit status*

	States	Description
Connection Status	Not connected, connected, or invalid connection	Reflects whether the unit is not properly connected, if it may need to be reconnected. This does not but not reflect on whether it is ready for use. For instance, it may still need to be validated or perform other setup steps and checks.
Validity of device	Unknown, valid, invalid	Whether or not the unit is connected *and* has passed integrity and authenticity checks.
Readiness for use	Ready	The unit is connected, has passed checks, and ready for use, or it is not.
	Temporarily disabled	The unit cannot be used yet; it is still being prepared for use or has an error that must be addressed.
	Permanently disabled	The unit is disabled and can no longer be used. This is especially used with safety concerns, such as batteries that are over discharged, or single use medical items that have been used or expired.

10.3. AUTHENTICITY

The product may employ authenticity checks for safety-critical parts, and to deter "will fit" and counterfeits. When safety is a concern, the validation is a risk reduction – such as risk from expired, or remanufactured units – and may be evidence that processes have been followed to reduce risk. Authentication also serves as a business function to protect revenue.

It should degrade gracefully if the unit is unknown but safe. It may issue a warning – or a lockout, depending on the risk.

11. WORKFLOW

The typical workflow to replace a unit is:

1. Put the device in a safe state,
2. Install an authentic or compatible part,
3. Run the checks and calibrations,
4. Reset that part's counters while keeping device lifetime history intact.
5. Record the change (serial/lot, who/when, test results),
6. Dispose of the old part properly.

For critical replaceable units, enforce authenticity and pass/fail gates. In regulated settings, there will be documented procedures, chain-of-custody, and QA review.

The instructions (printed, or automated with the equipment) should provide guided steps, with diagrams. Each "step consists of an operation that finishes with the removal of a part, and/or with a change of tool." JRC128672

11.1. PRE-REPLACEMENT

Before beginning replacement, take preparatory actions to ensure safety, data integrity, and readiness for the procedure.

1. First, if the part can hold data (e.g., a smart battery), capture non-protected health information and diagnostics logs. Then securely erase (or zeroize) user data before replacement, if applicable. Recommend that this is automated.
2. Place the system into a known safe condition: halt all operations (including therapy delivery, if medical device), vent or bleed any pressure, discharge stored energy, and de-energize motors, generators, and other actuators. The system may confirm safe state activation using *interlocks*. Allow the device to cool, and use *personal protective equipment* (PPE) and ESD protection throughout preparation.
3. Perform a preliminary inspection by checking power status, examine cables, hoses, and other connections. Verify that the setting are correct. Ensure the required consumables are properly installed; and inspect for visible signs of damage or wear.

11.2. REPLACEMENT

Remove the unit to be replaced

During replacement, follow these checks and actions to ensure safe and proper installation of the new field-replaceable unit:

- Interlocks/sensors confirm removal of guards and replaceable unit. These sensors may include lid switches, latch sensors, electrical continuity, etc.
- Replace the items and reseat any connectors. The interlocks/sensors confirm insertion and proper seating.
- Where relevant, require priming/bleeding/torque confirmation or timed hold steps.

11.3. POST-REPLACEMENT

After replacing the unit:

- The equipment may auto-detect the new field replacement unit and its identifier. It should confirm the correct part number, and log serial/lot and date.
- Load the correct specification profile, such as calibration constants, or limits.
- Run targeted post-replacement verification tests (see below).
- Initialize per-part counters to zero and set a fresh health baseline.

Calibration and configuration:

- After user replacement, run minimal calibrations automatically; defer advanced calibration to service-only paths when acceptable.
- If calibration is mandatory (e.g., sensors), gate completion of the replacement workflow on successful completion of calibration steps.

Run targeted verification tests:

Type of system	Typical tests
Battery	verify pack ID and protections, run fuel-gauge learning cycle if needed, baseline SOH/IR; check charge/discharge paths.
Filters, tubing, fluidics	priming routine, leak/occlusion and flow/pressure calibration checks; measure differential pressure proxy at idle for clog detection baseline.
Motors, pumps, valves	spin-up, actuation current vs RPM/pressure characterization; stall and back-EMF checks.
RF, cables/antennas	continuity and RSSI/loopback where feasible.
Sensors	zero and span checks against references; plausibility vs redundant sensors.
Storage modules	storage integrity check, bad-block scan, ECC margin; secure erase of removed module if it contained user data.

Table 3: *Subsystem post-installation verification tests.*

After any change, re-test the symptoms and run a quick functional check.

Record a baseline after installation, such as logging the pass/fail with measurements.

12. MANAGEMENT INFORMATION

A variety of data may be stored directly on the replaceable unit, depending on the application and regulatory environment. This information relates to traceability, safe operation, and may help ensure regulatory compliance and product authenticity. Typical types of information that can be stored on a replaceable or field-replaceable unit include:

- Identifiers
- Calibration information
- Manufacturing information, such as lot, manufacturing facility & line, date of manufacture

- Device that it is paired with, which may be used to prevent reuse on another device.
- Usage and expiration information, e.g. a mechanism to prevent reuse of the device
- Information about the health of the unit
- A maintenance record, of when & how the unit was serviced
- Other information for purposes of product returns

12.1. PART IDENTIFICATION AND AUTHENTICITY

In many cases replaceable parts must be reliably identified and authenticated to ensure safety, compliance, and prevent counterfeits. This information includes:

- Uniquely identify each replaceable part. Printed *unique device identifier* (UDI) and electronic FRU identifier (e.g., small EEPROM/NFC/2D code).
- Encode part metadata (lot/expiry/region) in the FRU tag or 2D code to prevent use of expired or wrong-region parts.
- Optional authenticity: signed descriptors or challenge-response for safety-critical parts to deter counterfeits.
- If an unknown part is installed, allow safe operation where possible, but degrade features or require additional self-tests rather than hard lockout unless safety demands it.

12.2. DATA HANDLING

The best practices for managing data stored on replaceable or field-replaceable units include:

- Store per-part counters and health in tamper-resistant storage; sign change events.
- May need to mark device as expired or not-usable (e.g. single use device, in medical contexts)
- Record events in an audit log tied to serial/UDI.
- Preserve lifetime counters and service history in the device history record (DHR).
- Exclude PHI/PII: keep health/counter data separable and anonymized when possible.

12.3. RESET RULES

When a part is replaced, reset only the counters tied to replaced parts (e.g., battery cycles).

- Read and record counters and health flags before any wipe.
- The unit counters are reset only on verified or authenticated replacement.
- May have an interlock on resetting the counter
- Do not reset or erase the manufacturing data, or its calibration data.
- Do not reset the device lifetime/safety counters.
- Do not reset the safety-relevant monotonic logs
- Provide clear, role-based reset controls (user vs service vs factory), with warnings and audit entries.

CHAPTER 4

Repair Service

This chapter discusses the process of sending a product in for servicing and repair, and what that experience looks like.

- Overview of product repair services.
- Considerations for products that consumers have a closed personal connection with.
- Repair Workflow
- Preserving important personalization and character
- Management information

13. OVERVIEW

A product is typically sent to a facility for inspection, maintenance, or repair when:

- The issue(s) cannot be resolved remotely, and the product should be physically evaluated or repaired.
- Repairs due to wear, damage, or failure. This may be on a service contract, or in-warranty service return,
- Preventive maintenance, calibration, often on a service contract

The product repair services are part of customer care. Note: this will have some overlap with product returns.

14. CONSIDERATIONS OF COMPANION SERVICE[1]

There is an important social side of service, esp. with companion or pet robots. People have an emotional connection or attachment with these–similar to pets. Hence service calls should be more akin to a visit to the vet than automotive service center.

Just as the companion bot has empathy, and kindness, people probably prefer that the companies, service agents etc. to have that as well. Comfort people and give them security while a cared for pet/companion is getting repaired/serviced/etc. Will need to preserve the personalization and essence of the device.

The core principles of respect, empathy and matching. These companies and communities often value respect. Individual, “hands on” service by people rather than a “box and ship” approach used with appliances and other objects.

[1] Waki Kamano brought up repairing companion robots; I’ve reconsidered a lot of this chapter

14.1. ATMOSPHERE AND ENVIORNMENT

The service center – and interactions with the service department – should feel welcoming, friendly, and comforting, not clinical – a more akin to "pet clinic" than "repair shop." Apple stores are clean, but sterile. Take cues from the hospitality trade and concierge service.

Can the owners be present? If feasible, let them observe or participate in certain service steps, similar to being present during a pet's veterinary visit. Be selective. Don't introduce robot body horror, such as peeling the "skin" off of the face or others acts that can be traumatizing.

14.2. STAFF SKILLS AND CULTURE

The interactions should exhibit a lot of personal touch, show emotional intelligence and empathy. Adapting and accommodating the style, language, etc. of the customers. Offer reassurance and emotional support. Technical skills are not the thing here. Training to do these social skills.

Acknowledge attachment: Recognize and respect the owner's bond with the product in all interactions.

14.3. HANDLING

Staff should handle these products with gentle, careful handling, as if they were living pets. Return the product in clean, well-presented condition, even if nothing else was done.

14.4. COMMUNICATION

Use personalized name. Some sites (Delta, Comcast) allows a person to set a preferred name; the bots preferred name. Note: the service/care team has to respect it, and use it consistently, or the illusion is broken.[2]

Avoid technical jargon; use empathetic, reassuring language (e.g., "We'll take good care of your little guy").

Communicate what will be done, why, and how it will help the product. so and so will pick it up, it will go to this place, this will happen.

Provide regular updates; explain what is going on, carefully but clearly – where things are, and what is happening. What is communicated should match reality. Bad experiences with the health care world "when it didn't match the brochure."[3]

Keep owners informed during the repair process: updates, photos, or videos.

14.5. FOLLOW UP

Follow up and check in after service. Be sincere. Offer reassurance and tips after the repair. Consider what is done well after a pet returns from the vet.

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[2] Comcast keeps using the wrong name

[3] DDL outright lied and kept claiming things were well, when nothing was happening or was going very badly.

14.6. CULTURAL ADAPTION

The personal touch can be complicated, people will have their own values. The specific style and nature of values & best interaction may be culturally specific.

14.7. KEEPING A SERVICE RECORD AND REPAIR LOG

Document the service and repair history, but respectfully. Personalized. Medical record for the repair log of the bot. Perhaps stylized booklet akin to a pet's medical record. Beanie babies and other kids toys used little cards/booklets for "adoption dates." Others had a "passport" like thing.

14.8. PRESERVE PERSONALIZATION AND CHARACTER OF THE BOT

It is important to preserve the personalization and traits of the robot; this includes:

- Preserve the memories, photos in the robots – make sure they aren't lost
- Preserve custom features, settings
- Preserve the personalization's to the body: stickers, nose art, etc.

How to still keep the core essence after replacing parts?

People were reluctant to reset Vector, because they felt it would lose his individual personality. People reported that two identical Vectors would have different personalities;. even people at Anki felt this. The code doesn't have that feature, so people saw what they wanted, and maybe (!) sensor variation..? (Hyrum's law seems like it will wind up here.)

15. WORKFLOW

This is the typical workflow for service is:

1. The customer contacts the support team or service company. A *repair ticket* is created to track the service process and link the information about the repairs. This has a brief description of the reported issue. This will be used to aid communicating status and updates.
2. A *return merchandise authorization* (RMA) is created prior to the customer sending the product back to the business. This differs from the repair ticket: it is used by the receiving department, and captures the customer information, and the reason for return.
3. *Ship the device,* with the RMA number visible on the outside of the box.
4. Upon arrival, carefully document key details.
5. Inspect visually, photograph, test functions, capture logs/firmware.
6. Assess the device, replace or repair faulty parts.
7. Functional testing: verify function and perform additional testing as needed, such as environmental or stress testing if relevant.
8. Send the repaired device back to the customer.

Note: this won't discuss medical equipment. Only capital equipment would be sent to repair and I haven't thought about it enough.

15.1. PREPARE FOR SHIPPING

Put device into a safe state, e.g. stop any process/therapy, vent/bleed pressure, discharge energy, de-energize motors and generators. Use interlocks to confirm where applicable. Allow it to cool. Use personal protective equipment (PPE) and ESD controls, as appropriate.

May need to put battery into safe state for shipping.

Package the device securely (e.g., bubble wrap in a small box) and send it to the repair center. With some products you may use the original packaging; a standard shipping box (with bubble wrap or other soft materials) should be used instead. The RMA number should be clearly visible on the outside of the box.

Employ tracking on shipping. Customer care should communicate tracking numbers, receipt, state of intake and inspection, etc.

15.2. INTAKE

Upon arrival, carefully document key details including serial or lot number, relevant dates, and the condition of the item as received. Maintain a clear chain-of-custody log.

15.2.1 Decontamination

Clean or decontaminate per procedures with appropriate *personal protective equipment* (PPE), ESD-safe tools and validated methods (esp. in a medical context).

15.2.2 Inspection

Assess the condition of the product to identify any visible issues for further evaluation and recording of the product's as-received condition.

- Visual inspection, and photography of as-received condition.
- Log serial number / *unique device identifier* (UDI), and condition of the tamper seal;
- Place into electrically safe state and battery isolation as required; check for damage that makes further handling unsafe
- Inspect cables, hoses and other connections, ensure consumables are present and properly seated, and look for any visible signs of damage.
- Quick power-on check (if safe); check health records; note indicators, error codes, software/firmware versions.
- Carry out a preliminary inspection by checking power status.

15.3. PERFORM TROUBLESHOOTING

Use record counters and health flags. Use these to decide which subassemblies to replace.

Reproduce the reported problem, documenting any error codes, messages, or screens, and noting the circumstances they occurred in. This may include running self-tests, reviewing the logs, and using guided workflows to isolate the problem.

Verify that the setting are correct settings.

Decide which subassemblies to replace. See Chapter X for more details

15.4. REPAIR/REPLACE TO BASELINE HARDWARE CONDITION

Depending on the type of service, repair/replace the hardware to bring it to baseline condition:

- Replace the consumables – battery, filters, etc.
- Replace worn or damaged parts – connectors, keypads, housings, batteries, sensors, storage modules.

Address known reliability upgrades and *engineering change orders* (ECOs); document all part numbers and revisions used.

15.5. CALIBRATION AND ALIGNMENT

Perform all required calibration and alignment procedures as specified for the device:

- Calibrate sensors, actuators, RF subsystems, and power measurement. Include all required by the *device master record* (DMR).
- Seal calibration points; record calibration data with traceability to standards/equipment.
- Re-run full tests

15.6. PERFORM FUNCTIONAL, SAFETY, AND CYBERSECURITY TESTING

Conduct comprehensive functional, safety, and cybersecurity testing as required

15.7. COSMETIC REFURBISHMENT

Restore the device's external appearance:

- Clean and sanitize;
- replace worn bezels, keycaps, labels, and tamper seals.

After any change: re-test the symptom and run a quick functional check.

Front staff will need skills, attitudes that are non-technical. have background, and not usual electronics customer support. Geeks/techs are often brusque, not very good with connection, often male, often messy appearance, personal care, and social posture.

+trying to think of the ideal of my customer service interactions+

15.8. OUTCOMES

Continue with emotional support:

- Environment and atmosphere
- Staff skills and culture
- Empathy, and supportive communication. Other reassurance and emotional support. Be sincere
- Handling
- Communicate the status, and updates
- Preserve personalization
- Follow up

CHAPTER 5

Product Return and Refurbishment

This chapter discusses the process of returning a product and optional refurbishment for reuse by other customers.

- Return Product Workflow
- Management information
- Refurbishment of the product
- Support needed for returned product and refurbishment.

16. OVERVIEW

A customer may send an item back to the seller or manufacturer. This is often due to reasons such as the product being defective, not meeting expectations, having been ordered incorrectly, or for service. The product may be examined and, in some cases, refurbished or later reused.

return product analysis

Return Product Analysis (RPA) is the structured, documented investigation of devices or components that are sent back from the field. The goal is to:

- *Confirm* (or refute) the reported complaint
- *Identify failure mode(s)* and probable root cause(s)
- *Assess risk* and potential patient/user impact
- *Provide evidence* for regulatory vigilance, complaint handling and/or quality system records
- Drive changes to design & process (e.g. *corrective and preventive actions*), supplier actions, manufacturing improvements, and design / process changes
- Feed reliability metrics and product performance trending

Typical scope includes:

- Product returns where customers may not have used or minimally used the item and returned for unknown reasons
- Field returns due to suspected malfunction, damage, recall participation, service exchanges, or (in the case of medical devices) explants
- Hardware, software, accessories, disposables, and packaging as relevant

Refurbishment is the process of restoring a used or worn product, device, or component to a like-new condition through cleaning, repair, and replacement of necessary parts. This must be done without compromising safety or (in stringent product tiers) traceability.

This chapter has some similarities – *parallels* – with the process of sending a product in for repair.

17. RETURN PRODUCT HANDLING

There are different workflows for non-medical and medical products; and if they are safety vs non-safety related returns.

17.1. CORE WORKFLOW FOR NON-MEDICAL PRODUCTS

The typical process for handling returns of non-medical products is as follows:

1. Initiate a *return merchandise authorization* (RMA), prior to the customer sending the product back to the business. This form captures the customer information, and the reason for return. *Note: An RMA is often also used for sending a product in for service.*
2. Ship device**,** using original packaging where possible. The RMA number should be clearly visible on the outside of the box.
3. Intake and chain of custody: Record serial/lot, complaint ID, event description, dates, and condition-as-received.
4. Inspect visually, photograph, test functions, capture logs/firmware.
5. Functional testing: bench tests against specifications; environmental or stress testing if relevant.
6. Determine disposition: servicing/repair and refurbish vs. scrap. If tied to a complaint or investigation, capture evidence before any wipe.

17.2. CORE WORKFLOW FOR MEDICAL PRODUCTS

The typical process for handling returns of medical products includes the following key steps:

1. Initiate a *return merchandise authorization* (RMA). In addition to capturing the customer information, and the reason for return, as well as any biohazard declaration.
2. Intake and chain of custody: receive and quarantine, decontaminate if needed (biohazard); record serial/lot, complaint ID, event description, dates, and condition-as-received.
3. Safety and triage: prioritize safety-critical complaints and potential reportable events; ensure data preservation (logs, memory, seals).
4. Verification of the complaint: attempt to reproduce the reported symptom under controlled conditions.
5. Non-destructive evaluation: visual inspection, photograph, metrology, X-ray/CT, check electrical function, capture log/firmware.
6. Functional testing: bench tests against specifications; environmental or stress testing if relevant.

7. Destructive analysis (if needed): teardown, cross-sections, microscopy (optical/SEM), chemical/FTIR/EDS, metallurgy, fluid/contamination analysis
8. Root cause analysis: map failure mode to likely cause using methods like 5 Whys, fishbone, fault tree analysis (FTA); compare with FMEA; identify detection/escape points
9. Disposition and documentation: repair, scrap, quarantine for CAPA evidence; issue a formal analysis report stating whether the complaint was confirmed, the failure mode, probable cause, and recommendations
10. Feedback and closure: feed results to CAPA/design changes, supplier corrective action, labeling/IFU or training updates; update complaint file and trending.

17.3. SAFETY AND DECONTAMINATION

Perform essential safety steps and decontamination procedures for returned products, especially in medical products.

- (Esp. in medical context) Perform a biohazard assessment; decontaminate per SOP with appropriate PPE and validated methods.
- Electrically safe state: battery isolation as required; check for damage that makes further handling unsafe.
- Quarantine in a controlled area (ESD-safe, access-controlled).

Maintain chain-of-custody records throughout.

17.4. INSPECTION

Assess the condition of the product to document any visible issues and gather objective evidence for further evaluation and analysis.

17.4.1 Initial triage and documentation

The initial triage and documentation phase begins with a thorough assessment and recording of the product's as-received condition.

- Visual inspection, photography, and metrology of as-received condition. Note the accessories returned: cables, chargers, disposables
- Log serial/UDI, lot, and seal condition;
- Quick power-on check only if safe; Check health records, counters; note indicators, error codes, software/firmware versions.

17.4.2 Capture Evidence (before any erasure)

Preserve all the relevant technical details so that they are available for accurate root cause analysis, regulatory investigation, and to support any complaint or CAPA follow-up. Without preserving this information, it may be impossible to verify, reproduce, or fully understand the reported issue.

1. If associated with a complaint/CAPA or potential reportable event, harvest the diagnostic logs, crash dumps, configuration snapshots, and software versions.
2. Preserve data in a secure quality repository; avoid extracting or retaining cryptographic private keys. Document limitations if evidence cannot be obtained.

3. Read and record counters and health information before any wipe;
4. Preserve lifetime counters and service history in the DHR.
5. Decide refurbishment eligibility once analysis needs are met.

17.5. OUTCOMES

The outcomes of the analysis typically may be one of:

- *Confirmed device failure* with identified root cause and corrective actions
- *Use/environmental/handling issue* with mitigations (training, labeling)
- *No-fault-found* with recommendations for improved diagnostics or data capture
- *Indeterminate* due to damage or insufficient evidence, with follow-up actions

The process will typically produce reports and update metrics:

- Analysis report with confirmation status, failure mode, probable root cause, and risk assessment.
- Disposition the returned product based on inspection, determine if the product will be repackaged, serviced/repair, or scrapped.
- Update the trends: failure rates, top failure modes, use error rates, no-fault-found rate, time-to-root-cause, and actions initiated.

18. REFURBISHMENT

Returned products – esp. unused ones – may be refurbished for use by other consumers, depending on the type of product family, its age, hours of use, cosmetic condition, contamination status, and regulatory allowances in the destination market. The typical refurbishment workflow is:

- Preparation
- Inspect, and capture key information
- Replace replaceable parts, esp. single use elements
- Wipe logs, memory, other stuff
- Install software
- Update security credentials
- Run the checks/calibrations, -- more than the wimpy built in ones
- Record the change (serial/lot, who/when, test results),
- Dispose of the old part properly.

18.1. PREPARATION

Clean or decontaminate per procedures (SOP) with appropriate *personal protective equipment* (PPE) and validated methods.

Read and record counters and health flags. Use these to decide eligibility and which subassemblies to replace.

18.2. REPAIR/REPLACE TO BASELINE HARDWARE CONDITION

Repair/replace to baseline hardware condition:

- Replace the consumables – battery, fans, filters, etc.
- Replace worn or damaged parts – connectors, keypads, housings, batteries, sensors, storage modules.

Address known reliability upgrades and *engineering change orders* (ECOs); document all part numbers and revisions used.

18.3. DATA (AND KEY) SANITIZATION

If tied to a complaint or investigation, capture evidence before any wipe.

18.3.1 Expire or rotate cryptographic material

Erase or rotate cryptographic material and destroy *data-encryption keys* (DEKs) using so user data becomes unrecoverable. This includes:

- Device-issued operational certificates/keys,
- Bluetooth LE keys: *identify-resolving keys* (IRK), *connection signature resolving keys* (CSRK), *long-term keys* (LTK),
- TLS client keys,
- API tokens,
- VPN/IPsec material,
- Debug/auth tokens.
- Revoke old certificates in PKI, *certificate revocation list* (CRL), OCSP as applicable
- Erase secure elements/TPMs if appropriate

Note: Do not attempt to alter immutable hardware root keys/fuses. Only rotate operational keys above the root of trust.

Remove or sever server-side cloud/app registrations.

18.3.2 Wipe all data

Wipe all user/environmental data. Use erasure method based on design, such as multi-pass overwrite erase for memories without crypto (flash, EEPROM, NVRAM):

- Patient/user data, logs containing PHI/PII,
- Network configs: Wi-Fi SSIDs/PSKs,
- Bluetooth pairings,
- Phone/host associations,
- Clinical server URLs/tokens,
- Cloud account bindings.
- Application settings, localization, time/time zone,
- Change identifiers
- Alarms/profiles, clear *real-time clock* (RTC)
- Any residual files on removable media.

- Caches
- Internal & external flash, EEPROM, etc.

Do not reset the device lifetime/safety counters, safety-relevant monotonic logs.

18.3.3 Verify erase

Post-wipe check that:

- No user accounts remain,
- No known SSIDs/pairings,
- No tokens in keystores, and
- Data partitions are blank or re-encrypted with new keys.

18.4. INSTALL SOFTWARE

Reflash to the approved current released software (including firmware and bootloader) using validated tools.

Apply mandatory patches and cybersecurity updates per SBOM and patch policy.

18.5. RE-PROVISION FRESH OPERATIONAL CREDENTIALS

Re-provision with fresh operational credentials:

- Generate new device certs/keys using a *true random number generator* (TRNG)-backed process;
- Enroll to PKI or license server;
- Bind to correct product SKU/region.
- Configure default device profile and regulatory domain;
- Clear all optional features unless licensed.

18.6. CALIBRATION AND ALIGNMENT

Perform all required calibration and alignment procedures as specified for the device:

- Calibrate sensors, actuators, RF subsystems, and power measurement. Include all required by the *device master record* (DMR).
- Seal calibration points; record calibration data with traceability to standards/equipment.
- Re-run full tests

18.7. PERFORM FUNCTIONAL, SAFETY, AND CYBERSECURITY TESTING

Conduct comprehensive functional, safety, and cybersecurity testing as required:

- Run the complete functional test suite (DFT) including boundary and stress cases.
- Electrical safety and leakage tests; battery health, charge/discharge, and protection checks.
- RF/EMC self-tests, where applicable.

- Cybersecurity self-tests: secure boot, key presence/absence validation, FIPS module power-on self-tests if applicable, penetration smoke tests (e.g., ensure debug ports closed, default credentials disabled).
- Perform secure boot and integrity checks; ensure attestation passes and anti-tamper fuses/flags are in the expected state.

One tool for evaluating SBC security configuration is CHIPSEC. This can be used to check for misconfiguration, lower-layer features settings, other non-SBOM approved software items. It may be used during the manufacturing process as well.

18.8. FINAL STEPS

18.8.1 Cosmetic refurbishment and labeling

Restore the device's external appearance and required markings:

- Clean and sanitize;
- Replace worn bezels, keycaps, labels, and tamper seals.
- Apply “Refurbished” markings if required;
- Ensure UDI/serial visibility and accuracy.

18.8.2 Documentation and traceability

Update the *Device History Record* (DHR) with the parts replaced, firmware versions, calibrations, test results, cryptographic re-provisioning events, certificates’ serials, and wipe verification logs.

Link to the originating complaint/CAPA and the RMA record; record who performed each step and on what date, with tool versions.

Maintain evidence that PHI/PII was removed or anonymized, and that any necessary diagnostic data was preserved off-device.

18.8.3 Quality review and release

Independent QA review of the DHR packet; confirm conformance to DMR/SOPs.

Issue a Certificate of Refurbishment/Conformance; set warranty and service intervals per policy.

18.8.4 Packaging and distribution

Package with current IFU, power accessories, updated labeling/UDI if required by the destination market.

Record shipment, destination market controls, and any re-registration steps needed for connectivity services.

18.8.5 Post-refurb monitoring

Track early-life performance of refurbished units; feed metrics to reliability and CAPA.

Ensure revoked certificates/tokens remain invalid and old cloud associations stay severed.

19. REFERENCES & RESOURCES

CHIPSEC Platform Security Assessment Framework,
https://chipsec.github.io
https://github.com/chipsec/chipsec

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Part II

Personalization

Personalizing of a product:

- Settings, Preferences, and Feature Flags. How preferences, settings and are stored and loaded.
- First Use. Discusses setting up and using the product the first, as well as setting up for each session.

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CHAPTER 6

Personalization and Setup

This chapter describes the personalization of a product, such as its setting provides and the role of the first use of a product:

- Personalization thru settings (aka preferences) and entitlements
- Feature flags
- First use, when the product is first turned on and must prepare for subsequent uses.
- Onboarding, serving dual roles to initial settings, and introducing the features to the user.

20. PERSONALIZATION, CONFIGURATION AND SETTINGS

Personalization provides the ability to tailor the system to the user's settings and preferences. This includes the ability to change the system's behavior, and appearance.

Table 4: *Components of personalization*

Component	Description
Birthdays and Holiday	Tracks anniversaries, holidays, and the birthdays of the robot and human, to celebrate special days.
Entitlements	Entitlements allow the user to access certain features or services, based on achievements or purchases.
Robot Name	The name of the robot, or its designation is unique and gives the robot its identity.
Settings	Allows the user to customize the settings of the robot for their taste and preferences.

Special days – birthdays and holidays – are used to trigger little celebrations to make the product more engaging. With robots, these might be animations; they may be positive messages from customer care. Devices with cameras create special photo albums.

20.1. THE KINDS OF CONFIGURATION AND SETTINGS

There are many kinds of settings and configurations. They include:

- *Preferences and settings*. The user app may choose or choose not to make the settings available to be set by the user. Some of these settings are synchronized with a server by communication services.

- *Feature Flags*. These flags allow enabling and disabling granular application features. Typically, the features are a subset of those mentioned in [The Overview], as well as other developmental features.
- *Owner Entitlements.* An entitlement is a family of features or resources that the program or owner is allowed to use. The entitlement settings associated with the account (as opposed to the per-device settings). The signal-flow graph treats these as settings, delegating to communication services to handle the synchronization with a server.
- *Programmable configuration*. This is a configuration loaded from a file at start-up. The user isn't expected to change the configuration. These are something allowing the developer to tweak the set up as extension hardware is added or tuned.

20.2. FEATURE FLAGS

feature flags

Granular application features can be enabled and disabled using feature flags. Feature flags allow the code to be deployed and selectively enabled. As a software engineering practice, a feature is usually disabled because the feature is:

- Not yet fully developed, or
- Specific to a customer, or
- Mostly developed and being tested in some groups, or
- Only enabled when there is some error occurs or other functionality is not working intended, or
- A special/premium function sold at a cost or reward (like an entitlement).

The feature flags are implemented as *boolean* configuration variables, that are loaded from a file separate from the preference settings. In many cases, a feature flag may be set to override the default or build setting. The feature flags have extra information that tracks their default setting, allowing an overridden value to be reset.

20.3. SETTINGS

The settings that can be customized include:

- Display settings (brightness, contrast)
- Sound settings (volume, tone)
- Notification preferences (alerts, reminders)
- Theme customization (light/dark mode), eye colors

21. FIRST USE AND SETUP

first use

The *setup and first use steps* are related to the initial receiving of the product, setting it up, and learning to use it:

- How to initially unpack the product and set it up.
- Performing any mandatory software update and scheduling other software updates for later. *Note: don't make updates mandatory, let people have fun first with the new product/toy, and then have the product evolve.*

- An interactive walk to demonstrate features. This provides awareness of features to explore, and some training on the basic interactions, both of which serve as the foundation for discovering other features.

The *setup and first use steps* are setting up the product: *first use*

- Directing to install the mobile phone application and setting up account
- Pairing with phone app and setting up network access
- Giving the robot a name
- Performing any mandatory software update and scheduling other software updates for later. *Note: don't make updates mandatory, let people have fun first with the new product/toy, and then have the product evolve.*
- The first use interaction to demonstrate features. This provides awareness of features to explore, and some training on the basic interactions, both of which serve as the foundation for discovering the other features.

21.1. ONBOARDING

Onboarding is the process of introducing a new user to a product, ensuring they understand its features and functionalities. The goal is to help users feel comfortable and confident in using the product. This typically involves a series of guided steps designed to familiarize them with the interface and capabilities of the product.

The onboarding process can vary in complexity depending on the product, but it generally includes the following phases or state of onboarding:

1. *Introduction*: A warm welcome to the user, often accompanied by a brief overview of what to expect during the onboarding process.
2. *User Information Collection*: Gathering essential information from the user, such as their name, preferences, WiFi settings, and any necessary configurations.
3. *Account Setup*: Assisting users in creating or linking their accounts, which may involve verification steps or complex configurations.
4. *Set up connection to the robot*, setting up its WiFi and pairing the mobile device with the robot.
5. *Give a name to the robot* and other personalization.
6. *Feature Exploration*: Guiding users through the product's key features, highlighting functionalities that enhance their experience and engagement:
    a. Practice feeding;
    b. Stimulate pet with petting, talking to it, play music.
7. *Support and Resources*: Providing access to help resources, FAQs, and customer support to assist users as they navigate the product.

Onboarding has a business reason too: they want people to enjoy and keep the product, rather than return it. This helps ensure awareness of the features – esp. if the features tend not to be discovered.. no matter how much effort is put into making them easy to discover.

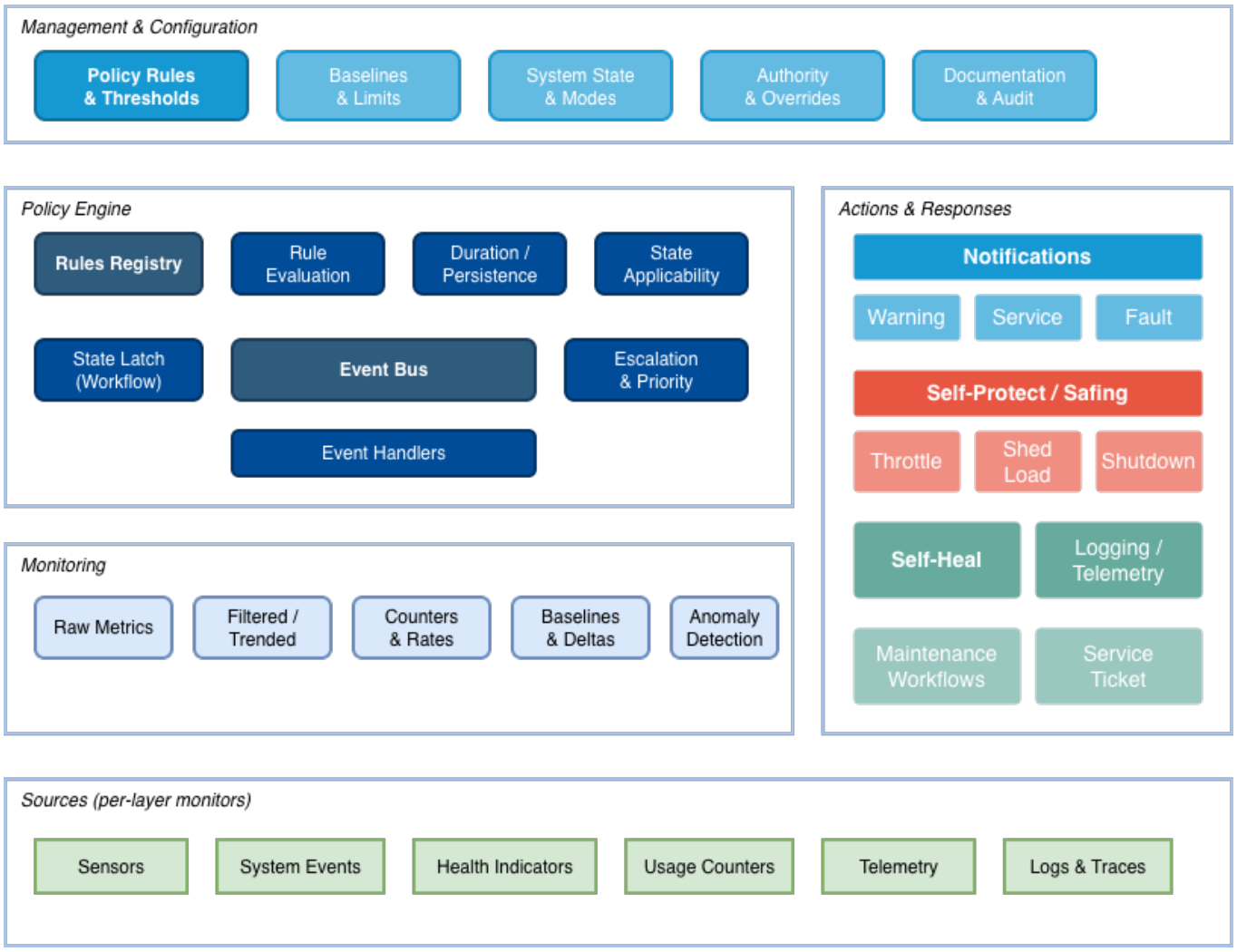
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Part III

Serviceability Architecture

This part describes architecture and processes to support servicing the product:

- Serviceability. A look at serviceability over the lifecycle of the product such as design features, and customer support.
- Monitoring. A detailed look at how a product is monitored and managed, including crash logs, etc.
- Policies. The framework for deciding how to respond to events and conditions.
- Error and Service Notifications. How error information and service notices are presented to a person.
- Maintenance Services. Support for maintaining a device, including management console, and remote management.


Management & Configuration
Policy Rules & Thresholds
Baselines & Limits
System State & Modes
Authority & Overrides
Documentation & Audit
Policy Engine
Rules Registry
Rule Evaluation
Duration / Persistence
State Applicability
State Latch (Workflow)
Event Bus
Escalation & Priority
Event Handlers
Actions & Responses
Notifications
Warning
Service
Fault
Self-Protect / Safing
Throttle
Shed Load
Shutdown
Self-Heal
Logging / Telemetry
Maintenance Workflows
Service Ticket
Monitoring
Raw Metrics
Filtered / Trended
Counters & Rates
Baselines & Deltas
Anomaly Detection
Sources (per-layer monitors)
Sensors
System Events
Health Indicators
Usage Counters
Telemetry
Logs & Traces

CHAPTER 7

Serviceability

This chapter discusses service interactions, design and culture to support, monitor the products function and health, to maintain it, repair it, and otherwise keep it loving life.

- Overview of customer care and the ability to service a product.
- The range of features and customer services available in products for serviceability
- Dividing up the features and services up by different product tiers

22. SERVICEABILITY

Design for serviceability

Products can be designed with features that facilitate serviceability, such as

- *Modular design*, with replaceable modules or components that simplify troubleshooting and replacement.
- *Health Tracking*: The usage, health, and environment of components.
- *Built-in diagnostics* such as self-test routines or diagnostics that run automatically or on demand to identify faults.
- *Trouble codes and logs* present information about the problems encountered and a record of information about the events leading up to the problem.
- *Visual Indicators* such as lights, status displays, or audible alarms to indicate faults and guide users.
- *Guided Troubleshooting,* such as interactive troubleshooting guides or apps that walk users through fault finding.
- *Remote Monitoring* allows service personnel to access diagnostic data remotely
- *Clear markings* for fasteners, connectors, lift points, etc. esp. for set up, preparation for use, and servicing.
- Fasteners and cable assemblies that are easy to undo and redo.
- *Captive fasteners* to prevent them from being lost when loosened, simplifying management of disassembled parts.

And design with qualities to ease service, such as:

- Avoid needing tools, or design to allow inexpensive, widely available tools for common maintenance such as attaching (and detaching) accessories, replacing consumables and field replaceable units.
- Make it accessible, and straightforward to nondestructively disassemble and reassemble esp. for these common maintenance and repairs.
- Focus on ease of access to those accessories, consumables, and commonly replace items. Should require removing few fasteners and other assemblies to get at these.

23. TIERING

To aid discussion, let's employ the tiered model for product serviceability. This will give some insight into the design and culture elements. The dimensions that we'll look at are:

- *Serviceable parts*: The parts that the consumer or technician can safely replace or maintain, and any authenticity checks tied to those parts.
- *Fault finding:* how issues or malfunctions are identified and diagnosed.
- *Tracking*: The usage, health, and environment data the device records (device-level vs per-part), plus how it's protected.
- *Diagnostics and logs*: Built-in tests, health checks, the depth of event logging.
- *Resetting Tracking Info*: What information can be reset, by whom, under what conditions, and how integrity (anti-rollback) is enforced.
- *Processes/resources*: The depth of procedures, training, documentation, tools, and governance supporting serviceability.

23.1. TIERS

In planning, one can pick a tiering level, select the desired (or relevant) features, and serviceability, skipping the rest. (Its senseless to overburden a value product with expenses.)

Tier 1: Low-Cost Consumer	
Serviceable parts	Limited to swapping simple consumables, e.g. a filter or battery; products are mostly disposable or return-for-replacement.
Fault-finding	Troubleshooting by manually checking power, connections, settings; rely on status lights/error codes; trial and error replacing simple consumables, and performing factory reset to see if it resolves the issues.
Tracking	Basic indicators (e.g., overheat, stall events), little or no usage tracking.
Diagnostics/logs	Minimal; simple status lights or trouble codes. Minimal on-device checks.
Resets	Might have full reset clearing user settings
Processes/resources	Quick-start manual; no formal service documents; minimal tooling.

Table 5: *Tier 1, Low-Cost Consumer*

Tier 2: Premium Consumer	
Serviceable parts	User-replaceable consumables & field replaceable units, such as fans, filters, basic sensors, and battery packs. Replacement parts, such as screws, flaps, etc.
Fault-finding	A service menu or companion app with quick self-tests, and guided workflows to isolate issues, and auto checks after replacing consumables/low-risk field replaceable units (FRUs). May capture error codes/screenshots for support. Trial and error replacing consumables, changing settings and performing factory reset to resolve the issues.
Tracking	Usage tracking and per-consumable change counters (e.g., filter changes). Used

Table 6: *Tier 2, Premium Consumer*

	with preventative maintenance. Environmental exposure: e.g. time above temp threshold.
Diagnostics/logs	Built-in diagnostics that run self-tests and report detailed error codes. Simple test routines after replacement. On-device service menu and rolling logs for tracking operational anomalies and identifying fault patterns.
Resets	Device usage tracking persists across user resets. Per-consumable counters reset on replacement. Users can clear non-critical logs (not lifetime counters).
Processes/resources	Better instruction guides with maintenance steps; guided troubleshooting steps and replacement workflow (via manuals, or companion apps). Basic web/app support.

Tier 3: Products for Contractor/Professional market	
Serviceable parts	Modular design with quick field replacement of complex components such as motors, pumps, sensors, storage modules, batteries, and cables.
Fault-finding	Built-in diagnostic modes, sensor readouts and test routines, and service logs. Automatic checks after replacing consumables/low-risk field replaceable units (FRUs). Follow guided workflows and checklists to isolate subsystems; use targeted measurements (multimeter, leak/flow tests, motor current vs RPM).
Tracking	Per-replaceable unit usage counters and environmental history: hours, cycles (e.g. engine starts), stress, error counts. Used with preventative maintenance and predictive failure. Environmental histograms for temperature, shock events.
Diagnostics/logs	Built-in diagnostic modes, sensor readouts and test routines (voltage rails, leak/flow, motor current) Targeted post-replacement self-tests (leak/flow, motor current vs RPM, flash bad-block scan). Detailed reporting, including timestamps and operational conditions leading to faults. Exportable service bundles. May have remote access capabilities to analyze fault data and provide solutions without physical inspection.
Resets	Per-part counters reset when a part with a new ID is installed. Device lifetime counters never reset. Audit log is append-only, track items replaced (and time) with id of new item.
Processes/resources	Instructions also include field-service playbook, with calibration steps, service tooling and checklists. Part authenticity checks (barcodes/NFC)

Table 7: *Tier 3, Contractor / Professional market*

Tier 4: Enterprise & Industrial	
Serviceability	Extensive field replaceable units. Modular design with quick and safe field replacement of complex components such as motors, pumps, sensors, storage modules, batteries, and cables. Redundancy, to handle failure while being serviced. Hot sparing. Failover. Many support large, autonomous facilities.
Fault-finding	Predictive maintenance and health models, automated resolution support, and remote diagnostic. Compatibility guards and change management.
Tracking	Full fleet remote monitoring (telemetry): usage hours, cycles, environmental stress, error-rate slopes. Health metrics: battery state of health, flash ECC rates and bad-block growth, motor current signature analysis, etc. These integrate with network management tools. Planned and condition-based maintenance.
Diagnostics/logs	Predictive maintenance models (thresholds + trend analysis). Rich logs with event classifications; remote retrieval and incident bundles for investigation.
Resets	Only reset per-part tracking after authenticated replacement. Lifetime and safety counters never reset. Anti-rollback protection for counters and software versions.
Processes/resources	Service procedures, training; spare-parts governance; service windows. Part authenticity checks (barcodes/NFC)

Table 8: *Tier 4, Enterprise & Industrial*

Tier 5: Regulated (e.g. medical/aerospace)	
Serviceability	Clearly delineated user-replaceable vs service-only parts based on risk analysis.
Tracking	Device lifetime and safety-relevant counters protected in secure memory. Per-part counters and health baselines with serial/lot traceability. Audit logs: when/what replaced (perhaps "who", depending on privacy), pre/post-replacement test results.
Diagnostics/logs	Validated self-tests and calibrations; evidence capture before any wipe. Attestation of software/firmware state.
Resets	Only per-part counters reset on verified replacement. Device lifetime, safety, and audit logs never reset. Cryptographic anti-rollback; signed state snapshots.
Processes/resources	Extended process with written service procedures, training, role-based access; may have spare-parts governance and service windows. Privacy controls, e.g. PHI excluded from telemetry

Table 9: *Tier 5, regulated, e.g. medical & aerospace.*

	Compliance: May be governed by standards and institutional procedures.

24. DESIGN CONSIDERATIONS

Design products for ease of setup, servicing, disassembly and reassembly. Serviceability enables maintenance, reduces downtime, and helps provide consistent operation over the products life. To enhance serviceability, include:

- Simple unpacking, intuitive installation, and streamlined setup processes
- Accessible and safe replacement of consumables,
- Straightforward cleaning procedures,
- Provisions for jam and fault recovery, and
- Practical, actionable troubleshooting guidance.

24.1. PHYSICAL ACCESS AND COMPONENT MOUNTING

For serviceable or replaceable components (for in-field or workshop maintenance):

- Arrange serviceable elements – such as heaters, cooling modules, sensors, and thermal fuses – so they can be accessed with minimal disassembly and using standard service tools.
- Position connectors, fasteners, and thermal interfaces to be easy to access during inspection, cleaning, and replacement tasks. The documentation should specify intervals for the reapplication of thermal interface materials (e.g., pads, pastes).
- Use durable, repeatable mounting methods for critical parts. Design for repeated removal and reattachment. Avoid using adhesives or permanent bonding unless strictly necessary.
- Important regions and connection points should be visibly marked and labeled directly on the product.

24.2. STRUCTURAL DAMAGE TOLERANT DESIGN

Damage-tolerant design limits the spread of a localized failure. A crack, impact, component fault, or assembly error should not automatically cascade into total product loss, safety hazard, or unrecoverable damage. The goal is to keep the product operational – or at least stable and diagnosable – long enough for maintenance, repair, or controlled shutdown. Suggested practices:

- Avoid single points of failure for safety-critical or hard-to-service functions. Where risk analysis justifies it, use redundancy, alternate paths, or graceful degradation rather than relying on one component or connection.
- Segregate subsystems so damage or failure in one area does not propagate. Examples include separate power domains, isolated hydraulic or fluid circuits, fire-rated compartments, and routing critical wiring away from high-stress or high-heat regions.
- Contain structural damage with crack-arresting features, overlapping layers, sacrificial elements, or replaceable wear items in structures subject to fatigue, vibration, or impact.

- Apply design margins and inspection intervals for wear-prone or safety-critical structures, aligned with expected service life and operating environment.
- Use modular assemblies so a failed module can be removed and replaced without disturbing adjacent healthy subsystems.
- Protect critical components with shields, bump stops, strain relief on cables and connectors, and guards against common field damage (drops, over-torque, mis-mating).
- Support detection and handoff to service – localized damage should be observable through diagnostics, containable, and repairable through accessible, labeled replaceable units.

Higher serviceability tiers generally invest more in redundancy, segregation, and post-failure operability (for example, limp modes, hot sparing, or failover) so that service can be scheduled rather than forced by catastrophic failure.

24.3. CONNECTORIZATION AND LABELING

Label all important regions directly on the product, preferably using silkscreen or other durable marking methods that ensure clear and lasting identification.

Clearly label all wiring, emphasizing high-voltage (mains) connections. Favor commonly available connectors.

Choose connectors designed to mate only one way, to prevent incorrect connections that could damage components, or short circuits. This can be achieved through two main approaches:

Polarization prevents improper connections, such as attempts to connect a plug 180 degrees off. This uses features like tabs, notches, or asymmetrical housing shapes that only allow mating in the correct orientation.

Keying is a slight modification to a connectors' physical structure, such as adding a pin or notch at a unique location; or blocking a pin. This ensures that each specific cable can be connected only to its corresponding port. This is used when multiple identical connectors are used in the same assembly, as with servers or automobiles.

24.4. DIAGNOSTICS AND SELF-TEST FUNCTIONS

Well-designed systems assist operators and technicians in quickly identifying and resolving issues – such as jams, calibration issues, or low consumable levels – while avoiding added disruption.

Integrate self-test and diagnostic capabilities into the product whenever possible. These features should support monitoring and reporting of parameters like element resistance, detection of open or shorted sensors, over-temperature conditions, and operational statistics (e.g. heater duty cycle, actuator run times). The automatic self-tests are performed for reasons:

- They verify that the risk control mechanisms are present and functional,
- To detect tampering and other unauthorized modifications
- Maintain data integrity,
- To ensure all features work as advertised, prevent unexpected failures during use,

- To identify gradual failures early and allowing repairs to be scheduled before failures occur
- To support regulatory compliance, as required by legal or industry standards, and provide documentation of system health and performance over time.

When abnormal conditions, failures, or trends are detected, the system should provide immediate notification or logging. Design diagnostic interfaces and logs to clearly indicate the source and nature of detected issues, using standardized codes or descriptions whenever possible. When practical, support remote retrieval of diagnostic data to support troubleshooting and service without needing on-site visits.

Include guided troubleshooting workflows that cover common sensor failure modes and troubleshooting procedures.

24.5. SELF-HEALING

self-healing

Self-healing extends these detection capabilities with the ability to automatically diagnose, and correct faults – such as software errors, driver failures, or data corruption. The techniques include:

- Design components so that if one fails (for example, a device driver crash), it does not compromise the entire system.
- The system detects failures and automatically restarts only the affected component or switches to an alternate, in a way that minimizing disruption.
- The system regularly saves checkpoints points, allowing it to revert to a known-good configuration if a new update or configuration change causes issues.
- Updates are first applied to a snapshot or copy of the folder. If the update fails, the system simply reverts to the previous snapshot or copy.
- The policy engines can monitor for anomalies and isolate or take components off-line before they cause damage or major system failures.

24.6. SELF-PROTECT SAFING ROUTINES

self-protect

Self-protect behavior safeguards equipment and users when operating limits are approached or exceeded. It complements diagnostics and self-healing, prioritizing safety and preservation over continued operation. Hardware self-protect – fuses, current limits, thermal cutoffs, power supervisors – should be considered for safety-critical paths.

safing routine

A *safing routine* is a sequence of autonomous actions equipment may take to protect itself when it detects a major problem. These routines transition the equipment to a safe, stable state – they may disable high-voltage power supplies, close valves, or shed load non-essential loads. In munitions, safing mechanisms prevent accidental detonation. These are emergency fallback plans designed to keep the equipment stable until an operator or service team can investigate. The priorities generally follow this order:

- Stop non-essential tasks, canceling or suspending non-critical operations to reduce load, heat, and further risk.
- Establish a safe operating posture – restore or maintain acceptable power, thermal, and mechanical states. For example, vent pressure, engage brakes, park actuators, or open cooling paths.

- Establish contact – alert the user, log the event, and notify remote monitoring or support systems when available.
- Wait for help, entering a quiescent or benign state, minimizing activity and staying in a known, stable configuration until reset, service, or new instructions arrive.

Typical triggers include overheating, over-current or short circuit, loss of control authority, failure of safety interlocks or critical sensors, repeated faults, power or thermal extremes, and loss of required communications.

Staged response should escalate before a hard shutdown, where safety allows:

1. Warn the operator and surface actionable fault information.
2. Degrade service: reduce throughput, limit actuator range, or switch to an alternate path or mode.
3. Disable automatic retries, log the event, and capture diagnostic context for service.
4. Execute self-protections, or shutdown of the affected subsystem when limits are exceeded or instability persists.

Record what triggered safing, the state at entry, and actions taken; expose standardized fault codes and recovery prerequisites; define clear service reset and inspection steps; and align depth of automation with product tier (a value product may simply shut off, while a professional or enterprise product may shed load, failover, and retain rich logs for remote analysis).

24.7. REMOTE AND APPLIANCE SYSTEM MANAGEMENT

When feasible, support remote retrieval of diagnostic data to assist with troubleshooting and maintenance without requiring on-site intervention. The helps customers and automates software updates (to embedded controllers, or firmware), "soft" hardware updates (e.g. FPGA, timing and other mechanical calibrations), and potential for enabling or updating new features and operational modes (e.g., a gentler wash cycle deployed remotely). Any remote console is recommended to be restricted focused on the task area, with security restrictions.

High-end tiers can apply advanced statistical methods (or machine learning) to identify usage trends, common failure modes, and detect emerging anomalies. This informs product improvements and allows for earlier response to potential issues.[4] Most individual units (at lower tiers) don't need statistical data analytics; but massive fleet sizes do, and the results driven from them can improve the results for others.

24.8. CUSTOMER SERVICE

"White glove" services – dispatching knowledgeable staff to engage directly with customers (which put sales guys in front of customer) – benefits from preventative maintenance and replacing consumables.

25. MANAGEMENT

The management responsibilities include:

- Ability to configure the system, such as elements, resources, and policies.

[4] It seems likely that automated software repair based on field information is possible and will come.

- Resource management: provisions system resources to serve the workload
- Monitors function, performance, and consumption of resources
- Handles life cycle events, including attaching and removing hardware, on-boarding, relocating, and off-boarding.
- Telemetry: tracks each tenant's use of your features and the performance of the system.
- Authentication and identity functions of the control plane include validating device identities, maintaining access control policies, tracking device registration and removal, and managing device authorization states.
- Scope: all layers of the abstraction stack and other parts of the system.

25.1. INTERFACE AND CONFIGURATION

Jeanine Adkisson

A well-designed management interface should prioritize clarity, reliability, and ease of use. This applies both the interactive portion and the long-term configuration (such as files or persistent settings). The following design criteria are particularly relevant:

- *Interpretability*: The system should accurately interpret valid input and reject invalid input.
- *Predictability:* Users should be able to anticipate how the system will interpret or process their input.
- *Discoverability*: Users should be able to learn how to specify new goals within the system's constraints.

25.2. MAINTENANCE

Logging records system events and errors. These logs assist with debugging as well as product returns and customer support. Logging helps track and document system issues, making it easier to report problems and facilitate troubleshooting.

26. REFERENCES & RESOURCES

Jeanine Adkisson, *How to make programming terrible for everyone*, https://jneen.ca/posts/2026-03-27-how-to-make-programming-terrible-for-everyone/

CHAPTER 8

Management Information

This chapter discusses tracking the health of equipment, logging, and remote monitoring of system health.

- Tracking equipment status, and their health indicators, used to prepare for use and to identify ailing parts.
- Logs as a chronological record of events
- Telemetry and Analytics, remote information to support customers, allowing product maintenance and providing better understanding of customers for improved products.

The next chapter will discuss presenting fault messages and service notifications to operators.

27. OVERVIEW

Management information typically includes the type, configuration, and current health status of the device. This information is typically exposed through the system report or diagnostics tools. It is a useful resource for customer support and field service engineers to troubleshoot emerging issues. A system's health can change from hardware degradation & malfunctions, evolving usage patterns, operator errors or inexperience, and security incidents or abuses.

observability

Observability is the overall ability to understand what is happening inside a system, to administer, improve it (e.g. debug) or justify its behaviour. It is the framework to understand the health. Patterns and recurring issues can be surfaced by monitoring errors and usage anomalies over time.

Problems are typically reported with alert and fault messages. To understand these problems, it takes knowing the context, sequence of events and activities leading up to each event. This is achieved by systematically collecting and analyzing a variety of signals, such as metrics, logs, and traces. Tracking usage errors, faults and anomalies can assist with identifying patterns and areas for improvement, enhanced stability and user experience.

monitoring

Monitoring is this process of continuously assessing a system to detect malfunctions, degraded performance, or failures. It is built upon collecting measurements and event counts, and evaluating them to to classify a subsystem as operating normally, impaired, faulty, etc.

anomaly detection

Anomaly detection is the process of identifying potential faults, misuse, or emerging failures. It is done by looking for data points, events, or patterns that do not conform to expected behavior – unusual measurements, outlier counters, or unexpected trends. Anomaly detection is used in predictive maintenance to prompt early investigation and intervention before larger issues arise.

28. MANAGEMENT INFORMATION

Management information – such as current conditions, and health indicators – is used to drive communication, preventative and triggered maintenance. *Health indicators* are abstractions that represent aging (and potential degradation), the ability to meet service expectations and potential for outages. Metrics should be tracked for the system, each subsystem, component, interconnection, and resource.

The health indicators are used to drive communication and guiding preventative and triggered maintenance actions. These messages and actions include:

- Warning messages
- Schedule service notifications
- Transition into safe or reduced-functionality modes if risks are detected
- Lockout if safety margins are exceeded.
- Fault messages alert when an item has failed or worn out, and describe the actions needed to bring it back to working order.

28.1. TYPES OF HEALTH INDICATORS

Health indicators can be organized according to their nature and how they change over time.

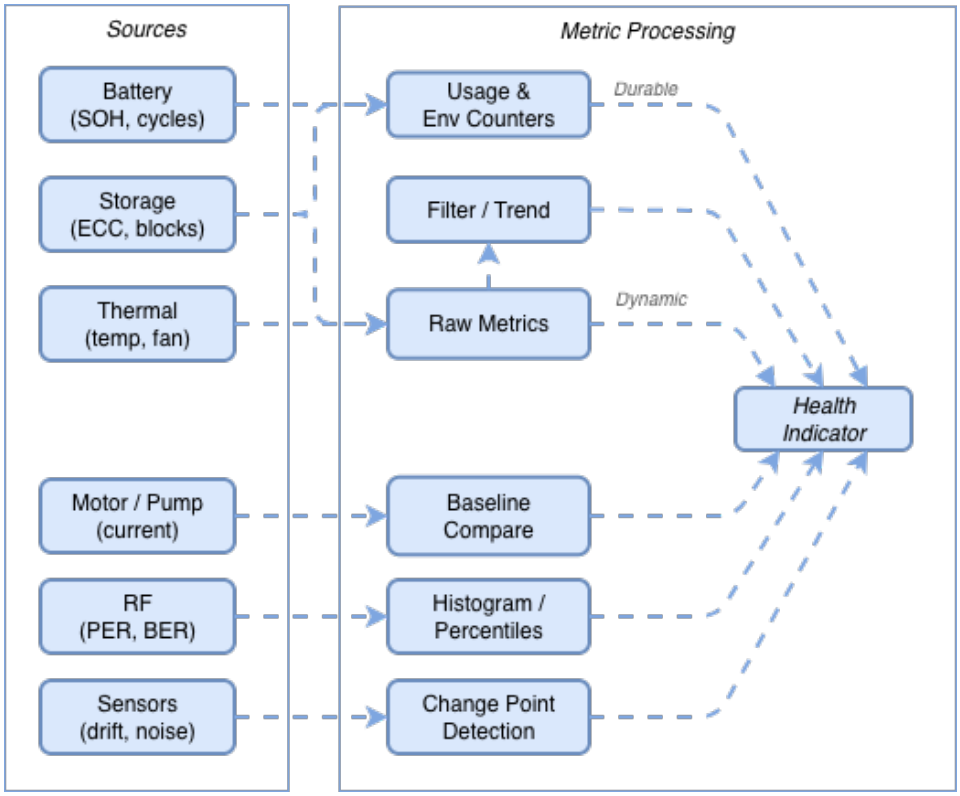


Figure 1: *Health indicators data flow*

DURABLE VALUES are persistent, slow-changing metrics, reflecting cumulative or long-term states of the system. These often increase (or sometimes decrease) steadily over equipment life and represent underlying wear or exposure. They retain their value across restarts.

- *Usage and environment counters* track total operational hours, accumulated cycle counts, total start/stop events, and accumulated environmental exposures. For

example, time above/below temperature thresholds, accumulated vibration, humidity exposure, shock events. They can be used to predict degradation and enable preventative maintenance.

- *Capacity* is relevant for resources such as batteries, storage remaining, or consumables. For example, if a resource is running low, it may need replacing; or if storage is filling up, added storage may be needed.
- *Cumulative errors* (including faults, resets, etc.) reveal ongoing or worsening problems, and point to deeper or chronic issues. Tracking them supports routine maintenance, and finding the root cause of persistent faults.

DYNAMIC VALUES are the live metrics of system health. These are often tended etc.

- *Errors*, or the count or rate of failures. This metric helps identify changes in behaviour, and potential system failures. Spikes or sustained elevation may signal new or worsening issues, shifts in usage, or systemic failures.
- *Utilization* is the percentage of time that the resource is actively in use. High, sustained utilization can indicate load issues or stress.
- *Latency* is the duration it takes to perform an operation. High or increasing latency may indicate degradation, or bottlenecks (esp. with changes in usage). Latency should be tracked separately for successful and failed operations.
- *Saturation* indicates how close a resource is to capacity – how much more it can handle before performance degrades. Monitoring saturation helps identify system bottlenecks. It is commonly measured by the length of a queue waiting on a resource. Persistent high or rising saturation is an early warning of performance degradation or outage.
- *Traffic* is the volume of requests or activity, representing the demand being placed on the system. This is used for capacity planning and anomaly detection.

OTHER groupings: A subsystem or component may report values about its health (e.g., "healthy/use within expected range" vs. "approaching service threshold"). These may be a composite derived from several measures. The exact may depend on the system architecture, risk model, or maintenance strategy.

28.2. CAPACITY, LOAD, AND INTEGRITY

At a system or product level, it is useful to consider three interrelated perspectives when interpreting status indicators and operational issues. Problems like reduced performance, restricted functionality, maintenance notifications, or slow response can arise from any of these, so they should be evaluated together:

Table 10: *Health related views*

View	Description
Capacity	Remaining useful life and headroom, such as the state of health or charge, storage or consumable life, hours or cycles to the next inspection, reserve margin on batteries or actuators.
Load	Stress from duty, such as over-temperature events, sustained high utilization or torque, cycle counts under work, charge or energy sessions, pressure or thermal peaks.
Integrity	Whether the product is still sound. Examples: trouble code count and severity, isolation or interlock trips, wear indicators,

	network or gateway health, calibration or readiness state.

The metrics are typically trended or captured in histograms (distribution patterns). The metrics, trends, and statistics are used in preventative maintenance by comparing the values against thresholds or identifying shifts in minimum/maximum values or histograms (distribution patterns) to indicate degradation.

28.3. INDICATORS OF POTENTIAL LOSS OF CONTROL AUTHORITY

control authority

Control authority describes the ability of a controller, actuator, or system to influence a process variable – such as temperature, pressure, or flow – over its full required range under all expected conditions. We want to watch that the components and overall system has sufficient capacity, and adequate actuator dynamic range, so that the system can consistently reach and maintain its setpoint, respond effectively to disturbances, and recover from transient events.

Loss of control authority can occur if system components degrade, external demands increase beyond the system's rated capability, or if faults prevent the system from acting as commanded. When control authority is lost, even a correctly tuned controller will be unable to restore the process variable to its target, leading to persistent errors and possible fault states.

The following indicators paint a picture of where and how control authority may be lost or compromised, empowering better diagnosis, trend detection, and preemptive mitigation.

Table 11: *Example of feedback controller health metrics*

Parameter	Description
Actuator travel / utilization ratio	Percentage of time/channel spent at (or near) actuation limits versus within active range. This indicates available margin. **Units:** %
Control loop delay / lag ramp time	The measured lag between command and effect (response time), indicating sluggishness or excessive deadtime. **Units:** Seconds
Control output saturation	The total duration or frequency controllers with actuators at min/max command (output). This signals limited control reserve. **Units:** Seconds, count
Cumulative off-nominal time	Total time the results (process variable) are outside allowable range (the target band). **Units:** Seconds
Dead zone / actuator constraint count	The number of times that a control output is clipped or constrained (e.g., output "stuck" at limit). **Units:** Count, percent of control cycles
Disturbance rejection error	Magnitude and duration of deviation from setpoint following a transient or known disturbance. **Units:** delta, second
Failure to meet	The number of instances and duration that the system is unable to reach or maintain

setpoint	targets within the acceptable setpoint/tolerance band. **Units:** count, seconds above tolerance
Frequency of control interventions chattering	Frequency of control output (e.g., pump speed, valve position, heater power) changes. High rates may indicate instability or chasing noise. **Units:** adjustments per second
Integral Windup Events	The duration or count of controller integral term being "maxed out." This indicates persistent error correction attempts. **Units:** Seconds, count
Out of range events	Incidents where process variable is outside of safe range or specified bounds, regardless of setpoint. **Units:** count
Overshoot / undershoot	The maximum deviation above/below setpoint during transitions or disturbances. **Units:** Δ°C, % deviation
Reserve margin reduction	The difference between available actuator capacity and required correction. This indicates the erosion of available control reserve. **Units:** %, Δ (e.g. % actuator travel remaining)
Sensor / reference discrepancy	The maximum or average difference between redundant sensors or reference readings for the same parameters. **Units:** Δ (e.g. °C, %, PSI)
Time-to-Settle (Process Recovery)	The duration required for the process variable to reach and stay within target band after a setpoint or disturbance. **Units:** Seconds

28.4. HARDWARE HEALTH INDICATORS

Analyzing hardware health metrics is easier to understand, as they are tangible and well-defined.

Hardware health indicators are derived from measurements of aging, usage, performance trends and failure counts. They include metrics such as rates of change (e.g., slopes of rising temperatures or error frequencies) and histograms (such as how often measurement values fall into specific bins, or the distribution of operational cycles).

Type of system	Typical health indicators
Battery	Capacity fade and internal resistance trend; cycle count and depth-of-discharge histogram; high-temp exposure time.
Fans, filters, thermals	Fan tach vs command slope changes; time at near-thermal-limit; clogged-filter inferred from delta-T.
Storage flash, EEPROM, hard drive	Bad-block and ECC rate slope; program/erase counts per block; refresh/scrub success; uncorrectable error flags.

Table 12: *Hardware health indicators by subsystem*

Motors, pumps	Rising current for same load, longer spin-up, speed ripple/FFT indicating bearing wear; pressure/flow calibration drift.
RF	Gradual Tx power drop vs calibration, rising *packet error* rate (PER) / *bit error rate* (BER) at fixed conditions; antenna path intermittency.
Sensors	Zero/gain drift vs reference, increasing noise/variance; saturation/railing occurrences; difference between complementary sensors

28.4.1 The Usage and Environment Counters

Usage and environmental counters provide quantitative measures of the duration and intensity that a subsystem has been used, as well as the environmental conditions it has experienced in its operational life. These counters can be used to assess the health of the subsystems, and to identify trends and patterns indicating emerging issues or inform maintenance schedules.

***Table 13:** Counter types*

Type	Examples
Time-based	A duration from some point (e.g. opening sealed package), or use. ▪ Power-on hours, active/therapy active/therapy hours, time-in-high-load mode, time above temperature thresholds.
Cycle/operation-based	These count the number of engine starts, flash erases, operation-specific pump strokes, valve actuations, connection/disconnection cycles, charge/discharge cycles, sterilization cycles.
Cumulative stress	Total energy throughput (battery mAh/Wh), motor revolutions, total flash erases, shock/vibration events, humidity/condensation exposure.
Quality-of-service	▪ Reboot/crash counts, watchdog resets, link dropouts, retransmission rates. ▪ Memory: CRCs, ECC scrub stats. ▪ Storage: bad-block growth, ECC correctable/uncorrectable error rates, wear-level counters. ▪ Watchdogs, brownouts, tamper, ESD event counters. ▪ Environmental exposure histograms: temperature bins, shock events.

The metrics for hardware health may be derived from device reporting, periodic diagnostic tests, and monitoring during operation. Many powered subsystems – such as batteries, chargers, storage devices, communication devices, and motor drivers – provide temperature & power measurements, and metrics tracking the environmental exposure. These metrics can be trended over time:

***Table 14:** Common health metrics*

Parameter	Description
Controller resets	Total number of times the device controller was reset for recovery. **Units:** count
Endurance remaining	Indication of estimated remaining usable life based on vendor algorithms. **Units:** percent (%)
Error correction	Total number of error correction actions performed.

count	**Units:** count
Interface error count	Total number of errors encountered on the host interface (e.g., SATA, PCIe). **Units:** count
Power cycles	Total number of times the device was turned on. **Units:** count
Power on hours	Total time the storage device has been powered on. **Units:** hours
Temperature (max)	Highest operating temperature measured by the device. **Units:** °C
Temperature (min)	Highest operating temperature measured by the device. **Units:** °C
Temperature excursions	Number of times the device has operated outside recommended temperature limits. **Units:** count
Time above critical temperature (Cumulative)	Total time spent operating above the device's recommended maximum temperature threshold. **Units:** seconds
Uncorrectable error count	Total number uncorrectable error events encountered. **Units:** count

Note: each kind of hardware will have many specific measures that are often abstracted into the health or used to further supplement it.

In high end tiers – usually in rugged environments – may track information around the connector (and button switches, etc.):

Parameter	Description
Insert cycles	Total number of times the connector was disconnected/connected. **Units:** count

Table 15: *Common connector metrics*

28.5. ABSTRACTING THE VALUES

Health indicator metrics should be continuously tracked, trended, and used to compute statistics like minimum, maximum, and histograms of observed values. These smooth noisy or fluctuating readings, making it easier to detect patterns and apply policies.

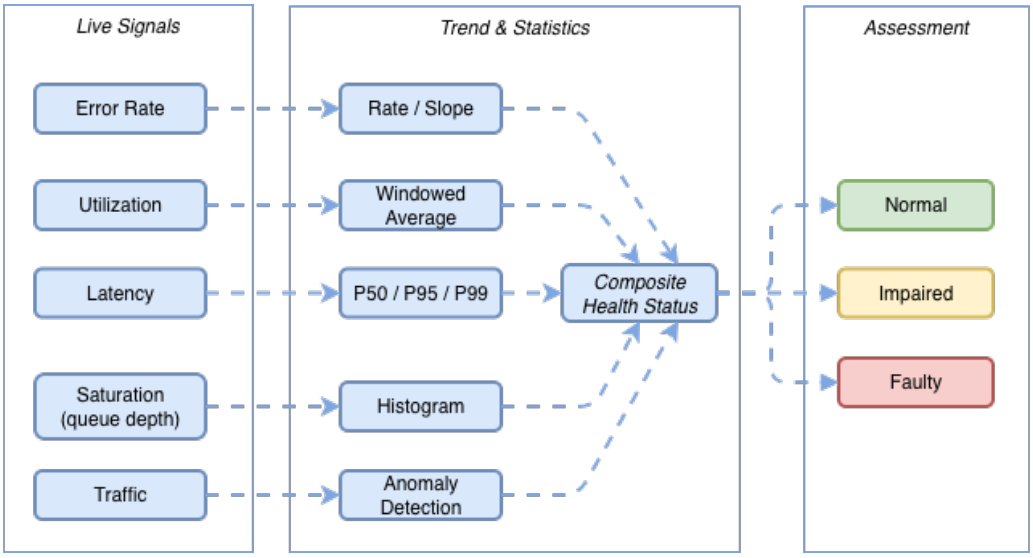


Figure 2: *Abstracting trends and into health status*

28.5.1 Trending and changes

Trending is the rolling history (windowed or summary statistics) and moving averages of the metrics. Trend data helps identify gradual degradation and is used by predictive maintenance, to report when replacement or service is likely needed.

Detecting deviations from a system's normal behavior enables early identification of faults or performance degradation. Shifts in key metrics can signal the onset of problems or deterioration processes, allowing for proactive alerts before hard limits are reached.

A *baseline* is a reference level of a metric, established under normal conditions. The current measures (or their trend) can be compared against a baseline to detect deviations (e.g., rising error rates or dropping signal quality). A baseline may be defined by

- The vendor performing a design characterization,
- In the field with tools that perform characterization tests to establish values, or
- Automatically, using characterization tests, under representative and healthy operating conditions.

The management interface should provide option to establish such a baseline. And should automatically establish baselines, where possible.

Change point detection automatically identifies significant shifts or jumps in monitored variables, possibly from a fault onset or change in operating condition. This is commonly used where the statistical properties (like mean, variance, or trend) change abruptly. This approach doesn't rely on establishing a baseline, making it less invasive, less manual and more dynamic.

28.5.2 Latency Bands – Response times

Not every metric is suitable for trending. With system response times (latency), percentiles are preferred. These better reflect user-perceived performance. It is possible for the system or a subsystem to not meet its goals, even though the individual hardware is in spec. Commonly used latency percentiles are:

Band	Description
50th Percentile P50, median	Half of all requests are completed at or below this response time. This is the response time for typical user experiences.
95th Percentile P95	95% of requests complete as quickly as or faster than this value. This provides a view of the performance experienced and *perceived* by almost all users, discounting occasional delays.
99th Percentile P99	All but the slowest 1% of requests fall within this band. This value draws attention to rare performance outliers and tail latencies.

Table 16: *Common latency percentiles*

Note: not all changes in performance are signs of wear or damage. Systems do slow down with usage and its health, unrelated to media change. This dimension is often due to the data structures, and other organization at the scale: classic Big-O concerns. This is separate from a derating for power or the media has changed, slowing operations.

28.6. POLICIES AND THRESHOLDS

Polices are rules that says what action to take, by whom, and under what conditions. They evaluate overall device health status and determine when to issue notifications (e.g., warning, scheduling maintenance) *Warnings* are issued for developing conditions that do not yet interrupt service but suggest an emerging issue. For example, if the rate of errors, retries or time outs increases above baseline over a defined period, a warning can be issued. Similarly, exceeding time limits for temperature or power outside of allowed ranges should also prompt warnings.

- A *service notification* is issued when a maintenance interval is reached; when an error or fault rate exceeds acceptable limits for a sustained period; or when a remaining usable capacity has fallen below a threshold; or when cumulative time outside temperature or power ranges exceeds policy limits.
- *A fault* is raised for serious issues – when component health has deteriorated to where continued operation could risk hardware damage, or safety hazard. For example, this may be triggered by persistent uncorrectable errors or recurring recovery failures.

Policies can take automated responses, by workloads, retiring the device, shutting down. When a threshold is crossed, the system should attempt a graceful degradation of service before forcing a shutdown or hard lockout – unless immediate stop is required for safety reasons. For instance, it should reduce throughput, lower the data rate, or switch to an alternate channel or path.

For these, the conditions of the policy look to see if a resource is over loaded or underutilized. This can be a combination of:

- Is the unit capable of the performance necessary to fulfill the intended use, or is the demand on the unit too high?
- Can load be shifted to another unit?

Is the unit degrading? The conditions are typically the health measure compared against a threshold. For example:

- Battery SOH < 80%,

- ECC uncorrectable count > 0,
- Bad-block growth > X/month,
- Fan unable to meet speed RPM,
- Pump current +Y%.

Use models where helpful to establish thresholds for hardware replacement:

- Arrhenius for temp-driven aging,
- Battery cycle life vs *depth of discharge* (DoD), vendor NAND wear specs.
- Calibrate what "normal" and expected/acceptable drift are

Some thresholds may be a relative change from an established baseline.

28.7. TRACKING EQUIPMENT STATUS

Better systems allow tracking the status of the equipment at a per replaceable unit/part level:

- The counters, trends and other health indicators
- Derived health and status. The resource might have an over status: OK, warn (X% life remaining), replace now, not usable or prohibited (e.g. expired/incompatible)
- The last verification test date & result; next due date or threshold.

These values should be made accessible to customer support, or on a service screen.

29. FAULT AND TROUBLE CODES

trouble codes

Fault and *trouble codes* are predefined identifiers that represent specific failures or issues encountered by the application. They serve as a standardized way to communicate the nature of the problem, easing the diagnosis and resolution of problems by support teams. Typically, each code is associated with a detailed description that explains the circumstances under which the error occurred and potential solutions. The code is often displayed with supplemental information which should be logged, and the documentation should describe how to interpret them. The use of the standardized code allows localization, and easy searching.

faults

Faults are typically "catastrophic" in that service is halted and a restart is required to recover. They are not expected to occur under normal circumstances. These may include hardware, and software faults, as well as other unexpected conditions.

30. LOGGING, A RECORD OF UNFORTUNATE EVENTS

Logging is a chronological record of events, actions, or changes within a system. A log is used to provide critical information after the fact, to trace activity for review or investigation. Or a log may be used to hold events if there is no user interface for fault notification. Logs may also be used to monitor system performance and provide insights into user interactions.

- A *fault lo*g records diagnostic information. It stores the malfunctions, or abnormal conditions detected during system startup or operation. It includes a way to refer to the type of problem and a timestamp. Alerts/faults would also be stored in the fault log.
- May include support for auditing, suitable to verify compliance.

- Tiers increase detail and depth of information, ability to bundle the logs and supporting information. All service reminders, warning, configuration changes and changes in operation state or status could have a corresponding log entry.

Avoid high-frequency raw logs of measurements; instead maintain histograms (e.g., temperature bins, DoD bins) of these.

This will now look at the different kinds of events and information to log, and the *longevity* of each. That is, how long it should be retained for, sometimes a minimum, sometimes a max.

30.1. FAULTS, WARNINGS & ALERTS

Faults, Warnings & Alerts	
Role	To understand the occurrence of alternate and exception workflows of the system.
Information to log	▪ Identifier for the type of issue, such as a fault code ▪ Identify the system element(s), e.g. by subsystem and part ▪ Is it a warning or fault? Can it continue with primary function albeit with inefficient/non-standard operation? ▪ The state (identifier) of the system at the time of the event, and resulting state ▪ The diagnostic – why did the event happen? ▪ It may include operating parameters or measured values to aid identifying and troubleshooting issues. e.g. battery level. ▪ In the case of incompatible versions, the version identifiers. ▪ The time that the event occurred, when the information was displayed, when it was closed (and it if was automatically closed)
Track acknowledgement	Recommended: track the time the information was closed/disabled, and if it was acknowledged.
Longevity	forever
Examples	▪ Sensor mismatch ▪ Sensors not calibrated ▪ Over temperature ▪ Out of specification ▪ Critically low battery, ▪ Detection of incompatibility with a system element ▪ Catalytic converter failure ▪ Unexpected fault

Table 17: *Faults, Warnings & Alerts*

30.2. CONFIGURATION / SETTINGS CHANGES

Configuration & Settings Changes	
Role	Provides context to understand the system, and settings

Table 18: *Configuration & Settings changes*

Information to log	▪ The configuration setting or feature flag that has been changed ▪ The value before and after the change was applied ▪ The status of change (e.g. unsuccessful) if applicable
Track acknowledgement	If not initiated by user
Longevity	e.g. xyz KB or 3 months, whichever is longer
Examples	Brightness set to 50%

30.3. EQUIPMENT CHANGES

Equipment Changes	
Role	Log every part change, insert/attachment/detachment of accessory or removable item; magazine/door events
Information to log	▪ Install/remove timestamps ▪ Identifiers (serial number, UID), bar code reads ▪ New items part type and number; manufacturing information such as lot, expiry. ▪ Authenticity status. ▪ Operator/user ID (if available), ▪ Reason for change (scheduled, fault, user-initiated). ▪ Verification test results.
Track acknowledgement	
Longevity	E.g. xyz KB or 3 months, whichever is longer
Examples	Replaced air filter Accessory attach

Table 19: *Equipment changes*

30.4. INTEROPERABILITY, DATA CONNECTION AND TRANSMISSION EVENTS

Interoperability, Data Connection and Transmission events	
Role	Aid understanding of how the software/device interacts with entities (excluding UI)
Information to log	▪ What it is connected to ▪ The transport connection, e.g. I^2C, BLE, TCP/IP, USB, etc. ▪ The nature and success or interruption of the relevant communication. E.g. timeouts, excessive error rates ▪ The version identifiers for each party ▪ Checking data and connections such as related to identity, incompatible versions,

Table 20: *Interoperability, data connection, and transmission events*

	integrity, and so on. ▪ Other associated information, values
Track acknowledgement	
Longevity	e.g. xyz KB or 3 months, whichever is longer
Examples	USB connected, or disconnected

30.5. OPERATIONAL STATUS

Operational Status	
Role	Provides context to understand the system and its states.
Information to log	▪ Change in status of a system element: battery, attachment, component ▪ The attempt, success, or failure to fulfill a User Request ▪ A change in the state or mode, including power mode
Track acknowledgement	Not necessary
Longevity	E.g. xyz KB or 3 months, whichever is longer
Examples	System startup Battery end of service Switched from peripheral to central mode Entering sleep mode

Table 21: *Operational status*

30.6. SECURITY EVENTS

Security events	
Role	To understand the security and related use, and support audits
Information to log	▪ Events such attempted, successful, or unsuccessful logon, logoff, security provisioning, request for elevated privileges ▪ Changes in security related configuration
Track acknowledgement	Recommended
Longevity	forever
Examples	Unsuccessful login attempt Administrator privileges granted

Table 22: *Security events*

30.7. SERVICE REMINDERS

Service Reminders	
Role	Provide early reminders to service unit, to prevent degraded operation or failure, such as when health indicators trigger a notification.
Information to log	▪ Identify the system element(s) to be serviced, e.g. by subsystem and part ▪ The service to apply to the part ▪ If a lock out was enforced ▪ Time reminder was initiated
Track acknowledgement	Track if user elected to continue without carrying out the recommended service; and time that the reminder was closed.
Longevity	E.g. xyz KB or 3 months, whichever is longer
Examples	Change air filter Change oil

Table 23: *Service reminders*

30.8. SOFTWARE UPDATES

Software Updates	
Role	Provide context to understand the system behaviors.
Information to log	▪ The attempt to update the software ▪ The version updating from, and version updating to ▪ The update status (whether update was successful or not)
Track acknowledgement	
Longevity	e.g. xyz KB or 3 months, whichever is longer
Examples	

Table 24: *Software updates*

30.9. USER REQUESTS

User Requests	
Role	Aid understanding how users interacted with the system
Information to log	▪ Request to boot, sleep, reset, turn off ▪ Navigate to screen, ▪ Select an option ▪ Change setting ▪ Choose action

Table 25: *User requests*

Track acknowledgement	n/a
Longevity	e.g. xyz KB or 3 months, whichever is longer
Examples	Go to a home screen Cancel

30.10. BEWARE OF LOG SPEW

Log spew is excessive, unhelpful, and/or noisy log entries. These hide important information in the logs, making it hard to process or analysis; it also fills storage. Signs of spew include:

- The entry is not directly or clearly categorized (e.g. per the logging guidance).
- The entry is too verbose
- Repetitive messages.
- The entry lacks relevancy. Log entries by programmers are often traces in disguise and not relevant to the operator or service team.
- The entry is hard to interpret its subject or meaning. Entries by programmers are often idiosyncratic or cryptic. Or lack consistent structure, naming, and style.
- It references the objects of the language: the variable, procedure or source code file name; chain of what calls what, etc.

31. OPERATIONAL TELEMETRY AND ANALYTICS

Russ Cox, 2023 Feb 8

"How do [people] understand which parts of their [system] are being used and whether they are performing as expected? The modern answer is *telemetry*, which means software sending data to answer those questions back to a collection server." *Telemetry* is this automated collection of data and transmission to a remote destination. It provides insights into system health, performance, user behavior, and function.

- At the professional & enterprise tiers, it is critical for operations. Traditionally applied "to big ticket items where unexpected breakdowns can cost a company tens or hundreds of thousands of dollars." With support subscriptions, this can put sales and support into contact early, with "white glove" treatment.
- At consumer tiers, remote monitoring is used for support and to understand usage, what the users do, the features & resources used, and what is not working. These usage logs are typically very narrow set of information, anonymized and aggregated.

By tracking usage errors and faults over time, developers and owners can identify patterns and areas for improvement.

remote monitoring

Remote monitoring allows administrators to oversee device status, activity, and service uptime on centralized dashboards. This is typically used at professional and enterprise tiers:

- Send periodic device summaries to remote services: hours, cycles, health deltas, and exceptions (e.g., first uncorrectable ECC).
- User-visible warnings for approaching limits, and alert when outside of bounds; remote notifications for fleet monitoring.

- Trending: per-model dashboards of sate of health (SOH), failure precursors, “no-fault-found” correlations, early-life vs wear-out phases.
- Synchronize part-change events and anonymized health summaries; keep local copies for servicing and return product analysis.

“New statistical techniques that allow researchers to distill useful information from mountains.”

In higher end tiers, the service management might allow for remote control, such as performing changes to configuration, software updates, and directly restarts.

31.1. OPERATIONS CENTER

An *operations center* is a centralized facility/team; “When a task is so big in scope it needs multiple teams to complete it; an operation center is needed to coordinate the efforts:

- “Bring in relevant information from all corners of organization
- “Analysis review info and make recommendations to leader
- “Leadership makes decision

Operations Center coordinates deployment out to individual teams to execute.

31.2. CRASH REPORTING

crash reporting

A *crash reporting* system automatically captures and transmits diagnostic information when a failure (such as a crash) occurs. This includes the logs to provide sufficient context to understand or explain the fault or crash, such as the events leading up to the fault/crash.

The logs that are automatically posted typically have reduced information (for security/privacy reasons) and are only sent if approved and there was a crash or user request to post the logs.

31.3. ANALYTICS

analytics

Analytics leverages the data collected via telemetry and crash reports to uncover trends,

- Enhance system reliability and availability
- Predict failures, reduce downtime and maintenance costs
- Improve user experience and satisfaction
- Improve performance,

In consumer products, the analytics allow the developers to understand customer use, product crashes, etc. In professional and enterprise tiers the owner may consolidate information from a site (or multiple sites) to use in managing its equipment.

Advanced analytics may apply machine learning models to identify anomalies or forecast future issues, identify root causes, prioritize fixes, and improve overall reliability.

31.4. SECURITY CONSIDERATIONS

What should not be logged?

- Don’t log secrets, such as keys, and passwords

- As always exclude or anonymize identifying information such as personally identifying information, geolocation, machines, MAC/GUIDs. This may include time stamps.
- Use hashed identifier, so it can be used to link structures within single report

32. OTHER

Document the source each measurement and counter and provide traceability. Document the meaning of each value, and how to use them appropriately. The management interface should provide access to these values.

33. REFERENCES & RESOURCES

Wikipedia, *Exponential Moving Averages*,
https://en.wikipedia.org/wiki/Moving_average#Exponential_moving_average

CHAPTER 9

Policy

This chapter looks at policy decision / response framework:

- Overview of policies, and their types
- How policy rules are defined and drive consistent, predictable actions.
- When to issue warnings, escalate to service notifications.
- Discussion on how rules are structured and expressed

34. OVERVIEW

Polices are high-level governing principles used to manage access, security, or broader organizational goals. They specify what action to take, by whom, and under what conditions. Prescriptive "if-then" *rules* are derived from these policies to automate evaluation of overall device health status and determine when to issue notifications (e.g., warning, or scheduling maintenance), initiate automated responses (such as retiring the device, shutting down), or other self-healing operations. Note: a policy system is not the same as a feedback control system; its aim is not to continuously regulate towards a target value.

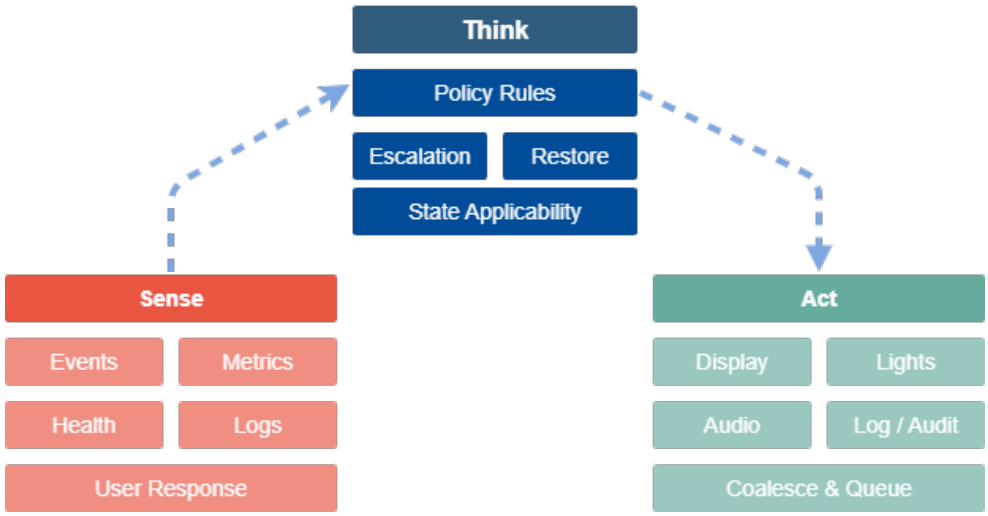


Figure 3: *The policy system integrates sensing and actions*

This discussion covers general concepts and structure of policy, and less a specific rule system. In practice, many specialized systems handle different domains – such as network addressing, email filters, system health monitoring, hardware interactions, and environmental management. There is little universal agreement on policy frameworks.

An example of a policy might be:

> *If an error occurs, first raise a warning for the operator, then prompt a guided cleaning, disable auto-retry after two failed attempts, and create a service ticket if the issue persists.*

34.1. TYPES OF POLICY

There are several types of policy rules, encompassing a variety of objectives, each designed to address specific needs. Common categories of policy rules include:

Table 26: *Common types of policy*

Policy Area	Description
Environment management	*Role*: Manage temperature, humidity, and air quality to ensure reliable equipment operation, protect components, and safeguard people. *Triggers*: Temperature, humidity, pressure; hardware over-temperature fault signals *Actions*: Issue warnings and alerts for out-of-range conditions; manage fans, heaters, coolers, adjust power levels, pause or resume charging and other operations.
Maintenance and service	*Role*: Facilitate maintenance and supports servicing. Manages schedule and triggered maintenance. Provides guided service workflows. *Triggers*: Upcoming maintenance requirements, specific service events, deteriorating health and other deviation from normal operating parameters. Actions: Reminders for scheduled maintenance, escalate alerts when error rates rise, initiate automated diagnostics.
Network communication	*Role*: Ensure reliable, secure flow of data and control across networks. Adapts to network health changes and service outages. *Triggers*: Network connectivity, packet error/delivery/rate thresholds, authentication failures, traffic patterns, protocol errors, service endpoint availability/unavailability, and detection of anomalous or suspect network behavior. *Actions*: Warn on transient failures; log and notify on sustained outages; escalate if critical services are impaired; adjust retries or switch to backup paths; restrict nonessential functions if bandwidth or security is at risk; start diagnostics and guide operator troubleshooting.
Power management	*Role*: Manages use of energy resources to make device available. *Triggers*: Power source, battery capacity, system load, thermal state, schedules, user activity & commands, environmental conditions (e.g., ambient light, temperature, power loss or restoration). *Actions*: Changing power source, throttling performance, reducing or shutting down features & unneeded components (e.g. backlight and display), or system (such as with critically low battery.)
Safety	*Role*: Prevent harm to people. Manages workflows for use. *Triggers*: Unsafe state, hazard detected (e.g., mechanical fault, failed self-test, overtemperature, obstruction, opened guard or interlock), operational state (such as setup), accessory attached or removed Actions: Issue notifications, perform workflow to enter a safe state (e.g., disengage actuators, isolate voltages, close guards, secure energy), halt operation, log, require explicit steps (e.g. acknowledgments and safety checks) before resuming operation.
Security	*Role*: Protect operation of equipment, sensitive data and maintaining regulatory compliance. Governs access control, data privacy, and protection against unauthorized actions or breaches. *Triggers*: Authentication failures, suspicious activity, unauthorized access, malware, abnormal data flows, tampering with system functions. *Actions*: Raising alerts, disabling certain functionalities (including lockdown), requiring multi-factor authentication.

Self-protect	*Role*: Safeguard hardware and software before and during failure. *Triggers*: Overheating, voltage out of range, excessive error rates. *Actions*: Throttling performance, reducing service levels, or shutting down components
User Experience	*Role*: Create positive user experience; ensure the system responds to user actions in intuitive or helpful ways. *Triggers*: Usability-impacting errors, user action is required. *Actions*: Provide notifications or guidance, alert users of critical issues or proactively warn about actions that could impact usability.

Other domains may also have policy rules tailored to address regulatory compliance, specialized operational constraints, or the behaviour and character of the product. In many cases, a single policy rule could span multiple categories, such as a rule that both protects data and ensures user safety when a critical fault is detected.

35. POLICY RULES

M Perry & M Bauer

"A policy defines or specifies a desired state – one that should be maintained or one that should be achieved… [It] is a set of conditions on one or more states associated with a particular system, with, perhaps, associated elements defining what constitutes 'the system', how those states are determined and how elements of the conditions are measured."

Use thresholds and policies so health indicators drive consistent service actions – warn, schedule service, restrict operation, or retire a device. Use separate trigger and restore thresholds with hold-off periods to avoid oscillation between warn and fault states.

35.1. KEY PARTS OF A POLICY RULE

A policy rule includes the following core elements:

Table 27: *Core parts of a policy rule.*

Part	Description
Trigger	The condition that initiates the rule. This might be a specific event from the subsystem or system, or it could be when a metric crosses a defined baseline (optional) and threshold.
Duration/Persistence	Specifies how long the metric must stay beyond the threshold (e.g., for a certain number of minutes or cycles) before the rule is triggered.
Action	The specific step or response to be taken when the trigger and persistence conditions are met. This may involve issuing a warning, notifying an operator, reducing service levels, disabling automated retries, initiating a shutdown, etc.
Escalation	Defines the next steps if the issue recurs or remains unresolved, such as raising the response level or initiating additional actions.

A threshold-based trigger – a conditions where the health measure is compared against a threshold – involves three components:

Part	Description
Metric	The parameter being monitored. For example, jam count, bit error rate (BER), or O_2 response time. It is often is the filtered, trended value.
Baseline value *optional*	A reference value or typical range for a metric under normal or healthy operating conditions, used as a comparison point. This value is typically determined empirically or set using historical data; and is used to detect deviations.
Threshold	The value or boundary that the metric must reach or exceed to activate the rule.

Table 28: *Comparing a metric against a threshold*

hysteresis

Threshold-based policies often require more nuanced handling than simply acting whenever a value crosses a set point. Policies may define separate conditions for triggering an action and for restoring normal operation. For example, if temperature rises above 65°C, the system might halt to protect itself, but service will be allowed to resume only once the temperature has dropped below 60°C and has stayed there for at least two minutes. This approach prevents rapid toggling between states (oscillation), giving a more predictable, stable behaviour.

The activation of policy rules often depend on the current system state or operational mode. Some rules may only apply during specific conditions, such as when the device is running on battery power. For example, certain power-saving policies might activate only when the device is operating on battery and will automatically be disabled once mains power is restored. Likewise, rules related to safety or security might only apply during active operation, system startup, maintenance windows, or when particular subsystems are engaged or offline.

The system may define and track major states – such as standby, active operation, charging, or fault modes – and policy applicability will be determined based on these overarching states or sub-states. This approach ensures policies are context-sensitive and respond appropriately to the system's operational environment and user needs.

35.2. ESCALATION

Policies should apply a staged escalation process before enforcing a shutdown or hard lockout, whenever possible – except when an immediate stop is necessary (such as for safety reasons).

1. First, raise a warning to the operator, possibly providing guidance or recommended service actions. This may include a link to prompted, guided service.
2. If the issue continues, gradually degrade service: reduce throughput, lower data rates, or switch to an alternate channel, mode, or path.
3. If the problem persists, disable automatic retries or generate a service ticket for required intervention.
4. If previous steps fail to restore normal operation, escalate to a shutdown or a lockout of the affected subsystem.

35.3. WARNING AND NOTIFICATIONS

Warnings are issued for developing conditions that do not yet interrupt service but suggest an emerging issue. For example, if the rate of errors, retries or time outs increases above baseline

over a defined period, a warning can be issued. Similarly, exceeding time limits for temperature or power outside of allowed ranges should also prompt warnings.

- A *service notification* is issued when a maintenance interval is reached; when an error or fault rate exceeds acceptable limits for a sustained period; or when a remaining usable capacity has fallen below a threshold; or when cumulative time outside temperature or power ranges exceeds policy limits.
- *A fault* is raised for serious issues – when component health has deteriorated to where continued operation could risk hardware damage, or safety hazard. For example, this may be triggered by persistent uncorrectable errors or recurring recovery failures.

Issue a *warning* for issues is not yet severe enough to risk damaging the equipment; and to schedule service:

- Small but increasing mis-pick or jam count,
- Inventory reconciliation errors that rise above normal variance,
- Increased drive load/unload failures for a given medium type,
- Increasing filament feed stalls/mis-feeds, rising material runout events, or elevated extruder clog precursor counts. (3D printers)

Escalate to a *service notification* when health problems become significant:

- Pick/place success rate drops below a policy baseline,
- Recurring jams in the same slot region or repeatable pick misses indicate a mechanical/calibration issue,
- Cleaning overdue and correlates with optical read errors or tape path issues,
- Repeated extruder clogs/nozzle blockage events or persistent filament routing/mount mismatches. (3D printers)

A fault is raised for serious issues – when component health has deteriorated to where continued operation is a risk. This includes situations such as:

- Repeated jam/breakage events indicating mechanical failure or unsafe operation,
- Persistent inventory identity failures that risk mounting the wrong medium,
- Persistent clogs or repeated failed material switches that prevent safe operation or risk wasted/unsafe thermal cycles. (3D printers)

Include context in the notification. The metric(s) that triggered the alert, the observed trend (slope/rate), timestamps, and any recent maintenance events (firmware update, scrubs, drive swaps) so the field team can reproduce and isolate the issue.

35.4. SMALL, LIMITED NUMBER OF RULES

Many products do not support adding custom policies, relying on a core set of built-in rules. User control is generally limited to threshold adjustments, or the ability to enable/disable certain rules. There isn't an inherent benefit from increased rules or flexibility of policy construction; these can harm system effectiveness.

Policy systems have a problem. Committees and less experienced operators reflexively add more alerts. This creates "alert fatigue," with operators delay responding or ignore them. When everything is labeled as critical, nothing stands out as truly important.

35.5. ASSESSING RULE EFFECTIVENESS

To verify that the rules work as intended requires confirming that each rule triggers under the correct conditions and behaves as expected. This process involves both static checks and simulation or observation during operation. There isn't a clear set of industry practices.

35.5.1 Static Testing

Check for obvious mismatches or invalid rule definitions. This may involve checking for

- Identifying references to events, actions, or states that do not exist in the system,
- Ensuring all referenced actions can be performed by the system,
- Verifying that requested state transitions are valid.

These checks catch configuration or logic errors before deployment.

35.5.2 Dynamic testing

Dynamic testing involves simulating or inducing system events and metric changes to observe the rules engine response. This is typically a combination of automated validation and manual testing. Note: We'll assume that the event generation and the actions are tested elsewhere.

The rule engine will need to make the following observable:

- Track the number of times a rule was invoked, per state
- The counts of events and aggregated metrics, including trends over time.

Conduct functional tests to verify effectiveness:

- Stress tests, the natural workflow scenarios, and deliberate triggering of edge cases, watching how rules activate and whether actions match expectations,
- Identifying rules that never fire (potentially obsolete or misconfigured) and system states that are never entered.
- Analyze for conflicting rules, and runaway feedback loops, where one rule's actions might unintentionally keep another rule triggered or escalate undesirably.

Observe the system during real-world operation as well ("natural experiments") to catch unanticipated issues.

35.6. CONSIDERATIONS IN A POLICY SYSTEM

Assess policy system effectiveness and usability using these core criteria:

- *Predictability:* Are the policy rules and metrics understandable to users? Can they clearly grasp what each rule means, how the system will interpret events & conditions, and its response?
- *Discoverability:* Are users able to identify their goals and express them within the policy system? Is the process of working with policies intuitive?
- *Interpretability.* When configuration files or policy rules contain mistakes, does the system provide clear feedback and reject invalid input, or does it risk misinterpreting the rules?

I recommend the process of crafting policy rules begin with writing in descriptive, qualitative language that conveys the underlying intent – these are easier for people to review and evaluate.

Once the intent is clear, these rules can be translated into precise, quantitative criteria with necessary technical parameters. These forms are complex, hard to understand and review. But an objective, actionable format is essential for implementation and verification.

35.7. SELF-PROTECTION POLICIES

Self-protection policies apply protective actions to maintain overall service continuity and minimize risk to hardware, data, or safety. They are typically crafted to minimize user impact wherever possible. They typically:

1. Monitor critical measurements – such as temperature, current, or utilization – and enforce limits to settings, reduce activity, or disable operations when operating thresholds are approached or exceeded.
2. Isolate failed or degraded components to prevent propagation of faults. For example, shutting down a failing drive, switching to backup units, or disabling features.
3. When persistent errors or instability is detected, attempt automated recovery by restarting the system or subsystem, downgrading (or reverting) software versions, or performing other recovery actions.

35.8. FEEDBACK CONTROL SYSTEM POLICIES

Control authority policies ensures that feedback control is both effective and appropriately bounded, preventing overreach or unsafe operation. These policies:

- Define and enforce the upper and lower operating bounds for actuators and outputs. These limits are based on safety margins, physical properties, and regulatory or design specifications. For example, a heater should not raise temperature above a safe maximum, or a motor should not be driven past its rated speed. These have an overlap with the policies related to self-protection; those are covered above.
- Specify constraints on the rate and magnitude of control adjustments. These prevent instability or overreaction. This includes ramp limits, maximum correction per cycle.
- Detect and address when the control system no longer has effective authority – such as actuator faults, loss of feedback/control signal, or when the process variable is unresponsive to control efforts. These policies should mandate clear notification and, if appropriate, trigger fallback or safe-state behaviors.
- Specify the conditions that automated control is permitted to intervene, and when authority must be escalated to an operator. For example, routine adjustments may be automatic, but critical overrides (e.g. emergency shutdowns) may require human confirmation.
- Define rules for transferring control authority between subsystems or operators, ensuring seamless and predictable transitions. For example, from automatic to manual control during maintenance, or when system health is degraded.

Policy Example:

- Limit setpoint changes to within defined bounds unless a credentialed operator has overridden.
- Restrict actuator adjustments so that no more than 10% of the total range may be altered per minute except in the case of emergency override.

- If the controller issues maximum output for N consecutive cycles and the process variable has not moved closer to target, raise an alert for loss of authority.
- When feedback loss is detected (e.g., sensor failure, communications timeout), hold the last command, notify an operator, and transition to a preconfigured safe state if the feedback signal is not restored within a specified grace period.
- Require explicit logging and audit trail whenever control authority changes, overrides or escalations occur. This is used in audits to ensure policies remain effective and are not routinely circumvented.
- If a subsystem manually disabled by an operator is re-enabled, require confirmation that pre-conditions (safety, calibration, readiness) are met before restoring automatic authority.

These policies should be adjustable, such as modifiable parameters and limits. Process dynamics and operational risks evolve over time. They should be adjusted as systems age, are reconfigured, or as new hazards are identified.

35.9. SERVICEABILITY POLICIES

Serviceability policies ensure systems remain maintainable and supportable throughout their lifecycle. They guide behavior around diagnostics, maintenance, and field service activities:

- Track wear, usage hours, self-test results, and scheduled maintenance intervals. Automatically notify operators or service teams when maintenance actions – such as cleaning, component replacement, or calibration – are due or overdue.
- Record all maintenance actions, including what was performed, when, and by whom. This creates an audit trail for issue investigation, compliance, and continuous improvement.
- Ensuring that policy-driven actions do not impede safe and convenient access to field-replaceable units or service points (e.g., keeping locks disengaged when service is required, providing step-by-step instructions).

35.10. PERFORMANCE MANGEMENT POLICIES

Performance management policies focus on resources that are overloaded, underutilized, or otherwise unable to meet operational requirements:

- Is the unit capable of the performance necessary to fulfill the intended use, or is the demand exceeding its capacity, thus creating operational risk?
- Can workloads be shifted to alternate units to balance demand and prevent overload?

Monitoring systems assess multiple, complementary metrics to form a comprehensive view of resource performance:

- *Utilization*: The percentage of time a processor or resource is actively handling work; this expresses how fully the resource is being used. High, sustained utilization can indicate load issues or stress.
- *Queue Length* (or saturation): The size of the work queue, representing the number of tasks waiting to be processed or use the resource. This shows if tasks are accumulating, indicating that processing can't keep pace with incoming work.

- *Throughput Rates*: The rate of operations or data transfer (e.g., read/write operations per second, network throughput).
- *Performance Trends*: Monitoring for trends such as increasing queue length, declining throughput, or consistently high utilization signals potential bottlenecks.

Some policies use simple threshold-based rules. While easy to implement, these provide a coarse view and don't always reflect whether a resource is truly over capacity. These should be used sparingly, when a better automated response is not available, and it is something the operator can address. They are best supplemented with additional context. For example:

- Trigger an alert if the read rate > 500 operations/sec for 5 consecutive cycles
- Trigger an alert if write rate > 200 operations/sec for 5 consecutive cycles
- Trigger an alert if download data rate > 10 MB/s for 5 consecutive cycles
- Notify if total uploaded > 1 GB within last hour

35.10.1 Assessing Processing Load and Saturation

Assess if a system is keeping up with its workload by monitoring the load, e.g. damped trends of the work queue length, and utilization of the processing resources. Performance analysis is usually iterative: addressing one limiting factor may reveal a new bottleneck elsewhere.

Watch for bottlenecks or overload due to rising demand (e.g. increasing queue length), and signs of declining throughput (which may indicate resource degradation). If an overload is detected, take redistribute the load, upgrade hardware, and/or tune the software. Overload conditions may result from increased demand, system degradation, or a combination of both.

Gather context information and cross-check multiple indicators. Consider confounding factors such as dynamic thermal limits that may be constraining processing speed or throughput:

- Temperature measurements (to catch thermal throttling)
- Power consumption
- CPU core and thread statistics

35.11. USER EXPERIENCE POLICIES

User experience policies focus on creating a positive experience. They should ensure that the system responds to user actions in intuitive and helpful ways. These behaviours should positive, predictable, making interactions as effortless as possible.

Responses to events, errors, and system state should be communicated clearly through meaningful notifications, status displays, and logs.

If notifications are needed, the use of alerts, warnings, and prompts should be limited to only those that *require* user awareness or intervention.

When user action is required, the notification should provide specific guidance, indicate the root cause (where possible), recommended next steps, and relevant context (e.g. recent changes or operations).

35.12. NETWORK POLICIES

Network policies provide a framework for governing how devices, users, and data interact over a network. They should set clear rules for connectivity, usage, and response to both normal and exceptional network events. The common topic areas are:

Table 29: *Network policy topic areas*

Role of Policy	Description
Access controls	Defines who or what can access specific network resources, segments, or services. These are typically enforced through authentication, authorization, permissions, firewall rules, and segmentation.
Security enforcement	Focuses on the detection of abnormal or unauthorized activity, isolation of compromised devices, enforcement of encryption, and monitoring for policy violations. This can include restrictions on protocols, websites or remote networks.
Network routing and configuration management	Covers management of subnet routing rules, exit node configurations, and other network topology adjustments.
Traffic management	Describes how traffic is prioritized, rate-limited, routed, paced, or shaped. These tune performance, maintain reliability, and ensure critical operations are not disrupted by congestion or malicious behavior.
Service resiliency	Aims to maintain connectivity and protect service integrity. Automated responses to failures include rerouting traffic or enabling backup connections.
Compliance and auditing	Logging and audit trails that demonstrate adherence to regulatory requirements, internal standards, and best practices for network operation and data protection.

35.13. AUTHORIZATION POLICIES

Authority policies clarify decision-making boundaries, specifying who or what entities are allowed to enact changes, override automated behavior, or perform specific actions within the system. They safeguard against accidental or malicious disruption that could compromise safety, reliability, security, or intended operation.

These should define who or what (e.g. subsystems, or automated routines) can access specific resources, or may initiate actions (such as system resets, configuration changes, overrides, or escalations). These are typically enforced through authentication, authorization, permissions, etc.

Certain actions may require explicit escalation, or multi-factor authentication. Examples: disabling safety interlocks, unlocking critical settings, or bypassing self-protection mechanisms.

Authority and permissions may need to adapt in certain operational states (for example, maintenance mode vs. normal operation), and policies should specify how transitions between modes are handled and secured.

All actions taken under elevated authority should be logged, later investigation and audits.

If a policy allows some to override of automated protections or warnings, it should define:

a. under what conditions the override is permitted;

b. what mitigations are enforced (such as time limits, temporary escalation, or required reset after manual action); and

c. explicit confirmation prompts or justifications.

36. MANAGING AND IMPROVING POLICY

Policy management ensures that system behavior meets organizational goals, regulatory obligations, and allows for ongoing operational improvement.

- Automated Monitoring: All critical metrics should be instrumented with real-time telemetry and regularly compared to configured thresholds.
- Alerting and Escalation: Breaches should cause automated alerts, with escalation paths depending on severity and affected components.
- Automated Safeguards: Some policies can trigger automated corrective actions (e.g., scaling, process restart) before manual intervention is required.
- Review and Adjustment: Policies and thresholds must be reviewed regularly and adjusted based on historical performance data, deployment environments, business priorities, and evolving risk assessments.
- Policy Documentation. All effective policies and their thresholds should be clearly documented and made accessible for operations, development, and audit purposes. Documentation must include:

Documenting policies alongside health indicators facilitates smooth handoff between teams, and supports compliance in regulated environments.

37. REFERENCES & RESOURCES

Perry, M., & Bauer, M. (2004). “Policies, rules and their engines: What do they mean for SLAs?” In M.Gh. Negoita, R.J. Howlett, & R.L. Jensen (Eds.), *Knowledge-Based Intelligent Information and Engineering Systems (KES 2004), Lecture Notes in Artificial Intelligence* (LNAI 3213, pp. 1164–1170). Springer-Verlag Berlin Heidelberg.

CHAPTER 10

Error and Service Notifications

This chapter is about v*isual indicators* such as lights, status displays, or audible alarms to indicate faults, service notifications, and to guide users:

- Interactions to provide notifications, warnings and errors
- Service notifications
- Error messages
- Crash reporting

Note this about the messaging, not too much about alarm systems for safety, which are also an interesting topic.

38. INTERACTIONS

A key feature is that the application report problems (aka faults and errors) and may present information about needed service. The display of information may be only status lights, at the lowest tiers, progressing to more advanced interactions. Higher tiers may include more sophisticated display and controls. The interaction may also be logged, acknowledging that the user was presented with the error, and how it was handled.

error

An *error* is a communication of something wrong that a person can fix. The interface should provide a graceful way to recover from these:

- Communicate the abstract information and relevant details to put the information into context, a unique identifier (e.g. a code that can be clearly articulated over a telephone call). For instance, it might communicate a needed removable attachment is needed,
- Provide options for the user. One option may be for the user to attach an item.
- If it is possible to detect that the user took actions on the issue, use this as a signal to clear communication, provide new communication, and/or resume, as appropriate.

Note: The term *error* has another technical use at low level, where it is always as a qualifier: ECC *error* rate. That use is only for standard, pre-existing phrases.

faults

Faults are “catastrophic”: service is halted and the system must be restarted. They are not expected to occur under normal circumstances. These may include hardware and software faults, as well as other unexpected conditions.

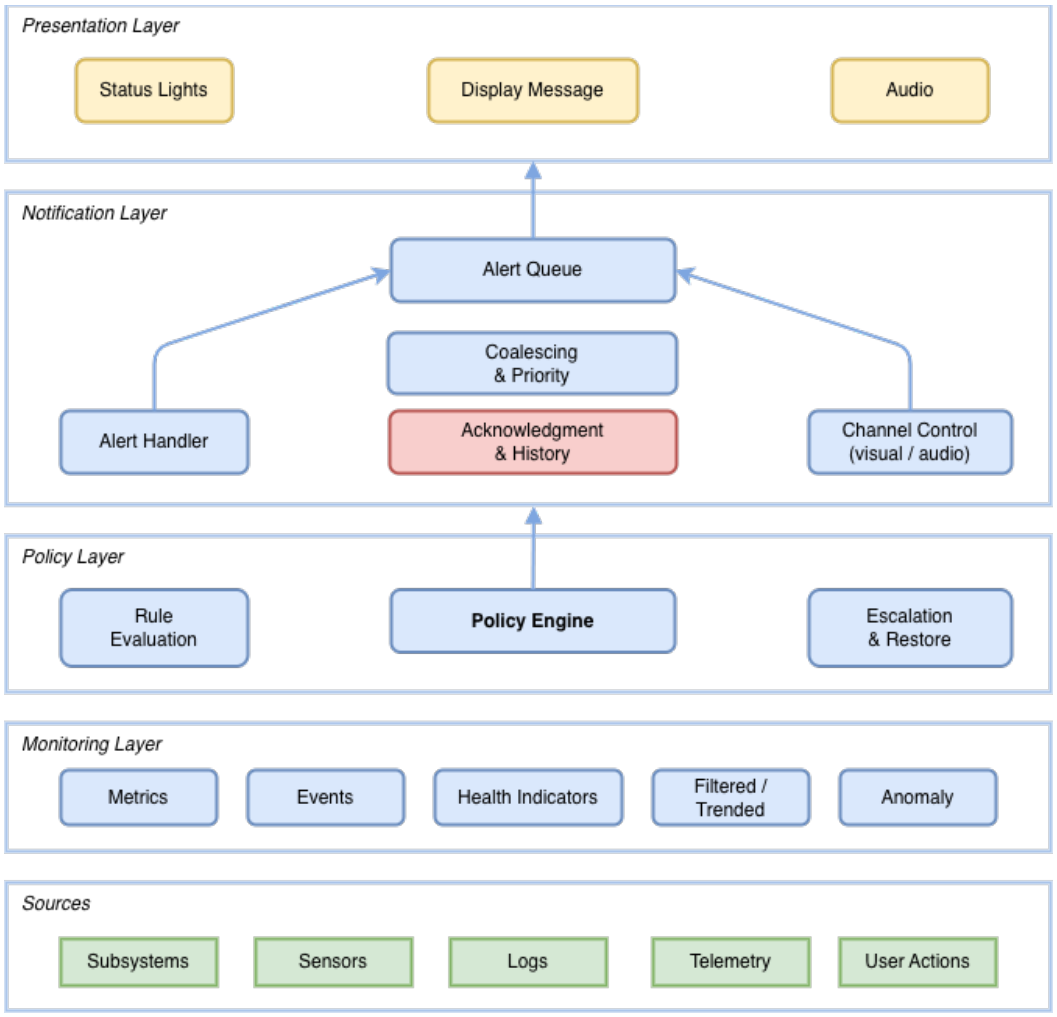


Figure 4: *The error and messaging system*

38.1. ON THE ORIGIN OF INDICATOR ACTIONS

Policy rules evaluate health indicators, events, and system state, and then specify what action to take, for whom, and under what conditions.[5] Whether to light an indicator, play a sound, or show a message is a policy decision – not something decided by the presentation layer its own.

A single underlying condition may drive several outputs. For example, a temperature excursion might update a status light, log an event, and queue a user-facing warning.

Indicators should automatically clear when exit criteria are met (if there is one defined). For example, when the measured value returns to an acceptable range and remains there for a defined hold-off period. Use separate trigger and restore thresholds to avoid rapid on/off oscillation.

With significant indicators, log when the condition occurred, when the user was notified, when the user acknowledged or responded, and when the indicator cleared (automatically or by service action). Higher tiers should expose an alarm or event history for operators and support staff.

[5] See the Policy chapter for rule structure: triggers, persistence, actions, escalation, and state applicability.

38.2. ALARM FATIGUE AND OTHER CONCERNS

alarm fatigue

Alarm fatigue is frequent, low-value, or ambiguous alerts train users to ignore them. There can be serious implications of ignoring alarms; when they indicate real safety or equipment risk. Over-notification devalues paying attention, and may distract operators (further contributing to risk). To reduce fatigue while preserving safety:

- Issue warnings only when policy confirms a sustained or meaningful deviation.
- Escalate severity gradually (inform → warn → service notification → fault/shutdown) rather than jumping straight to the loudest alert.
- Make each alarm actionable: state what is wrong, what risk exists, and what the user or technician should do next.
- Distinguish acknowledged-but-unresolved conditions from new or worsening ones.
- Restrict blanket silencing of safety-critical alarms without addressing the underlying condition. Allow temporary acknowledgment or snooze only where risk analysis permits, with audit logging.
- Separate informational reminders from fault and safety alerts so routine maintenance prompts do not compete with urgent warnings.

Implement strict policies that prohibit users from muting or disabling all alarms without first resolving the root cause. Critical alerts should continue to be visible and actionable until safety or service risks are resolved. Persistent problems are a major contributor to alarm overload, where important signals are missed because unresolved issues remain unaddressed. These policies prevent alarm muting from hiding persistent problems.

38.3. MANAGING MULTIPLE NOTIFICATIONS

When several conditions are active at once, the system must decide what to show, in what order, and through which channels – without overwhelming the user or hiding the highest-severity issue.

Assign each message a priority band (typically a small number, such as three: informational, warning, and critical). Within a band, order messages first-in-first-out or by time stamp. Higher-priority items preempt lower-priority presentation: a critical fault should not wait behind a routine service reminder.

When there are multiple notifications, what can be presented simultaneously? Limit how many distinct alerts are active at once on each channel (display, status lights, audio). If capacity is exceeded, queue or summarize lower-priority items and surface a count or summary (“3 additional warnings”) rather than cycling through every message.

Channel controls. Support independent enable, pause, and off states for visual alarms and audio where product tier and regulations allow. For example, silencing audible alerts during maintenance while keeping visual indicators active. Safety-critical paths may require that certain alarms cannot be fully disabled without credentials or service tools.

38.3.1 Coalescing alarms and notifications

Coalescing combines multiple related alerts into a single notification so the operator sees one clear message instead of a burst of duplicates. This reduces noise from transitory or redundant events while keeping critical issues visible. This is appropriate when alerts share a root cause, affect the same subsystem, or arrive in rapid succession (high-frequency

"chattering") from the same source. Combine them into one entry that reports the count, first and last occurrence, and the highest severity in the group. Retain individual events in the log even when the user interface shows a single coalesced alert.

Coalescing rules should define:

- Eligibility – which event types, subjects, or codes may be merged. Alerts on independent indicators (for example, separate status lights for different subsystems) may remain distinct even when coalesced in a message list.
- Grouping key – shared subject, fault code family, or policy rule identifier.
- Time window – events within a defined interval are merged; after the window closes, a new occurrence starts a fresh notification.
- Presentation – one message, optional repeat count, and escalation if the coalesced count or duration crosses a threshold.

It must not suppress a higher-severity alert behind a lower one or hide unrelated faults that require separate response.

39. PRESENTATION

When presenting notification, the kind should be distinct, easily recognized, in proportion to urgency and importance. Avoid "generic" (which would render them imperceptible).

With audible notifications, utilize a range of frequencies (typically between 150 Hz and 1,000 Hz). Each alert should be designed with a distinct audible signal or melody specifically associated with its condition. Avoid generic beeps; uniqueness ensures that critical alarms are never lost in a sea of indistinguishable sounds.

Incorporate haptic (vibration) feedback and spoken messages where appropriate, to provide additional or redundant notification channels.

39.1. COLORS

Colors should be distinct, with appropriate contrast, blink rates and duty cycles, as well as adjustable brightness and clear visibility to effectively communicate alert severity and status.

A *color system* to indicate the severity of notifications and issues:

Table 30: *Color*

Color	Meaning
Green, White, Blue	These colors are used to indicate that the (sub) systems are functioning correctly.
Amber, Yellow	Suggest that something needs to be checked but is not urgent. Monitor the situation and have it checked at your convenience. This could include the traction control light or oil pressure warning.
Red	Indicate serious problems that require immediate attention. Example: stop driving immediately and seek professional help. Other examples include the engine temperature warning and brake warning lights.

Colors are very culturally specific. They likely need localization. They may change with time.

40. SERVICE REMINDERS

service reminders

Provide early service reminders based on predicted life (e.g. hours/cycles/health trend), not just fixed schedules.

- Proved option to "continue anyway" while explaining risk clearly; offer grace periods, where safe.
- Enforce lockout only when risk analysis indicates unacceptable hazard; provide an emergency override if required, with prominent on-screen warnings and audit logging.

41. ERROR MESSAGE[6]

global harmonized system

Error messages are part of a structured approach to understanding that there is a problem, what caused it, and how to address it. Borrowing from the *Global Harmonized System* (GHS) model the core elements of an error or warning message include:

- Signal word: Error, Warning, or Information indicating the severity of the problem.
- The event identifier: identities of the cause of the error, preferably with a unique identifier (such as an error code number)
- Pictogram: conveying different types of hazards or information about the message.
- Hazard statements: these describe the problem encountered, in a human readable way. These statements should use standardized and assigned phrases that describe the hazard(s).
- Precautionary statements: these describe what will, should, or needs to happen – or suggestions on how to address the problem. These statements should use standardized phrases that describe measures to minimize or prevent adverse effects.
- Supplier identification: The name, address, telephone number and/or other contact information for support.

OSH3636

"Precautionary statements provide standardized wording and/or pictograms to describe recommended measures that should be taken to minimize or prevent adverse effects." These include:

- A general precautionary statement.
- A statement to prevent or minimize problems.
- A statement to address or minimize problems once the item has turned off
- A statement around storing them item.
- A disposal precautionary statement.

Optionally, extra information might be placed into the log or displayed including:

- Identify the module(s) (or subsystem or feature) that generated the text. This can include manufacturer & type of module.
- Specifics about the locus of failure (or trouble) such as the module, subsystem or file.
- Relevant statistics regarding the problem, subsystems, etc.

When creating the errors, should create a guide and run book that:

- Explains the meaning of the message

[6] Before getting into messages, consider: is the system designed to be meaningful at its errors? That is, are the messages given a useful form, provide clear information where to go, and how to test/diagnose the cause?

- Explains why the crash or message happened
- Explains why the error happens only under certain circumstances – machines, versions, users, etc.
- Provides steps to correct the problem; should include reassurance why it won't happen again
- Explains how to prevent the problem, and provide steps to do so
- Provides a mechanism to suppress the message

41.1. THOUGHTS ON WRITING THE MESSAGE

Some tips on writing the message and statements:

"This application has requested the Runtime to terminate it in an unusual way." – An actual Microsoft error message

- Don't be cute. Just state the problem
- Use the same message as in other applications. Having multiple different messages for the same thing makes it hard to support, both by machine and by hand.
- Use empathy for the reader and the frustrated user
- Use standardized phrases when explaining the context and actions
- Use consistent capitalization for naming events and other nouns

One technique is to employ an "empathy map" of the user in each situation, to better craft useful messages:

- What they were *doing* and what were their *goals* when the error (or crash, or …) happened?
- What the user *thinks* when seeing this message?
- How would the user repeat or *describe* this message on the phone, e.g. to customer support?

Identifying the kinds of problems, what triggered it

The thing that the person should have done instead

- Treat the different ways that things can fall, how one would analyze the results of an "experiment" for the "hypothesis" and what evidence we'd need to support it. The system should gather that relevant information and log it.

42. CRASH REPORTING

When the application "crashes," or is unable to start, the *fault handler* displays an error to a person. The code should be one that can be read on the phone.

A run book looks like:

1. If you see this *XXX* in the log, do this:

The crash logs and logging in general is useful for:

- Providing sufficient context to understand or explain the error or crash. This includes capturing the events leading up to the error/crash,
- To understand usage, what resources are used and what the users does/what feature the user uses. (e.g. DAS)
- Experiments to test hypothesis about what the users use, how things work, etc.

43. REFERENCES & RESOURCES

ANSI/AAMI/IEC 60601-1-8:2006 & A1:2012 & A2:2020 *Medical Electrical Equipment – Part 1-8: General requirements for basic safety and essential performance – Collateral Standard: General requirements, tests and guidance for alarm systems in medical electrical equipment and medical electrical systems*

Although specific to safety related interventions, this has interesting considerations on conveying intent thru colors, contrast, blinking, melody, speech and other sound.

OSHA, *Hazard Communication Standard: Labels and Pictogram*, pub OSH3636, https://www.osha.gov/Publications/OSHA3636.pdf

IEC 60417-5307 (2002-10)

United Nations Economic Commission for Europe, *Globally Harmonized System of Classification and Labelling of Chemicals (GHS)*
https://unece.org/transport/dangerous-goods/ghs-rev10-2023

Key academic citations regarding the standard's implementation and limitations include:

Edworthy, J., Reid, S., McDougall, S., Edworthy, J., & Hall, S. (2016). *The Design and Testing of Novel IEC 60601-1-8 Audible Alarms*. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 60(1), 641-642.

Bolton, M. L., Edworthy, J. R., & Boyd, A. D. (2020). *Masking Between Reserved Alarm Sounds of the IEC 60601-1-8 International Medical Alarm Standard: A Systematic, Formal Analysis*. Human Factors: The Journal of the Human Factors and Ergonomics Society.

https://en.wikipedia.org/wiki/Standards_for_Alarm_Systems,_Installation,_and_Monitoring

These are big scope alarms, for facilities

CHAPTER 11

Maintenance Services

This chapter discusses the functional support for maintaining a device, including management console, and remote management.

- Overview of ability to manage a product
- The main interactions with a user of the product
- Supporting different configurations and configuration management.
- Troubleshooting or fault finding to identify the cause of a problem and solve it

44. OVERVIEW

maintenance services

The *maintenance services* are activities to prevent equipment failure, extend its lifespan, and repair equipment in the event of failure. They typically allow managing the equipment, tracking its health, and reporting any need for service:

- Installation, and update of system software components
- Perform run-up and stress tests
- Examine the performance, usage, health and maintenance information of a product allows feedback to tune and service it.
- Support for user serviceable parts (field replaceable units or modules). Information and diagnostics, suitable for customer support call.
- Support for technician serviceable parts

This needs a structured approach to know when the system is not performing to expectations, that there is a potential problem, or requires immediate maintenance. Reporting of problems typically is done with alerts and fault messages. To understand a bug or crash, it often benefits from (or even needs) to know the context and activities that lead up to the event. These are done thru user interface, logging, and remote management.

45. MAJOR INTERACTIONS

main maintenance features

The *maintenance features* are used to perform tests, update management information, and to update the software:

- Services for to allow the users to change settings, preferences and other personalization's
- Update or remove the software components and application software
- Field update of modules, firmware of controllers, etc. This feature is often internal and coordinated with the application and system component updates.
- An operation to clean out the settings and other potentially corrupted values.
- A “factory reset” that erases all user settings, data, etc.

45.1. SOFTWARE UPDATE

Software updates provides the ability to install software updates, providing fixes and adding new features. The update feature may be part of the customer service tool, or integral to the system, with automatic updates and on demand in several of the subsystems.

Dual A/B partitions for over-the-air (OTA) updates, with a recovery partition allow safe updates with automatic rollback to a known-good image if an update fails. This minimizes the likelihood of device bricking.

45.2. MANAGING SECURITY

Security services: key management tools; token exchange not described here

- Password entry
- Reset password
- Account registration
- Set up WIFI
- Accounts
- Token management

45.3. ERASING SETTINGS OR PRIVATE DATA

Support operations to clear out settings and data.

45.3.1 Reset settings or account

reset settings

Reset settings is an operation to clean out the settings and other potentially corrupted values. This typically doesn't remove the linkage to the user account and critical security credentials/tokens, or the device name. Instead, it tends to erase log and other files, reset settings to default, etc.

reset accounts

Reset accounts is a similar operation to reset any accounts or pairing, without wiping the whole device, or removing the device name.

45.3.2 "Factory Reset"

factory reset

A *"factory reset"* fully wipes the device, erasing all user settings, data, including pictures, faces, and API certificates & tokens. This also clears out personalization data, including the robot's name. This is typically used as a last resort to address problems, during refurbishment or when intended to resell or dispose of the device.

45.3.3 Erasing the private data

Erasing user data private data implementation:

- If the data is in a file system subtree, this subtree can be renamed and removed.
- If located on a separate partition, this makes it easier to erase the data. A secure erase can be performed; there are standards on how this is done

45.4. CUSTOMER SUPPORT TOOLS

customer support tools

The *customer support tools* are built-in tools, intended for use by the customer, esp. while interacting with customer support personnel. They are optional and intended to be used by exception. Support personnel may direct a user to this tool as part of gathering information:

- Display configuration information, such IP address and the current network
- Device the information about the hardware models, and software versions
- Push logs, crash information, and perhaps performance information to remote servers.
- Perform diagnostics or report state of hardware, to be used in conjunction with manual actions.
- System Reports provide the ability to report system configuration, versions of software & hardware modules to customer services for support.

This information is used to help the customer over the phone, or email, esp. when network access is not available. This typically drives display features to aid communicating – such as codes to identify what screen is being showed, what errors and other status is encountered.

45.4.1 System Report

system report

A *system report* is a detailed summary of the system's configuration and current state. It is used to provide customer care with information to assist with diagnosing and fixing problems. This information is often automatically transmitted to the customer support office. This information may also be displayed on the screen.

The report typically includes the following:

- *System Information*, providing basic information about the system, such as the system's name, version, and build date.
- *Electronic identifiers* the serial #s and unique identifiers (e.g. EDID's) of the electronics modules. This may include the System on Module (SOM) manufacturer, model, and serial number; MAC address (Bluetooth, WiFi, Ethernet, etc.).
- *Memory Information*, providing detailed information about the system's memory, including the amount of RAM and ROM available.
- *Storage Information*, providing detailed information about the system's storage, including the amount of flash memory available and the type of storage used.
- *USB Information,* providing detailed information about the system's USB ports, including the type of USB ports available and the speed of the USB ports.
- *WiFi Information*, providing detailed information from the WiFi module, including the type of WiFi module used, the WiFi signal strength, and the WiFi network name.
- *Partition Information*, detailing the file-system's partitions, including the type of partitions, the size of the partitions, and the amount of free space available.
- Manufacturing/build info, such as the date of manufacture, facility, batch, etc.

45.5. RECOVERY MODE

recovery mode

Recovery mode is intended to help with issues such as failing to boot using the regular software. The recovery mode attempts to download and reinstall the latest software. This is

done under the assumption that the firmware may be corrupted, or not the latest, and that a check for corruption isn't possible with the read-only filesystems of production software.

The recovery mode is usually implemented as a separate set of software (sometimes just a very early version of the main software. The recovery mode will not automatically delete any user data or software previously installed via Over-The-Air (OTA) update.

45.6. SERVICE TOOLS

service tools

Tools to support *user serviceable parts* typically include features such as:

- Check batteries, their state of charge and health. Display charger status, to confirm connectivity
- Perform run up tests
- Install/uninstall field replaceable units, accessories
- Bandwidth tests, cable tests
- Ability to reset counters, esp. on a replaced unit. Not all counters can be reset.

Support for *technician serviceable part*s may include items that would not be meaningful to the end user – or likely to be misunderstood or cause unnecessary support calls.

- Check function of sensors
- Motor run-up
- Display test patterns
- Input tests
- Radio stats, metrics, strengths

45.7. NOTE ON THE TOOL IMPLEMENTATION

These tools and support applications are included for pragmatic reasons.

- Many are not part of the main application. Is the tool separate or built-in?
- Are the tools 3rd party or bespoke?
- How cohesive are the service tools with the rest of the product? Are the user interactions consistent between the two.
- How cohesive and integrated is the implementation?

46. CONFIGURATION MANAGEMENT

Most modern systems employ some degree of automatic configuration to simplify device configuration. This may include technologies like RFID, NFC, or barcodes on attached material or consumables to aid automatic setup. Devices can also exchange descriptors that define acceptable configuration options, allowing the system to automatically negotiate or enforce valid configurations, facilitating easier setup through plug-and-play approaches.

Nonetheless, a common cause of problems are misconfiguration and incompatibilities between different units. Provide troubleshooting tools to help identify and resolve conflicts.

- Range check individual settings
- Constraint checks of multiple settings

- When using a file or text, perform syntax checks, and identifying errors before using the file (esp. if used during startup or other hard to recover time).
- The ability to save a configuration, potentially including the software executables.
- The ability to rollback to known good configuration. Automatically, where possible.
- Where possible, provide tools to help understand why a configuration did not work, perhaps with log entries.

Where possible, provide tools to help understand why a configuration did not work:

- Providing relevant log entries or explanations.
- Provide utilities to identify and highlight changes or differences to configuration. (Mere text file comparison is incorrect, as it can shows differences outside of what changed to the configuration – often to a degree that hides those changes.)
- Include tools that analyze and communicate the impact of detected configuration differences on system behavior or compatibility.

47. FAULT FINDING AND TROUBLESHOOTING

fault finding

Troubleshooting or *fault finding* is the process of identifying and diagnosing issues or malfunctions during corrective maintenance. This process goes thru several stages:

First, *detect* that something is wrong. This could be a customer complaint, triggered by automated notification, or decreased performance.

Next *identify symptoms,* the signs of a malfunction. Gather information about unusual behavior, and the context of use when the problem occurred, such as the health metrics.

Diagnose the root cause using structured methods to isolate the issue. This may involve step-by-step testing, elimination processes, or/and guided workflows.

- *Localize*, identify the component where the fault exists, such as by narrowing down to a subsystem or specific part.
- Determine the exact nature of the fault (e.g., electrical short, software bug, mechanical wear, etc.). This may involve testing functions of the product to identify where it fails.
- *Trial and error* may be necessary, testing potential solutions and replacing suspected faulty components with known good ones one at a time to resolve the issue.
- This may require specialized tools like diagnostic software, multimeters, oscilloscopes, in conjunction with built-in product diagnostics to identify faults.

Correct the problem e.g., replace a faulty part, update software, or repair connections.

Verification, testing the product after the fix to ensure the fault has been resolved and the system is functioning correctly.

47.1. PREPARATION

Ensure that the device is in a *safe condition*: halt all processes (and therapy delivery, if medical device), vent or bleed any pressure, discharge stored energy, and de-energize motors and generators. The system should confirm safe state activation. Allow the device to cool, and always use *personal protective equipment* (PPE) and ESD protection throughout.

Gather information about unusual behavior and context of use when the problem occurred and operating context. With a customer complaint, note the exact wording, whether the

symptom/problem is constant, intermittent. Note any symbols displayed, gather trouble codes, freeze-frame and snapshot data at the time of fault. Retrieve the health metrics, and key parameter values.

Inspect for visible signs of damage. Check power status, examine cables, hoses, and other connections. Verify that the setting are correct settings. Ensure all consumables are properly installed.

For safety, identify symptoms that require stopping before doing any further diagnosis.

Validate the issue by replicating the reported problem, documenting any error codes, messages, or screens, and noting the circumstances they occurred in. This may include running self-tests, reviewing the logs, and using guided workflows to isolate the problem.

47.2. LOCALIZING THE PROBLEM

isolating causes

Review the health information to confirm exposure to adverse environmental conditions or identify error trend patterns. Then work thru the troubleshooting steps in the instruction manuals; verify power sources and connections. Ideally, the guides provide checklist-based troubleshooting techniques, such as measuring power rails, checking for leaks or proper fluid flow, monitoring motor current versus RPM, and verifying sensor zero and span values.

Some equipment automate troubleshooting – this is a particularly valuable feature often found in premium models. These are often guided workflows to assist in narrowing down the faulty subsystem – such as power, fluidics, sensors, or communications. If the equipment does not provide these features, users can still perform a manual troubleshooting process:

1. Map symptoms to possible causes, likely subsystems
2. Test for root cause, try other things outside of that.
3. Replace (only after isolation and verification)
4. Escalation: if triage points to a subsystem, move to guided diagnostics and service tests for that subsystem; if codes are absent, use logs and repeatable conditions to reproduce.

isolating causes by binary search

Problems can be systematically isolated by restarting with different combinations enabled. A typical iterative (or binary search) test process is:

1. Disable half of the equipment, or configuration settings; or replace user-replaceable *field replaceable units* (battery packs, fan, etc.).
2. Try to reproduce the problem or use the built-in self-tests and recent logs. If the problem goes away, the culprit is among the disabled group.
3. If the problem remained, the culprit was among the enabled group.
4. Repeat, enabling or disabling smaller and smaller groups until homed in on the specific setting or fault unit causing the issue.

47.3. GUIDED DIAGNOSTICS AND SERVICE TESTS

Guided diagnostics is a structured application of this process using a service tool or procedure to lead a person through step-by-step checks based on symptoms, health metrics, and measured results. The process is designed to use a repeatable test sequence for isolating root causes, to reduce unnecessary part replacements. It designs each step with expected value ranges, and clear pass/fail criteria.

The guided diagnostic process applies the above trouble shooting as these stages:

1. First, it collects evidence to identify symptoms. These are used to select the first subsystem test.
2. The tests are ordered to prefer performing quick, low-intrusion checks first, progressing to deeper, more involved functional tests (run-up, waveform or response-time checks).
3. If a diagnostic step fails, it defines a follow up branch for that specific symptom or failure mode; if it passes, continue to the next likely cause.
4. After each correction action, it confirms that the issue has resolved. The trouble codes(s) is clear, then operates the device thru run a representative cycle, and confirms all observed symptoms and diagnostic metrics have returned to baseline or normal operating range.

47.4. REPLACEABLE UNIT CORE TROUBLESHOOTING WORKFLOW

Diving into a more specific detail, replaceable units have a troubleshooting workflow:

1. Verify connections and power. Look for damaged connectors or bent pins. Check that the data and power cables are firmly connected; try different cables or connection points.
2. Check that the device is detected and recognized by the host system. If not detected, test the unit in a test station, or a known working system to isolate the issue.
3. Review system and device logs. Look for correction or recovery counts, uncorrectable error counts, sudden increases in errors, temperature or power cycle anomalies.
4. Run diagnostic tests, check for degraded or recovery states.
5. Assess performance and symptoms. Slow performance or pauses can indicate physical issues, or pending failure.
6. Check for environmental issues. Overheating, vibration, shock, or high humidity can accelerate failures. Review environmental sensor and exposure logs if available.
7. Isolate and replace if needed. If diagnostics point to device-level failure, replace the unit, preferably after backing up salvageable data.
8. Run functional checks or representative cycle, restore service, and monitor the system.

47.5. POOR CABLE OR CONNECTOR TROUBLESHOOTING

Connectors and cables are very common across all hardware types, so it is worth examining the workflow for troubleshooting. When troubleshooting a suspected cabling or connector issue, use a workflow like the following:

1. Check the connectors for bent or pushed-back pins, contamination, oxidation, cracked housings, loose retention latches, seal damage or moisture ingress.
2. Verify seating and strain relief; reseat both ends and confirm correct connector/key orientation, and that pins fit into mating connector.
3. Check the cable for chafing, melted insulation, rodent damage, pinched harnesses, broken locks, or loose back-shells.
4. Check physical stress points: tight bend radius, pinch points, vibration-prone mounts, improper strain relief near movement points, or cable pull tension.

5. Check the cable's electrical integrity.
    a. Power/ground and shield path quality: Measure voltage drop under load for supply and ground paths rather than relying only on static continuity checks.
    b. Perform loopback tests of the adapter, including use of loop back cable, to isolate if the adapter, the connector or external, such as the cable.
    c. Use a *cable tester* to check the cable, to verify all wires and shield paths, then re-test after cleaning, rewiring, or replacing it. For critical or high-frequency links, choose a tester that can validate both wiring and signal performance. If a cable tester isn't available, use a digital voltmeter (DVM) to check contacts and electrical connectivity; check continuity, short-to-ground/short-to-power checks, and resistance checks against OEM limits; for communication lines, verify expected bus levels/termination.
6. Clean the connectors. Swap cable with a known-good cable/adapter/transceiver to isolate whether the fault follows the cable path or stays with the endpoint. (If a replacement cable is readily available)
7. Compare *link metrics* before/after reseating or swapping. These metrics will vary with class of device that the cable connects.
8. Run continuity/impedance or loopback tests where supported; verify shield/ground integrity for noise-sensitive links.
9. Perform a controlled wiggle/vibration test while monitoring live error counters and logs. Note: this is not always applicable, such as with weak connectors; be careful not to destroy data or equipment, using a dummy load instead of live end points.
10. Replace the suspect cable and re-run post-repair functional and stability checks.

In concept some connector/cable tests could be automatic (short, open, etc.) but not done. Perhaps too many false results, perhaps just not knowing and focusing on primary function.

47.6. FEEDBACK CONTROL ISSUES

Feedback control issues can originate at any level and propagate through the control hierarchy. Troubleshooting may be required at each layer, from user interface down to physical sensors and actuators.

Layer	Description
Human Machine Interface	High-level user setting the setpoint (e.g., thermostat setting) Symptoms: HMI screen frozen, user login errors, or incorrect setpoint values.
State Machine & Mode Control	The machine's state machine (Idle, Startup, Running, Emergency Stop). This layer ensures that control commands are only sent when the system is in the correct mode. Symptoms: The system stuck in "Idle" mode, or failing to switch from "Manual" to "Auto"
PID Control & Flow Control	The *proportional integral derivative* (PID) control loop layer ensures reliable control actions. "Flow control" in this context is the controller deciding how much power to send to an actuator based on speed limits, matching the calculated demand to the actuator's capability. Troubleshooting: Tune PID gains (P, I, D) to prevent instability, fix runaway control signals.
The Local	Local PLC (Programmable Logic Controller) or microcontroller code. This layer decides

Table 31: *Troubleshooting across feedback control layers*

Controller Logic	where the sensor data belongs and routing it to the appropriate control algorithm.
	Hardware interfacing, such as Fieldbus (CAN bus, Modbus), ADC/DAC sampling, and error detection (checksums) to ensure the raw sensor signal is accurately read. Firmware tasks, and interrupts for communication. Integrates drivers. Troubleshooting: Verify the microcontroller is receiving inputs, checking I/O mapping
The Signal Conditioning	This layer converts electrical signals (voltage/current) into raw digital data or raw physical action. Interface to software. Firmware uses interrupts, and GPIO interactions. Troubleshooting: Check for noise on the signal lines; verify communication framing on the serial bus.
The Physical Layer Actuators & Sensors	The mechanical, and electronics design. The actual hardware sensors (thermocouples, encoders) and actuators (motors, valves, heaters). Troubleshooting: Is the motor plugged in? Is the sensor reading 0V?

47.6.1 The PID Control Loop

PID controllers may require retuning if it loses control authority, or exhibits symptoms like persistent overshoot, sluggish response or failure to meet target band, deviation outside target bands, excessive oscillation, unstable output, or. These often signify that the controller's parameters need adjustment. As components age, the system's response may drift, and retuning the PI/PID loop can help restore stable performance.

Many modern control systems offer automatic routines for re-identifying or auto-tuning control parameters. Make use of these features when available.

The system should log each re-tuning. Repeated retuning may indicate underlying hardware degradation or mechanical wear; the system should recommend component replacement instead of continual recalibration.

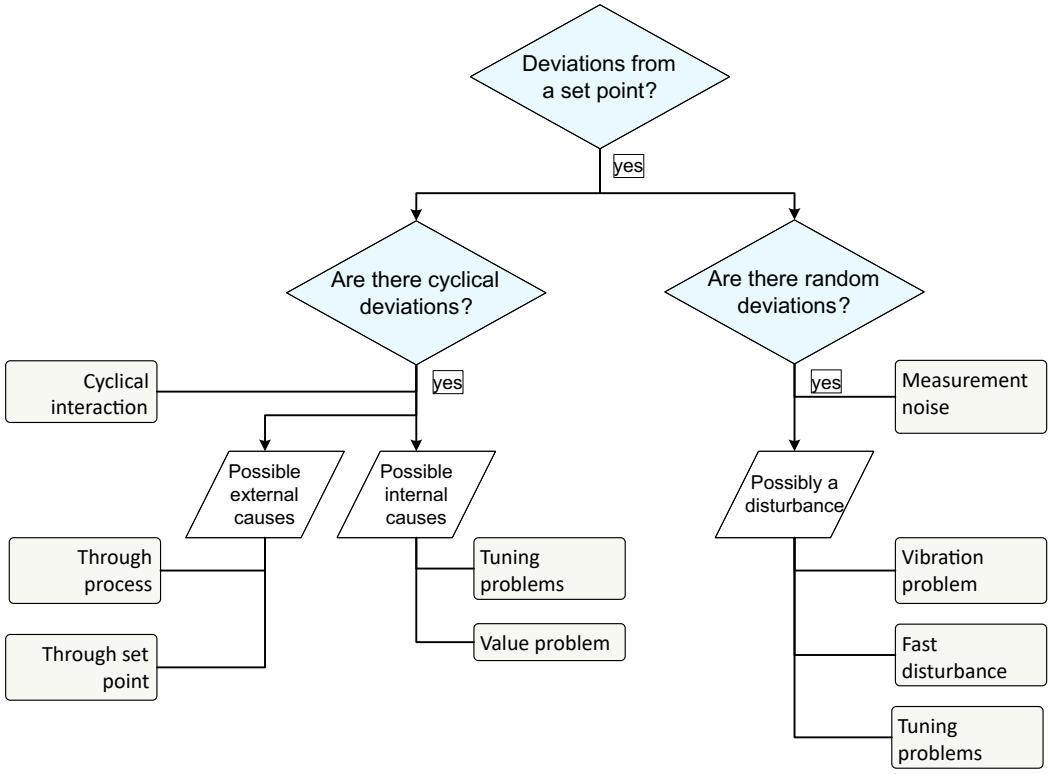


Figure 5: *Diagnosing a control loop (National Instruments)*

47.7. CHALLENGES IN FAULT FINDING

There can be several challenges to troubleshooting.

- *Intermittent faults* occur sporadically can be difficult to diagnose because they may not be present during testing.
- *Complex Systems*: In highly integrated systems, faults in one area can cascade and appear as issues elsewhere.
- *Limited Information*: some products don't provide sufficient diagnostic data or feedback, making the fault finding more challenging.
- In cases where customers perform fault finding, the process must be simple and intuitive to avoid errors.

Part IV

Hardware Serviceability

This part describes design and processes to support servicing the product hardware:

- Hardware & Software Serviceability Support. Common hardware and software designs to facilitate serviceability.
- Power Management. A detailed look at the architecture for power management, battery monitoring, and charging.
- Storage Management. A look at the monitoring the health of storage, such as hard drives, flash, EEPROM and RAM.
- Communication Management. Communication, health metrics.
- Automatic Loaders. Systems to manage tools, media, and consumables that can be automatically loaded into a tool head.
- Heaters and Chiller. Subsystems to intentionally add heat or cool, to hold a target operating range for a component, zone, or process.

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CHAPTER 12

Hardware & Software Serviceability Support

This chapter discusses common functional hardware and software designs to facilitate serviceability.

- Overview of functional support
- Control unit hierarchy (supervisor, manager, controller, driver)
- Configuring hardware
- Regulating the internal environment
- Remote Operation Support
- Software functions

48. OVERVIEW

Tiers above consumer grade often include a management or service subsystem. This provides self-regulation, allows monitoring of the equipment's condition, and servicing it.

- Keeps the system within operating parameters, esp. temperature and power levels.
- Chassis intrusion detection, such as opening of a door, or removal of a panel or other guard. Typically, these are guards, so may trigger risk controls. Unauthorized opening or tampering might be logged, e.g. for security reasons.
- Protects the main processor, aids its startup, be independent of its faults (i.e. to record crashes), and restarts it as needed.
- Tracks simple management information: component identifiers, serial numbers, asset tags, firmware versions.
- Monitor environment conditions (e.g. temperature, voltage, and fan status), a tracks system health, such as equipment age and condition.
- Provides remote access for monitoring/telemetry, diagnostics, troubleshooting, and system management, such as power and boot control (i.e., boot device, image), file transfer & firmware/software update.
- Event logging and alerting or gathering information when the main processor or host OS is not operating.

The management subsystem typically operates independently of the main processor(s), performing tasks even when the main system is powered off or unresponsive. In some products, this may be its only significant processor.

Note: there are many different management controllers based on the domain:

- For engine-based equipment, there are hour meters (premium consumer and contractor), engine controller or engine management unit (professional and industrial).
- On servers, this is called a *service processor* or (on Intel servers) the *Baseboard Management Controller (BMC).*

49. HARDWARE SUPPORT

This will look at functional behavior, rather than mechanical fittings etc. (Those were covered in the serviceability chapter.)

1.1. CONTROL UNIT HIERARCHY

Distributed products stack control by *who owns the loop* – from policy down to physics. The same pattern appears on road vehicles, building automation, industrial PLCs, and many embedded platforms (with different names).

***Table 32:** Control unit layers*

Layer	Description
Supervisor	Maintains the product or fleet-level policy and commands peers: ready/not-ready, derate, mode selection, "schedule service vs stop now." This may be a distinct module, or folded into another controller.
Manager	Domain or function coordination: energy, thermal, scheduling, body or building zone. Sometimes a separate processor; often a software component on the supervisor. A thermal manager may request derate from several controllers.
Controller	Closed loop on one plant. Most trouble codes, alarms and freeze-frame live here. This is the layer a service tool usually names.
Driver	Actuation and smart I/O: gate drivers, injector and coil drivers, LIN slaves, H-bridges. Often no diagnostic address of its own; faults surface as controller codes (open load, short to ground, implausible current).
Sensors and Actuators	The engine, motor, pack, valves, belt, wheels, damper.

49.1. CONFIGURING HARDWARE

The service subsystem may have a role in configuring some of the hardware. If misconfigured, the main processor may not start or even damage the hardware. Many components include configuration information; examples include:

- Clock speeds
- Voltage levels
- Timing, e.g. refresh. DIM Training
- JEDEC: Memory sizes, timing, discovery
- The ability to enable and disable features

BIOS / UEFI firmware often applies aggressive settings as a trial rather than writing them as the new default. A typical flow is:

1. Operator or policy selects candidate settings, i.e. clocks, voltages, memory timings including XMP/EXPO profiles, feature enables.
2. Firmware stores them as pending (or one-boot), then reboots.
3. If the platform makes it past a defined gate (e.g. POST, memory training, watchdog, a firmware self-test, or an operator "confirm" prompt) the settings are committed as the new known-good configuration.
4. If the boot does not complete that gate (hang, training failure, watchdog timeout, power loss, or the operator declines), firmware rolls back to the last committed set, typically the JEDEC / OEM baseline.

This keeps a failed overclock setting or mismatched DIMM profile from leaving the system unbootable. The system should log the pending vs committed profile, the gate that passed or failed, and the rollback so service can tell a bad setting from a failed module.

49.2. HARDWARE FOR REGULATING INTERNAL ENVIRONMENT

These controllers keep the system within operating parameters:

- Temperature regulation, monitoring temperature and coolant, regulating fan speeds. See chapter on chillers for more information
- Power management: monitoring voltages, power supply (PSU) status, power draw, and controlling subsystems
- Battery management: This is a subsystem of the power management, monitoring the health of batteries, charging and temperature

Table 33: *Temperature regulation*

Temperature Regulation	
Role	Temperature regulation measures temperature and can control fan speeds to increase cooling. In some cases, it may include a heater or chiller. A *System Management Bus* (SMBus) is typically used.
Management interactions	*Thermal throttling*: The regulator controls or provides feedback to reduce performance (e.g. clock speed) and power consumption. Sends alerts when a measurement is outside of set bounds (typically a warning threshold, and then protective stop threshold.) Safety shutdown, to protect equipment, if temperature threshold is exceeded for a duration.
Sensing	Temperature, humidity, air pressure
Actuators	Fans, Chillers & cooling pumps, Heaters
Tracked Metrics	Max, and min measure temperature Track time above temperature

The power and battery management will be discussed in a following chapter.

49.3. MANAGEMENT NETWORK

Sophisticated management processors often need ethernet like communication interface. The concern is that users will place the management network on the data/service network, which

would risk a “denial of service” for the management processor, causing intermittent issues. The answer is *Network Channel Side Interface* (NCSI), which allows using the primary NIC, instead of requiring a 2nd network for service processors.

50. SOFTWARE FUNCTIONS

The service subsystem may provide software support for the main processor and other processors (including the service processor). This can include:

- Change firmware (BIOS) settings,
- Update software, including ability to download BIOS/firmware and apply updates
- Ability to disable / de-energize equipment; ability to cease using replaceable unit and detect when complete (Safe to turn off/remove)
- Detect hardware change / insert/removal of FRU
- Interact with external memory and other components (if any) on Field Replaceable Unit,
- Validate authenticity for critical FRUs,
- Support Calibration: perform calibration, store/retrieve calibration data
- Track usage, and estimate health
- Logs
- Perform test, and/or actions used in testing
- Ability to detect part change. probably separate system
- Sequences to follow: state flow with errors represented
- Safety shutdown, to protect equipment

50.1. TECHNIQUES FOR IMPLEMENTING COUNTERS

Many (not all) counters are non-volatile. They should monotonically increase, until the counter is manually reset.

- Store usage counters and health per replaceable item separate from others, and keep these separate from the overall device lifetime.
- Periodically checkpoint counters to nonvolatile memory. Use secure non-volatile memory in more stringent applications.
- Use power-fail-safe updates. Use a dual-slot or journaled write with monotonically increasing sequence numbers and checksums. When restoring a value (such as after restart) use the one with highest valid sequence.
- If working with flash-based storage use wear-friendly techniques, such as wear leveling.

Some counters should have increased protection or security. Anti-rollback techniques to ensure counters are monotonic:

- Some chips have fuse bits: once blown, they can no longer be changed. These can be used to count usage (or usage remaining). Secure elements (e.g. TPM), if present, have “non-volatile” counters that can only be incremented.
- Include an install-index for replacement replaceable parts, include per-part monotonic install index or signed state to prevent spoofing “new” status.
- Keep redundant copies and majority vote to resist partial rollbacks.

- Use append-only logs, with delta-encoded updates.
- Use secure storage.
- Signed snapshots

Policy/hardening: Only allow counter updates via signed, measured code paths; lock debug/programming interfaces; rate-limit increments; continuously audit and alarm on impossible deltas (e.g., hours decreasing while total cycles increase).

Mitigate pitfalls with protected counters, gated test flows, compatibility checks, authenticity validation, and automated logging.

50.2. SECURITY CONSIDERATIONS

Privacy/security considerations:

- Never import secrets via accessories, field replaceable or consumables. Or migrate old secrets. Operational keys stay in the device.
- Include part state in attestation only if needed for trust decisions.
- For replaceable parts, include per-part monotonic install index or signed state to prevent spoofing "new" status. If parts carry identifiers, validate signatures.
- Keep unit health/usage data free of PHI/PII. Keep health/counter data separable and anonymized when possible. Include part state in health/attestation reports when helpful (e.g., battery age, state of health, cycle count) without exposing traceable PHI.
- If a part can store data, wipe before disposal.

51. SPECIAL REMOTE OPERATION SUPPORT

In higher end tiers, the service management might be accessed remotely, and to support lights out operations. Implementations of *Simple Network Management Protocol*, can be used to get current values, counts, and could set alarms.

Remote consoles are also possible. Depending on the tiering this may range from serial, network keyboard video mouse (KVM), SSH, web, and other levels. Some of these are more sophisticated – such as requiring complex TCP/IP stacks, and security, which puts those service processors on a certain level of processor capability. Remote access may support the restarting the main system, or otherwise controlling its power.

With Intel type processors using a baseband management controller, *intelligent platform management interface* (IPMI) provides the remote management.

52. RESOURCES AND REFERENCES

https://en.wikipedia.org/wiki/Fleet_telematics_system

https://en.wikipedia.org/wiki/Intelligent_Platform_Management_Interface

https://en.wikipedia.org/wiki/On-board_diagnostics

https://en.wikipedia.org/wiki/Self-Monitoring,_Analysis_and_Reporting_Technology

https://en.wikipedia.org/wiki/System_Service_Processor

CHAPTER 13

Power Management

Power management is responsible for managing the power sources, controlling the power distribution and consumption of the system, including the battery and its charging.

- Monitoring the status and information about power sources, and consumption.
- Control distribution and use of the power consumption of the system.
- Monitors the battery's state of charge, health, status and control the charging.

53. OVERVIEW

Power Management	
Role	The power management monitors voltages & power draw, power supply (PSU) status, and the health of batteries. It controls subsystems (on demand) or signaling shutdown if there is brown, or mains power interruption. The manager is partially active while the rest of the system is completely shut down. This allows buttons, the clock, and remote activities to start the system. A *Power Management Bus* (PMBus) is typically used.
Management Interactions	Sends alerts when a measurement is outside of set bounds (typically a warning threshold, and then protective stop threshold.) *Safety shutdown*, to protect equipment, if power source is interrupted; or if power draw threshold is exceeded for a duration. Integration with the *battery management subsystem*. Interacts with real time clock to schedule time to automatic system restart
Sensing	Power supply status: Monitoring power connections, the source(s) of power, and the voltages of power source and battery. UPS's, and power distribution units may provide information to the power management system. Battery management, such as tracking battery state of charge, charging, and health. Power usage, voltages & current levels
Controls	Control the source of power, power distribution, shutting down unnecessary system components when they are left idle. Turning the power/on off. Controlling low-power states such as sleep and stand by. Can wake device via button press.
Tracked Metrics	The power usage, or histogram, max and min of power usage. The duration in each state Things initiating changes to power state or preventing changes in power state.

Table 34: *Power management*

54. POWER MANAGEMENT

The power management system typically looks like:

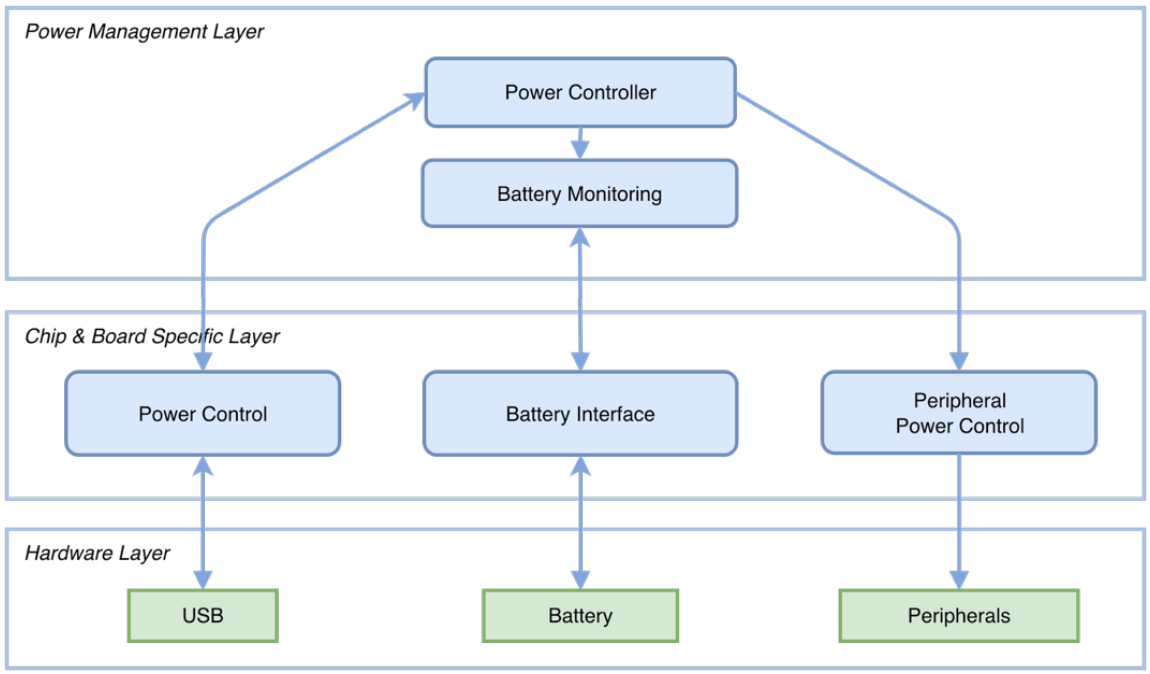


Figure 6: *The power management subsystem within the architecture*

Power management policies govern how the system monitors, distributes, and reacts to power supply and consumption across its components. Their aim is to maintain reliable operation, efficient energy use, protect equipment, and safeguard data and user safety during normal operation and exceptional scenarios such as brownouts, blackouts, and power surges.

54.1. POWER SOURCES

Power management policies define how the system selects among available power sources and defines the conditions for switching from one source to another. The goal is to ensure orderly transitions to backup or emergency power whenever the power is disrupted or capacity drops.

Typical policy examples include:

- If the main power supply voltage remains below threshold (X volts) for longer than a specified duration (Y seconds), automatically switch to backup power and alert operators.
- When projected power reserves fall below the minimum required for continued operation for a defined period (Z minutes), begin staged shutdown of non-essential subsystems.

Possible *power sources* for a device may include:

- *None*. The device is not powered.
- *Internal Battery*. The device is powered by a battery; more in a section below.
- *Charger*. If one is defined for the product
- *USB* powered by USB power delivery.

- *AC Mains*
- *Wireless recharge*
- *Generator*
- *Backup battery*

Multiple power sources may be present simultaneously. Their statuses can be independent or in parallel with the overall power management state. The power manager must monitor and coordinate between them.

54.1.1 External power sources

A typical device supports operation when connected to an appropriate USB-connected power source, regardless of battery state. The USB power may be supplied from USB chargers, USB battery packs, etc. The USB interface circuit is responsible for managing the power delivery in this mode, negotiating between external USB or mains, and battery; charging and powering the device.

54.2. POWER STATES

Products may be designed for continuous operation, but they inevitably transition through distinct power states depending on usage, low battery level, storage, shipping, or other operational requirements. The typical set of power states for an electronic device includes:

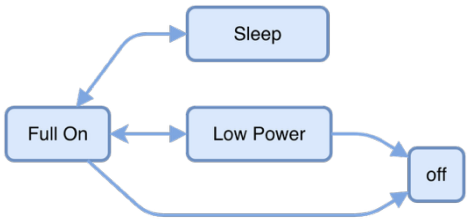


Figure 7: *The power states*

The enumeration of power states abstract the state of the subsystem(s). The states restrict what actions are allowed and may drive some actions/interactions:

State	Description
Full On	Processor on, with everything on (or allowed to be): display, lights, Wifi, BLE, etc.
Low power	The device is in the lowest power state but not off: display is off, lights are off, and communication (Wifi, BLE, etc.) is off. The processors may be stopped or reduced performance but are not required to. be. The electronics are put into a lower power. Touch screen or button press can restart the system.
Off	Power is electrically disconnected, everything off. This is the initial device state. Exit: reconnect power or recharge.
Sleep	A low activity mode where lights and optionally the display are turned off. The processor remains on, with WiFi & BLE active, running minimal app functions without vision or audio output.

Table 35: *Battery quantitative state of charge values*

54.2.1 Power load management strategies

Power management should include dynamically reducing or redistributing load to maintain operation of essential functions when available power is constrained. Often this is accomplished by ranking subsystems according to their criticality and employing policies staged shutdown, throttling, placed into lower power modes, or load-shedding non-essential components.

Policies should clearly define both the triggers for entering power-constrained modes (such as low battery, unstable or insufficient input voltage, or thermal constraints) and the procedures for implementing these changes. Example triggers and management actions include:

- Shedding non-critical loads, preserving core functionality
- Disconnect battery (open the switch)
- Turning off and on manually
- Turning on initiated by attaching to a charger or external power supply
- Turning off spontaneously due to brown-out or significant loss of power
- Automatic power-down due to brown-out or severe voltage drop
- Shedding load.

54.3. MANAGEMENT INFORMATION AND DIAGNOSTICS

54.3.1 Monitoring, and notifications

Power management systems often continuously monitors power status, consumption, and event history. It will typically provide information about the power state(s), including states of battery and charger. Notifications should be provided to users or administrators in response to important events, such as power degradation, imminent shutdown, or automatic switching to backup sources. All relevant power events, including state transitions, overrides, and abnormal supply or consumption patterns, should be recorded and stored for post-incident review and auditing.

Rresponsibilities of the power management system often include:

- Record charging information, errors and any detected faults.
- Log charging activity and fault or error events.
- Notify users of errors, faults, or critical states.
- Advising the user to charge or replace the battery or automatically taking actions to connect to external power.
- Trigger a graceful shutdown (sometimes called a clean shutdown).
- Initiating a clean (graceful) shutdown when required by battery level or fault conditions.

54.3.2 Power Measurement

There are often many points within the power tree where quantities are measured. With power subsystems it is best to use *filtered,* low frequency values to prevent sudden jumps, values jumping back, and nuisance trips. In some power systems the battery charge can increase due to regeneration, such as from motor braking.

Parameter	Description
Coulombs	Measures the amount of energy used over a duration. Often this is provided by the battery management system, it may also be part of motor drivers (to detect stalls). **Units:** Coulombs (C) or milliamp-hours (mAh)
Current	The current presently applied to the system **Units:** mA
Voltage	The voltage. **Units:** mV
Temperature	The temperature of the unit. **Units:** °C

Table 36: *Power usage measures*

The instantaneous values will be those as above, the tracked health metrics typically will include the common health indicators (see chapter 11), and:

Parameter	Description
Incidents of voltage/current excursions	Number of times supply voltage/current exceeded recommended limits. **Units:** count
Power consumption (average)	Mean electrical power drawn during operation. **Units:** watts (W)
Power consumption (peak)	Highest instantaneous power usage recorded. **Units:** watts (W)
Time above critical power (Cumulative)	Total time spent operating above the device's specified maximum power threshold. **Units:** seconds

Table 37: *Power health indicators*

54.3.3 Diagnostic support

Consider troubleshooting charger connections and power sources. The management information around the voltage and current provided externally allows checking connector connectivity and if the source is applying suitable power. For example, solar recharger (such as on wireless cameras) should gather information to help check if the solar cell is place properly – too much shade, and the current will tend to be low; too much direct sun, the system goes into protect mode too often, and so on.

54.3.4 Power information from the accessories

The power management system should monitor the power condition of attached and remote items, especially accessories that it acts a hub for. It should provide access to power information in a manner that is consistent with how it provides information elsewhere.

54.4. SELF PROTECT

Electronics often include self-protect mechanisms: fuses, current limits, and power supervisors (to reset electronics who state may have been scrambled in a brown out). The software may have a supervisor function to monitor conditions and respond to operating outside of specified conditions.

Some products may include an operator-controlled safety power shutoff – such as an ESTOP.

Protection from Power Events: Set requirements for surge protection, voltage regulation, and fault detection (e.g., breaker trips, fuses, overcurrent sensors). Specify monitoring thresholds for anomalous events such as brownouts, surges, or frequency/power factor excursions, and the corresponding automated or manual responses.

54.4.1 Thermal management

The software often has thermal management responsibilities, to prevent exposing people to excess temperatures and to promote a longer service life of the batteries, motors, and product. Temperature is typically measured at key points such as the power switch MOSFETs, the charging circuitry, battery cells and motor driver bridge.

When the temperature rises above one or more thresholds (e.g. 50°C), the product can be slowed down (derated), or activities and charging can be stopped to allow the battery cool. This is typically done at several levels – automatically in the electronics (as a last resort), software control loops to moderate activities and automatically turn on environmental cooling/heating, and the application can be notified to take more aggressive actions as the temperature rises, to protect the battery and let the chips cool down.

54.4.2 Graceful Shutdown and Recovery

Often subsystems or the whole system should be safely shut down when power cannot be sustained, to ensure system integrity and avoiding data loss. There should also be policies for safe resumption of operations, including checks for system health and power stability before reactivation.

54.5. MANAGEMENT CONTROLS

The application can set management parameters, and command the power distribution and functions, overriding automatic management:

- Electronically disconnect the battery from powering the device. This is typically used to put the battery into a “ship” mode, or if the battery power is very low.
- Disable (and re-enable) charging.
- Enable and disable major subsystems, esp. output systems such as motors, lights, displays, RF generators, heaters and chillers, etc. This may include controlling the applied voltage.

54.6. REPLACEABLE UNITS

Power supply units may be field replaceable. The external wall warts are replaceable at consumer level and built in PSUs at the contractor/professional level and above. At the enterprise level, the built in PSUs may be hot-swappable: these need an M+1 design (or M+N, where N is the number that can fail or be removed at a time without interrupting service).

Removable battery packs are often hot-swappable if the unit is plugged in, but that may not be an intended part of design. There may be sparks.

55. BATTERY MANAGEMENT

Battery management oversees a battery pack.

Table 38: *Battery management*

Battery Management	
Role	The battery management is an important, common power management subsystem. It monitors states such as state of power, state of charge, controls recharging, and state of health. It addresses several the risks associated with batteries (over-discharge and overcharge).
Sensing & Measurement	Battery state of charge, charging current Battery temperature and temperature of charging circuit
Controls	Charging: slow, fast, disable, modulate to prolong battery life
Tracked Metrics	Battery identifiers Battery State of Health (SoH): its capacity relative to when it was new; this reflects the battery's overall condition. Remaining Useful Life (RUL) estimate The number of Charge/Discharge cycles, and histogram of durations The Max, and min temperature measured Charge duration Time above temperature thresholds, voltage limits, high current draws.

55.1. REMOVABLE BATTERY PACK PRESENCE DETECT

For removable battery packs, the *battery presence states* include:

- *Not present*: The battery is not connected or detected.
- *Present*: The battery is detected (e.g. via I^2C).

If the battery is present, the *battery connection states* include:

- *Battery Off*: The battery is present, but electrically disconnected, providing no energy to the load. This state is useful for shipping and storing the device. To exit this state, typically applying a charger or USB power is required, or pushing a specific button that is (weakly) powered the battery management and used to trigger a startup sequence.
- *Connected*: The battery is present; it may be charging, discharging or neither.
    - *Charging*: The battery is currently being charged.
    - *Discharging*: The battery is currently discharging (in use).
    - *Neither*: *The* battery is connected, the charger is not applying energy to the battery, but the battery is not providing power to the product.

55.2. MANAGEMENT CIRCUITRY

Many lithium-ion (Li-ion) and lithium-polymer (LiPo) battery packs include an integrated circuit (IC) that provides fuel gauge, battery charger, and protection functions:

- *Fuel Gauge*, to provide accurate state-of-charge (SOC) and state-of-health (SOH) information. A coulomb counter tracks the charge entering or exiting the battery.
- *Battery Protection*, which includes over-voltage, under-voltage, over-current, and short-circuit protection.
- *Battery Authentication* to ensure battery pack authenticity.
- *Temperature Monitoring*: Monitors battery temperature to prevent overheating.
- *Alerts*: Can be configured to signal various battery conditions.
- *Battery charging*. Typically, the charging is automatic but may be disabled by software.

Note: batteries with small number of cells or capacity may not include these.

The battery management IC(s) typically holds the battery charging parameters, capabilities, state, and health; and provides this provide information via communication bus, such as I^2C.

Not all batteries are "smart" – many toys, small electronics, and home power tools lack management circuitry. Small electronics have small batteries, designed for low cost, small size and weight, and to reduce the need for battery management. These tend to need one or two cells, and the battery can be flat, and thin. The battery management for these depends on battery voltage, and timers.

55.3. BATTERY STATE OF CHARGE

The batteries state of charge is often represented as combination of enumerative and quantitative values. The enumerative state is the easiest for the application to interpret in its decision logic.

55.3.1 Quantitative state of charge

The q*uantitative battery state of charge* are the most direct measurement of the current battery state, often provided by the battery management IC. They include:

State Parameter	Description
Battery voltage	The battery voltage. **Units:** volts
State of charge (SOC)	The current battery charge level – the amount of energy stored in the battery. *Note this requires a "fuel gauge."* **Units:** Coulombs (C) or milliamp-hours (mAh)
Percent charge	The current state of charge as a percentage. **Units:** percent (%)

Table 39: *Battery quantitative state of charge values*

55.3.2 Abstract state

The enumerative state of the battery is an abstract categorization of its state often based on using the fuel-gauge's state of charge (coulomb count) with a decision tree, thresholds, and hysteresis. Typically, the states are arranged like:

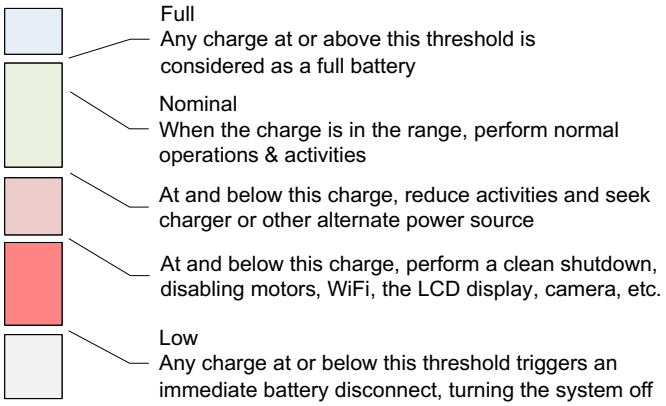


Figure 8: *The battery state of charge and thresholds*

The *battery state of charge enumeration*, including fault/error states of charge would be:

State	Description
Full	The battery is fully charged. (The battery typically will be discharging, but may be charging)
Empty	The battery is completely discharged. The system should shutdown gracefully.
Critically Low	The battery charge is critically low. The firmware should shut down the system, skipping graceful shutdown.
Nominal	The battery charge is in the desired range and can support the product's normal operations.
Over Charged	The battery has been charged beyond its safe limit. The battery must be replaced. This state cannot be exited.
Over Discharged	The battery has been discharged below its safe limit. The battery must be replaced. This state cannot be exited.

Table 40: *Battery state of charge enumeration*

55.3.3 State of Charge thresholds

Note: The thresholds in this section use units of Coulombs or milliamp-hours if working with a coulomb counter; and volts if not.

The initial battery state of charge is found by comparing filtered battery state of charge or voltage with the thresholds to find the region containing the value. Thereafter, a hysteresis is applied: if the filtered state of charge is greater than the upper threshold for the current level, it must exceed it by at least the hysteresis value before the upper regions can be considered. Similarly, if it is less than the lower threshold by at least the hysteresis value for a lower region can be considered. The battery state of charge thresholds are:

Threshold	Description
Critically Low	When the state of charge (or voltage) is at or below this (but above the over

Table 41: *Battery thresholds*

threshold	discharged threshold), the battery is in the critically low state. **Units:** Coulombs (C) or milliamp-hours (mAh) **Units:** Voltage (V)
Empty threshold	When the state of charge (or voltage) is at or below this (but above the critically low threshold), the battery is in the empty state. **Units:** Coulombs (C) or milliamp-hours (mAh) **Units:** Voltage (V)
Full threshold	When the state of charge (or voltage) is at or above this (but below the over charged), the battery is in the full state. **Units:** Coulombs (C) or milliamp-hours (mAh) **Units:** Voltage (V)
Over-charged threshold	When the state of charge (or voltage) is at or above this, the battery is in the over-charged state. This threshold takes precedence. **Units:** Coulombs (C) or milliamp-hours (mAh) **Units:** Voltage (V)
Over discharged threshold	When the state of charge (or voltage) is at or below this, the battery is in the over discharged state. This threshold takes precedence. **Units:** Coulombs (C) or milliamp-hours (mAh) **Units:** Voltage (V)

55.3.4 Without a fuel gauge

Without a fuel gauge, the state of charge is estimated by mapping a filtered battery voltage to a battery level (with hysteresis) as described earlier. The filtering is necessary as the voltage will bounce around with activity the motors, lights, driving the speaker and processors. This noise increases with battery impedance – at lower temperatures and as the battery ages.

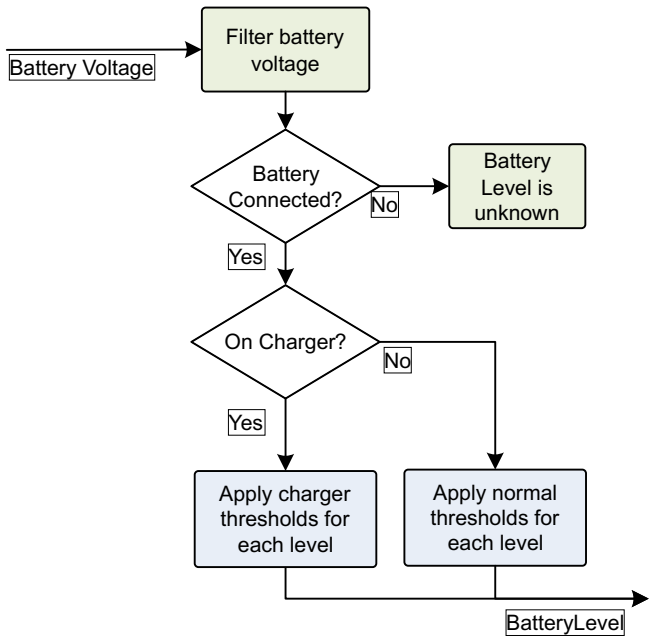


Figure 9: *The battery level classification logic*

55.4. FAULT STATES, ERRORS AND SELF-PROTECT

Other battery errors (these can happen concurrently) include:

- *Overheated*: The battery temperature is too high. To self-protect, the system should shut down and not perform any charging.
- *Over current* or *short circuit*: A short circuit has been detected in the battery. To self-protect, the system should shut down and not perform any charging.

The application may react to battery fault conditions overcurrent, under/overvoltage, and temperature extremes to protect the user and battery.

- The electronics take precedence, and may disconnect the battery at any time, without warning.
- The software may, based on the fault/error states of charge and other battery errors, etc. disconnect the battery (to self-protect).

55.5. BATTERY HEALTH AND REPLACEMENT

The battery's quantitative health values – in conjunction with the state of charge enumeration and values, and fault states & errors – are used by the application. These may drive service notifications, such as advising to replace the battery. The health values include:

Health Measure	Description
Capacity (max)	The maximum battery capacity when fully charged. This decreases as the battery ages. **Units:** Coulombs (C) or milliamp-hours (mAh)
Capacity (original)	The original capacity of the battery when it was new (i.e. time of manufacture). **Units:** Coulombs (C) or milliamp-hours (mAh)
Health	The current health of the battery as a percentage of its original capacity. This may be calibrated for the battery chemistry. **Units:** percent (%)
Storage State of Charge for Layup	The system should track (and encourage) placement of the battery into a recommended state-of-charge for seasonal or long-term storage ("layup") to minimize calendar aging. **Units:** percent (%) target; last stored value; layup timer
Usable Capacity	The amount of the battery capacity (when fully charge) that can be practically used; this excludes an amount for low state of charge (to protect the battery), and lower max capacity (to reduce charge time by reducing the 'top off charge') **Units:** Coulombs (C) or milliamp-hours (mAh)

Table 42: *Battery quantitative health values*

Note: these require a fuel-gauge feature to track the capacity and state of charge of the battery.

For lead-acid batteries, health metrics should extend beyond capacity and state-of-charge:

Health Measure	Description
Equalization Events	Tracks when the battery has undergone an equalization charge (overcharge to balance cells and reduce sulfation) and intervals between these events. **Units:** count of events, time since last equalization
Sulfation Indicator	Diagnostic indicating evidence of chronic undercharge or extended inactivity, which may degrade capacity. **Units:** Severity score, or flag
Terminal Corrosion Index	A qualitative or semi-quantitative assessment of corrosion at battery terminals. **Units:** measured resistance (mΩ) or qualitative (e.g. none, minor, moderate, severe)
Watering Interval	The elapsed time or number of charge/discharge cycles since the last electrolyte replenishment (watering event). Timely watering is required for longevity and safety. **Units:** days or cycles

Table 43: *Lead-acid battery health measures*

Note: These measures enable the power manager to advise or automate service reminders (watering, desulfation, terminal cleaning, seasonal storage prep) and can drive safety-critical interventions. Equalization and watering metrics mainly apply to lead-acid chemistries, while detailed SoH/SoC and lay-up preparation apply chiefly to lithium chemistries.

[Todo add health thresholds to the battery]

55.6. CHARGING PARAMETERS

The charger goes through different states as it charges the battery. Each state pulls a different amount of current from the charging pads and treats the battery differently.

charging states

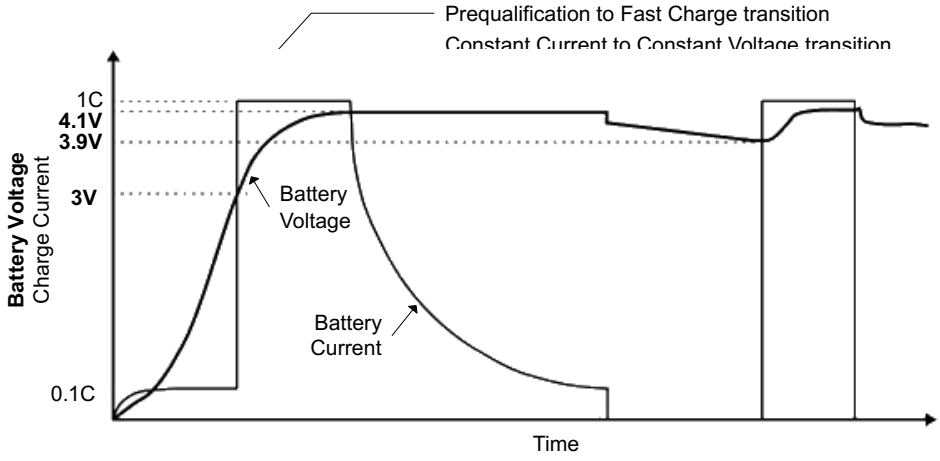


Figure 10: *Charging profile* (adapted from Texas Instruments)

The basic idea is that the charger first applies a low current to the battery to bring it up to a threshold; this is called *prequalification* in the diagram. Then it applies a high current, called *constant current*. Once the battery voltage has risen to a threshold, the charger switches to *constant voltage,* and the current into the battery tapers off.

constant current
constant voltage

Battery charging parameters are intended to tune for fast battery charging while prolonging the battery service life:

Parameter	Description
Charge Target Voltage	The target voltage for the battery, at which point it is intended to cease charging. **Units:** mV
Constant Charge Current	The current used during the pre-charge phase. **Units:** mA
Pre-Charge Current	The constant current used during the main charging phase. **Units:** mA

Table 44: *Battery charging parameters*

Might also have temperature thresholds and duration to cool down

55.7. CHARGER STATE

The charger state is often represented as combination of enumerative and quantitative values. The battery charger may be in one of the following states. (Note: the charger electronics typically controls the states independent of software.)

State	Description
Not Charging	The charger is not actively charging the battery.
Pre-Charge	The charger is in pre-charge mode, typically used for deeply discharged batteries. Sometimes called *zero-volt charging*.
Constant Current	The charger is in constant current mode, the main charging phase. Sometimes called *fast charging*.
Constant Voltage	The charger is in constant voltage mode, the final charging phase.
Trickle charge	The charger is in trickle charge mode, maintaining a full charge. Sometimes called *top-off charging*.
Fault	The charger has encountered an error or fault condition. This may include an open or short with the charger circuit, checksum failure of the non-volatile configuration or it has reached a thermal limit and needs to cool.

Table 45: *Battery charger states*

The charger state has measured quantities such as:

Parameter	Description
Charging voltage	The voltage of the charger applied to the battery. **Units:** mV
Charger temperature	The current temperature of the charging circuit. **Units:** °C

Table 46: *Battery charging quantitative state*

Charing current	The current inflow into the battery. **Units:** mA

Might also have times/durations, such as time in a state or time to cool down

55.8. THERMAL MANAGEMENT

Battery heat is the most significant source of battery "aging," reducing its effective service life. The design often is made to promote cooling or prevent the cells heating.

55.8.1 Passive Cooling

There are several passive approaches suitable to all products to increases the battery service life:

- Physically separating and isolating the battery from the power electronics (e.g. motor drivers, inverters, and charger FETs), and the heat generated.
- Large, flat, thin batteries have a high surface area to volume ratio allowing them shed heat faster, reducing the internal heating from charging and heavy loads. This form factor usually has a low internal resistance, reducing the internal heating and allowing it to usefully deliver higher currents without resulting in a brown-out.
- Mount cells on metal (e.g. aluminum or copper) plates and frames that conduct heat to an outer shell or to a liquid channel.

55.8.2 Active Cooling

Large battery packs – such as tool packs, vehicles, UPS banks, and home storage – often need more sophisticate cooling. These packs are expected to have high discharge and fast charge (yet charging takes longer due to size). To move heat away from cells and charger circuitry:

- Fans (forced air) blow air across fins, plate edges, or ducted channels inside the pack. These are more common on larger tool packs, server UPS modules, and some home-storage enclosures. Fan failure or blocked vents reduce effective cooling. This may be detected using tachometer feedback, airflow proxies, or rising cell temperature under known load.
- Liquid coolant circulated through cold plates or jacketed modules; common in electric vehicles and power grid, occasionally in high-power stationary storage. Adds pump, reservoir, and leak-detection serviceability concerns but handles sustained high charge and discharge rates.
- Heat exchangers (such as metal plates, fins, or tubing) transfer heat from battery cells or coolant loop to the environment.

55.8.3 Charging management

High recharge rates internally heat the cells, causing them to age (deteriorate). Typically, temperature is monitored, and cool down periods are used when the temperature exceeds a threshold. When cell or charger temperature crosses a warning threshold, firmware may reduce charge current, pause charging for a cool-down interval, or derate discharge – patterns already described for charger fault and over-temperature states.

The charger may also run fans to cool the batteries, a feature of larger (and more expensive) battery packs. This active cooling extends the operating envelope, allowing fast charge to resume once temperatures fall.

55.8.4 Serviceability

Log time above temperature bands, fan run hours, and cool-down events; sustained operation near limits often predicts accelerated capacity loss (state of health decline) before a hard fault appears. Record which cooling path is active, whether fans or pumps report fault, and whether thermal limits are currently constraining charge or output power. A pack that charges slowly in an otherwise cool room may indicate a failed fan, blocked vent, or degraded thermal interface.

56. RESOURCES AND REFERENCES

Advanced Power Management (APM) (https://en.wikipedia.org/wiki/Advanced_Power_Management)

Battery University, *Charging at high and low temperatures (thermal limits, cool-down, and charge derating)* https://www.batteryuniversity.com/article/bu-410-charging-at-high-and-low-temperatures/

Power Distribution Unit (PDU) (https://en.wikipedia.org/wiki/ Power_distribution_unit)

Power Management Unit (PMU) (https://en.wikipedia.org/wiki/Power_Management_Unit)

CHAPTER 14

Storage

This chapter discusses the storage subsystems that support serviceability, including how they are organized, monitored, and erased.

- Data integrity, and error handling
- Data partitioning for serviceability
- Erasing and sanitizing data for repair, return, and refurbishment
- Storage interconnects: adapters, buses/links, timeouts, aborts, and path health
- Health indicators, thresholds for media reliability, interconnect and other management information
- Troubleshooting

57. OVERVIEW

Storage management monitors the status and health of the underlying media, and how do we service or replace it.

Tier	Common features
Low-cost consumer	Limited or no error checking, usage or storage health tracking; minimal diagnostics, simple reset/reformat workflows.
Premium consumer	Basic usage counters (e.g., power-on hours), simple storage health (such as SMART status).
Contractor / professional	Supports secure elements for storage (e.g., TPMs), device-level protection features (e.g., one-time programmable fuses, ROMs) Enhanced monitoring for storage health events (e.g., bad-block scans), structured replacement workflows.
Enterprise / industrial fleets	Storage fault tolerance mechanisms, such as redundant storage (e.g., RAID, dual storage modules), with removable or hot-swappable storage modules to maximize availability and serviceability. Encryption and other security; secure erase procedures for safe removal and replacement. Higher endurance media (for extended storage life); scheduled and predictive maintenance.

Table 47: *Serviceability tiers for storage*

In the large – more complex systems – is a layered stack consisting of storage management, media management, drivers, and the underlying hardware and physical media:

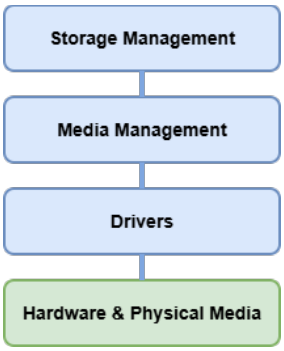


Figure 11: *The storage subsystem*

The layers:

Layer	Description
Block layer / I/O scheduler	Queues, merges, and reorders requests to the device.
Device / controller cache	Volatile or battery-backed DRAM on the drive or HBA. Whether write-through is forced is often a per-request attribute.
Media	Flash or magnetic storage, with its own internal caching and write amplification.

Table 48: *Storage system layers*

57.1. STORAGE MANAGEMENT

storage manager

The *storage manager* is a component or module responsible for orchestrating the use and allocation of available storage resources. Its responsibilities include:

- Provide a logical abstraction such as a file system to the application layer. This often includes protocol-like layers to use well-defined storage formats.
- Allocate storage space according to application, system, and policy requirements. It manages logical storage constructs (volumes, pools, or file-systems) that may span multiple physical devices; it may rely on hardware controller or other subsystems to perform this.
- Enforces permissions, quotas, or access rights on media and devices, and managing secure access to storage.
- Monitor the operational status and health indicators of the storage resources, reporting anomalies or predictive failures. This integrates the media state, as well as other factors from the storage system, such as reduced redundancy, errors, latency, and so on.
- Data Protection: Facilitating backup, snapshot, replication, and restore operations to ensure data resiliency and recoverability.
- Implementing data migration between storage tiers, according to *hierarchical storage management* (HSM) policies.

57.1.1 Hierarchical storage management (HSM)

hierarchical storage manager

Hierarchical storage management is a policy-driven approach placing data on different storage tiers according to cost, performance, access frequency, and retention requirements:

- Online storage – SSD, flash, primary disk
- Near-line storage is removable media, but cannot be managed without operator involvement – such as with tape libraries, optical jukeboxes, and other loaders
- Offline storage where an operator is required, such as to load a tape, disc, or other media.

Data may migrate "down" the hierarchy as it ages or becomes cold and restore "up" when it is accessed again or when policy requires it.

57.2. MEDIA MANAGER

media manager

Beneath the storage manager is the media manager. The *media manager* is responsible for providing a consistent interface to storage devices and/or their media. It monitors the status and health of the underlying physical media, and the devices that hold or access it.

- Detect and catalog all accessible storage devices and media, whether fixed or removable. It handles identification (and validation) of a storage device or media, allowing them to be moved across location. (This doesn't necessarily validate the contents)
- Continuously assess the operational status and health indicators of storage resources, reporting anomalies or predictive failures.
- Track the age, wear, capacity utilization, and service history of storage media; issue alerts for media approaching end-of-life or requiring replacement.

Drivers bridge with adapters – which may provide interfaces for specific technologies, such as SCSI, or IDE. Drivers can also interact with media loaders, managing and tracking available resources, the locations of media within loaders, and the health status of each media asset.

57.3. TYPES OF STORAGE DEVICES AND MEDIA

The storage devices are the physical peripherals, including host adapters. They have a gradation of attachment and accessibility:

Accessibility	Example devices:
Embedded non-removable storage	NOR/NAND flash chips, EEPROMs, NVRAM, SRAM, *universal flash storage* (UFS) secure elements, trusted platform modules (TPMs), one-time programmable fuses, ROMs
Internal, removable storage	hard drives, SSDs, eMMC, *universal flash storage* (UFS)
External	hard drives, SSDs At enterprise level in drive bays and racks for serviceability RAID or similar schemes for availability. tape drives (tapes, as well as audio/video or legacy data cassettes where still used), optical drives (for CD, DVD, Blu-ray discs), Automated loaders/changers, devices that hold banks of removable media in slots or magazines and use a robotic picker or shuttle to mount media into drives (tape

Table 49: *Roles of different types of storage system accessibility.*

	libraries, optical jukeboxes, multi-slot changers).

Some storage units are sophisticated subsystems. SSD's are the later. The addresses are logical, and internally there is management of bad block lists and some automatic remapping of blocks. They include an internal garbage collection process to reclaim blocks by moving valid data and erasing old blocks. During heavy garbage collection, write operations can slow down and latency can spike (especially write latency), because the drive is doing extra background work.

57.3.1 Working memory modules: SPD and XMP / EXPO

DIMMs and similar DRAM modules carry a small non-volatile *Serial Presence Detect* (SPD) EEPROM. The SPD holds identity and timing metadata the firmware or memory controller reads at boot: manufacturer, part number, density and organization, JEDEC baseline timings and voltages, and optional performance profiles.

JEDEC baseline is the guaranteed interoperable timing/voltage set. XMP (*Extreme Memory Profile*; Intel) and EXPO (AMD) are alternate profiles stored in SPD that raise frequency and tighten timings (often with a higher DRAM voltage). Firmware selects which profile to apply, i.e. JEDEC default, an XMP/EXPO profile, or a manual override.

57.4. STORAGE ORGANIZATION AND PARTITIONING

It is good practice to partition into different areas for software, logs, crash reports, configuration, and user content, etc. This is based on the different security levels, whether the region is modifiable (and when), and selective erasure (or preservation) during servicing. For instance, the core executable software may be protected with read-only, securely signed and are checked for authenticity; it would change only during OS updates, not by the user. The user has their own data they can change – or erase at will, such as when preparing the device for resale.

57.5. STORAGE INTERCONNECTS

Every command, write and read (eventually) must go through an interconnect: the host adapter (HBA), bus or fabric (SCSI, IDE/ATA, SATA, SAS, NVMe/PCIe, USB, Fibre Channel, etc.), cables, connectors, expanders or bridges. The system buffering can hide short bursts of path latency but not stuck queues, flapping links, or chronic CRC/parity errors. When diagnosing a "slow disk" always ask whether the path is healthy. See Chapter 14 for guidance on checking interconnect health.

The following are some symptoms that may occur when the interconnect path is not healthy:

- *Timeouts* occur when an IO is aborted rather than letting writes or reads block indefinitely. Timeout storms with a full dirty list are often caused by the link or device failing to complete operations. Timeouts can result in increased latency and retries on the interface.
- Adapters may freeze a device queue on error and require explicit recovery actions such as issuing a sense, reset, unfreeze. Drivers should not freeze due to dirty-aborted requests. Bus or device resets may invalidate in-flight work, forcing cache invalidation or replay.

Host, HBA, or RAID controller caches may acknowledge a write *before* it has been committed to media. With battery- or capacitor-backed cache, write-back is often acceptable; without it, prefer write-through for durability-critical commits. When diagnosing latency or data-loss after power failure, check cache policy and battery/capacitor health as well as the interconnect and media.

57.6. MEDIA OPERATIONS

Table 50: *Actions for a storage device*

Operations	Description
Authenticate	Read media ID and authentication of an individual item.
Eject	Physically remove a media from a drive, port, or slot, typically making it available for manual retrieval. For removable drives (e.g., optical/tape), this may trigger mechanical ejection. For virtualized or software-controlled media, 'eject' may mean 'unload' – logically detaching and making media inaccessible.
Load	Prepares a media for access (read/write/scan operations). It may request the operator to load an item into storage; with an automated it loader, the work flow involves pick and mount; potentially requested import.
Mount	Make media accessible to a system by inserting and initializing it in a drive, and/or logically attaching a file system or image. For file-based media (e.g., disk images), mounting makes the data accessible at a mount point.
Retension tape	Actions to prepare cassette, exercise tape, retension it.
Secure Erase	Overwrite media contents using approved sanitization methods (per policy/standards) to prevent data recovery.
Status check	Report hardware health, error codes, sensor states, and maintenance due.
Unload	Make media inaccessible by logically detaching the associated file system or logical volume; the volume is not necessarily made inaccessible. This makes the media available for export/ejection, manual removal or (with a loader) return it to a designated storage location (slot, magazine).
Unmount	Make logically detaches the associated file system or logical volume; this often makes the media eligible for unloading the drives (see Unload/Eject for hardware action). Ensures data integrity before removal.
Update configuration or firmware	Apply new firmware or loader settings (as permitted).

58. STORAGE INTEGRITY AND ERROR HANDLING

Storage devices degrade with temperature, use and shock, and many will wear out over time, requiring replacement. Fault tolerance mechanisms detect and recover from data and file-system corruption, esp. as the storage device ages. This includes the following features:

Media tests and scrubs are periodic bad-block scans, read scrubs, and short/extended self-tests to detect weak or failing areas of the media before they cause uncorrectable errors.

Wear leveling is a preventative technique that extends the usable lifespan of flash memory. It distributes erase cycles evenly across all memory blocks to prevent premature wear of any individual block.

Error detection such as *error correction codes* (ECC), CRCs, checksums, and periodic scrubbing automatically identifies data corruption during read/write operations.

Error correction recovers data from partially corrupted blocks.

Bad block management maintains a list of known bad blocks. When an bad block is identified (such as from the error detection above), it is added to the list to protect against potential data loss. A recovery attempt is made, and the faulty sector or block is remapped to the recovered one.

Over-provisioning with spare capacity prolongs device life by allowing them to perform these wear-leveling and bad block management techniques. Drive sparing – reserving spare drives (in RAID or similar arrays) – and spare blocks/pages within a drive allow data to be rebuilt or remapped onto the spare when a drive or region fails (or is predicted to fail).

59. ERASING AND SANITIZING DATA

This section outlines approaches and design considerations for erasing and sanitizing data. If the system design segregates data onto different media or partition based on security and lifetime, selectively erasing a specific partition of the storage device is a good way to erase data for product returns, refurbishment, and resale. It allows removing PHI/PII by erasing user and clinical data partitions while preserving system, calibration, and lifetime safety counters.

Table 51: *Erase modes*

	Secure	Description
Soft erase	No	A logical deletion of data; quick and easy.
Secure erase	Yes	A vendor-provided or standards-based erase of non-volatile media.
Crypto-erase	yes	Destroying or rotating encryption keys so encrypted data becomes unreadable.

60. MANAGEMENT INFORMATION AND DIAGNOSTICS

Management information about the storage device includes the following:

- The type and configuration of the storage device (interface, geometry, partitions)
- The capacity of the storage device and remaining spare blocks, and any configured spare drives in arrays
- The health of the storage device – e.g., SMART/NVMe health status, recent media tests or bad-block scans, correctable and uncorrectable error count, wear-level indicators, etc.
- Backup and restore status, the last successful backup time, last failed attempt, and the scope of data included in backups (e.g., user content and configuration but excluding certain calibration or safety logs)

- For working memory modules: SPD identity (vendor, part, density), SPD readability, JEDEC baseline vs enabled XMP/EXPO profile index, configured DRAM voltage and frequency, and DDR training outcome

These values are typically exposed through the system report or diagnostics tools and can be used by customer support and field service to troubleshoot emerging storage issues.

60.1. SUSBYSTEM STATE AND METRICS

Drivers, HBAs, and storage devices expose operational state so service can tell a stalled queue from an offline or failed unit.

Table 52: *Storage subsystem state*

Parameter	Description
Buffers available	Number of I/O buffers (or descriptors) available for use. **Units:** count
Buffers used	Number of buffers that are currently in use. **Units:** count
Device / media state	Operational condition: present, online, offline, degraded, failed, or recovering. **Units:** status
Memory profile in use	JEDEC baseline vs XMP/EXPO (or OEM) profile index selected by firmware **Units**: enum / index
SPD readable	Whether each installed module's SPD EEPROM responded with valid identity and timing tables **Units:** status (per slot)
Write cache / dirty state	Whether write-back cache holds uncommitted data (when reported). **Units:** status; bytes or count

Table 53: *Storage subsystem metrics*

Parameter	Description
Resource allocation failures	Number of times that a buffer, descriptor or other resource could not be allocated. **Units:** count

60.2. METRICS, ERROR RATES

Statistics in the context of storage are a collection of counters and metrics maintained by drivers or storage devices to monitor activity. These statistics help in diagnosing issues, tuning performance, understanding the behavior of the software and hardware, and abstracted into a health indicator. These metrics and other subsystem signals can be trended over time to detect degrading media conditions.

Common statistics include:

Table 54: *Example storage metrics*

Parameter	Description
Abort rate (clean / unclean)	Cancelled before vs after device issue. **Units:** count/sec
Adapter error / hot-plug events	Device failure, unload, path loss. **Units:** count
Double-buffer events	Extra copies forced by scatter-gather/alignment limits. **Units:** count/sec
DDR training duration	Total time spent during DDR training **Units:** milliseconds (ms)
Direct I/O volume	Bytes transferred bypassing the elevator. **Units:** MB/s, %of total
Elevator effectiveness	Merge ratio; avg transfer size on the wire. **Units:** ratio, KB
Elevator depth / threshold	Requests held before reorder/merge **Units:** count
IO in/out rate	Blocks move to/from the storage device per unit time. **Units:** Blocks/sec, Operations/second (IOPs)
IO read rate	Rate of completed read operations over time; used to detect workload changes and throughput constraints. **Units:** Operations/second (IOPs)
IO write rate	Rate of completed write operations over time; useful for identifying sustained write pressure and potential wear/latency impacts. **Units:** Operations/second (IOPs)
Merge ratio	Fraction of logical I/O operations combined into a single device request, indicating the effectiveness of request merging in reducing transport-level I/O. **Units:** ratio
Overall IO latency (Average)	Average end-to-end completion times across both read and write I/O requests over the reporting interval. **Units:** milliseconds (ms)
Overall IO latency (Max)	Maximum end-to-end completion time across read/write I/O requests in the reporting interval. **Units:** milliseconds (ms)
Overall IO read/write rate	Combined rate of completed read and write I/O operations over time, useful as a top-level load indicator for capacity and saturation monitoring. **Units:** Operations/second (IOPs)

Overall I/O size average	Mean size of IO transfers, after elevator. **Units:** KB/MB
Overall Throughput	Combined read + write data volume per unit time, top-level indicator of storage bandwidth utilization. **Units:** MB/s (or GB/s)
Queue freeze / bus reset count	Path recovery events. **Units:** count
Read latency (average)	Mean time to complete read requests; rising values can indicate queuing, media stress, or controller contention. **Units:** milliseconds (ms)
Read latency (max)	Maximum time to complete read operations in the interval; highlights worst-case stalls and tail-latency events. **Units:** milliseconds (ms)
Read Throughput	Sustained data volume read per unit time; useful for bandwidth planning and detecting read-path bottlenecks. **Units:** MB/s (or GB/s)
Service time	Completion time, or End-to-end I/O latency (device) **Units:** ms (p50, p95)
Total bytes written	Cumulative amount of data written to the device. **Units:** bytes
Write latency (average)	Mean time to complete write operations; useful for tracking write amplification, GC impact (SSD), and sustained write pressure. **Units:** milliseconds (ms)
Write latency (max)	Maximum time to complete write operations in the interval; captures worst-case write stalls and QoS violations. **Units:** milliseconds (ms)
Write throughput	Sustained data volume written per unit time; useful for bandwidth planning and detecting write-path bottlenecks. **Units:** MB/s (or GB/s)

See NTFS.com for many more interesting metrics.

Note: there are *many* more potential measures that are specific to hardware that can be abstracted into the health, such as head-seek time, seek errors, spin up time, etc.

60.3. HEALTH INDICATORS AND ERROR RATES

Storage health has three related views. The same symptoms (deep queues, rising latency, timeouts) can come from any view. Read them together rather than in isolation:

- Capacity and spare headroom: Is there adequate free space and spare capacity?

- Utilization and contention: Is the demand exceeding what the path can deliver, or is one consumer starving others?
- Media error rates and wear: Is the medium failing or wearing out?

60.3.1 Capacity and spare headroom

Capacity metrics track whether the storage system is running out of usable space or rebuild/remap headroom. These are distinct from media health, such as wear-out though both can force service. Capacity exhaustion and contention often appear together: a nearly full volume or exhausted SSD spare raises latency and queue depth under the same I/O rate.

Table 55: *Storage capacity metrics*

Health Measure	Description
Available spare (SMART / NVMe)	Vendor "available spare" or similar attribute relative to the critical threshold; complements bad-block and reallocated-block health indicators. **Units:** %
Capacity growth rate / time-to-full	Trend of used space over time; estimate when free space will cross policy thresholds under current duty. **Units:** bytes/day or days remaining
Hot-spare status	Whether a RAID or similar array still has an unused spare drive; running without a spare reduces ability to absorb the next failure. **Units:** status, count of available spares
Log / scratch / spill partitions	Free space on partitions reserved for logs, crash dumps, or spill files; filling these often breaks serviceability (no room for diagnostics) before the data volume is full. **Units:** bytes, % free
Over-provisioning / spare-block remaining	Spare capacity left for wear leveling and remapping. Dropping spare blocks increases write amplification and latency. **Units:** %, count
Rebuild / resync progress and ETA	During degraded operation, percent complete and estimated time remaining; long rebuilds extend the window of reduced redundancy. **Units:** %, time
Snapshot / thin-provision over-commit	Provisioned vs written capacity for thin provisioning volumes and snapshot chains; over-commit can exhaust the pool while each volume appears under quota. **Units:** bytes, % over-commit
Usable capacity remaining	Free space on volumes, pools, or file-systems relative to total usable size. Warn before absolute full so backups, logs, and updates can still complete. **Units:** bytes, % free

60.3.2 Utilization and contention

High utilization and contention show up when the storage path cannot keep up with demand. Trend these alongside the I/O rate and latency metrics; sustained saturation with rising tail latency is a serviceability signal, as well as a performance one.

Parameter	Description
Command timeout rate	Frequency at which issued commands to the storage device exceed their allotted timeout interval and are aborted. Timeouts and unclean aborts under load often mean a hung or overloaded path, or indicate device, path, or controller issues. **Units:** count/sec
Controller or HBA CPU / queue saturation	Host-bus adapter or RAID controller busy time and internal queue fill; many drives may look "slow" when the shared controller is the contended resource. **Units:** %, count
I/O wait / blocked time	Fraction of time tasks spend waiting on storage rather than computing. Elevated wait with high queue depth points to contention rather than CPU-bound work. **Units:** %, seconds
IOPS vs rated IOPS	Sustained operations rate near device or array limits; small-block random workloads often hit IOPS before bandwidth.. **Units:** IOPS or % of rated
Latency under load (average and max / p95)	Rising average with a growing max or p95 while the workload is steady indicates queueing, GC pressure (SSD), or controller contention. Pair with Read/Write Latency metrics. **Units:** %, seconds
Queue depth / Outstanding I/O	Commands in flight per device/adapter. **Units:** count
Throughput vs rated bandwidth	Sustained read/write (or aggregate) throughput near the media, link, or controller ceiling. Useful with Overall Throughput and Read/Write Throughput from the metrics table. **Units:** MB/s or % of rated

Table 56: *Storage utilization and contention metrics*

Contention among consumers (multiple volumes, virtual machines, or tenants on shared media) typically shows as one consumer's IOPS/latency spikes while others starve or a noisy neighbor drives queue depth for the pool.

60.3.3 Media error rates and wear metrics

The metrics for storage are typically derived from device reporting (e.g., S.M.A.R.T. or NVMe health checks), and periodic media tests, e.g. background scrubs and bad-block scans. The units have the common health indicators and environmental exposure tracking (see chapter 8), temperature and power measurements (see chapter 12.). The device specific metrics may include:

Health Measure	Description
Bad-block and ECC rate slope	Rate of bad blocks and ECC errors increasing over time. **Units:** count per time period
Charge pump erase count	Total number of times a charge pump was used to erase pages or sectors. **Units:** count

Table 57: *Example storage health indicators*

DDR training retries	Number of DDR training retries required **Units:** count
DDR training success rate	Percentage of successful DDR training operations **Units:** %
Error correction count	Total number of error correction (ECC) actions performed. **Units:** count
Pending sector count	Number of sectors currently pending reallocation due to read/write errors. **Units:** count
Program/erase counts per block	Number of times each block has undergone a program/erase cycle. **Units:** cycles
Read errors (cumulative)	Total number of times where data was lost due to retention failures. **Units:** count
Read error rate	Number of data read operations resulting in errors. **Units:** count per time period
Reallocated block count	Total number of blocks that have been reallocated due to being found unreliable. **Units:** count
Refresh/scrub success	Percentage of successful refresh or scrub operations. **Units:** percent (%)
Wear-level counts	Indicator of the distribution and uniformity of wear across storage blocks. **Units:** Count or index

60.4. ABSTRACT HEALTH INDICATORS

Health indicators fuse the earlier counters, rates, and measurements in signals that policies can employ. They do not replace the capacity, utilization, or media metrics. They are normalized, trended and combined so one noisy metric does not dominate.

Indicator	Patterns
Consumer contention	One volume, VM, or tenant's IOPS/latency spikes while others starve, or a noisy neighbor drives queue growth.
Corrective pressure rising	ECC correction growth; bad-block / ECC rate slope; pending or reallocated sector/block counts rising above baseline.
Interconnect / controller saturated	High bus, link or HBA/RAID controller utilization with moderate queue depths.
Media failed	Persistent uncorrectable errors; repeated self-test failure; RAID degraded without redundancy.

Table 58: *Abstract storage health indicators*

Path recovery unhealthy	Command timeout and unclean-abort rate elevated; queue freeze or bus reset under load, without media-error signature.
Path saturated (oversubscribed)	Queue depth pegged; IOPS or throughput near rated; latency under load (average and p95) and I/O wait elevated while demand grows, without a matching media-error signature.
Scrub / refresh unhealthy	Refresh or scrub success rate declining at steady conditions.
Serviceability space low	Log, crash-dump, or spill partitions near full.
SMART / NVMe health declining	Vendor health summary trending toward failure.
Thin provisioning / snapshot over-commit	Provisioned much greater than written; pool pressure while each volume still looks under quota.
Usable space low	Free capacity or time-to-full crossing policy.
Wear / endurance high	Program/erase cycles, wear-level uniformity, or charge-pump/endurance counters approaching limits.

60.5. POLICIES AND THRESHOLDS

Use thresholds and policies against health indicators to drive consistent service actions, such as warn, schedule service, or retire a device.

Issue a *warning* when:

- The free space or number available spares crosses the early threshold; plan cleanup, restrain thin provisioning, or expansion.
- When the correctable errors (e.g., ECC correction , bad-block, or pending sector) count trend deviate from baseline. Schedule closer monitoring or a scrub/self-test.
- If there is consumer contention, request the operator isolate the noisy volume/virtual machine/tenant.
- If the path, interconnect, or controller is saturated, reduce load or move work; confirm the bottleneck is not media errors.
- If the path recovery is unhealthy, check interconnect and adapter logs.
- If the thin provisioning / snapshot over-commit approaches policy, stop unchecked provisioning growth.
- If the serviceability space is low, reclaim log/crash/spill space before diagnostics fail.

Escalate to a *service notification* when these metrics become more serious:

- If the free space or number available spares crosses the threshold; notify to expand capacity, migrate data, or schedule drive/array service before absolute full or spare exhaustion.

- If the serviceability space is low, expand or repartition log/crash/spill space so diagnostics and dumps can complete; schedule service if the operator cannot reclaim space.
- A RAID array is running without an active hot spare. Require action such as adding drive to reinstate spare capacity.
- If the corrective pressure is rising, the scrub / refresh is unhealthy, or SMART / NVMe health declining continues after the warning window. Schedule drive or array service.
- If the wear / endurance is high, schedule retirement or replacement before hard failure.
- If the usable space is low or over-commit persists after operator cleanup, expand capacity or migrate data.
- If the path is saturated or path recovery is unhealthy persists after load reduction, perform path/controller diagnostics.
- Retire or replace drives if critical health metrics (such as persistent uncorrectable errors, rapid increase in bad blocks, or failure of SMART/NVMe health checks) exceed limits. These signify media degradation, and indicate failure is imminent.

A fault is raised for serious issues – when component health has deteriorated to where continued operation is a risk. When:

- Uncorrectable errors persist, drives repeatedly fail self-tests, or RAID arrays degrade without redundancy; advise urgent service, limit destructive operations. Where supported by the system, switch to a degraded/safer mode until hardware is replaced.
- A storage unit, such as a drive, has failed, remove from service; fail over or rebuild if redundancy remains.

60.6. NOTE

Not all changes in performance are signs of wear damage: healthy systems do slow with use. This component of slowdown is due to data structure performance – the classic big-O classification.

61. SERVICE TOOLS

Typical service tools and interfaces for storage devices include:

Reporting and Status Utilities: Tools and utilities (such as manufacturer-provided dashboard applications or command-line tools) allow users to view storage device status, health summaries, error logs, and configuration details. Most drives implement SMART (Self-Monitoring, Analysis, and Reporting Technology) functionality; and NVMe drives expose detailed logs, such as health and error Information, in addition to SMART info.

On-Demand Diagnostic Tests

- Many drives support built-in self-tests (e.g., short/long diagnostics, read/write surface scans), ability to update the firmware. These often requires a software tool from the vendor. The vendor may have also supported storage controllers to access these features.
- RAID and Storage Controllers provide their own service interfaces for viewing array health, failed drives, rebuild status, and predictive failure events.

Remote Management Interfaces: For datacenter and enterprise storage, remote monitoring interfaces (such as SNMP, REST APIs, or proprietary dashboards) can aggregate health and error events across fleets of drives.

61.1. PHYSICAL LAYER TEST TOOLS

Physical layer test tools are specialized tools to assess the integrity and performance of the underlying signal paths to a storage device. These testers are used on electrical cables, connectors to identify issues like opens, shorts, poor contacts, crosstalk, or signal degradation.

- Memory (DIMM) testers can validate DDR/DDR4/DDR5 DIMMs, isolating marginal modules, perform burn-in testing, and capturing error logs for RMA/repair decisions. These employ controlled patterns, such as address/data line faults, timing/refresh behavior, and error-rate under load. Pair with SPD/XMP readers (or firmware setup) to confirm profile metadata matches the timings under test.

SATA and NVMe testers validate and diagnose data/power connections, as well as analyze signal integrity and device health. Common testers include:

- *Power Supply testers* verify that the SATA or NVMe connection delivers the correct voltages (to the drive). Out-of-spec or missing voltages that could lead to drive malfunction or start-up issues.
- *Cable and Continuity testers* check individual data or power cables for electrical continuity and wiring integrity. They help detect faults such as opens, shorts, or mis-wired conductors.
- *Compliance and Signal Analyzers* connect directly to data paths to check compliance with specifications. These assess bit error rate, eye patterns, signal integrity, timing, and protocol compliance. These are professional test equipment primarily for hardware development, and validation.

61.2. STORAGE PROTOCOL TEST TOOLS (LINK LAYER)

Protocol testing tools used to verify, diagnose, and validate the operation and interoperability of storage device interfaces and protocols. Many are intended for design tested, to validate conformance, and improve interoperability between hosts, storage devices, and controllers at the protocol layer. The types of tools include:

Table 59: *Storage link layer tools.*

Type of Tool	Description
Protocol Analyzers	These are specialized hardware tools capture and decode low-level command and data exchanges on storage buses. These troubleshoot timing, protocol errors, command/response sequences, and compliance with protocol standards (e.g. SCSI, SATA, NVMe).
Bus Protocol Exercisers and Traffic Generators	These tools simulate initiator and target behaviors, sending protocol-compliant or intentionally malformed requests to test device or system responses under various conditions.
Compliance and Certification Test Suites	Many standards have protocol compliance test suites and checklists, sometimes required for product certification. Official ones are often provided by the industry standard organizations (e.g., NVM Express, SATA-IO, SCSI Trade Association).
Drive test stations	interface directly with SATA hard drives or SSDs. They test drive health, run surface

	diagnostics, measure performance, validate data transfers, and may allow viewing of SMART information or logs outside a host system. These are primarily used by repair and QA environments to quickly screen or refurbish drives.

Note: this document is not about design testing. Rather about testing to identify if no longer meets design specification.

62. FAULT FINDING AND TROUBLESHOOTING

Effective troubleshooting requires a structured approach that addresses common failure modes, leverages built-in diagnostics, and uses available error logs and reporting data.

The core troubleshooting workflow is:

1. Verify connections and power. Look for damaged connectors or bent pins. Check that the data and power cables are firmly connected; try a different cable or connection point.
2. Confirm that the host system recognizes the device. If not detected, test the storage device in a tester, or a known working system to isolate the issue. For RAID or network-attached storage, ensure controllers & network connections are healthy.
3. Review system and device logs. Look for storage errors, disconnections, or timeout messages; reallocated/bad sectors, pending or uncorrectable sector counts, sudden increases in read/write errors, temperature or power cycle anomalies.
4. Run diagnostic tests. On complex controllers, run array and member disk diagnostics, check for degraded or rebuilding states.
5. Assess performance and symptoms. Slow performance or pauses can indicate physical issues, excessive write amplification, or pending media failure. Frequent file-system checks or “drive needs to be repaired” warnings indicate pending media or controller failure.
6. Check for environmental issues. Overheating, vibration, shock, or high humidity can accelerate storage failures. Review environmental sensor and exposure logs if available.
7. Isolate and replace if needed. If diagnostics point to device-level failure, replace the drive, back up salvageable data first, if possible. For intermittent problems, swap cables/ports and repeat tests to identify the faulty link. In RAID, follow the controller’s procedure for replacing and rebuilding failed drives.
8. Restore and Monitor. After repair or replacement, restore data from backups if necessary. Continue monitoring SMART and error statistics for early signs of recurrence.

62.1. RAM

For triage of no-boot, training failure, or ECC storms, fall back to JEDEC baseline first. Confirm SPD readability. Retest with a known-good module if instability persists.

After DIMM replacement, confirm the new module’s SPD is readable and that the intended profile (JEDEC vs XMP/EXPO profile index) matches platform policy. Mixed kits or a module without the expected profile fall back to JEDEC or fail training.

Record part number, SPD revision, enabled profile name/index, DRAM voltage, and trained frequency in the service bundle. Intermittent faults after a “memory upgrade” are often profile or training mismatches, not bad media elsewhere.

Do not enable XMP/EXPO to eek our performance on regulated or fleet platforms unless the OEM documents the profile as a supported configuration.

See Chapter 12 for BIOS/UEFI trial of XMP/EXPO and other overclock settings: pending apply, reboot, commit if POST/training/watchdog pass, else rollback to the last known-good (typically JEDEC).

62.2. SPECIAL CONSIDERATIONS

For enterprise or datacenter storage, review controller logs, redundant pathing, and remote management dashboards.

RAID and redundant setups require careful handling to avoid data loss—verify rebuild progress and consistency after hardware changes.

For SSDs, keep firmware up to date, and be aware of drive’s TBW/endurance limits.

If data is irreplaceable and the device is inaccessible/unreadable, consider data recovery services—further write attempts can exacerbate data loss.

Always maintain recent backups and periodically review storage health status to prevent unanticipated failures.

63. REFERENCES & RESOURCES

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S.M.A.R.T. Attributes https://ntfs.com/disk-monitor-smart-attributes.htm
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CHAPTER 15

Communication

This chapter discusses the serviceability of communication subsystems, including how they are organized, monitored, and diagnosed.

- Overview of interface types and serviceability tiers
- Management information and Health indicators: link state, utilization, contention, capacity headroom, error rates/deterioration metrics plus policies and service notifications
- Service tools (built-in, physical layer, protocol, and network testers)
- Fault finding and troubleshooting (cables, RF, Wi-Fi)

64. OVERVIEW

Communication management monitors the status, health, and errors of the communication subsystems.

Tier	Description
Low-cost consumer	Basic built-in ports (e.g., USB, Ethernet, Wi-Fi, Bluetooth); minimal diagnostics or monitoring.
Premium consumer	Enhanced wired/wireless interfaces; basic counters; more connectors (HDMI, DisplayPort). Replaceable adapters.
Contractor / professional	Field-replaceable communication interfaces; advanced diagnostics and error reporting, loopback/self-test features.
Enterprise / industrial fleets	Hot-swappable modules (e.g., pluggable transceivers: SFP/SFP+/QSFP, FC/AL). Dual/redundant channels (e.g. Ethernet/Fiber/WiFi); remote monitoring, logging, and analytics; robust connectors.
Regulated / safety-critical	Ruggedized connectors (e.g. environmentally sealed), electromagnetic shielding for industrial reliability; authenticated and secure communications; continuous health and integrity monitoring; security audit trails.

Table 60: *Serviceability tiers for communication devices*

Communication subsystems are typically configured with a range of parameters – for example, window sizes, timeout durations, queue management strategies, and more. These parameters directly influence performance, reliability, and error recovery behaviors.

64.1. DATA PROCESSING UNITS

A *data processing unit* (DPU) – aka Smart NIC – is a programmable adapter to offload networking, security, and related I/O work from the host CPU. Responsibilities often include

data processing unit

routing and VPN termination, packet inspection and encryption, and storage or RDMA offload. High-performance designs use network processors, FPGAs, or dedicated accelerators, and are common in data centers and other high-throughput environments.

Fabric / Interface	Description
Ethernet	Smart NIC / DPU; host bus adapter (HBA)
Fibre Channel	Host bus adapter (HBA)
InfiniBand	Host channel adapter (HCA)
SCSI / SAS	Host bus adapter (HBA)

Table 61: *Offload adapter types*

A DPU or comparable offload adapter are typically used when:

1. The main board does not provide enough network interfaces or ports.
2. The main processing doesn't have the performance needed to perform network and security functions along with the application, or the network access must be isolated so the main computer can keep its primary task.
3. The communication expansion is optional and better supplied as a field-replaceable module.
4. The platform needs an embedded switch or other capabilities that are impractical on the host alone.

65. MANAGEMENT INFORMATION AND DIAGNOSTICS

Communication management employs the state, counters, and derived health signals. Communication devices also have the common health indicators and environmental exposure tracking (see chapter 8), temperature and power measurements (see chapter 13).

The adapters, drivers, and communication subsystem maintain this information. It typically includes link-layer events (training, retries, recovery), activity & usage rates, integrity checks such as CRC or frame errors, and measurements. Trended these metrics help in diagnosing issues, tuning performance, understanding the behavior of the software and hardware, and its health.

65.1. SUBSYSTEM STATE AND METRICS

State of communication subsystem

Parameter	Description
Buffers available	Number of buffers available for use. **Units:** count
Buffers used	Number of buffers that are currently in use. **Units:** count

Table 62: *Communication subsystem state*

Parameter	Description
Resource allocation failures	Number of times that a buffer or other resource could not be allocated. **Units:** count

Table 63: *Communication subsystem metrics*

65.2. LINK STATE

Link state describes whether a connection is up, down, training, or recovering, and the reasons it left a stable operational state. Drivers and adapters expose these fields so service can distinguish a steady fault from an intermittent one.

link flapping

Link flapping is when a connection (cable, switch port, transceiver, bus link or network transport) rapidly alternates between an operational (up) and non-operational (down) state. The cycles of loss and restoration can cause repeated device resets, timeouts, retrains or error bursts. They degrade throughput and reliability and can be mistaken for application or endpoint failure. These are often caused by faulty cables, connectors, marginal signal integrity, power/ground issues, or failing PHY/transceiver hardware. Detect flapping from up/down transition rate and short time-in-state –not from error counters alone.

Parameter	Description
Baseline Privacy (BPI+) Status	Status of link-layer encryption, important for security diagnostics.
Channel Bonding	Bonding enables higher data rates.
Channel Modulation Type	Quadrature Amplitude Modulation (QAM) level and profile for both downstream and upstream channels (e.g., 64-QAM, 256-QAM, OFDMA).
CMTS MAC Address	Identifies which Cable Modem Termination System is serving the modem.
Internet Connectivity Status	Indicates successful acquisition of DHCP, TFTP, and registration processes.
Link / bus state	Operational condition of the link. up, down, degraded, training, or recovering, and the reasons it left a stable operational state
Negotiated speed / width	Current vs capable link rate **Units:** Gb/s; lane count
Upstream Channel Lock Status	Indicates if upstream channels are locked and operational.

Table 64: *Link state*

Driver state (There may be multiple drivers for the same link)

Parameter	Description
Receive buffer used	The number of data bytes received and currently held in receive buffers.

Table 65: *Communication node state*

Receiver condition	The condition of receive buffer e.g. whether input buffers have filled and overrun is possible.
Receive queue depth	The depth of the receive queue, waiting for processing.
Transmit buffer used	The number of data bytes currently held in transmit buffers.
Transmit condition	The condition of transmit buffer e.g. whether output buffers have filled, whether output is stopped due to flow control.
Transmit queue depth	The depth of the transmit queue, waiting for sending.

65.3. HEALTH INDICATORS

Three related views of the link provide its health:

- Capacity and spare headroom – Are there sufficient local transceiver resources?
- Utilization and contention – Is demand exceeding what the path can deliver? Is the problem overload or a single noisy consumer?
- Error or deterioration, indicating the medium or path is failing. Do error and link-state trends point to PHY, cable, connector, or RF path failure?

65.3.1 Abstract health indicators

A single abstract indicator may fuse several metrics. Abstracted health indicators are built from the counters, rates, and measurements to form stable signal. These normalize units and intervals, trend over time (slopes, baselines, percentiles), and combine related measures so one noisy counter does not dominate the picture.

Table 66: *Abstract health indicators*

Health Measure	Patterns
Consumer contention	One session, VM, application, or QoS traffic class (on a shared NIC or access point) sees rising latency or drops while others starve, or class queues fill unevenly.
Link failed	Link down, repeated recovery failure, or flap/error storm after isolation attempts.
Link integrity degrading	Error-rate slope, retry/timeout counts, BER/PER, or CRC/frame errors rising above baseline; dropped or lost frames; retraining or link-recovery failures sustained over the reporting window—often with retrain count rising under modest demand.
Link instability (flapping)	Rapid up/down transitions with short dwell in current state (see Link State) – distinct from a one-time link-down fault.
Path quality degrading	Frequent channel switching or retraining above baseline, or persistently high dropped frames; intermittent errors often mean poor cable, connector seating or contamination, power, or RF path conditions.
Path recovery unhealthy	Command timeouts, unclean aborts, queue freezes, or bus/link resets without a matching endpoint failure. Check adapter firmware, termination, and expander or switch health
Path saturated	Demand exceeds what the path can deliver: negotiated speed below policy minimum; queue depth pegged at max with rising latency and tail latency;

(oversubscribed)	throughput or PPS near rated ceiling while demand still grows; rising retry/drop/timeout counts—without a matching PHY-error signature (utilization or capacity pressure, not media failure).
Remote congestion	High latency (for example p95) while local transmit/receive queues and buffers stay below pressure – remote peer, switch, or RF congestion rather than local backlog.
Resource exhaustion	Lack of free buffers, descriptors, credits, or session/ID pool entries. This may lead to longer queue depths, increased retries, latency, and timeouts; new work fails while existing flows may still look healthy.
Transport vs medium fault	Error mix dominated by transport or hardware errors rather than endpoint or medium errors – interconnect or adapter more likely than cable/PHY alone.

Note: Wireless and other noisy media usually need longer windows or baselines: temporary error bursts are common. Deterioration appears as a gradual rise in BER/PER at fixed conditions, transmit power dropping versus calibration, or intermittent antenna path behavior.

These indicators should be read together rather than in isolation. Many of these areas often appear together, or share similar symptoms – deep queues, retries, timeouts, drops, and rising latency. Physical-layer health issues often have longer queue depths, reduced throughput and higher latency even with modest demand.

65.3.2 Resource capacity and spare headroom

Capacity metrics track whether the system is running out of link, buffer, or session headroom, distinct from the physical-layer health.

Table 67: *Capacity health indicators*

Parameter	Description
Address / lease / ID pool remaining	DHCP leases, device IDs, or modem registration capacity on shared infrastructure; pool exhaustion blocks on-boarding, not existing links. **Units:** count or percent free
Available buffer / descriptor / mailbox capacity	Free DMA rings, buffer lists, or adapter descriptors relative to pool size. Exhaustion causes drops and allocation failures even when the media is idle. **Units:** count or percent free
Connection / session table free entries	Remaining slots for TCP/UDP connections, VPNs, or modem sessions; filling the table rejects new work while existing flows continue.. **Units:** count per time period
ISR / interrupt-off time	Time spent in interrupt context or with interrupts disabled. Approaching platform budgets (for example sustained near a 25 ms disable limit) makes the path look slow when the driver is the bottleneck. **Units:** milliseconds or percent
Log / capture / diagnostic scratch space	Free space for packet captures, modem event logs, or diagnostic bundles; filling these breaks serviceability before the data path is full. **Units:** bytes or percent free
Negotiated speed / width vs policy minimum	Current link rate relative to the required minimum; a silent downshift reduces capacity without a hard link-down.

	Units: Mb/s, lanes, or % of policy
Redundant path availability	Whether a failover or secondary path is available; running without a spare path extends the outage window on the next flap or timeout storm. **Units:** status; count of available paths
Window / credit headroom	Protocol receive window, Fibre Channel credits, or similar; collapsed windows under load look like low throughput with low local CPU. **Units:** bytes or credits remaining

65.3.3 Utilization and contention

Is demand exceeding what the path can deliver? Sustained saturation with rising retries, drops, or tail latency may indicate that the link or stack cannot keep up with demand.

Table 68: Utilization and contention health indicators

Health Measure	Description
Buffer utilization	Fraction of subsystem or DMA buffers in use (buffers used vs available). Sustained high use with allocation failures points to under-sized pools or a consumer not draining. **Units:** % or count
Channel / airtime utilization (wireless)	Fraction of the channel busy (Wi-Fi CCA, cellular resource blocks). High utilization with modest local queues can mean RF contention, not a bad radio. **Units:** % or count
Flow-control / pause events	Pause frames, credit exhaustion, or backpressure events; frequent pauses with full transmit queues indicate downstream congestion. **Units:** count per time period
Latency under load (average and max / p95)	Rising average with a growing max or p95 while offered load is steady indicates queueing, retransmission backoff, or remote saturation. Distinguish end-to-end link latency from local ISR delay. Units: milliseconds
Packet / message rate vs capacity	Sustained PPS or message rate near adapter, CPU, or interrupt limits; small frames often hit PPS before bandwidth. **Units:** packets/sec or % of rated
Retransmission, retry, and timeout rate	Elevated retries or timeouts under load often mean congestion or a recovering peer; pair with CRC/PHY errors to separate lossy media from overload. **Units:** count per time period
Throughput vs negotiated / rated bandwidth	Sustained bytes or frames per second near the link, channel, or bonded-group ceiling. Compare to negotiated speed/width and policy minimums. **Units:** MB/s, frames/s, or % of rated
Transmit / receive queue depth (trend)	Length of driver or adapter queues. Persistent growth or a queue pegged at its max with rising latency or overflow counts indicates saturation or a stalled peer. **Units:** count
Utilization	Fraction of the interconnect consumed. High link utilization with moderate device

	queues can mean the bottleneck is the fabric, not the end device. **Units:** %

65.3.4 Link Error Rates

Link error rates measure integrity and deterioration of the medium or path – whether bits, frames, or packets arrive intact, and whether recovery activity is rising over time. Trend BER, PER, CRC/frame errors, loss, retries, timeouts, and retrain or recovery counts with link state: a rising error slope under modest demand points to cable, connector, PHY, RF, or peer-path faults, while the same symptoms under heavy load may be overload rather than a failing medium.

Common metrics include:

Table 69: *Example communication link metrics*

Health Measure	Description
Bit error rate (BER)	The ratio of incorrectly received bits to the total bits transmitted over a communication channel. **Units:** % or errors per bit
Bytes received (total)	Cumulative amount of data received. **Units:** bytes
Bytes sent (total)	Cumulative amount of data sent. **Units:** bytes
Channel quality (SNR/RSSI)	Signal-to-noise ratio or received signal strength indicator / link margin indicator **Units:** dB or unitless index
Channel recovery events	Number of reset, re-sync, or flush events performed after error bursts. **Units:** count
Channel switch events	Number of drop/reconnect or failover/switchover events. **Units:** count
CRC/frame errors	Integrity check failures (e.g., CRC mismatches, framing/parity errors). **Units:** bytes
Fragmentation & reassembly count	Number of times data is fragmented into smaller pieces for transmission and then reassembled upon reception. **Units:** count
Last state-transition timestamp	When the link last changed state; used with time-in-state to spot rapid cycling. **Units:** timestamp
Link up/down transition count	Number of transitions between operational and non-operational states over the reporting interval or since reset. **Units:** count
Link up/down	Transitions per unit time; the primary flap detector when it exceeds policy over a

transition rate	hold-off window. **Units:** count per time period
Loss/missing frames, messages	Count of lost frames or sequence gaps detected by the protocol. **Units:** count
Messages received (total)	Cumulative amount of data received. **Units:** message count
Messages sent (total)	Cumulative amount of data sent. **Units:** message count
Overflow / underflow count	Number of times an overflow or underflow occurred. **Units:** count
Packet error rate (PER)	The ratio of packets with errors to the total number of packets transmitted or received **Units:** % or ratio
Packet loss	The number of packets (or frames) lost. **Units:** message count
Path failover / multipath events	Active path changes (where applicable) **Units:** count
Retraining root cause	Reason or trigger for link recovery/retraining
Retraining or renegotiation count	Number of retraining, re-sync, speed or width renegotiations events. **Units:** count, count/hour
Retransmission count	Number of retry (or retransmission) attempts before success or final failure. **Units:** bytes
Time in current state	How long the link has remained in its present state; short dwell with a high transition rate indicates flapping.
Timeouts	Number of request/response timeouts. **Units:** count
Uncorrectable errors	Total number of uncorrectable error events encountered. **Units:** count

65.3.5 DOCSIS Health indicators and error rates

DOCSIS also exposes link-state fields beyond the general table above. Use these with the general up/down and retrain measures when the path is a cable modem link.

Parameter	Description
Bonded /	LAG, or multipath, unused members or channels still available; loss of members cuts

Table 70: *Example DOCSIS cable modem metrics*

aggregated member headroom	capacity before total failure. **Units:** count of active members; % of aggregate
Correctable codewords per channel	Count of uncorrectable codeword errors on a per-channel basis for both downstream and upstream. **Units:** count
Downstream Power	The signal strength transmitted from the upstream network to the modem. The ideal range is -5 to +5 dBmV, with an acceptable range of -15 to +15 dBmV. **Units:** dBmV
Micro-reflections (Echo)	Detected line reflections can indicate impedance mismatches or damaged cabling; sometimes reported as "In-channel Frequency Response" (ICFR). **Units:** decibels (dB), or qualitative ICFR metric
Multicast Traffic Counters	Counts for various types of packet flows, useful for diagnosing congestion or abnormal usage. **Units:** count
Power Level per Channel	Downstream and upstream power levels on a per-channel basis, useful for identifying marginal channels. **Units:** dBmV
Receive Symbol Rate	Indicates the speed and stability of data transmission. **Units:** symbols/second (baud)
Signal-to-Noise Ratio (SNR)	SNR values greater than 33 dB are generally considered good, with values above 35 dB being excellent. Higher SNR indicates better signal quality. **Units:** decibels (dB)
T3/T4 Timeout Counts	Number of ranging request (T3) and ranging response (T4) timeouts, which indicate communication trouble between CPE and CMTS. **Units:** count
Transmit Symbol Rate	Indicates the speed and stability of data transmission. **Units:** symbols/second (baud)
Uncorrectable codewords per channel	Count of uncorrectable codeword errors on a per-channel basis for both downstream and upstream. A substantial number of uncorrectable codewords indicate excessive line noise or possible cable damage. **Units:** count
Unicast Traffic Counters	Counts for various types of packet flows, useful for diagnosing congestion or abnormal usage. **Units:** packet count
Upstream Power	The signal strength transmitted from the modem to the upstream network. Levels significantly outside the range 40 to 50 dBmV (especially above 55 dBmV) can result in connection drops. **Units:** dBmV

65.4. POLICIES AND THRESHOLDS

Communication policies apply thresholds and policies against the health indicators to drive consistent service actions, such as warn, schedule service, or retire a device. The subsections below spell out warning, service-notification, and fault triggers for communication links.

65.4.1 Automated response

Automated responses may run before or alongside notification. It is preferred to degrade and isolate before hard lockout unless safety requires an immediate stop. Require a hold-off window and clear exit criteria for each response so noisy media do not oscillate the automation.

Typical responses include:

- When a path is saturated, or congested, it should gracefully degrade, reducing throughput or data rate, disabling nonessential traffic, or switching to an alternate channel.
- If a link is unstable (flapping) or the path recovery is unhealthy, isolate or fail over the path; keep remaining paths in service when redundancy exists.
- If the link is not stable, or is recovering the path, suppress higher-layer retries that would amplify the fault; resume when link state is stable for a hold-off interval.
- If WiFi channel congestion detected; perform a scan to check if wireless channels are overcrowded.
- Escalating to service notification or fault, capturing a diagnostic bundle for service: link counters, retrain/flap history, adapter logs, negotiated speed.

65.4.2 Warnings

Issue a *warning* when an indicator is off baseline but the link is still usable, giving early notice before a service notification or fault. Always pair the condition with a next step for the operator.

- Environmental conditions are outside the allowed temperature or power operating range. Check cooling, power source; reduce load if nearing thermal or power limits. See Chapter 12 for other checks.
- Link integrity degrading (early): error-rate slope, retry/timeout counts, or BER/PER/CRC rising above baseline for a short window. Reseat /inspect the connector, verify cable routing, and compare counters after a settle interval.
- If a path is saturated or congested, such as the queue lengths are long, or throughput is flat against rising demand. Reduce nonessential traffic or scan/rechannel to check if wireless channels are overcrowded; confirm the bottleneck is not a silent speed downshift.
- If high latency (for example p95) while local queues and buffers stay below pressure, check the peer, switch, or RF path before replacing the local adapter.
- The buffer, descriptor, credit, or session pool headroom is low, risking resource exhaustion. Free or resize pools; isolate a noisy consumer if contention is visible.

- Path recovery unhealthy (brief): elevated command-timeout or unclean-abort rate without endpoint failure. Check link state and adapter recovery logs; avoid amplifying with aggressive higher-layer retries.

65.4.3 Service Notifications and Reminders

Escalate to a *service notification* when link health problems become significant, e.g. sustained past the warning window, recurs after operator mitigation, or clearly needs physical path work or scheduled service. For example, give a service notification when the cable (wiring) may be poorly connected, disconnected or broken. Include trigger metrics, trend, timestamps, and recent changes (firmware, cable swaps) so service can reproduce the issue. Reminders prompt planned diagnostics or replacement before a hard fault.

Issue a *service notification* when:

- The dropped/lost frames count, retraining events count, retry rate, timeouts, BER/PER trend, number of uncorrectable communication errors, number of the link recovery failures, or CRC/frame errors, dropped/lost frames, channel switching or retraining events remain above limits for a sustained interval. This may be poor connection, disconnect, or broken wiring. Give a service notification to inspect/reseat/replace cable or connector, run a cable or link test, and compare counters before/after.
- Sustained command timeouts, unclean aborts, or queue freezes; or error mix dominated by transport or hardware errors– require physical path inspection (cable/connector/power) and adapter, switch, or expander logs.
- If the link is unstable, such as rapid link up/down (flapping) persists across a hold-off window, notify service to stabilize or isolate the link before further higher-layer work (see Link State).
- If the is path saturated or remote congestion persists after load reduction or rechannel: Negotiated speed/width remains below policy, or latency stays high while local queues are not saturated – schedule link/path diagnostics and capacity or failover review.
- If resource exhaustion persists after pool resize or consumer isolation, schedule capacity review or hardware upgrade before new sessions fail continuously.

Generate *service reminders* when link health approaches policy limits:

- Recommend diagnostics when training/recovery events rise or channel switching becomes frequent.
- Schedule service or replacement when uncorrectable errors exceed limits or recovery attempts fail.

65.4.4 Faults

When there is a continued operation risks hardware damage, safety, or loss of the channel, raise a *fault,* mark the device or communication channel failed and take urgent corrective action, such as safe/degraded mode or lockout. This includes situations such as:

- If the link is down, the path is lost, or there are repeated bus/link resets with no recovery, take the channel out of service; fail over if a redundant path exists; dispatch service to restore or replace the path.
- If there are persistent uncorrectable communication errors after service-level mitigation, stop destructive or high-rate traffic on the path; replace cable, transceiver, or adapter as indicated by diagnostics.

- If there are repeated failures in link recovery procedures, isolate the path; do not leave automated retrains thrashing; schedule immediate service.
- Flapping or transport-error storms that continue after isolation attempts and automated failover, so the channel must be marked failed and taken out of service.
- If the link is unstable (flapping) or there are transport-error storms that continue after isolation attempts and automated failover, mark the channel failed and remove it from service until repaired.

66. SERVICE TOOLS

This section discusses tools used to diagnose, monitoring, and maintaining communication links, whether wired (Ethernet, USB) or wireless (Wi-Fi, cellular, Bluetooth, RF). These include built-in diagnostics, management utilities, and external test equipment that show signal quality, error rates, and configuration status.

66.1. BUILT-IN TOOLS

Many communication controllers or system-on-chips include built-in self-test routines, and special diagnostic modes. These self-tests typically generate test patterns, monitor internal loopbacks, and stress-tests. The controllers and system software may also log information to aid diagnosing. Examples of built-in diagnostic and reporting capabilities in communication systems:

- DOCSIS modems provide signal quality checks and channel diagnostics (e.g., signal-to-noise ratio, power levels, error rates).
- Wi-Fi interfaces monitor channel utilization and interference, reporting when channels are congested or busy.
- Dedicated cable testers assess integrity, continuity, and signal quality of cabling and connectors, and identify wiring faults.

66.2. PHYSICAL LAYER TEST TOOLS

Physical layer test tools assess and troubleshoot the integrity and performance of the underlying signal paths that connect devices. These testers are used on electrical, optical, and wireless links—including cables (such as twisted pair, coaxial, and fiber optic), connectors, and wireless transmission hardware. This includes cable testers and certifiers, time-domain reflectometers (TDRs) for pinpointing faults or signal loss, optical power meters, and wireless spectrum analyzers. These instruments help identify issues like opens, shorts, poor contacts, crosstalk, or signal degradation.

66.2.1 Loopback Plugs and Fixtures

Loopback adapters and plugs provide a simple, effective means of verifying that the transmit and receive channels of a device or port are working correctly. Internal loop back is used to check the internal device, and does not requiring a second device. An external loopback cable can be used to check if the connector is working as expected.

66.2.2 Cable Testers

Cable testers are specialized devices used to check the integrity and performance of cables used in high-speed digital interfaces such as Ethernet, USB, HDMI, and other connection standards:

- Check for continuity, open/short circuits, mis-wires, split pairs, and connector pinout / wire-map errors.

If a cable tester is not available, a *digital multimeter* (DVM) to check for simple shorts/opens in cable (and connectors), check for mis-wires, and measure voltage or resistance.

Advanced cable tester models (e.g. network analyzers) can

- Measure signal quality, near-end crosstalk (NEXT), attenuation, cable length, load-carrying capability, perform signal integrity analysis.
- Perform *time-domain reflectometry* (TDR) or advanced logging to identify intermittent issues such as flaky contacts or marginal cable faults.
- To ensure that the cable can carry required signal rates for high-speed interfaces (Ethernet, HDMI), advanced testers support certification against standards (e.g. TIA/EIA). High-end HDMI testers perform signal quality checks at various HDMI versions, test for support of high-resolution modes (4K/8K), and measure eye patterns or bit error rates.

In some cases, applying using a signal or pattern generators to supply known waveforms or protocol transactions can be used for margin testing, and stress-testing noise immunity.

Testers for optical fiber communication include:

- *Optical time-domain reflectometers* (OTDR) are used to check the integrity of fiber-optic cables, detect connection losses, and localize faults.
- Optical power meters measure signal strength/end-to-end light loss and verify cable termination quality.

66.2.3 RF Tools

With RF (radio frequency) communication links, a spectrum analyzer and field meter may be used to verify installation quality and troubleshoot sources of interference, especially when new high-power transmitters are deployed nearby. RF test tools include:

Table 71: *RF tools*

Type of Tool	Description
Cellular Test Tools	Tools such as signal sniffers or apps that analyze signal strength (RSRP/RSRQ), quality, handover behavior, and neighboring cell info – used in LTE/5G site deployment and troubleshooting.
Field Strength Meters	Measure RF signal strength at various locations to optimize antenna placement or detect dead spots.
Handheld Spectrum Analyzers	Portable devices for visualizing and measuring RF signals over a wide range of frequencies (commonly used for cellular, radio, IoT, and custom wireless). Identify interference, spurious emissions, or weak signals.
Network analyzers	Instruments to test the properties of RF components/cables/antennas (S-parameters, VSWR, return loss).

66.3. LINK AND NETWORK LAYER TEST TOOLS

Link and network layer test tools are used to verify, analyze, and troubleshoot data transmission and addressing between devices. At the link layer, these tools identifying and diagnosing issues such as frame loss, incorrectly addressed packets, and data flow congestion.

Layer 2/3 testers validate network setup, detect transmission errors, and monitor traffic patterns.

66.3.1 Bluetooth

Smartphone-based test apps and Bluetooth testers can scan, connect, and measure signal quality. These can help validate pairing, RSSI, packet loss, and service advertisement/connection correctness.

66.3.2 WiFi

Wi-Fi software and hardware tools help identify weak signal paths, crowded or noisy channels, and reconfigure APs for best coverage and throughput. They can be used in conjunction with spectrum tools to prevent or solve problems:

Table 72: *WiFi tools*

Type of Tool	Examples	Description
Built-in Diagnostics		Health and diagnostics, such as channel utilization stats, error counts, and client activity logs. Many computers and Wi-Fi routers provide these.
Sniffers and Packet Capture	*Wireshark*	Capture Wi-Fi packets for protocol-level troubleshooting, security auditing, and interoperability tests. May use dedicated adapters.
Spectrum Analyzers	*MetaGeek Chanalyzer)*	Visualize the entire 2.4/5/6 GHz bands to detect non-Wi-Fi sources of RF interference (microwave ovens, cordless phones, Bluetooth).
Wi-Fi Analyzers	*Ekahau, NetAlly AirCheck, WiFi Explorer*	Visualize Wi-Fi spectrum usage, signal strength (RSSI), interference, and overlapping channels. Useful for site surveys, placement optimization, and diagnosing dead zones or congestion.

66.4. PROTOCOL ANALYZERS

Protocol analyzers are specialized tools used to identify protocol errors (framing, parity, arbitration loss), timing violations, and unexpected message patterns. They typically support trigger/record features to capture data, filtering by address/packet type, and detailed frame-by-frame decoding, , and display of the data exchanged. They are invaluable for troubleshooting protocol-level issues, debugging interoperability, and validating compliance with communication standards.

These exist for many protocols: USB, I2C, SPI, CAN, FlexRay, LIN, UART/RS-232, Ethernet, Bluetooth, and other serial or packet-based protocols.

66.5. NETWORK (AS A SYSTEM) TESTERS

Network test and validation tools are used to verify that an infrastructure can meet required performance and resilience goals before deployment. This is especially useful for complex networks, mixing WAN, WiFi, cellular, IoT systems, and when validating service-level agreements (SLAs) or compliance requirements. These tools can simulate, benchmark, and validate the performance and scalability of wired and wireless communication networks, including Ethernet, Wi-Fi, 5G, cloud, and virtualized environments.

- Generates application and network-level traffic to test network infrastructure, security solutions, or cloud deployments under real-world conditions.
- Measures metrics such as latency, throughput, packet loss, jitter, and responsiveness during stress or scale testing.
- Supports validating security controls (firewalls, IPS/IDS) and Quality of Service (QoS) policies.
- Operates in both traditional hardware appliance and containerized/cloud-native deployment models, making it suitable for lab, remote, and distributed/cloud-based network validation.
- Enables "virtual test agents" to be placed in different parts of a network (on-premises, cloud, edge) to simulate end-user/application traffic and monitor experience.

66.6. REMOTE MANAGEMENT

The communication elements may also be available for remote management (see chapter X). Service interfaces typically expose information such as frame/bit error rates, symbol statistics, link negotiation outcomes, spectrum data, and environmental status (e.g., temperature, power, antenna connectivity).

67. FAULT FINDING AND TROUBLESHOOTING

Fault finding for communication focuses on the physical and link layers: the medium, connectors, PHY/transceiver, and adapter. Adapters and drivers report many of these faults directly; the host also reports when it can no longer reach the hardware. Cables and external connectors are frequent failure points and deserve early attention whenever integrity or link-state indicators degrade.

Start from symptoms and the established health indicators to choose the next step:

1. Do the physical layer error trends point to PHY, cable or connector failure?
2. Is the path saturated? Is demand exceeding what the path can deliver?
3. Is the free capacity and spare path headroom adequate?
4. If multiple applications, traffic classes, or virtual machines share the link and one is starving others, isolate that consumer or class before replacing cables or radios.

Stabilize or isolate a flapping or timeout-bound link first, then use symptom triage, service tools, and the guided checks below before chasing application or endpoint failures.

67.1. SYMPTOM-BASED TRIAGE

Higher layers may supply symptoms; decide whether the fault is the path, the adapter, or something above the link before retuning applications or replacing endpoint equipment.

Symptom	Possible causes
Bandwidth persistently below target, or performance declining under steady demand.	This indicates that the link is under-provisioned. Compare throughput and packets/sec to negotiated rate and policy; verify capacity headroom.
Command-timeout count is elevated or	Hung or recovering path; check link state, stability, and adapter

Table 73: *Bucketing symptoms with likely causes*

unclean-abort rate is rising.	recovery logs.
CRC/parity/PHY errors or frequent retrains	Cable, connector, power, or PHY/transceiver fault; verify the physical path.
Dropped/lost frames, rising BER/PER, uncorrectable errors, or link-recovery failures sustained above threshold	The link integrity is degrading. Inspect/reseat/replace cable or connector; run a cable or link test; compare counters before/after.
Error mix dominated by transport or hardware errors rather than endpoint or medium errors	Check adapter, interconnect, termination, expander or switch health.
ISR / interrupt-off time approaches or exceeds platform budgets (for example disable for >25 ms)	Device driver has problem. Profile driver/interrupt load before replacing the peer (capacity / ISR budget).
Lack of free buffers, descriptors, credits, or session/ID pool entries; new work fails	Resource exhaustion. Free or resize pools.
Latency Is high (e.g. p95) while local queues or buffers remains good	Remote congestion – suspect remote peer, switch, or RF congestion.
Link/bus reset without a matching endpoint failure	transport recovery; check adapter firmware, termination, and expander or switch health.
Link flapping (rapid up/down)	Stabilize or isolate the link first.
Negotiated speed/width below policy;	Path saturated. Confirm negotiation, headroom, and whether demand exceeds what the path can deliver.
Performance declines	This indicates that the link is under-provisioned.
Queue depth pegged at max with rising latency	Path saturated. Confirm negotiation, headroom, and whether demand exceeds what the path can deliver.
Queue freeze without a matching endpoint failure	*Transport recovery; check adapter firmware, termination, and expander or switch health.*

67.2. POOR CABLE OR CONNECTOR TROUBLESHOOTING

Use the structured isolate-and-verify cable troubleshooting work flow (Chapter 11) to troubleshoot a suspected cabling issue. Compare link metrics before/after reseat or swap; typical link metrics to use are CRC/frame errors, retries, timeouts, BER/PER, link speed renegotiation events. For critical or high-frequency links, choose a tester that can validate both wiring and signal performance.

67.3. RF

Radio environment can change due to competing equipment, weather, and even building occupancy. Always test at several physical locations and times of day. Check the link up/down and retrain rates first to stabilize or isolate a flapping link.

The following indicate underlying hardware or connection faults; inspect antenna seating, cabling, and RF path if they occur.

- The transmit (Tx) power has dropped, relative to initial calibration
- Sustained increases in packet error rate (PER) or bit error rate (BER),
- Intermittent issues in antenna or signal path

67.4. WI-FI TROUBLESHOOTING

With Wi-Fi issues start with the radio and higher layers reports, then narrow down RF, configuration, and coexistence problems:

1. Define the problem: cannot join, intermittent drops, low throughput, or high latency/jitter. Note which band (2.4 / 5 / 6 GHz), SSID, security mode, and whether this problem affects one client, one AP, or many endpoints.
2. Check the signal measures and other health indicators: Review RSSI (or equivalent), SNR or link margin, channel utilization, retry rate, PER/BER trends, reassociation or retraining counts, and DHCP/DNS timeouts. Compare against a known-good location or time window.
3. Check the security: Verify correct SSID, passphrase or 802.1X credentials, supported cipher suites, and that the client and AP firmware are current. Mismatched WPA versions, expired certificates, or captive-portal failures can look like “bad Wi-Fi” when the radio is fine.
4. Reduce RF variables: Move the client closer to the AP (or vice versa) to see if symptoms improve; if they do, suspect path loss, obstruction, antenna orientation, or local interference. Repeat at more than one location and time of day – RF conditions change with competing devices, weather, and even building occupancy.
5. Channel and congestion, perform a scan to check if wireless channels are overcrowded. Use a Wi-Fi analyzer or built-in diagnostics to see overlapping BSSIDs, co-channel interference, and non-Wi-Fi noise (microwave ovens, cordless phones, Bluetooth-heavy environments). Prefer clearer channels or band-steering to 5/6 GHz when policy allows; verify automatic channel selection is not thrashing.
6. Power saving and sleep: Rule out aggressive client power management or AP-side features that disconnect idle clients. Test with power save disabled briefly on the client if supported, and compare stability.
7. Isolate the failing element: Swap or compare against a known-good client adapter, a wired connection on the same LAN segment, or a second AP/SSID. If the fault follows one device, focus on its driver, antenna, or settings; if it follows a location, focus on RF and installation.
8. Document and verify: After changes (channel, placement, firmware, credentials), re-check the same metrics as in step 2 and confirm stability over an operational period.

For suspected coexistence or regulatory issues (DFS radar events on 5 GHz, neighboring high-power transmitters), combine Wi-Fi analyzers with spectrum tools; packet capture helps when the issue is authentication, roaming, or upper-layer timeouts rather than raw signal strength.

68. REFERENCES & RESOURCES

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Universal Serial Bus Specification, Revision 2.0 https://www.usb.org/document-library/usb-20-specification

Universal Serial Bus 3.2 Specification https://www.usb.org/document-library/usb-32-specification

CHAPTER 16

Automated Loaders

This chapter looks at automated loader subsystems:

- Overview of loaders, their features, key operations and interactions
- Management and health information
- Logging
- Fault finding and Troubleshooting

69. OVERVIEW

Automated loaders move media/resources between a storage slots and a drive or tool head that will use them. These combine:

- A magazine or rack of slots (sometimes dozens or hundreds) that holds media (optical disc, tape), tools or resources.
- A *toolhead* to use the tools, *printhead* to use the material, *drive* to access the media, or other mechanism.
- A mechanism (robot arm, carousel, shuttle) that moves a resource, media, or tools between slots and the drive or tool head.

There are many different kinds of loaders depending on the type of resource they manage:

Type of Loader	Description
3D printing	filament spools or cartridges, with a consumable filament.
Magnetic tape	Tape held on cartridges or cassettes; treat as sequential-access archive/backup media
Optical disc	CD, DVD, Blu-ray
Pen plotter and vinyl cutter	Pens/ink cartridges, cutting tools (knives/punches), and paper/media supplies
Other tools	Electronic components for pick-and-place machines, nuts and bolts. microfiche

Table 74: *Types of resources managed by an automated loader*

69.1. FEATURES

An automated loader may include features such as:

- The ability to choose which media (e.g., tape, disc, microfiche) is loaded into the active drive; or, with materials, to select which filament spool is routed to the active extruder

via a feed path; which pen/ink cartridge is active and which paper/roll/sheet media is loaded; or (with pick and place machines) which electronic components is supplied to the tool head.

- Performs inventory management tracking the contents of each slot.
- Automatically selects an alternate or known-good backup when a material runs out or is unavailable.
- Detects empty or missing media, or material (e.g. empty spool, low ink/pen/paper levels), and weak feed; and tracks thee amount of material (or components) remaining.
- Detect pickup failures, misalignment, filament breaks and jams or obstructions using sensors and motion monitoring; motor stall current.
- Monitor environmental conditions (e.g., temperature and humidity) of the loader for material integrity, and safe operation.
- Enforce interlocks and safe shutdown during active moves.
- Scheduled cleaning cycles and media retirement policies based on health indicators.
- Multi-stage material handling for complex workflows that may require materials to be conditioned (such as baking/drying) prior to loading into the feed or tool.

69.2. INVENTORY MANAGEMENT

The loader should report detailed information about its configuration and capabilities, such as:

- The number of slots it has
- Supported features
- The types of materials or media it supports
- Presence of elevators, and sizing of buffers, or caching mechanisms

The inventory management includes:

- Identifying the contents of each slot using barcode, RFID, or NFC scanning. This can include information that uniquely identifies the item, its manufacturer, usage parameters, quantity etc.
- Tracking and reporting the quantity of each resource remaining where possible
- Ability to switch to alternate or back up material if depleted or missing.
- Generating requests for missing or depleted media or materials
- Detect if replacement items have been added, if the substituted items are correct and suitable for use.
- Marking slots or items as offline if unavailable or out of specification

69.3. ENVIRONMENT MANAGEMENT

The loader may continuously monitor and, where appropriate, controls the enclosed environment. This may involve:

- Monitoring environmental conditions such as humidity, temperature, and the status of doors or slots (open/closed, sealed).

- Managing the internal climate using temperature controllers, fans, heaters, or dryers to maintain target conditions for stored media and components.
- Generating alerts for environmental excursions or unsafe conditions.
- Enforcing locking of doors or slots to secure contents or maintain environmental integrity.

See the heater/chiller chapter for additional details on environmental control.

69.4. SPECIAL OPERATIONS

Operation	Description
Emergency stop / safe state	Halt all motion and secure picker for maintenance or error.
Export	Move media from the loader via a slot accessible to an operator, updating inventory. This allows removing media without opening the main door, preserving the internal environment, and isolating the operator from tools, etc.
Import	Move media into the loader via a slot accessible to an operator, update inventory and authentication. This allows adding media without opening the main door. These are called mail slot (as they resemble that of a residential door's mail slot) or import/export port (I/E port).
Initialize / Homing	Calibrate or reset mechanisms to a known-good state at power-on or after faults, ensuring accurate position tracking.
Inventory scan	Detect and enumerate items in all available slots, updating system record of slot contents. The items might be detected with sensors, and (uniquely) identified with barcode, or RFID; it may validate authenticity (e.g. in security/trust context).
Load	The work flow involves pick, mounts the media or puts the material into a print head or tool into tool head; potentially requested import.
Lock / unlock door / magazine	Control access to magazines and loader doors.
Move	Transfer media between two locations (slot, drive, export slot).
Mount	Insert media into a drive for reading/writing.
Pick	Grasp and lift a specified item from a slot or magazine. Typically part of Load or Export workflows).
Place	Deposit a held item into a designated slot, drive, tool, or magazine. (part of Unload or Import workflows).
Scan barcode / RFID	Read media ID/barcode/RFID/UID and authentication of an individual inventory item.
Status check	Report hardware health, error codes, sensor states, and maintenance due.
Unload	Remove a media from a drive; it may make it available for export/ejection.

Table 75: *Actions for a loader / magazine controller*

Unmount	Remove media from a drive and return it to a designated storage location (slot, magazine).
Update configuration or firmware	Apply new firmware or loader settings (as permitted).

69.5. MAJOR INTERACTIONS

69.5.1 Adding new media or materials

New media or materials may be added by operator initiative or in response to an automated system request. When the loader detects that an item is missing, depleted, or otherwise requires replacement, it can notify the operator supply the required item.

If the loader is equipped with input/output slots (also called mail slots), the operator would place the item into the slot, the system would detect the item and attempt to place it. This is all done without opening the main door, preserving the internal environment and security, and protecting the operator from exposure to potentially harmful mechanisms and conditions.

For loaders with locking doors that restrict physical access, the operator begins by requesting the door to be unlocked. Once the door is unlocked, the loader confirms to the operator that the loader is ready for access. The operator can place the item material into an available slot. The loader detects the item, scans its barcode or RFID tag, authenticates it as appropriate, and updates the inventory records to reflect the change.

69.5.2 Remove media or material cartridges

The removal of media or materials can be initiated either automatically by the system or manually by the operator, and generally a reverse of the steps in the addition process.

For manual removal, the operator may need to request the door to be unlocked first. The operator opens the cover or door and removes the desired item. The loader detects when the door is closed or when the item has been removed, conducts an inventory scan, and updates the inventory records to reflect any changes.

In an automated workflow, the system can unmount an item from a drive, transfer it to the output mail slot, and eject it for removal. The inventory records are updated accordingly to document these changes.

70. MANAGEMENT INFORMATION AND DIAGNOSTICS

70.1. HEALTH INDICATORS

Loader health indicators should capture both mechanical reliability and inventory:

Parameter	Description
Breakage count	Count of media breakages attributed to loader mechanism. **Units:** count
Door/magazine	Door open and magazine change events, and whether they occurred during active

Table 76: *Example loader health indicators*

event count	moves. **Units:** count
Filament feed success rate	Percent of material feed moves that complete without step loss, stalls, or clog-related errors. **Units:** %
Inventory reconciliation error count	Mismatch count: expected vs scanned slots after refresh/magazine change/door events. **Units:** count
Inventory scan failures	Barcode/RFID/NFC read failures or ambiguous identity events. **Units:** count
Jam count	A count of jam/obstruction detections in the delivery path. **Units:** count
Material runout events	A count of spool empty/runout detections. **Units:** count
Mechanical cycle wear	The number of operating hours/cycles for belts/gears/picker travel mechanisms (where available). **Units:** count
Mis-pick count	The number of failed picks requiring retries, operator intervention, or abort. **Units:** count
Pick/place success rate	Percent of pick/place operations that complete without error. **Units:** %

Drive, toolhead, and printhead health indicators:

Parameter	Description
Extrusion calibration drift	The number of deviation of commanded vs measured extrusion amount over time (where available). **Units:** count
Extruder clog / jam count	A count of detected clogs, filament jams, or nozzle blockage events. **Units:** count
Ink starvation / pen clog count	A count of detected ink starvation, pen clogs, or print/no-ink events (pen plotters). **Units:** count
Load/unload failures	The number of failures to mounting/unmounting selected media or material. **Units:** count
Positioning error count	A count of positioning/step-loss events affecting plot quality (pen plotters), or print quality (3D printers).

Table 77: *Example drive / toolhead health indicators*

	Units: count

70.2. POLICIES AND THRESHOLDS

Policies guide key workflows and interactions, using health indicators to drive consistent actions such as issuing warnings, scheduling maintenance, or retiring a device.

The following actions are triggered automatically in response to specific events:

- After any manual magazine change or door event, perform an inventory scan and reconcile actual slot contents with expected inventory.
- On pick failures, check for mis-seated media, barcode occlusion, RFID obstruction, slot blockages, and robot calibration; avoid repeated retries that could damage media.
- Verify environmental parameters (temperature, humidity) are within OEM-specified limits after relevant events.

Issue a *warning* for conditions not yet severe enough to threaten equipment safety or operation; and prompt scheduled maintenance actions:

- Increasing mis-pick or jam count,
- Inventory reconciliation errors that exceed normal variance,
- Increased drive load or unload errors for a given medium type,
- Increasing filament feed stalls/mis-feeds, material runout events, or early signs of extruder clog. (3D printers)
- Increasing ink/pen depletion events, paper feed jams, tears, or indicators of pending pen-clogs.

Escalate to a *service notification* when health indicators worsen or start to impact reliability or performance:

- The pick/place success rate drops below a defined policy threshold,
- Jams recur in the same slot region, or repeated pick failures. These suggest a mechanical or calibration problem.
- Cleaning overdue and correlates with optical read errors or tape path issues,
- Multiple extruder clogs, nozzle blockage events or persistent filament routing/mount mismatches. (3D printers)
- An increase in paper tears, feed jams, indicators of pending pen clogs, or repeated mismatches in media selection.

A fault is raised for serious issues – when the health of a component declines to where continued operation is a risk may be unsafe or unreliable. Scenarios include:

- Repeated jam or breakage events indicating mechanical failure or unsafe operation,
- Persistent inventory identification failures that risk mounting the wrong medium,
- Persistent clogs or repeated failed material switches that could prevent safe operation or risk damage from excess thermal cycles. (3D printers)
- Persistent clogging or carriage-positioning faults that prevent safe operation or legible output. (plotters)

71. SERVICEABILITY

This section covers recommended practices and design features that support serviceability in loader-equipped devices.

Service documentation should include:

- Clearly defined workflows for maintenance and jam removal
- Diagrams showing slot layouts, numbering schemes, and magazine types
- List the supported magazine/spool types
- Inventory label requirements (barcode/RFID standards) and instructions for handling unidentified media
- Cleaning/maintenance and safe shutdown procedures for each drive, toolhead and printhead type.
- For 3D printing loaders: guidance on filament handling, spool compatibility rules, feed-path inspection, and hot-end/extruder cleaning intervals.
- For pen plotters: recommended ink/pen cartridge handling, compatible paper/media, media feed-path inspection, and pen-head/nozzle cleaning intervals.

Preventative maintenance servicing activities generally include:

- Regular cleaning for tape heads and optical lenses, and addressing overdue cleaning tasks or resolving detected cleaning errors (e.g., dirty tape heads or optical lenses).
- Picker calibration drift, if repeatable positional misses or drift are observed.
- Addressing pick/place failures, misalignment, worn belts or gears, and recalibrating as needed.
- Verifying inventory accuracy, resolving occurrences of “unknown media” or wrong-slot assignments.
- Managing environmental factors that may impact operation, such as temperature and humidity levels for tape libraries and filament storage areas, and dust and vibration (which can affect optical or tape systems).
- Operational policy: cleaning cycles (tape heads, optical lens), media retirement by age or error history, and safe shutdown during active moves.
- Retiring media based on age or error logs,
- Ensuring safe shutdown procedures during active moves.

71.1. COMMON SITUATIONS THAT REQUIRE OPERATOR INTERVENTION

Operator intervention is generally needed to address issues such as

- Incorrect media selection (“mis-picks”) can result from media not being properly seated, being placed in the wrong slot, errors in slot-to-ID mapping, unscanned manual changes, or damaged/unreadable barcodes or RFID tags. For 3D printers, similar issues include filament feed stalls, tension errors, spool runout, or misrouted filament entering the extruder. An inventory scan should occur automatically. The operator should have the ability to manually move media, or perform a scan. The scan reports should identify “unknown media” or incorrect slot assignments.
- Pick and place errors, misalignment, worn belts or gears, and calibration drift. Dust or vibration can cause errors in optical/tape systems. Skipped or ineffective cleaning

cycles (such as neglecting to clean tape heads or optical lenses) can degrade system reliability. Persistent or repeating pick/place errors often indicate the system requires recalibration. Clear, documented steps for recalibration should be provided.

- Jams can result from physical obstructions, slot misalignment, debris, or inconsistent sensor readings. Operators can usually clear a jam by following documented procedures to stop the loader safely, open the service panel or door, remove any stuck media or debris, inspect for obvious obstructions or damage, and re-seat affected media before initiating an inventory scan or retrying the operation.
- Media breakage: Physical damage to media (discs, tapes, cartridges), filament breaks, feed jams, excessive tension, or running out of consumable material. Material mismatches or poorly routed supplies can also contribute.
- Toolhead issues: For 3D printers, clogging or jams of the extruder or nozzle; for pen plotters, ink starvation, clogged pens, or carriage movement faults can occur. Maintaining proper temperature and humidity is especially critical for tape libraries and filament storage.

72. LOGGING

The loader/media management system should maintain logs covering the following:

- Media handling events, such as insert, remove, load, unload, mount and unmount.
- Loader operation: slot changes, inventory scan activity (including unknown items, mismatches between expected and actual contents), pick & place success/canceled/failure, door events, magazine insert/removal.
- Material feed path events: spool load/unload, feed start/stop, extruder jam/clog alarms, paper jams/tears, job error summaries. Log which slot and which physical piece of media were involved (barcode/label ID).
- Ink/pen cartridge events: load/unload, low-ink/runout detected, clog/ink-missing alarms.
- Media identification: track the media identifiers, and authenticity/validation status.
- Drive health: integrity check results, any drive-reported error/fault codes, corrective actions taken (e.g. reimaging, error correction, quarantine, retirement).
- Operator and tool context: Record who performed actions (operator ID), the tool/software version(s), configuration profile used during imaging, restoration, or erasure operations.
- Service and maintenance: backup/export/import outcomes (if relevant) and resulting disposition (reused/reimaged/quarantined/retired).
- Log backup/export/import outcomes and validate transferred files or packages to trace the chain of custody and correctness.
- The recovery procedures taken after events

73. FAULT FINDING AND TROUBLESHOOTING

Troubleshoot issues using a systematic isolate-and-verify workflow:

1. First, ensure the system is in a safe state – stop all automated movements and prepare the device accordingly.
2. Gather relevant information, including:
    a. The loader's error code or the reported jam reason,
    b. Information on the last commanded move – the source slot, the destination, the media identifier(s), etc.
    c. Results from the latest inventory scan, comparing expected vs actual contents.
3. Determine if the issue is specifically with a certain media identifier (indicating a potential media problem), tool or if it is linked to a particular slot or section of the loader (suggesting a mechanical or handling issue).
4. For pick failures, investigate for mis-seated media, blocked or obscured barcodes, slot obstructions, and calibration issues with the robot/picker mechanism.
5. To distinguish between media and loader/drive path faults, use a known-good material or medial, or tool.
6. Inspect the picker's movement area and media path for debris, misalignment, worn parts, or damaged seals.
7. For optical drives: inspect and clean optics in accordance with the manufacturer's guidelines.
8. For tape drives: carry out scheduled head and path cleaning per instructions, and confirm the cartridge is compatible.
9. With repeated mis-picks in the same area, verify calibration before attempting additional retries.
10. Verify by running a representative movement cycle and confirm health/status indicators return to baseline (no recurring faults).

CHAPTER 17

Heaters and Chillers

This chapter looks at heaters and chillers items:

- Overview of their types, features, essential operations and interactions
- Management and health information
- Logging
- Fault finding and Troubleshooting

74. OVERVIEW

Heater/chiller subsystems intentionally add heat or cool, to hold a target operating range for a component, zone, or process. These are used where temperature directly affects safety, reliability, material properties, or output quality. Typical applications include:

- Heating a sensor to its required operating range
- Regulating print bed temperature
- Heating a “hot-end” while chilling the sounding regions so that the material doesn’t melt or change properties in the wrong place.
- Heating (or cooling) a chamber or enclosure
- Heating an applied part during process steps
- Removing localized heat from electronics, e.g. CPUs, motor drivers, or power modules.

Heater and chiller control systems typically consist of several coordinated components to regulate temperature, flow, and pressure within precise limits. These components may include:

- Heating elements, such as a lamp, resistive heater, RF generator and induction coil, or similar heat source
- Fans to circulate air and provide convection cooling or evenly distribute heat.
- Exhaust port
- Coolant circulation pumps, and fluid path components for liquid cooling, including fittings, tubes, sensors, and other leak-detection provisions
- Active cooling devices chillers,
- Temperature, flow, and pressure sensors used for thermal control and coolant circulation transport
- A vacuum pump
- Solenoid valves lead to either an air purge to get rid of the built-up vacuum or the vacuum pump that moves air in the reservoir out of the loop.
- Control logic (e.g. PI/PID) to maintain stable target temperature bands with minimal overshoot

74.1. FEATURES

To maintain precise temperature conditions, thermal regulation systems commonly include:

- Closed-loop temperature regulation using PI/PID control
- Multiple sensor inputs (e.g. includes a secondary) for cross-checking and safety validation
- Configurable setpoints, ramp rates, warm-up/cool-down profiles, and hold periods
- Zoned control. For example bed, chamber, nozzle, electronics bay, fluid loop segments.
- Protection logic such as over-temperature, under-temperature, and no-rise/no-cool detection
- Leak detection and management
- Health monitoring for heater current, fan/pump operation, and coolant flow/pressure
- Safe-state behavior on faults (disable heaters, force cooling, isolate affected zone)
- Event reporting for alarms, warnings, and threshold excursions

Some systems attachment to industrial vacuum cleaners or specialized evacuation pumps. In severe cases or during service, these can be activated to quickly clear air or unwanted coolant.

74.2. ENVIRONMENT MANAGEMENT

When responsible for managing the environment temperature and humidity, the subsystem may:

- Monitor environmental conditions such as humidity, temperature, and the status of doors or slots (open/closed, sealed).
- Manage the internal climate using temperature controllers, fans, heaters, or dryers.
- Generate alerts for environmental excursions or unsafe conditions.
- Enforce locking of doors or slots to secure contents or maintain environmental integrity.

74.3. MAIN OPERATION

Heater and chiller control systems operate by coordinating multiple components to maintain precise control over temperature.

The core is a temperature feedback control loop that continually monitors temperature sensors at strategically placed points – such as coolant supply, return lines, and zones critical to the process. The controller actively adjusts heater elements, chillers, fans, and fluid valves to keep system temperature within the specified operating band.

In systems using a liquid coolant, the loop pressure is monitored, and the pressure is regulated using pumps, variable-speed drives, and control valves.

When the device starts operation, the system allows warm-up or cooling stabilization before proceeding – it waits until the temperatures are within target limits and have remained steady for any required dwell time.

If an over-temperature, under-temperature, loss of flow, and leak or other fault is detected the controller can automatically initiate a safe shutdown, disable heating or cooling outputs, and/or raises alarms to inform operators and prevent equipment damage. Recovery typically requires acknowledgement, root-cause check, and controlled restart.

74.4. PREVENTING LEAKS

One common cause of coolant system failures is cracks or leaks at fittings the developing over time. To minimize the risk, systems should be constructed with reinforced, corrosion-resistant, low-permeability double-walled pipes or tubing; and robust, high-quality fittings.

The cooling loop pressure should be kept below ambient atmospheric pressure. In the event of a breach, air is drawn into the system rather than allowing coolant to escape. This minimizes fluid loss and external contamination.

The control logic should operate pumps and valves in a manner that reduces mechanical stresses on hoses and connections. These should be configured to prevent rapid pressure changes or spikes. These accelerate wear and increase the likelihood of leaks developing over time.

74.5. ANTI-LEAK SYSTEMS WORK

anti-leak

Anti-leak systems are designed to prevent coolant leaks, to quickly detect and manage leaks should they occur. This gives time for human intervention, and potentially automated shutdown.

First, the cooling system operates at below ambient atmospheric pressure. The loop pressure is continuously monitored. An increase in pressure indicates a leak, as the air ingress raises the pressure toward atmosphere pressure.

The coolant loop includes a reservoir with a headspace of gas (such as air) above the fluid. A vacuum pump maintains the desired negative pressure by drawing air from the reservoir as needed. Any air entering the system (e.g. from a leak or setup) is captured in the reservoir.

The reservoir is equipped with valves. One allows for air purging to release the vacuum. Another permits the vacuum pump to evacuate air from the reservoir, thereby maintaining the negative pressure in the loop.

74.6. SPECIAL OPERATIONS

Operation	Description
Cool-down (post-cycle)	Controlled cool-down after process completion or shutdown.
Enable / disable zone	Enable or disable heating/cooling for a specific zone or loop.
Leak check / flow validation	Verify coolant loop integrity and adequate flow before full-power operation.
PID tune	Run tuning routine to restore stable control after hardware changes or aging.

Table 78: *Common actions for heater / chiller controllers*

Safe shutdown	Transition to safe state during fault, emergency stop, or power-loss handling.
Sensor validation	Cross-check sensor readings; detect drift, open/short, or implausible values.
Set target temperature	Update setpoint and acceptable control band.
Update profile or firmware	Apply new firmware or control profile.
Warm-up / pre-condition	Bring subsystem to operating temperature before a process starts.

75. MANAGEMENT INFORMATION AND DIAGNOSTICS

75.1. HEALTH INDICATORS

In heater/chiller systems, control authority refers to having sufficient heating or cooling capacity, and adequate actuator dynamic range (e.g. the range of a proportional valve's opening, or heater current adjustment), so that the system can consistently reach and maintain the target temperature.

The health monitors factors that can impact the performance of environmental controls include:

- Component degradation, e.g., a heater element aging and producing less output, or a pump delivering less flow,
- Increased demands beyond the system's capability, such as excessive process heat load,
- Faults such as stuck valves or broken connections

Key metrics and indicators include:

Parameter	Description
Control output saturation	Duration or frequency controllers must run actuators at min/max output. **Units:** Seconds/Count
Coolant temperature	Temperature of coolant. **Units:** °C or °F
Failure to meet setpoint	Number/time of instances when the system is unable to reach or maintain temperature targets within specified tolerance. **Units:** count/min
Flow rate	Measured coolant flow through the loop. **Units:** L/min, GPM
Frequency of control interventions	Number of times control outputs (e.g., pump speed, valve position, heater power) are adjusted per unit time. **Units:** L/min, GPM

Table 79: *Example of chiller/heater health metrics*

Intake air temperature	Measured at intake or ambient. **Units:** °C or °F
Leak/vacuum status	Air ingress or vacuum loss detection status. **Units:** Status/Values
Loop pressure	Pressure within coolant circuit (gauge/absolute) **Units:** kPa, psi
Overshoot magnitude	Degree to which measured temperature exceeds (or falls below) setpoint during transitions. **Units:** °C or %
Over-temperature events	Incidents where temp exceeded safe range. **Units:** count
Power consumption	Total or per-unit heater/chiller power draw. **Units:** W
Pump/fan speed	Actual vs commanded speed; detects degraded or failed motors. **Units:** RPM or %
Ramp time actual vs target	Time taken to reach target temperature compared to expected/policy value, **Units:** Seconds, Minutes
Recovery time after event	Time required to return to normal operating range after alarm/fault. **Units:** Seconds, Minutes
Reservoir level	Fluid level in reservoir; detects low/empty state. **Units:** % or level sensor
Self-test result status	Recent results of system self-diagnostics, including failed checks. **Units:** Pass/Fail/Code
Sensor discrepancy	Maximum or average difference between redundant/independent sensors monitoring the same point. **Units:** °C or %
Service interval status	Time or cycles since last maintenance/service action. **Units:** Hours/Cycles
Supply voltage/current	Key electrical values for heaters/coolers and drives. **Units:** V, A
Time outside target band cumulative	Total duration when temperature is outside allowed control range. **Units:** Seconds, Minutes

75.2. POLICIES AND THRESHOLDS

Policies establish standardized workflows, using health indicators to drive consistent actions such as issuing warnings, scheduling maintenance, or retiring a device.

Log information if

- Recurring control output saturation (actuator repeatedly at max/min limit) is detected, as this may point to control loop tuning issues or undersized components.

Issue a *warning* for conditions not yet severe enough to threaten equipment safety or operation; and prompt attention or scheduled maintenance actions if:

- The coolant temperature remains above the allowable threshold beyond the permitted interval. For example, repeated over-temperature events or overheating during operation.
- The cumulative time spent outside the target temperature band surpasses policy-defined allowances, which may indicate control system instability or hardware degradation.
- The flow rate drops below minimum required levels for safe operation, or if significant fluctuations indicate possible blockages or pump issues.
- The loop (or vacuum) pressure is outside of established safe operating limits, signaling a possible leak, air ingress, vacuum loss or pump malfunction.
- The supply voltage or current is persistently outside the recommended range, suggesting power delivery problems or failing electrical components.

When health indicators decline further or show evidence of impacting reliability or performance, escalate to a *service notification*. This signals a more urgent requirement for corrective action:

- If service interval limits (time or cycles since last maintenance) are reached, ensuring preventative actions are scheduled.
- If repeated self-test failures or sensor discrepancies are detected, indicating the need for maintenance or component replacement.

A fault is raised for serious issues – when the health of a component declines to where continued operation is a risk may be unsafe or unreliable. It disable further heating/cooling as appropriate, enter a safe state, and log all relevant diagnostic data for post-mortem analysis. When any of these scenarios are detected, the system should declare a fault:

- If safety circuits are disabled or fail; the temperature may rapidly exceed safe thresholds, risking equipment or process damage.
- If there are electrical supply faults (e.g. undervoltage, blown fuses, or wiring faults) that could prevent heaters, chillers, or pumps from operating as required.
- If diagnostic routines repeatedly fail or reported heaters, compressor or pump faults.
- If critical temperature, flow, or pressure sensors send out-of-range signals, fail to respond, or diverge from redundant sources – which could potentially result in undetected over- or under-temperature conditions.
- If the coolant flow rate is too low (or zero), potentially due to pump failure, blockage, stuck or broken valves, or empty reservoir.
- If there is detected loss of coolant, air ingress, or pressure drop outside of tolerances.

- If the system fails to deliver sufficient heating or cooling, due to open-circuit heaters, compressor/pump failure, or degraded heat exchange surfaces.
- If the system fails to deliver sufficient heating or cooling. This may indicate degraded heat exchange surfaces.
- If there is a persistent inability to reach or maintain setpoint, especially after a known disturbance. This suggests a loss of control authority or a serious hardware problem (such as heaters stuck off, excessive heat load, or leaks).

75.3. SERVICE REMINDERS

Service reminders translate health indicators and usage counters into scheduled or prompted maintenance. The table suggests what types of health indicator would trigger the service.

Consumable	Triggers
Coolant fluid	Scheduled service interval; Reminders after repeated over-temperature or low-level events.
Filters/strainers	Scheduled service interval; Detection of flow restriction or pressure drop.
Fuses/relays	Scheduled service interval; Detection of electrical or actuator faults.
Heater elements	Cumulative usage hours; Detection of declining performance or output faults.
Pump/fan belts	Scheduled service interval; Abnormal speed readings; drive slippage detected.
Seals and gaskets	Scheduled service interval; Detection of leaks, loss of pressure, or maintenance cycle completion.
Sensors RTD, thermistor	Scheduled service interval; Abnormal readings; sensor discrepancy detected.

Table 80: *Preventative service notifications based on what to replace*

Consumables are principally on a schedule. Some can be inspected, with judgement passed. Few have automated measurement, or derived measurements before serious issues.

76. SERVICE TOOLS

Typical service tools include:

Type of Tool	Description
Multimeter	For checking voltage, current, continuity across heater elements, sensors, relays, and wiring.
Data logger	Captures time-stamped sensor and control data during operation or diagnostics; used to identify trends and intermittent issues.
Portable sensor display / logger	Allows for real-time observation and recording of temperature, flow, pressure, or level.

Table 81: *Service tools*

Flow meter clamp-on or inline	Verifies coolant flow rates and can help identify restrictions, blockages, or pump issues.
Hand pump or pressure / vacuum tester	For leak detection and validation of fluid circuit integrity.
Coolant refractometer or tester	Checks coolant mixture concentration and freeze/boil point.
Temperature reference calibrator / simulator	To verify sensor readings and recalibrating RTDs, thermistors, or thermocouples.
Thermal imager (infrared camera)	To detect hot/cold spots, insulation gaps, or malfunctioning heater/chiller zones.

76.1. BUILT-IN TOOLS

Reporting and Status Utilities: Tools and utilities (such as manufacturer-provided dashboard applications or command-line tools) allow users to read logs, updating settings, or triggering tests (if available). Should permit operators (subject to safety interlocks and access restrictions) to

- Directly operate pumps, heaters, and valves during servicing, calibration, or troubleshooting.
- Revise temperature, flow, and pressure targets, modify ramp rates, dwell periods, and change safety boundaries.

On-Demand Diagnostic Tests that can be triggered manually. These are to quickly assess the status and health of system components in real-time or during troubleshooting scenarios. Common on-demand diagnostics might include:

- Actuating relays or outputs individually to confirm electrical function and actuator response.
- Forcing the heater or chiller to cycle, so temperature rise or fall rate can be monitored and compared to specifications.
- Perform a sensor check, such as displaying instantaneous readings, or cross-checking sensor redundancy for mismatches.
- Running a test to measure coolant flow with all pumps operating, verifying that flow sensors respond correctly and expected rates are achieved.
- Checking pressure sensors, either statically (system at rest) or dynamically (during pump operation), to reveal hidden leaks or blockages.
- Executing self-tests for electrical insulation, ground continuity, and supply voltage levels.
- Triggering alarm or fault simulations to verify correct logging, display, and hardware safety responses.

77. SERVICEABILITY

Design heater, chiller, and other thermal regulation systems with practical service and maintenance in mind. (See earlier chapter for many basics.)

- Select heating cartridges, RTDs/thermistors, fans, and pumps that are field replaceable.
- Use robust mounting points for heating and cooling elements. Specify which thermal interface materials (pads or paste) to use and outline when and how they should be reapplied during service.
- Place temperature and over-temp sensors where they can be easily reached for checks and recalibration.
- With liquid chillers, include accessible drain and fill points, drip containment, and insulation that can be removed and reinstalled without damage during maintenance.
- Design so that if a heater or chiller fails, problems remain confined. Thermal fuses, over-temperature sensors, and coolant flow sensors can be used to catch issues and respond.

Aging equipment often experiences degradation in performance due to factors such as:

- The heating element not attaching as securely (leading to poor thermal coupling)
- Sensors coming loose or no longer making good contact with the surface
- Changes in heater or sensor characteristics over time

Preventative maintenance (physical reattachment, cleaning) along with regular control loop tuning, can maintain reliable performance in aging heater/chiller systems even as their components wear.

The operator should be able to change temperature targets, select or switch between these profiles as desired.

Should provide tools to validate settings. Following parameter updates or other changes, check that the system operates within defined limits. A run-up test can enforce ramp rate restrictions and monitor for overshoot or unstable response whenever changes are made.

78. CONFIGURATION MANAGEMENT

Heater and chiller configuration management tracks parameters & setpoints. Parameters are often grouped into profiles, defining how the system operates for different materials, process steps, or environmental conditions.

The device should have standard, “preset” profiles for common materials and processes. It should be possible to create and modify custom profiles, such as for different materials, processes, and environmental conditions. These may include temperature setpoints, ramp time, flow rates, etc.

After every modification, validate all configuration changes to ensure consistency and safety, checking the settings against defined constraints.

The changes should be recorded in an reversible manner, such as using version management, so previous states and profiles can be restored if needed. This may include the dates and who made the changes, so there is a reliable change history.

Restrict access to critical parameters using user permissions or other controls to prevent unauthorized changes.

Provide clear documentation for all configurable parameters, including default values, safety limits, and interdependencies, to support service, training, and compliance.

79. LOGGING

The management system should maintain logs covering the following:

- Events such as temperature excursions, sensor faults, alarms, system start/stop, changes to operational profiles, and maintenance actions. Each entry should include the time and the responsible user or process, when relevant,
- Record overshoot magnitude, time spent outside target temperature bands, and recovery times after disturbances or setpoint changes. These help identify emerging concerns.
- Service activities, such as controller retuning, recalibration, or maintenance actions (including reattaching heating elements or sensor replacements). Frequent or repeated retuning should be flagged, as it may suggest underlying mechanical issues that need further investigation.
- All configuration changes, including parameter adjustments, firmware or software updates, and profile modifications. Record hardware models, identifiers, versions, if available.

Poor control performance, frequent or repeated retuning should be flagged, as these suggest underlying mechanical issues that need further investigation. Replacement rather than repeated adjustment may be required.

If frequent retuning is necessary – chronic attachment issues may indicate mechanical wear requiring replacement rather than repeated adjustment.

80. FAULT FINDING AND TROUBLESHOOTING

First, inspect the elements, and consumables:

- Check that the heating element is firmly attached to the target surface; resecure with fresh thermal paste or proper mounting hardware if necessary.
- For temperature sensors, inspect the mounting; clean contact surfaces and ensure the sensor is pressed snugly against the measured area (replace adhesives or clips if degraded).
- Inspect all compression fittings, O-rings, and joints, paying particular attention to custom cooling or heating loops. Look for signs of wear, leaks, or degradation.
- Look for oxidation, corrosion, or wire fatigue at connection points; repair or replace as needed.
- Low coolant, wrong mix, or exhausted inhibitor can cause overheating, boil-over, or heater problems; verify level when cold, cap and hose condition, thermostat behavior.
- Check for detect blockages at exhaust or purge ports.

Verify sensor functionality:

- Cross-check sensors to identify any that may be faulty, drifting, or producing inconsistent readings; prioritize these for replacement or recalibration.. Verify sensor

accuracy by comparing system values to a reliable reference, such as a calibrated thermometer.

- Check heater and power/ground (most common preventable cause): Verify O_2 heater operation (time-to-light-off), check fuses/relays, sensor connector integrity, and ground paths. A weak heater can cause "slow response" codes even when the sensor element is otherwise healthy.

Common scenarios requiring attention include:

- It is unable to reach the set temperature, or reaching it significantly faster than expected.
- It cannot achieve or maintain stable temperatures, or display unusually rapid changes in temperature.

Potential underlying causes for these issues include system leaks, hardware malfunctions, or degraded thermal coupling.

80.1. PID CONTROLLER ISSUES

tuning temperature PID controllers

Temperature overshoots, unstable temperatures, or slow responses, indicate the controller needs retuning. As thermal coupling degrades, the system's heating response changes. Re-tuning the PI/PID control loop can help restore stable performance as these components age. Some control systems offer automatic routines for re-identifying or auto-tuning system dynamics; run these automated procedures when available.

If automate procedures are not available, manually adjust the proportional (P), integral (I), and derivative (D) parameters based on observed behavior:

1. Increase P (proportional) value if the system responds is slow to change temperature. Systems with high thermal mass (like heated beds or large hot ends) often require a higher P value to react quickly to temperature deviations.
2. Adjust the I (integral) parameter if the temperature is not changing, and there is a persistent difference between the system's actual temperature and the setpoint (steady-state errors).
3. Increase D (derivative) if new oscillations or overshoot are due to delayed heat transfer. Systems with low thermal mass and strong heaters are especially susceptible to oscillations and benefit from a higher D value to help dampen overshoot.
4. Perform step tests by making abrupt changes to the setpoint. Observe how quickly the system responds and how much it overshoots. Adjust the PID parameters accordingly.

If frequent retuning is necessary, log these interventions. This may indicate mechanical wear requiring replacement rather than repeated adjustment.

80.2. LEAK DETECTION

Replace damaged O-rings, seals, or degraded tubing. Tighten loose fittings or connections as needed and reapply clear RTV silicone sealant to any connections or joints as necessary.

After servicing, run the system and monitor closely for any new leaks.

81. REFERENCES AND RESOURCES

https://www.wildcircuits.com/2013/09/accurate-temperature-control.html

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PART V

Software Serviceability

This part describes design and processes to support servicing the product software:

- SOFTWARE MODULES. Software modules are often configurable and replaceable.
- THE ABSTRACT MACHINE SUPPORT. A detailed look at the abstract machines architecture, how it is monitored and serviced.
- STORAGE MANAGEMENT. A look at the monitoring the health of storage system features, beyond the hardware.

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CHAPTER 18

Software Modules

This chapter discusses software components within a system. Topics covered include:

- An overview of software modules, their features, and how they work within the system
- Monitoring module health and system thresholds
- Approaches to troubleshooting module-related issues

82. OVERVIEW

A software *module* is an distinct unit of software that encapsulates code and related data, designed to provide specific functionality within a larger system. In operating systems, modules typically refer to components (such as device drivers or file systems) that can be dynamically loaded or unloaded at runtime.

Software modules are often designed to be field-replaceable – meaning they can be swapped, often as part of an upgrade of features. The scope and granularity of what constitutes a replaceable module varies: sometimes a module is standalone, while other times it is embedded as part of a broader software package.

Like hardware, software modules have health metrics. They are evaluated using different metrics; they generally do not face concerns around environmental exposure or physical wear from cycles of use. However, some aspects of system health are relevant to both hardware and software.

82.1. SOFTWARE MODULE LIFECYCLE

lifecycle

The typical execution lifecycle of a software module includes:

- Loading the module into memory, preparing resources such as hardware access, registering features, etc.
- Entering an operational state where it can perform its designated functions, possibly indefinitely,
- Unloading the module from memory; this includes releasing all resources it has allocated. After unloading, an replacement may be subsequently loaded.

Some of the following sections touch on these.

82.2. QUEUEING MODEL

queue model

Many modules operate by processing work items conveyed by queues. Work is dispatched via messages that contain the necessary parameters for processing. Each message commonly includes information for scheduling, activation of operations, data being transferred, and a mechanism to acknowledge work completion back to the sender. It may also provide status

updates at various processing stages. The queue might support prioritization, servicing some items before others.

System management monitors these queues and workloads to measure operational health. Core areas monitored often include:

- Network activity, including transmission and reception of events, commands, and responses
- Storage operations: commands execution and responses,
- Communication subsystems, e.g. network, serial connections
- Windowing or graphical user interface systems

82.3. DRIVERS

Drivers are specialized software components that serve as the interface between hardware and the system. Drivers typically maintain their own work queues, managing incoming requests and returning results in a structured manner. They are responsible for exposing hardware-generated metrics – such as error rates and operational statistics – to the broader system.

They are often designed to be replaceable, either as standalone modules or embedded within larger replaceable software packages.

For drivers, the work items are often known as I/O requests. They orchestrate data movement or control operations. For I/O drivers it may be called an I/O request, being a request to an IO driver or processor rather than an request that is specifically input or output.[7] The main categories of work include:

- Read request, to read data from storage or receiving packets, gathered from lower layers
- Write or send, used for writing data to storage or transmitting packets, passed to lower layers.
- Power control
- Change communication parameters
- Perform hardware actions, such as ejecting media.

The data payload is often organized as a scatter-gather list, to efficiently transfer large volumes of data. The messages often include reference the target driver, current status, next processing steps, links to related messages, protocol headers, and related information.

82.4. RESOURCE TRACKING

The resources associated with each module are tracked. This is necessary to ensure that all resources are properly released when a module is unloaded. The module is responsible for cleanly releasing the resources; when it is unloaded, it should also put the hardware into an idle, off, or other quiescent state, and release the resources with the hardware. The OS may have an back up strategy to recover the tracked resources, esp. when a module is forcibly unloaded.

[7] On VMS and Windows NT, it is almost always called an I/O request packet (IRP); others called it an event control block.

The tracking information is also useful when tuning performance, understanding the behavior of the software and hardware, and diagnosing issues, such as buffer utilization and depletion.

The tracking records which modules use the resource, as well as metadata describing each resource – e.g. its purpose, usage context, and the internal unit that allocated it. This information helps administrators understand resource needs and usage patterns, correlates issues with specific usage contexts, and aids developers in diagnosing and resolving defects related to resource management.

Typical uses and benefits of resource tracking include:

- During testing, this is used to validate that when a module is unloaded, it correctly releases all resources. If resources are left allocated after a forceful unload, tracking enables their identification and cleanup.
- Additional descriptors (e.g., purpose, allocating unit) can be used in monitor to correlate with resource usage for a given purpose, or track down resource exhaustion or leaks.
- Developers can use detailed tracking data to identify and fix issues involving improper allocation or release of resources.

82.5. COUNTERS

Each driver maintains a set of counters that track key events, such as the number of successful and failed transmissions, packet overruns or drop events, buffer utilization and depletion, error conditions, and other critical runtime occurrences. These statistics are useful when tuning performance, understanding the behavior of the software and hardware, and diagnosing issues.

Each driver or protocol layer may also define additional counters pertaining to its specific functionality. These statistics are typically made available through control and status interfaces, supporting access for diagnostics, debugging, and management tools.

83. MANAGEMENT INFORMATION AND DIAGNOSTICS

83.1. HEALTH INDICATORS

This section outlines metrics used to evaluate the health, and performance of software items. These indicators address the following key questions:

- Is the software item functioning at all?
- Is it responding within acceptable time?
- Is the software item producing correct results?
- How effectives does it recover from issues or failures?
- Is the software item maintaining safe operational conditions?

Parameter	Description
Availability	Percentage of time service is operational. **Units:** %

Table 82: *Example of software health metrics*

Configuration Drift Events	Number of times configuration was out-of-sync or changed. **Units:** count
Crashes	Count of reboots/crashes, watchdog resets. **Units:** count/min
Dependency Health	Status of required upstream/downstream services **Units:** Pass/Fail/Status code
Disk I/O	Read/write rates to disk **Units:** MB/sec
Disk Usage	Disk space used **Units:** GB
Input/Output Validation Failures	Incidents of rejected or invalid data. **Units:** count
Invariant Violation Count	Failed assertions or invariant checks at runtime. **Units:** counts
Network I/O	Transmit/receive rate and error counts **Units:** MB/sec, counts
Queue depth backlog	Number of pending items in internal or external queues. **Units:** count
Request Failure Rate	Percentage/number of failed requests or operations. **Units:** %, count
Resource Leak Warnings	Unreleased handles/objects detected during or after operation. **Units:** count
Resource Usage	Consumption of CPU, memory, or other key hardware **Units:** %, MB, GB
Response Latency	Time to handle requests (p50, p95, p99 latency) **Units:** (milli)seconds
Timeout Rate	Operations that timed out as percentage of all attempts. **Units:** %, count
Uptime	Total time software/process has been running. **Units:** seconds

83.2. POLICIES AND THRESHOLDS

Policies and thresholds establish the operational boundaries (acceptable range of values for critical metrics) and define responses when these are exceeded or violated.

The types of policies are:

- Define minimum acceptable service uptime (e.g., 99.9%) and action plans for outages.
- Specify maximum acceptable response latency (e.g., p95 latency < 200ms), minimum throughput requirements, and allowable backlog size.
- Set upper limits for CPU, memory, disk, and network usage to prevent resource exhaustion.
- Establish error/crash rate thresholds triggering investigation, rollback, or module isolation.
- Mandate maximum authentication failure rates, required encryption standards, and allowable access attempt patterns.

Is the system functioning at all?

- If monthly *uptime / availability* < 99.9% page support, escalate
- If the *error rate* is 1% failures per transaction (over any rolling 10-minute window) or above, begin rollback/fallback procedure.
- If there are more than 3 *crashes*/hour, restart, investigate
- Failed job / transaction rate
- Service dependency availability

Is the system responding within acceptable time?

- Latency (p50, p95, p99). If p95 Latency < 250ms; trigger scaling automation if exceeded in 3 consecutive periods.
- If *response latency* (p95) > 500ms, alert on-call, scale up
- Request/transaction duration
- Throughput (requests per second)
- Queue depth / backlog size
- Timeout rate

Is the system running efficiently for the workload?

- If CPU utilization > 85% for >5 minutes, auto-scale, optimize
- GC pauses
- If memory usage > 90% total RAM, trigger garbage collect
- Disk I/O and disk space. If disk usage > 85% of partition, free space, alert
- Network throughput and errors. If network error rate > 1% packet loss, switch route, alert
- Thread pool / connection pool saturation

Is the system producing correct results?

- Data validation failures
- Invariant checks failing

- Mismatch between expected vs actual state
- Reconciliation errors

How well does the system recover from problems?

- Circuit breaker open rate
- Retry rate
- Fallback usage
- Mean time to recovery (MTTR)

Is the system operating safely?

- Authentication/authorization failures. If authentication failure rate > 10 failures minute/user, lock account, investigate.
- Suspicious access patterns
- Rate-limit violations
- Input sanitization failures
- Security policy violations

83.3. SERVICE REMINDERS

Service reminders translate health indicators and usage counters into scheduled or prompted maintenance, such as software updates, security reviews, backup operations, license renewals, or configuration audits. The table suggests what types of health indicator would trigger the service.

Module / Component Type	Triggers
Core Application	High error/crash rate, version outdated, failed self-checks
License / Compliance	License expiration nearing, compliance scan failed.

Table 83: *Preventative service notifications based on what to replace*

84. SERVICE TOOLS

A suite of service tools is often provided for use by system administrators, developers, or support staff. These tools can perform diagnostic checks, health monitoring, verify interfaces, and automate common service tasks. They assist in the maintenance, troubleshooting, and system management. The tools typically include:

- Diagnostic utilities to identify or localize faults, assess module status, and check for configuration errors.
- Programs that report resource consumption, active connections, and operational statistics for modules and related hardware. E.g. process monitors (ps, procmon, etc.), registry monitors

- Package management tools to install, update or remove software. These tools verify the integrity, authenticity, and proper installation of software updates and modules. The package manager may perform some of these automatically, such as with security updates.
- Automated recovery scripts to reset, reload, or reinitialize modules or drivers in response to faults or after updates.
- Test harnesses and simulators to exercise module interfaces, generate test traffic, or simulate client or hardware interactions to validate correct behavior.

84.1. MANUAL TOOLS

These tools are intended for hands-on investigation by experienced administrators and developers, complementing automated diagnostics. They are used in targeted intervention during complex or unusual system events for in-depth system analysis, troubleshooting, and configuration.

- System monitor utilities to track CPU, memory, network, and disk usage.
- Process Monitor / Task Manager, to tracks and logs real-time process.
- Procmon Tracks and logs real-time processes, file system, and registry activity.
- Registry monitoring tools observe changes to the system registry and diagnose configuration problems.
- Sysinternals is an suite set of utilities for process management, file and disk monitoring, networking, and diagnostics.
- Automation platforms, like Microsoft Power Automate to script, schedule, and orchestrate system tasks and remediation workflows.

84.2. INTEGRATED AND FLEET TOOLS

In enterprise environments, tools integrate health monitoring with automated incident response. Administrators and support teams are able to continuously monitor system health, verify configuration and integrity, and automate maintenance, troubleshooting, and compliance tasks.

Sense. Tools continuously collects telemetry from every critical subsystem – process & thread activity, program loading, registry changes, network events, file activity, malware scans, logs, etc. Advanced content inspection tools dynamically analyze scripts, running code, and other content.

Predictive failure and trend analysis. This data is consolidated and analyzed for patterns or anomalies that indicate impending failures. Security analysis combines real-time behavioral analysis, machine learning, and policy-based rule sets to identify threats, as well as abnormal or unauthorized behaviors as they occur.

Automatic, policy-driven responses to recurring issues, as well as manage scheduled maintenance.

Reporting features present a unified dashboard for monitoring, alerts, guided troubleshooting working flows, and support detailed report generation.

85. SERVICEABILITY

Serviceable software provides features and interfaces that make it easy to:

- Monitor system health and behavior with built-in logging, metrics, and status indicators.
- Perform regular maintenance tasks such as configuration changes, software updates, and patch management. Support replacement or hot-swapping of components where feasible.
- Quickly diagnose faults, anomalies, and performance issues, often via diagnostic tools, debug interfaces, and clear error reporting.
- Fault recovery features such as robust error handling, fast rollback mechanisms, and clear guidance for operators.
- Audit system changes and operational histories.

85.1. PACKAGES

packages

Software modules or components can be distributed, installed, updated, and replaced in the field. A *software package* is a collection of software components, resources, metadata, and instructions bundled together for this purpose. Distributing software as packages allows modules to be consistently deployed, authenticated, and validated across different environments. The format and structure of a package may vary depending on the packaging system and the operating environment. Packages usually include:

- Metadata, such as the package name, version, description, author, an list of all files contained within the package, and dependency requirements. This is used when verifying compatibility, ensure integrity, and facilitate updates or removal.
- One or more executable programs, libraries, configuration files, documentation, and installation scripts required to deploy and operate the software in a target environment.

Software packages are managed by *package management* systems, which automate the processes of installing, upgrading, configuring, and removing the software and its dependencies.

85.2. SOFTWARE AUTHENTICITY

Each package's authenticity must be verified before its installed or used to update. A structured update process ensures that every component can be uniquely identified, authenticated during installation and updates, and safely replaced when needed. The package manifest may include cryptographic digests of each contained file, used to confirm the integrity of individual installed files. The package itself is securely signed to verify its integrity and authenticity.

The package is validated by the following steps:

1. Checks that the manifest cryptographic signature is correct.
2. Verify that all required files are present within the package and that only permitted files are included, with path names adhering to specified conventions.
3. Ensure each file in the update package matches its declared digest/checksum in the manifest.

To verify that installed software matches the official package, the signature of each file (or resource) is checked against the one in the manifest. This confirms that all required modules and files are present and unmodified (that their contents have not been altered by accident or tampering), and identifies any unauthorized or unexpected files in protected locations.

Such integrity checking support compliance and security audits, and reliable recovery. They identify exactly which files require restoration or reinstallation if discrepancies are found.

85.3. UPDATES

The update procedure modifies persistent storage, e.g. files. Updating files and software that are not in use is straightforward. However, replacing actively used components – such as drivers, services, or other running programs – presents challenges.

An A/B update approach is often used to ensure high availability and reduce downtime. This method maintains at least two independent software versions ("A" and "B"). The update is applied to the inactive copy (for instance, updating slot B while the system operates from slot A), leaving the active environment undisturbed. Once the update is installed and verified, the running software and any related processes are stopped, and the resources in question unloaded from memory. The updated components are then restarted. This may require restarting the system. If any issues occur with the new version, the system can quickly revert to the earlier version.

Some environments provide support for replacing running drivers without interrupting service (hot swap). These dynamically replace an executing driver with an new one, without service interruption – even if the hardware itself (if any) is not hot swappable. The ongoing activities and resources are transferred to the new driver version. This feature is rare outside of specialized environments, mostly historical, such as NetWare systems. Most platforms rely on redundant hardware or service failover to ensure continuous operation during updates.

85.4. GOOD DOCUMENTATION

Software serviceability benefits from good documentation. Document:

- Configuration variables, their values, what they mean, and implications
- The source of each measurement and counter and provide traceability. The meaning of each value, and how to use them appropriately.
- Each status code, with a description that explains the circumstances under which the error occurred and potential solutions. The documentation should describe how to interpret them.

The management interface should provide access to these configuration and measurement values.

86. CONFIGURATION MANAGEMENT

Installed packages and package versions are handled thru the package manager. Most support access to public and private repositories of versioned package and patch sets. The manager allows coordinated updates, downgrades, and handles dependency management.

In basic systems, configuration often can be saved (exported) and maybe restored from these files.

Advanced systems utilize version control for configuration files, capturing the history of changes so that every modification is traceable and reversible. This makes it possible to reconstruct previous system configurations, which is useful for debugging or forensic analysis. This may not be eligible for configurations managed using non-file-based mechanisms (e.g. databases).

Following any configuration modification, the software tools should verify correct syntax, and consistency with system policies or schemas. Automated integration and smoke tests can be executed to confirm the system remains functional after changes.

Ideally, configuration formats should be compatible across software versions.

87. FAULT FINDING AND TROUBLESHOOTING

Troubleshooting gathers information about system activity, failures and responses from the event log, and diagnostic tools. Error messages, system status, log entries, performance counters, and self-checks assist identifying abnormal behavior or failing components.

The analysis typically involves examining recent configuration or software changes, analyzing module interactions, and reviewing logs to reconstruct what happened before the problem appeared. Reviewers can look for correlation with reported incidents and the timing of changes.

Systems with modular design allow software items to be replaced, reloaded, or rolled back. This allows engineers to temporarily replace specific components, adjust isolated items, verifying or ruling out suspected causes, and can help reveal whether the issue resides within a single area or emerges from more complex interactions.

87.1. SYMPTOM-BASED TRIAGE

Work from the evidence, narrowing the field using observable effects, before attempting repair or replacement. Gather and catalog all relevant warning indicators, error messages, alerts, abnormal measurements, user reports, and system logs. This is automatic is high-end tiers, and manual in low-end, like linux. Modern software supplements this technique with features such as structured error codes, self-describing status reports, and guided troubleshooting wizards.

Narrow the fault space before running deep tests: group the complaint, note when it appears, and tie it to likely subsystems.

Map the symptoms to possible causes. to connect each symptom to one or more potential sources or subsystems. High-end tiers will use a database, and rulesets cross-referencing with logs, historical incidents, and failure checklists. Low-end will combine past experience and just guessing.

Table 84: *Symptoms*

Module / Component	Symptoms and Health Indicators
Backup/Archiving	Missed backup jobs, failed verifications, storage nearing quota.
Configuration	Drift from baseline detected, checksum mismatches
Core Application	High error/crash rate, version outdated, failed self-checks.
Driver	Repeated I/O errors, hardware interface changes, deprecated driver

License/Compliance	License expiration nearing, compliance scan failed
Monitoring/Telemetry	Gaps in reporting, high latency, alert flood
Network Module	Excessive packet loss, connectivity drops, protocol errors
Scheduler/Task Runner	Repeated job failures, missed deadlines, queue backlog
Security Module	Policy violation detected, repeated authentication failures, expired certificates
Storage/Database	Disk space low, read/write errors, validation failures

Historically, less sophisticated systems depended on manual checks, ad hoc tools or scattered scripts, and basic status indicators. This provided little to no observability, no actionable guidance or insight into underlying problems. Tooling was reactive, and troubleshooting often relied on intuition or trial and error, witchcraft. Metrics were anecdotal

87.2. STATISTICS

Systems without the measures and health interpretation. Simpler, and often faster on some speed based performance. As consumer products evolved adopt some. But (usually/often) thru an path. It may be more ambient. Adapt for things meaningful to their audience, small number. Simplified, more abstract presentation of health to consumer.

So many ideas have been explored by different areas, some reinvented, harmonized, others can't be harmonized, incommensurable for some reason.

Many approaches to driver design have been proposed and implemented across different domains. Some have converged toward standardized models, while others remain distinct or incompatible due to technical or domain-specific limitations.

CHAPTER 19

Memory Tuning

This chapter discusses the virtual memory system and caching systems, how logical address spaces are mapped onto physical memory and backing store, how both memory and file data are cached for performance, how their behaviors are measured, and tuned.

- An overview of the abstract memory system, including virtual memory and demand paging, buffering and caching in the storage stack
- The mechanisms that must be monitored: address-translation caches, page faults, working/resident sets, page replacement, backing store
- Buffer cache, unified page cache, and mapped-file models
- Mechanisms: read-ahead, write-back, dirty limits, metadata caching, clustering
- Metrics and health indicators for cache and paging, along with policies, thresholds, and tuning knobs
- Tuning to the memory hierarchy, and the interaction between software and hardware.

88. OVERVIEW

Virtual memory lets each process use a large, contiguous address space, regardless of physical memory organization or availability. The operating system transparently maps virtual pages to physical frames, swapping pages between RAM and storage as needed so programs can use more memory than is physically installed.

At the same time, caching is used throughout the memory hierarchy to accelerate access to both anonymous (process) memory and file data. The system caches frequently used file blocks, metadata, and program code in RAM, reducing repeated I/O to slower storage and enabling faster data access for applications.

Careful tuning of this system is required to balance software and hardware performance. It requires adapting to the specific hardware resources, and workload patterns.

Because symptoms of memory pressure, cache contention, and storage path faults can overlap or be easily misattributed, this chapter also provides guidance for distinguishing between cache-related and interconnect-related issues. See also chapters 13 and 14 for monitoring the health of the storage device and interconnects. Cache symptoms and interconnect faults are often confused.

88.1. MEMORY ALLOCATION

Platforms typically provide more than one kind of physical memory, with different sizes, speeds, and usage restrictions:

- Fast RAM directly connects to the CPU, offering the lowest latency and highest bandwidth. Typically used for code execution and active data.

- Slow RAM, larger external memory modules such as *pseudo-static RAM* (PSRAM), provides megabytes of space but at a lower access speed.
- Tiered/Banked Memory or memory apertures, when the region is larger than the CPU can map at once. Embedded memory may use bank-switching to access more memory. Large flash, PCIe, or external RAM may be exposed through a movable aperture, where a smaller window is remapped to different segments of the device.
- Flash and related non-volatile memories (FRAM, MRAM, battery-backed SRAM) addressable directly by the processor. The processor can execute code from the devices. They are slower than RAM, with writes generally disallowed. Writes are very slow, and often need a programming protocol, with an erase-before-write.

This mix of memory raises several allocation concerns:

- Physical memory may be fragmented across chips or banks. The allocator tracks each region; virtual mapping can present a contiguous range even when physical frames are not.
- When using memory apertures or bank switching, moving the window to a new region must be carefully coordinated, usually granting exclusive access during the operation.
- Unified (or linear) addresses span RAM and memory-mapped devices so software need not know which chip backs a page.
- DMA buffers, ISR/critical code, and some device buffers may require fast or directly attached RAM; bulk caches and data can use slower pools.
- Permissions and cache policy (executable, uncacheable device registers, write-back vs write-through) are set per region via page tables / MMU.
- Flash regions are typically read-only or read-execute, with instruction cache enabled. Care must be taken so they cannot be used as heap or stack.

The memory manager is responsible for managing the RAM, using it for kernel resident code and data, available for use as page cache, file cache, and working data. It tracks each region's size and attributes, allocates by memory type, and balances speed, placement constraints, and capacity. Callers may explicitly request fast or slow memory, or the system may use automatic placement.

Critical code – ISRs, drivers, core OS, performance-sensitive code – and data structures should remain in fast memory. Some hardware-accessed buffers (e.g. DMA and LCD frame buffer) must be in direct RAM. Less frequently accessed or bulk data (such as network buffers, or file caches) can be offloaded to slower, larger external RAM. Sometimes code can only execute from fast RAM, while large data buffers, or caches are assigned to external PSRAM.

Code in flash, esp hot paths, may copied to fast RAM at boot or on demand.

88.2. MEMORY-MAPPED DEVICES AND MEMORY WINDOWS

Peripherals such as display controllers, NICs, custom blocks, frame buffers are often mapped onto the address space so drivers may read and write registers as memory. These mappings are typically uncached, unguarded or strongly ordered, and non-executable so speculative or out-of-order access cannot trigger side effects.

89. VIRTUAL MEMORY

Virtual memory gives the illusion of a large, contiguous address space decoupled from installed physical memory. The subsystem maps the pages of that address space onto physical page frames on demand, moving pages between primary memory and secondary storage (the backing store or paging store) as pressure requires. This allows lets a program, or a set of programs, use more memory than is physically present, without managing a large store themselves.

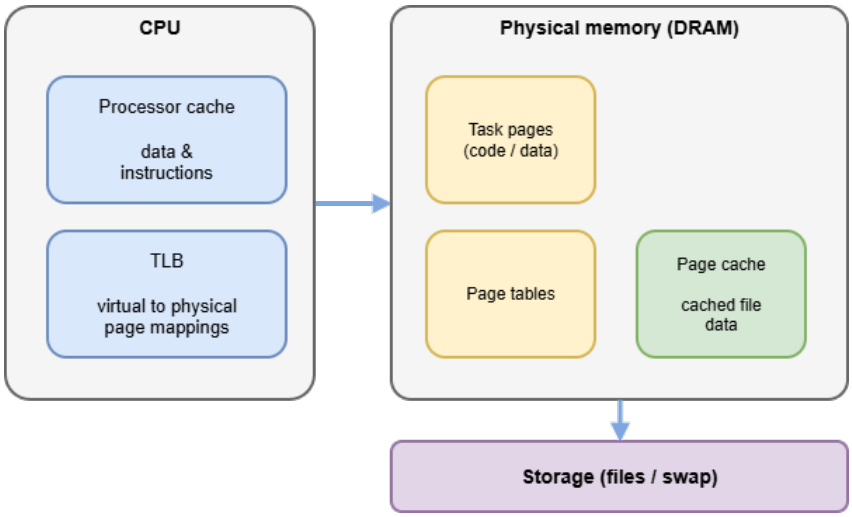


Figure 12: *The virtual memory subsystem*

89.1. ADDRESS TRANSLATION CACHES AND PAGE COLORING

memory management unit

RAM is often organized into logical page frames, with the *memory management unit* (MMU) translating the logical pages to physical addresses. Walking the page tables for each access would be costly, so the MMU caches translations and the OS lays out frames to keep those caches effective. Three mechanisms are used so a well-laid-out working set stays fast:

translation lookaside buffer

A *translation look-aside buffer* (TLB) is a small, fast hardware cache of recent logical-to-physical mappings (often split for instruction and data, sometimes with a second-level shared TLB). A TLB hit completes translation without a memory walk. A TLB miss walks the page tables (possibly using a software handler). Since each process (and the kernel too) uses a different logical address space, context switching could cause a performance hit to load the approach TLB for the next context. The CPU may attach small tags, such as *address space identifier* (ASID) or *process-context identifier* (PCID) to TLB entries so that different address spaces (processes) can coexist without a full TLB flush on every context switch.

page-table caching

Multi-level page tables are themselves ordinary pages. Intermediate entries are cached by hardware. Pinning or keeping hot page-table pages resident avoids turning a TLB miss into several DRAM references. Large or fragmented address spaces, frequent changes to memory mapping and pressure to reclaim pages all inflate walk cost.

page coloring

Set-associative processor caches typically choose the set from the low bits of the physical address. Nearby virtual pages are likely to be used together (spatial locality). If they map to frames with the same low page-number bits, they collide in the same cache sets and evict each other even when the working set would otherwise fit. The memory manager ‘colors’ frames so the low bits of the virtual page number match the low bits of the physical frame (commonly three or four bits of the page number). A small run of consecutive virtual pages then occupies distinct cache sets.

At this point, a system might stop here (especially if hardware features are limited or inaccessible). This provides memory isolation between processes, and decouples a program's address space from the underlying hardware. Demand paged virtual memory takes this further.

89.2. PAGING

paging

One of the functions of a virtual memory system is basic *paging*. When free physical memory is low, it reclaims frames. On many systems that means writing pages to secondary storage, and reading them back when those pages are referenced again. Some operating systems (e.g. iOS) reclaim by compressing pages in RAM, compacting and discarding clean file-backed pages. Tasks that use excessive memory many be terminated. Serviceability tracks pressure, reclaim cost, page I/O, compressor memory and process kill, as appropriate.

virtual memory manager

The *virtual memory manager (VMM)* allocates, maps, protects, and reclaims memory across all tasks. Its responsibilities include:

Table 85: *Virtual memory manager*

Responsibility	Function
Address space management	Create and maintain the per-task map from virtual ranges to memory objects, with protection and inheritance attributes.
Backing store management	Allocating, tracking, and reclaiming space in the paging store or compressor space where the platform provides it.
Cache management	Keep frequently used file and pages resident, and coordinating with pagers where present.
Demand paging	Populate pages only when they are first referenced, rather than up front.
Health monitoring	Exposing counters and rates (faults, scan activity, backing-store or compressor usage) to the broader system.
Page replacement	Select victim pages and writing dirty pages to backing store, compress in place, discard clean file pages, or terminate tasks when free memory is low.

The paging behavior is a sensitive indicator of overall system health: a system that is thrashing, leaking, or starved of backing store will show it in paging metrics long before it hard-fails. Its behavior is tunable at run time.

89.3. PAGE FAULTS

page fault

A *page fault* occurs when a task references a virtual address for which there is no valid physical mapping, or for which the access violates protection. The fault handler is the hub of the virtual memory system; on each fault it must locate the memory object and offset, obtain a physical page, and install the mapping. Faults fall into distinct classes that must be measured separately because their costs differ by orders of magnitude:

Table 86: *Page faults*

Fault type	Description
Copy-on-write (COW) fault	A write to a shared, read-only page triggers a private copy. This underlies efficient `fork`, shared libraries, message passing, and virtual copy operations.

Hard fault major	The page must be fetched from backing store or a file. This incurs secondary-storage I/O and dominates paging latency.
Invalid / demand-zero fault	Used to implement zero-fill-on-demand and demand paging of executables. The valid bit is left clear so that the first reference traps and the system fills or fetches the page. Lazy evaluation of this kind lets large regions be allocated at nearly zero cost until actually touched.
Protection fault	The mapping exists but the access is not permitted; may indicate a program error or a guard page.
Soft fault minor	The page is already resident (for example, on the inactive or free list, or shared by another task) and only a mapping must be established. No disk I/O is required.

89.4. RESIDENT AND WORKING SETS

resident set
working set

The *resident set* is the set of a task's pages currently in physical memory (its resident-set size, RSS). The *working set* is the set of pages a task has referenced within a recent time window – an estimate of what it needs resident to make progress. When the working set does not fit in the available resident set the task faults continuously. The system thrashes when many tasks are in this state simultaneously. Measure those with the fault, resident and working set indicators.

89.5. PAGE REPLACEMENT

The cache of resident pages eventually fills and the system must choose which pages to evict in order to free space. This is managed by page replacement algorithms such as FIFO (First-In, First-Out), LRU (Least Recently Used), second-chance / clock, working-set. The pages are organized into three queues:

Table 87: *Page queues*

Page List	Description
Free list	The frames available for immediate allocation.
Active list	The pages actively in use in some address space; newly allocated pages go here.
Inactive list	The page frames not currently mapped in any address space but recently used, effectively a second chance before replacement.

Whenever the free page count falls below a threshold, pages are moved from the active to the inactive list (removing their mappings), and pages on the inactive list are moved to the free list, writing the dirty ones to the backing store first (if the platform has a paging store), or compressing / discarding them where it does not. A page that is faulted while on the inactive list is simply reactivated. This yields FIFO-like replacement in constant time per victim, while the inactive list preserves genuinely hot pages. The rate at which scan is performed, and the ratio of pages scanned to pages actually freed, are primary health signals.

89.5.1 Backing store, mapped files, and clustering

Dirty anonymous pages are written to the paging store (if one exists), or compressed in RAM. Clean memory-mapped file pages can be discarded and re-read from the file; a "fault" triggers

an ordinary read/write operation on the file. Tasks may be terminated reclaim by terminating rather than paging them to flash.

The system typically clusters page-in and page-out operations, reading or writing several contiguous pages per operation to amortize IO cost. It may pre-fetch (read-ahead) based on access patterns. Clustering and pre-fetch effectiveness show up in the major-fault rate and average fault service time; on compressor-based systems, watch compressor size, memory pressure, and kill rates instead of page I/O.

89.6. SLEEP, HIBERNATION, AND POWER-STATE MEMORY

The virtual memory manager may support different product power states. So that resume restores a usable working set (without silent data loss), the page/file cache must preserve key management information to resume service after wake: the pressure band or watermarks, commit/backing-store (or compressor) usage, open mapped files, and recent fault/reclaim rates.

Sleep mode (*suspend to RAM*), leaves system memory powered via self-refresh, preserving address space maps and resident pages. Memory mapped IO regions and peripheral mappings must be re-established on wake if they do not remain valid. Before entering sleep, the dirty pages much be written to backing storage, and outstanding writes must complete, in order to meet the product's durability requirements so data isn't lost if power is interrupted during sleep. On wake, the system verifies the integrity of the free, active, and inactive queues, any compressor state, and the status of I/O interconnects and host bus adapters before resuming normal I/O. If an I/O path is stale or mis-configured after resume, cache misses may surface as inexplicable memory faults.

Hibernation (suspend-to-disk) will remove power from memory, so the contents (or a select subset) are written to non-volatile storage first. This maybe the page store if one exists or hibernation image. When resuming, the stored image is read back in, and memory mappings are reconstructed.

Policies should treat failed hibernation write, incomplete restore, or post-resume thrashing/major-fault storms as service conditions, distinct from steady-state pressure while fully on.

90. CACHING MODELS

When an application reads or writes a file, the operating system uses an in-memory cache to reduce the amount of device I/O. This section discusses the buffering and caching layer: how file and metadata I/O is staged in memory, how that cache is measured and tuned, and how those measurements drive service actions.

The buffer/page cache is often the largest consumer of free RAM and the dominant amortizer of disk and SSD I/O; its health, and the health of the path beneath it, are first-class serviceability concerns.

Such a design might look like:

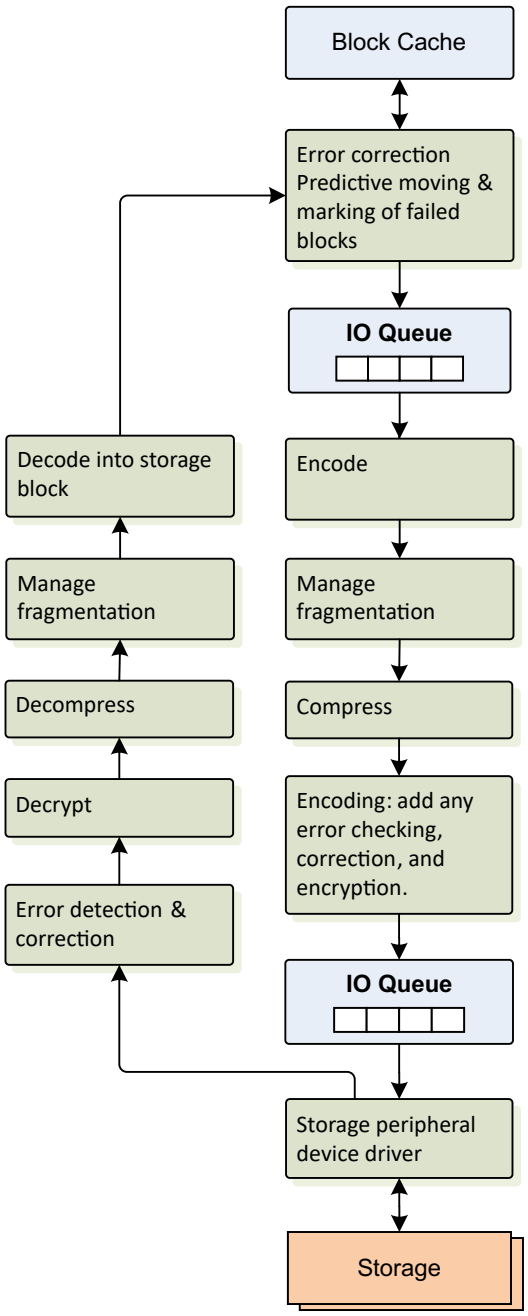


Figure 13: *Typical storage stack*

Elevator. Structure for each disk: a series of queues (the outgoing queue is owned by the driver). The storage peripheral driver is allowed to merge requests.

90.1. UNIFIED PAGE CACHE AND MAPPED FILES

Modern systems treat physical memory as a single cache of memory objects, including file-backed pages.

91. MANAGEMENT INFORMATION AND DIAGNOSTICS

91.1. HARDWARE CACHE METRICS

Metric	Description
Cache conflict-miss rate	L1/L2/L3 misses consistent with set collisions (coloring/alignment. **Units:** misses/sec; %
Huge-page / large-page coverage	Fraction of RSS mapped with large pages (fewer TLB entries). **Units:** % of resident set size
TLB miss rate	Translation look aside buffer misses **Units:** misses/sec, %

Table 88: *Example of processor cache metrics*

91.2. VIRTUAL MEMORY HEALTH INDICATORS

This section outlines evaluating virtual memory subsystem health using metrics. Memory pressure should be detected and addressed early, before it escalates to thrashing or allocation failures.

- Continuously track memory demand versus available resources, comparing trends to established baselines.
- Diagnose whether slowness is write-back stall, journal lag, device manage management issue (e.g. interconnect/path fault, device saturation), or anonymous memory pressure. Diagnose thrashing by separating soft from hard faults, attributing faults to tasks and regions, and correlating scan activity with free-memory watermarks.
- Respond with policy-driven responses reclaiming caches, throttling or isolating offending tasks, and, as a last resort, terminate “out of memory” processes to protect critical services.
- Tune at the working-set targets, watermarks, swap sizing to reduce the paging cost of the same useful work.

The trends, rates and other indicators address the following questions:

Is memory demand being met?

- Free memory relative to the low-memory threshold.
- Commit charge relative to the commit limit (physical RAM + backing store).
- Backing-store utilization and remaining capacity.

Is the system paging efficiently, or thrashing?

- A healthy, *well-cached workload* has negligible major (hard) faults, a low scan rate, but may have a high rate of minor (soft) faults.
- A *memory starved workload* will show rising major-fault and scan rates, decreased scan efficiency, and increasing page I/O activity.

- High major-fault and page-in rates signal that the working set is too large to fit in memory.
- A high reactivation rate signals that the workload is too large for physical memory. The lack of free memory is forcing page frames to be repeatedly reassigned to meet demand.
- Heavy memory pressure is indicated by a high scan rate with few pages freed per scan (scan efficiency) reveals that the page-out task is working hard with little effect.
- *Thrashing*, when the paging time dominates compute time and the system throughput drops sharply, is signaled by the fraction of CPU time spent blocked on paging I/O.

Is a task behaving correctly?

- Resident or working-set size growing without bound over time suggests a memory leak.
- Anomalous copy-on-write or protection-fault rates can indicate pathological access patterns or defects.

Is locality and translation good?

- Working-set size relative to resident set size, and page cache show whether the useful working set is resident.
- TLB miss rate (instruction vs data) and page-walk cycles or walks per miss show whether translations are hitting the TLB and whether page-table pages stay cached. High misses with low major-fault rate is layout, not shortage of RAM.
- L1/L2/L3 conflict-miss rate on sequential or strided access can indicate poor page coloring or alignment (nearby virtual pages sharing cache sets).
- Poor locality inflates the working set, TLB pressure, and the fault rate.

91.3. VIRTUAL MEMORY / PAGING METRICS

The virtual memory metrics should be trended over time as rates are more informative than raw counters. These help distinguish a healthy, well-cached workload from a memory-starved one.

Metric	Description
Backing store (swap) usage	Paging-store space in use. **Units:** KB/MB; % of capacity
Commit charge	Total committed (promised) virtual memory. **Units:** KB/MB; % of limit
Copy-on-write fault rate	Write faults triggering private copies. **Units:** faults/sec
Fault service time	Average time to resolve a page fault (by class) **Units:** ms

Table 89: *Example of virtual memory, paging metrics*

Free memory	Frames immediately available. **Units:** KB/MB; % of RAM
Hard fault rate	Hard faults requiring backing-store or file I/O. **Units:** faults/sec
Inactive/active list sizes	Pages on each replacement queue.. **Units:** pages; % of RAM
Memory pressure	Composite indicator of demand vs. availability. **Units:** index / level
OOM/allocation failure count	Allocations denied or OOM-kill events. **Units:** count
Page cache hit ratio	File pages served without I/O. **Units:** %
Page scan rate	Pages examined. **Units:** pages/sec
Page-in rate	Pages read from the backing store per unit time. **Units:** pages/sec (or MB/s)
Page in/out rate	Pages move to/from the backing store per unit time. Units: pages/sec (or MB/s)
Page-out rate	Pages written to the backing store/files per unit time. **Units:** pages/sec (or MB/s)
Page-table cache / walk-cache hit ratio	Intermediate PTE lookups satisfied without DRAM **Units:** %
Page-walk cost	Page-table walks or cycles spent walking after a TLB miss **Units:** walks/sec; cycles/walk
Reactivation rate	Inactive pages faulted back into the active set. **Units:** pages/sec
Resident set size (RSS)	Physical memory currently mapped by a task. **Units:** pages, MB
Scan efficiency	Pages freed per page scanned. **Units:** ratio [0-1]
Soft fault rate	Soft faults resolved without disk I/O. **Units:** faults/sec

Thrash Indicator	Fraction of CPU time spent waiting on paging I/O **Units:** %
Working set size estimate	Pages referenced within the recent window **Units:** pages, MB
Zero-fill fault rate	Demand-zero faults **Units:** faults/sec

Shared memory complicates memory accounting. Simply adding the shared pages to each participating task appears to inflate the total memory usage. A common approach is to apportion the shared memory, assigning each task a fraction based on how many tasks are sharing it.

91.4. FILE SYSTEM HEALTH INDICATORS

This section outlines evaluating file system health using metrics. The trends, rates and other indicators address the following questions:

Is the cache absorbing I/O effectively?

- Data hit ratio relative to workload baseline (sequential vs random differ).
- Miss rate and device read IOPS rising together without a workload change.
- Read-ahead waste high on supposedly sequential jobs (window mistuned).

Is write-back healthy?

- Dirty pages as a fraction of RAM vs configured watermarks.
- Write-back latency and `fsync` p95: rising values often mean the device or journal is saturated, or dirty limits are too permissive.
- Dirty throttle events, applications are waiting on the flusher; may indicate undersized storage, oversize dirty ratios, or a stuck write9back thread.

91.5. FILESYSTEM METRICS

These distinguish a healthy cache (high data and metadata hit ratios, modest dirty fraction, steady write-back, large merged I/Os) from a stressed one (rising misses, dirty near limits, throttle events, tiny unmerged I/Os, long `fsync` waits).

Metric	Description
Average I/O size	Mean size of cache-originated device I/Os. **Units:** KB
Buffer occupancy	User-level buffer pool fill if monitored. **Units:** %

Table 90: *Example filesystem buffering and caching metrics*

Cache buffer size	**Units:** blocks
Cache hit ratio	Reads satisfied without device I/O. **Units:** %
Cache miss rate	Misses per unit time or per I/O. **Units:** misses/sec; %
Cache reclaim rate	File pages scanned/freed under pressure. **Units:** pages/sec
Cache size	Memory held as file pages (active + inactive). **Units:** KB/MB; % of RAM
Dirty page count	Pages modified but not yet on media. **Units:** pages
Dirty throttle events	Times writers blocked on dirty limits. **Units:** count/sec
Dirty write-back rate	Pages flushed to media per unit time. **Units:** pages/sec, MB/s
Drop-behind / streaming detections	Sequential streams marked non-caching. **Units:** count
`fsync` latency (p50/p95)	The time a task waits for `fsync` (or equivalent) to push dirty file data and metadata through the page cache to durable media. The application is blocked during the process. *High p95 often means a full dirty cache, a slow or stuck storage path, or journal/commit delay.* **Units:** ms
Inode / dentry cache size	Entries and memory for metadata caches. Units: count; MB
Journal / log flush latency	Time to commit metadata transaction. **Units:** ms
Metadata cache hit ratio	Name/inode lookups served from cache. **Units:** %
Read-ahead pages	Pages filled by prefetch. **Units:** pages/sec
Read-ahead waste	Prefetched pages discarded unread. **Units:** % of read-ahead
Write-through	Writes with device cache disallowed.

override rate	**Units:** % of writes
Write-back latency (avg/max)	Time from dirty to durable on media **Units:** ms

91.6. POLICIES AND THRESHOLDS

Policies and thresholds govern how the paging subsystem operates, determining what actions are taken when limits are crossed. The primary policy categories include:

Table 91: *Types of policy*

Policy Area	Description
Availability	*The minimum free memory and minimum reserve of clean, reclaimable pages.*
Capacity	The maximum commit charge and maximum backing-store utilization before allocations are throttled or denied.
Containment	Per-task working/resident set size limits and out-of-memory (OOM) victim-selection rules so one task cannot starve the system.
Performance	The maximum acceptable major-fault rate and paging-wait fraction; minimum page cache hit ratio.
Power-state	The required dirty-data flush before sleep; hibernation image integrity and size limits; post-resume checks that maps, compressor/swap, and storage paths are healthy before admitting work.

Issue a *warning* when:

- Free memory drops below the low-memory watermark (for example, sustained below 10% for more than a few minutes), leading to a high page-out rate.
- Backing-store utilization crosses a soft threshold (for example, > 50%).
- The major-fault or page-in rate deviates persistently from baseline, or a task's resident or working set size trend indicates a probable leak.
- TLB miss or page-walk cost rises well above baseline while major faults stay low. Check layout, coloring/alignment, and whether large pages are available.

Issue a *service notification* when:

- Backing-store utilization approaches exhaustion (for example, > 80%), or commit charge approaches the commit limit.
- The paging-wait fraction (thrash indicator) stays elevated across multiple consecutive intervals. The system is spending significant time paging rather than doing useful work; recommend adding memory, rebalancing workloads, or tuning limits.
- Scan efficiency declines steadily while scan rate rises, indicating growing, unrelieved memory pressure.

Raise a *fault* when:

- Backing store is exhausted or the commit limit is reached and allocations begin to fail. To preserve the system, it much must terminate tasks. Advise urgent intervention and, where supported, switch to a degraded/safe mode.
- Sustained thrashing has driven useful throughput to near zero.
- Repeated task terminations (due to out-of-memory) of critical services.
- Hibernation image write or restore fails, or post-resume major-fault / thrashing indicate the working set or path was not restored intact.

92. SERVICE TOOLS

Service tooling for virtual memory reads the free/commit/page IO counters, separate soft from hard faults, attribute pressure to tasks and regions, size and monitor backing store, and distinguish leaks from genuine demand. The naming and access is OS-specific.

Typical tools for observing and diagnosing the virtual memory subsystem include:

Table 92: *Types of tools*

Tool Area	Description
Backing-store utilities	Tools to configure, size, and monitor swap/paging files and to report their utilization and I/O rates.
Fault and locality profilers	Hardware performance counters (TLB misses, cache misses), and page-fault tracing to attribute faults to code and data regions. Platform PMU tools where supported for TLB and cache misses.
Leak and allocation diagnostics	Tools that track resident/working set size growth and allocation lifetimes to distinguish genuine demand from leaks.
System monitors and counters	These report free memory, fault rates, scan activity, swap usage, and per-task resident/working set size.

92.1. LINUX

Table 93: *Linux tools*

Area	Description
Backing-store utilities	`swapon`/`swapoff`, `/proc/swaps`, and `free`/`vmstat` swap columns to size, enable, and watch swap utilization and I/O rates; watch for thrashing via sustained `si`/`so` with rising wait.
Fault and locality profilers	`perf` (page-fault,mem-load/store, , and TLB/page-walk PMU events such as `dTLB-load-misses`, `iTLB-load-misses`, `dtlb_load_misses.walk_completed`), hardware performance counters for L1/L2 conflict misses, TLB misses, and tracing (`ftrace`, `bpftrace`, or `perf script`) to attribute major faults to code and data.
Leak and allocation diagnostics	Track resident set size/PSS growth over time (`smaps`, `pmap`, or continuous sampling); heap tools (`valgrind`, `heaptrack`, sanitizer builds) when the growth is in anonymous memory rather than file cache.
System monitors and counters	`/proc/vmstat`, `vmstat` for system-wide free memory, swap in/out, and fault rates; `free`, `/proc/vmstat` and `/proc/meminfo` (Cached, Dirty, Writeback) for detailed reclaim, scan, and commit figures;

	`/proc/<pid>/status`, `smaps`, and `smaps_rollup` for per-task RSS, PSS, and mapped regions. `iostat -x`, `iotop`, `cachestat` / BCC/`bpftrace` hit-rate scripts, `fincore` / `vmtouch` (which pages of a file are resident), `slabtop` (dentry/inode caches),

92.2. MAC OS

Table 94: *MacOS tools*

Area	Description
Backing-store utilities	Observe swap usage through `vm_stat` and Activity Monitor; macOS manages swap files under the system service focus is utilization and pressure trends rather than manual swap device layout.
Fault and locality profilers	Instruments (Allocations, VM Tracker, System Trace) and sample-based tools to correlate faults and allocations with stacks; hardware counter access via Instruments or `dtrace` where permitted..
Leak and allocation diagnostics	Instruments Allocations/Leaks, `heap`/`leaks` command-line tools, and per-process memory graphs to separate retained objects from cache or shared mappings.
System monitors and counters	`vm_stat` for free, active, inactive, speculative, and swap-related rates; Activity Monitor (Memory pressure, swap used, per-process memory); `sysctl`/`memory_pressure` where available for pressure-band state.

92.3. WINDOWS

Table 95: *Windows tools*

Area	Description
Backing-store utilities	Configure and size the paging file (System Properties or policy); monitor `% Usage` and peak of `Paging File\*` counters and correlate with hard-fault rate under load.
Fault and locality profilers	Windows Performance Recorder/Analyzer (WPR/WPA) and ETW memory providers to attribute hard faults, standby-list behavior, and cache activity; platform PMU tools where supported for TLB and cache misses.
Leak and allocation diagnostics	Resource Monitor and Performance Counters for working-set growth; Debug Diagnostic Tool, Application Verifier, or heap-profiling in development builds when a single process commit climbs without bound.
System monitors and counters	Task Manager and Resource Monitor for commit charge, available memory, and per-process working set; Performance Monitor counters (`Memory\*`, `Paging File\*`, process working set and page-fault rates, Cache, Memory, and LogicalDisk) for baselines and alerts. Windows Performance Analyzer; `fsutil` behavior queries.

93. FAULT FINDING AND TROUBLESHOOTING

Start from symptoms and the health questions in Management Information. Separate memory shortage and thrashing from translation/layout cost and from storage-path stalls before changing watermarks, swap, or cache tunables. Use tools to confirm each step.

High translation or cache-conflict cost with a low major-fault rate is a locality or layout problem.

93.1. TRANSLATION AND PAGE-COLORING TRIAGE

Use when CPU time or latency is high but major-fault / page I/O stays low (see Address translation caches and page coloring).

1. Separate TLB/page-walk cost from major faults. High TLB misses and low page I/O means the working set is resident but translations or cache sets are thrashing.
2. Split instruction TLB vs data TLB, if the platform supports. Instruction-side misses point at code layout or I-cache/TLB coverage; data-side misses point at data stride, too many mappings, or missing large pages.
3. If page-walk cycles per miss are high, page-table pages are missing from cache: look for map/unmap storms, an oversized address space, or reclaim of page-table pages under pressure.
4. If L1/L2 conflict misses rise on sequential or strided access while occupancy looks fine, suspect coloring or alignment (virtual pages sharing physical color). Compare against a known-good build or allocator.
5. Expect a short TLB miss burst after context switch, mapping changes, or resume from sleep; a sustained rise is the service signal.
6. For large sequential regions, check huge/large-page coverage: fewer TLB entries for the same resident set size..

93.2. THRASHING VS MEMORY SHORTAGE

Use when free memory, scan, fault, or paging-wait indicators look bad.

1. Check free memory and commit against watermarks and the commit limit. Shortage shows low free and rising commit or backing-store use.
2. Separate soft from hard faults. Soft faults and high reactivation with low page I/O mean the scavenger is bouncing working-set pages under a too-small free pool. Hard faults and page-in mean the working set does not fit and is hitting backing store or files.
3. Read scan rate and scan efficiency. High scan with few pages freed per scan is unrelieved pressure: the page-out path is busy for little gain.
4. Check the thrash indicator (CPU time blocked on paging I/O). If paging wait dominates compute, throughput has collapsed; add memory, rebalance load, or isolate the offending task before retuning cache alone.
5. Attribute growth to a task (RSS/PSS or working set climbing) vs system-wide demand. Unbounded single-task growth suggests a leak; widespread pressure suggests undersized RAM or overcommits.

93.3. CACHE STALL VS INTERCONNECT PATH

Use when I/O latency, writeback, or `fsync` looks bad and it is unclear whether the page cache or the storage path is at fault (see also Chapter 13 and Chapter 14).

1. Check dirty fraction, write-back rate, and dirty throttle events. Dirty near limits with rising `fsync` or write-back latency can be cache pressure: reduce writers or flush policy, after ruling out the path.
2. Prefer link/HBA sense: timeouts, aborts, CRC/link errors, queue freezes, negotiated speed/width. Path errors with a modest dirty cache are interconnect or device faults, not needs more cache.
3. Compare cache hit ratio and device utilization. High misses plus a busy, error-free device suggest undersized cache or bad access pattern; idle device with long waits suggests queueing or a stuck path.
4. After cable/HBA or power-state resume changes, confirm the path is healthy before changing dirty ratios or read-ahead: stale path state often presents as cache or memory symptoms.
5. Only then tune dirty limits, read-ahead, or journal mode; document the product-tier durability choice.

94. TUNING

94.1. WORKLOAD-CLASS PROFILES

94.2. KERNEL AND MOUNT PARAMETERS (ILLUSTRATIVE)

- Dirty ratios / bytes** â€” lower values flush sooner (safer, more write traffic); higher values absorb bursts (risk longer recovery and throttle cliffs). Prefer absolute byte limits on large-memory hosts so percentages do not create multi-gigabyte dirty bombs.
- Dirty expire / writeback interval** â€” how old dirty data may become before background flush.
- Read-ahead size** â€” per-device or per-file; increase for large sequential media scans, decrease for random or mixed loads.
- Metadata commit interval / journal mode** â€” `data=ordered` vs `data=writeback` vs `data=journal` (and equivalents) trade durability for throughput; document the choice for the product's reliability tier.
- Atime updates `noatime` / `relatime` reduce write amplification from pure readers.

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PART VI

Vehicles and Facilities

This part is a think piece on describing vehicles and fixed structures that integrate many products from the prior chapters.

- VEHICLES. The elements, including controllers, sensors, actuators, and the networked communication.
- VEHICLE ENGINES. The elements of engine management systems, including controllers, sensors, actuators, and the networked communication that supports power train serviceability.
- RESIDENTIAL HOMES. Thermostats, furnaces, coolers, heat exchangers, heat pumps, solar panels and batteries, lights.
- LARGE INDUSTRIAL & MANUFACTURING BUILDINGS.

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CHAPTER 20

Vehicles

This chapter discusses road vehicles and mobile work equipment – automobiles, trucks, material-handling equipment, aerial work platforms, utility and golf-course carts, and related platforms.

- Vehicle types, including EV fleets, utility carts, and high-performance (e.g. racing)
- Electrical and high-voltage systems; hydraulics and mechanical subsystems
- Control unit hierarchy, in-vehicle networks and architecture, with segregation of sensing, control and entertainment.
- Configuration, logging, telemetry, and service tools
- Health indicators and service reminders, fault codes, troubleshooting, and verification

Power train-specific diagnostics (e.g. engine controllers, emissions sensors, fuel and air path), will be covered in the next chapter, *Vehicle Engines*.

96. OVERVIEW

Vehicles operate in environments harsher than other electronics products: vibration, temperature swings, moisture, dust, and duty cycles that wear mechanical and electrical parts. Many subsystems include consumables, calibrated components, and field-replaceable units. Serviceability spans power train, electrical, chassis, body, hydraulics, safety interlocks, and – on connected platforms – telematics and over-the-air updates.

Table 96: *Serviceability tiers for vehicles*

Tier	Common features
Low-cost consumer	Basic cars, small utility carts. Minimal status indicators: warning lamps; limited codes; manual checks. Little or no usage/health tracking: hour or distance counters. Owner manual; return-for-service.
Premium consumer	Passenger cars, light vans. OBD-II scan tools; app-linked diagnostics; guided checks. Trip/distance counters; limited logs (e.g. basic DTC history); some freeze-frame. Service schedules; web support; standard fasteners and filters.
Contractor / professional	Work trucks, forklifts, scissor lifts, fleet vans. Enhanced OEM modes; J1939 or UDS where applicable; Increased health tracking & diagnostics: Per-subsystem hours, lift cycles, battery cycles; exportable service bundles, subsystem self-tests. Replaceable modules. Structured workflows, calibration procedures, exportable service bundles, and service tools. Increased security.
Enterprise / industrial fleets	Bus fleets, rental lifts, delivery EV fleets. Remote diagnostics; fleet dashboards; predictive maintenance analytics. Centralized fleet-wide telemetry: operating hours, stress and duty monitoring, error-rate trends, geolocation. Replaceable modules and field-serviceable filters, pumps, and sensors designed for depot maintenance. Spare-parts governance; authenticated parts.

97. VEHICLE TYPES AND SERVICE CONTEXTS

The table below summarizes key vehicle categories (or operating roles) and example vehicles.

Table 97: *Vehicle types*

Vehicle Type	Examples
Light-duty automobiles	Passenger cars and light vans.
Electric vehicles	Add high-voltage systems, thermal management, charging interfaces.
Fleet electric and hybrid vehicles	Added serviceability requirements beyond a single-owner passenger car. Examples: Delivery vans, shuttle buses.
Utility carts and golf-course equipment	Simple propulsion, frequent daily cycles, outdoor storage. Examples: Low-speed electric utility carts, golf-course fleets, resort and campus transport, security patrol carts, and grounds-maintenance carts with tool beds or tow hitches.
Heavy-duty vehicles	Trucks, buses, and coach platforms
Off-road and construction equipment	Excavators, loaders, and other site equipment
Work vehicles	Include hydraulic actuation, structural wear items, and operator safety systems. May be internal-combustion, electric, or hybrid. Examples: Work trucks, forklifts, telehandlers, fleet vans.
Aerial work platform	Scissor lifts, boom lifts, and vertical mast lifts
High-performance / race	Track and motorsport platforms with dedicated logging, telemetry, and setup-oriented service.

97.1. LIGHT-DUTY AUTOMOBILES

Passenger cars and light vans typically expose service data through OBD-II and *controller area network* (CAN) networks. Technicians and scan tools read *diagnostic trouble codes* (DTCs), freeze-frame snapshots, and live parameters to isolate faults across power train, emissions, body, chassis, and electrical subsystems. Common serviceability features include plausibility-checked sensing on safety-related functions and service-interval tracking in the instrument cluster or infotainment system.

- Onboard self-tests for emissions-related components, including oxygen sensors, catalytic monitors, and evaporative-system checks.
- Reduced-power (limp-home) modes when a critical sensor or actuator fails, preserving mobility long enough to reach service.
- Redundant sensing or plausibility checks on safety-related functions such as braking and steering assist where implemented.

- Service intervals tracked by the instrument cluster or infotainment system for oil changes, inspections, and wear items.

Higher-end automotive and commercial platforms may also use Unified Diagnostic Services (UDS, ISO 14229), DoIP (ISO 13400), LIN for low-cost peripherals, FlexRay for timing-sensitive subsystems, and automotive Ethernet in newer architectures. These raise the bandwidth for cameras, ADAS, and infotainment. More on these protocols in the architecture section.

97.2. ELECTRIC AND HYBRID VEHICLES

Battery-electric and hybrid platforms add high-voltage systems, thermal management, charging interfaces, and battery management diagnostics.

limp home mode

Faults may automatically restrict acceleration, speed and power, or require immediate stop depending on severity; other features and accessories may be disabled. This protects the equipment while transporting the occupants to somewhere safe. This is triggered by:

- Sensor failures,
- Excess temperature (too high or too low) of the battery, motor(s), brakes, inverter or other power electronics.
- Low state of charge
- Brake / traction failures or alerts

Service requires trained personnel, appropriate personal protective equipment, and lockout/tagout before working on orange-cable or pack-related components. The considerations include:

- Battery state of health (SoH), state of charge (SoC), cell imbalance, and thermal derating history. See Chapter 13, *Power Management*.
- High-voltage interlock, isolation fault, and contactor health indicators
- Charging port, onboard charger, and DC fast-charge session logs
- Coolant loops for pack, power electronics, and cabin heat pumps where fitted
- Clear separation of user-replaceable items (cabin filter, 12 V battery) from service-only HV components

Documentation and tooling must distinguish HV service from conventional 12 V electrical work.

97.2.1 Fleet electric operations

Delivery vans, shuttle buses, rental pools, and depot-managed EV fleets add serviceability requirements beyond a single-owner passenger car:

- Charge planning and availability – track SoC by vehicle, assign routes to available range, and flag units that cannot meet the next scheduled duty without charging
- Depot and site charging – log session success, energy delivered, aborted sessions, and charger faults; distinguish vehicle-side vs infrastructure-side failures
- Pack health at fleet scale – trend SoH, cell imbalance, and thermal derate events across VINs to schedule pack service or retirement before roadside failure

- Duty and environment – correlate range loss with cold weather, HVAC load, payload, and topography; adjust dispatch policy rather than treating every range complaint as a defect
- Remote triage – prioritize shop visits using telematics (active HV faults, repeated charge failures, isolation warnings) instead of periodic inspection alone
- Swap and spare workflows – define when a driver receives a replacement vehicle vs waiting for depot repair; transfer trip and fault history to support root-cause analysis
- Warranty and asset tracking – tie pack serial, inverter, and charger replacements to vehicle ID, mileage, and energy-throughput counters

97.3. UTILITY CARTS AND GOLF-COURSE EQUIPMENT

Low-speed electric utility carts, golf cars, and campus or grounds-maintenance vehicles are a distinct vehicle class: simple propulsion, frequent daily cycles, outdoor storage, and minimal onboard diagnostics. They may use lead-acid or lithium packs, onboard or cart-side chargers, and basic motor controllers rather than full automotive networks. Serviceability is driven by uptime during daily rounds, predictable charging windows, and field-replaceable wear items.

Design and maintenance considerations:

- Battery systems – for lead-acid, track watering intervals, terminal corrosion, and equalization where specified; for lithium, track SoC/SoH, charger compatibility, and storage SoC for seasonal layup.
- Charging – verify charger cord, connector, and interlock; log failed or incomplete sessions in fleet-managed depots; distinguish “cart fault” from “charger fault” before replacing packs.
- Propulsion – motor, controller, solenoid or contactor, and brake drag are common causes of weak acceleration or shortened range; compare against baseline amp-hours per round
- Mechanical wear – tires, wheel bearings, steering linkage, brake shoes or discs, and seat/weight interlocks; inspect after heavy towing or off-path use.
- Environmental exposure – moisture, mud, and UV degrade connectors and harnesses; corrosion at battery terminals and motor leads is a frequent root cause.
- Operator checks – daily brake test, charge status, unusual noise, and fluid leaks on hydraulic bed lifts where fitted.
- Fleet rotation – balance duty across carts to even pack aging; retire units when range or charge time no longer meets course or shift requirements.

Many carts expose only indicator lamps, hour meters, or basic fault codes through a service port or manufacturer tool rather than OBD-II. Maintain a fleet log of rounds per charge, charge duration, and recurring symptoms to support replacement decisions without full scan-tool access.

97.4. HEAVY-DUTY VEHICLES

Trucks, buses, and coach platforms commonly use SAE J1939 over CAN for power train, chassis, brake, and body-controller communication. Fleet serviceability adds structured logging, often integrated with depot maintenance systems.

97.5. OFF-ROAD AND CONSTRUCTION EQUIPMENT

Excavators, loaders, and site equipment share patterns with heavy-duty J1939 platforms: modular controllers, field-replaceable sensors and actuators, harsh-environment connectors, and depot-based service. Dust ingress, hydraulic contamination, and structural fatigue drive health indicators and filter service intervals.

97.6. WORK VEHICLES AND MATERIAL-HANDLING EQUIPMENT

Forklifts, telehandlers, and similar equipment combine propulsion (internal-combustion, electric, or hybrid), hydraulic actuation, structural wear items, and operator safety systems. These vehicles are often governed by occupational safety standards and manufacturer inspection intervals in addition to onboard diagnostics. The designs typically include:

- Hour meters and lift/load cycle counters tied to maintenance policy
- Accessible hydraulic reservoirs, filters, and bleed points with clear labeling
- Replaceable wear items: tires, forks, chains, brushes, and cutting edges where applicable
- Pre-use and post-incident checklists supported by onboard interlocks (seat switch, hand brake, neutral/start interlock)
- Clear distinction between operator-level checks and technician-level service

97.7. AERIAL WORK PLATFORMS (SCISSOR LIFTS, BOOM LIFTS)

Scissor lifts, boom lifts, and vertical mast lifts are regulated work equipment. Serviceability combines mechanical, hydraulic, and electrical diagnostics with periodic inspection and proof-test requirements. Typical subsystems and service points:

- Hydraulic cylinders, hoses, and valve blocks; leak detection and pressure/temperature trends
- Platform controls, ground controls, and emergency lowering systems
- Tilt, overload, and limit switches; pothole and outrigger interlocks where fitted
- Battery or engine-driven power units; charger health on electric models
- Structural inspection points: pins, bushings, guard rails, and wear on scissor stacks or boom sections

Faults often result in platform disable, audible alarms, or restricted elevation until inspection clears the condition. Logs should capture fault codes, interlock trips, and hour/cycle counts for audit and scheduled maintenance.

97.8. HIGH-PERFORMANCE AND RACE VEHICLES

Track and motorsport platforms push sensors and controllers harder than street duty, and they treat data as a first-class tool. Beside the usual chassis and power-train controllers they carry dedicated *high-speed logging* (HSL), session metadata, and often live telemetry to the pits. The sample rates are orders of magnitude higher than OBD and fleet telematics, for the purpose of lap-to-lap performance and root-cause analysis.

Element	Description
High-speed logger (HSL)	Records many channels at high sample rates (often hundreds of Hz to multi-kHz on critical paths) to local media.
Telemetry	Live radio, cellular, or track Wi-Fi stream of a subset of channels to engineers.
Timing / GPS / beacon	Lap and sector marks so logs align to track position.
Video / dash sync	Optional camera or driver display time-aligned with the log.

Table 98: *Elements*

98. KEY ELEMENTS, ARCHITECTURE

The control-unit hierarchy in vehicle electronics is organized in multiple layers:

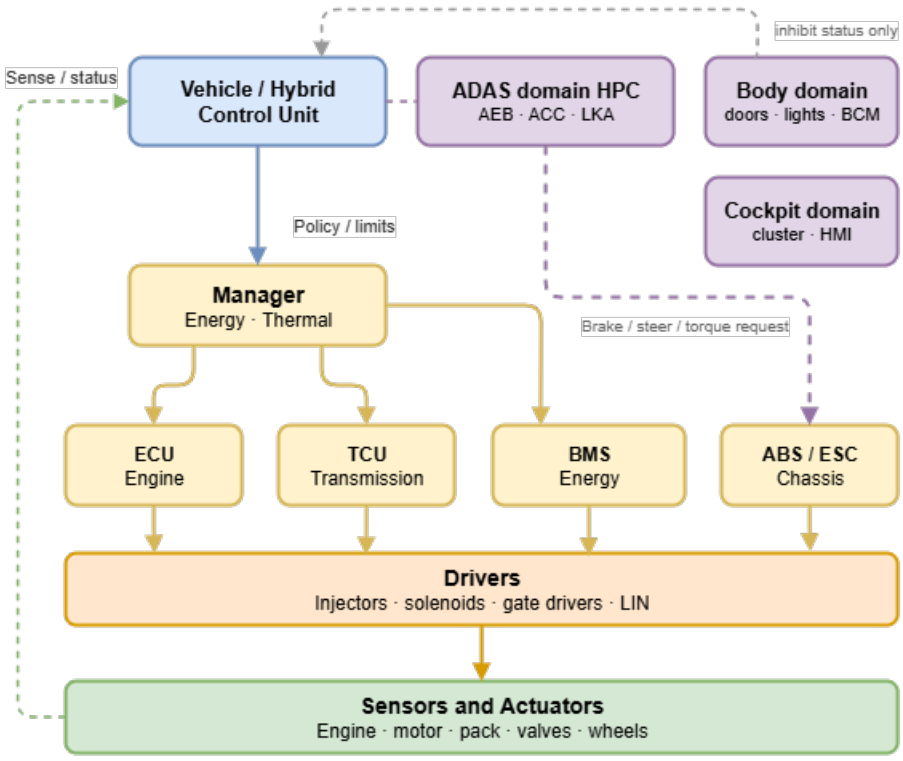


Figure 14: *Vehicle supervisor and controller layers*

Layer	Description
Supervisor	Maintains the vehicle-level policy and commands peers. Ready/not-ready, limp and derate, torque split, charge enable, "schedule service" vs "stop now." On many platforms this is the vehicle control unit (also HCU, power-train supervisory, or a domain HPC).
Manager	Domain or function coordination without owning that inner loop: energy, thermal, power-train, body. Sometimes a separate ECU; often a supervisor software component. A thermal manager may request derate from several controllers.
Controller	Closed loop on one plant. Engine ECU/ECM, TCU, BMS, inverter, ABS/ESC, body module. Most trouble codes and freeze-frame live here. This is the layer a scan tool usually names.
Driver	Actuation and smart I/O: gate drivers, injector and coil drivers, LIN slaves, H-bridges. Often no diagnostic address of its own; faults surface as controller codes.
Sensors and Actuators	The engine, motor, pack, valves, wheels.

Table 99: *Control unit layers*

On a golf cart or simple work machine the layers are merged: motor controller is supervisor, controller, and driver.

98.1. DOMAINS

A *domain* is a functional grouping of related vehicle electronics. Common domains include:

domain

Domain	Description
Advanced driver assist system ADAS	Covers: cameras, RADAR, LIDAR, perception, Automatic Emergency Braking (AEB) / Adaptive Cruise Control (ACC) / Lane Keep Assist (LKA)
Body	Comfort and convenience outside the power-train: doors, locks, windows, mirrors, seats, exterior lights, wipers, keyless entry (often a BCM on simpler platforms). Covers: Doors, locks, windows, seats, lights, wipers, keyless entry
Central compute	Several domains fused on one HPC on some platforms.
Chassis	Covers: ABS/ESC, steering, suspension (passive, adaptive, and active), tire-pressure monitoring.
Cockpit	Driver/passenger HMI: instrument cluster, HUD, center screen, infotainment. Covers: Instrument cluster, HUD, center screen, infotainment
Power-train / propulsion	Covers: Engine, transmission, VCU/HCU, motors, traction energy

Table 100: *Domains*

A *domain controller* or domain *high-performance computer* (HPC) supervises or manages one domain. Domain HPCs have more CPU/GPU and memory and usually run AUTOSAR Adaptive on POSIX. They supervise or manage their own domain. Only an ADAS (or

domain controller

safety) domain HPC peers with the propulsion VCU with authority to cut torque or request a stop.

Body and cockpit HPCs do not command braking or shifting. A body controller may only publish inhibit status (e.g. door open) that the VCU or ECU honors.

98.2. ZONES

zones

A *zone* groups vehicle electronics based on their physical location on the car. Typical zones are front, rear, left, right, or roof zones. A *central gateway* links the zones together and connecting them to the main vehicle network (such as Automotive Ethernet or CAN).

zone controller

A *zone controller* concentrates local power distribution, I/O, and often a network bridge for everything for that physical area. One zone may carry wiring, sensors, and actuators for several domains at once (e.g. body locks and radar on the same front zone).

Older platforms use many ECUs and little zoning. Newer ones use fewer domain and zone controllers to cut copper and weight.

98.3. IN-VEHICLE NETWORKS AND ARCHITECTURE

Modern vehicles are distributed systems. Multiple electronic control units (ECUs) – engine or motor controller, transmission, ABS/ESC, body control module, instrument cluster, HVAC, battery management, and accessory controllers – exchange data over in-vehicle networks.

Table 101: *Vehicle communication protocols*

Protocol	Description
Automotive Ethernet	A high bandwidth, low latency network for cameras, Advanced Driver Assistance Systems (ADAS), and infotainment. This is a version of Ethernet adapted for automotive networks.
CAN	A robust vehicle bus standard designed to allow microcontrollers and devices to communicate with each other within a vehicle.
CAN FD (Flexible Data-Rate)	A high throughput extension of CAN with longer data fields, for advanced automotive applications.
Diagnostics over Internet Protocol (DoIP)	A diagnostic communication bus, supporting high data volumes, parallel connections, and software updates. Based on Ethernet; ISO 13400. Common on passenger EVs and newer commercial platforms.
European On-Board Diagnostics (EOBD)	The European counterpart to OBD-II, ensuring emission control and compatibility with European standards.
FlexRay	A deterministic, high-speed communications protocol used in safety-critical or timing-sensitive automotive applications. Uses redundancy and time-triggered data transfer.
Local Interconnect Network (LIN)	Low-cost single-wire network for mirrors, window lifts, seat-controls, sensors, and other non-critical applications peripherals.
On-Board Diagnostics (OBD-II)	A standardized system in vehicles for self-diagnostics and reporting, providing the status through a standard connector. Light-duty hybrids and EVs may still expose emissions- or powertrain-related data through this interface.

SAE J1939	Standardizes how operating hours, fuel use, regeneration cycles, engine, transmission, and brake operating data and fault history are reported across manufacturers. Used with heavy-duty and many off-road platforms.
Unified Diagnostic Services (UDS)	A protocol for diagnostics and reprogramming across automotive ECUs, supporting advanced features beyond OBD-II. Advanced diagnostics and flashing on newer architectures. ISO 14229. Common on passenger electric vehicles and newer commercial platforms.

For protocol detail and communication-layer references, see chapter 15 on *Communication*.

98.4. VEHICLE CONTROL UNIT (VCU)

vehicle control unit

The *vehicle control unit* (VCU) is the supervisory controller for vehicle propulsion and energy management on electric and hybrid platforms. The VCU coordinates torque requests, drive modes, charging enable, thermal and power limits, and handoffs among subsystem – battery management (BMS), motor/inverter controllers, onboard charger, DC-DC converter, and often transmission or e-axle controllers. It enforces the "limp home," charge inhibit, and "schedule service" vs "stop now" policies for electric and hybrid vehicles.

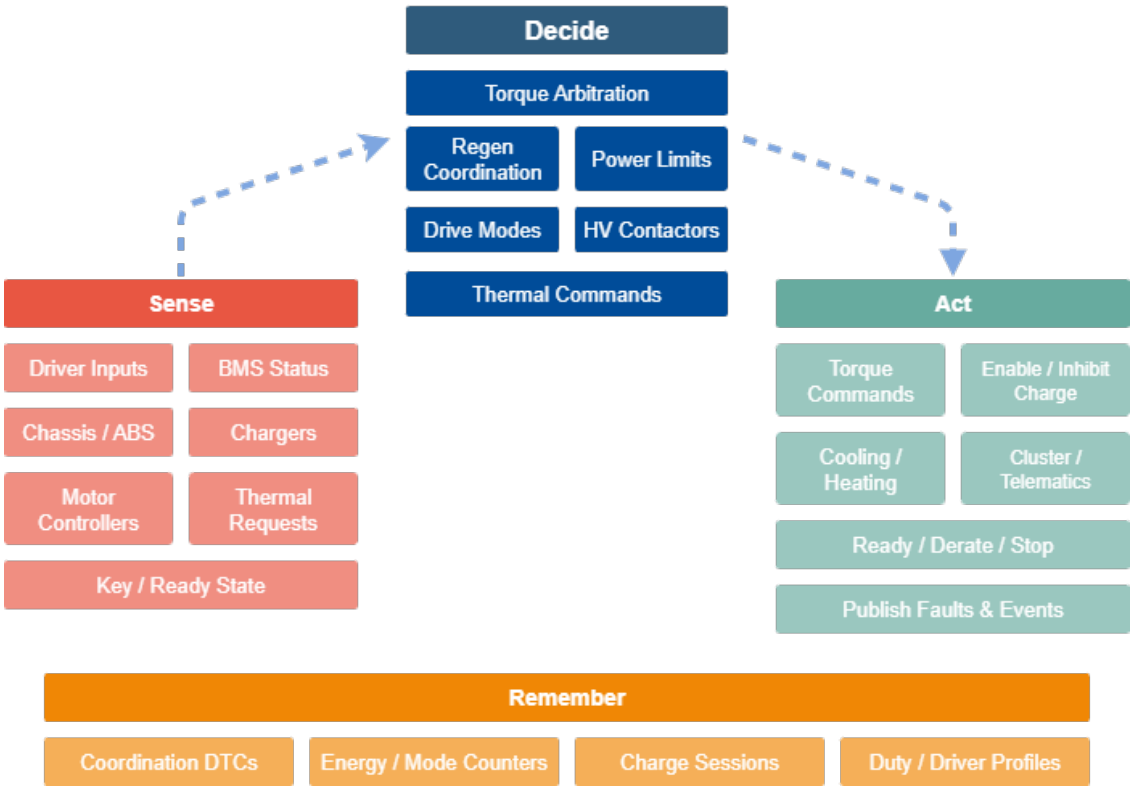


Figure 15: *The vehicle controller integrates sensing and actions*

Role	Function
Sense	Driver and operator inputs (accelerator, brake, gear/drive mode, key or ready state); network status from BMS (SoC, SoH, isolation, contactor state), motor controllers (speed, torque, temperatures), chargers, and chassis modules (ABS/ESC, park brake); ambient and cabin thermal requests where they affect propulsion limits
Decide	Arbitrate torque among axles or motors; apply regenerative braking coordination with friction brakes; enforce charge/discharge power limits; select drive modes (eco, tow, limp); initiate or inhibit HV contactor close; command thermal loops when propulsion or charge is constrained.
Act	Command torque and mode to motor controllers; enable/inhibit charging; request cooling or heating; set instrument-cluster and telematics status (ready, derate, stop safely); store vehicle-level faults and event history.
Remember	Retain coordination information, drive-mode and energy counters, charge-session outcomes, and sometimes learned driver or duty profiles across key cycles.

Table 102: *The vehicle controller's sense, decide, act and remember roles*

The VCU is the propulsion supervisor in the control-unit hierarchy. On some platforms the role is split across a zone or domain controller; on simpler carts it may be fused into a single motor controller. Body, lift, ABS/ESC, and HVAC controllers may request torque or power limits that the VCU must honor.

Note: controller naming varies by manufacturer and region. It may also be called a vehicle domain controller, power-train supervisory controller, or hybrid control unit (HCU).

Class	Examples and notes
Aux Power	12 V / 24 V DC-DC, LV battery status. Weak LV supply resets the VCU and floods spurious network codes.
Charge	Onboard charger, DC fast-charge interface, charge-port latch/interlock | Distinguish vehicle-side inhibit from infrastructure faults using session logs.
Energy	BMS (SoC, SoH, cell limits, isolation, contactors). When the contactor will not close: check HVIL, isolation, and BMS readiness before condemning the VCU.
Operator	Accelerator and brake pedals, PRND or drive-mode switches, key/ready. Implausible pedal signals or dual-sensor mismatch often force limp.
Safety and control	ABS/ESC, EPB, airbag or restraint status where shared. Stability or brake faults can request torque cut without a propulsion DTC.
Propulsion	Motor/inverter controllers, e-axle or transmission control. Torque-limit or communication loss may be VCU policy or peer-module fault.
Telematics	CAN, CAN FD, Ethernet gateway; fleet or OEM portals. Gateway or domain faults can isolate the VCU from peers.
Thermal	Pack, inverter, and cabin heat-pump requests. Thermal derate may look like "weak motor" until coolant and pump health are checked.

Table 103: *Common VCU-connected elements*

98.5. ADVANCED DRIVER ASSISTANCE SYSTEMS (ADAS)

advanced driver assist

Advanced driver assistance (ADAS) is a collection of features to assist the driver. It incorporates cameras, RADAR, LIDAR, and ultrasonics fed into an ADAS domain controller. It hosts perception, planning, and other services. The ADAS controller may request a stop, brake intervention via ABS/ESC, steering torque, or an engine/motor torque cut.

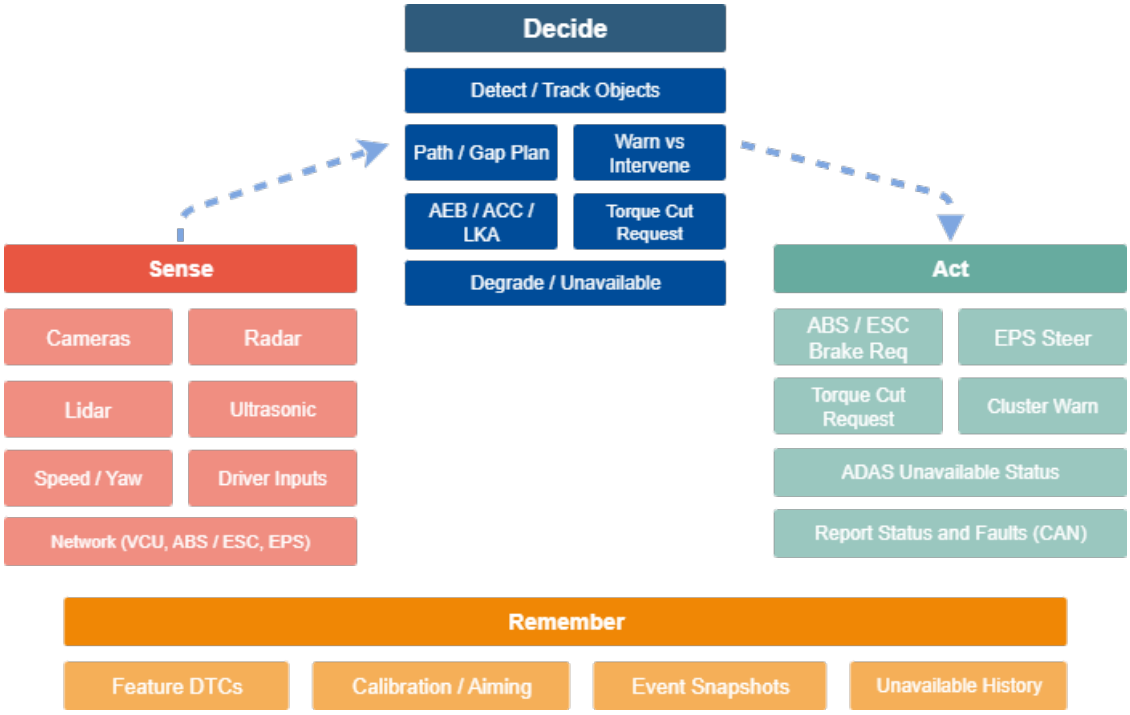


Figure 16: *The ADAS sensing and actions*

Role	Function
Sense	Forward and surround cameras, RADAR, LIDAR, ultrasonic; vehicle speed, yaw, and driver inputs from the chassis and power-train buses.
Decide	Object detection and tracking, path or gap planning, and whether to warn, brake, steer, or request torque reduction. Can detect and track people, cyclists, and other risks; recognize traffic signs. Blind spot detection.
Act	Commands to ESC, electric power steering, and propulsion (VCU or ECU). Cluster warnings and "ADAS unavailable" status. *Adaptive Cruise Control* (ACC) holds a set speed and gap to the vehicle ahead. Collision warning, and *Automatic Emergency Braking* (AEB), applying brakes if a collision is imminent and the driver does not. *Lane Keep Assist* (LKA) steering or warning to stay in lane (stronger than lane-departure warning alone). Blind-spot and parking assist.
Remember	Feature trouble codes, calibration / aiming state, and sometimes event snapshots for warranty or fleet review.

Table 104: *The ADAS sense, decide, act and remember roles*

98.6. ACTIVE SUSPENSION

active suspension

Active suspension is a chassis controller that adds force at the wheels to shape body motion (heave, pitch, and roll) rather than only reacting through fixed springs and dampers. A fault usually degrades ride and handling, not ready state, unless the platform also uses the same module for stability or brake coordination. There are four related systems:

***Table 105:** Types of suspension*

Suspension Type	Description
Passive	Uses fixed spring, conventional shock absorbers
Adaptive (semi-active)	Uses variable-orifice or magnetorheological dampers, accelerometers. This can only changes damping; it cannot push the body up.
Air / load leveling	Uses air springs, compressor, height sensors, valve block to change ride height and effective spring rate.
Fully active	Uses hydraulic, electrohydraulic, or electromechanical actuators per corner, along with high-rate sensors to apply force into the suspension on a fast loop, including against body motion.

Adaptive damping and air leveling are common on passenger and light commercial vehicles. Fully active systems are less common and draw more power (often a dedicated pump, 48 V supply, or a high-voltage actuator drive). Some combine air springs for height with fast actuators or adaptive dampers for body control. A few use a forward camera to preview the road and pre-position the actuators; that preview path is an ADAS sensor feed into the chassis controller.

***Table 106:** The active suspension sense, decide, act and remember roles*

Role	Function
Sense	Ride-height and body or wheel accelerometers at the corners; vehicle speed, steering angle, brake and throttle, and sometimes yaw from the chassis bus; compressor or actuator current and temperature; on preview systems, a road profile from the forward camera.
Decide	Choose mode (comfort, sport, off-road, load-level, access height) and compute corner forces or damper commands. Hold the body flatter in cornering and braking, and keep ride height inside limits under load. Apply fail-soft policy when a sensor, pump, or actuator is implausible.
Act	Command damper valves, air valves, compressor, or corner actuators. Publish height, mode, and "suspension fault / limited" status to the cluster. May request a modest power budget from the LV or HV system; it does not command ABS or the VCU except where OEM integration shares a stability request.
Remember	Height calibrations, mode and option coding, compressor run-time, and suspension trouble codes.

Fail-soft is part of the design. On a detected fault the controller typically drops to a passive or firm-damper mode, parks air springs at a safe height, and sets a warning, while the vehicle remains drivable.

98.7. TIRE-PRESSURE MONITORING

Proper tire inflation is fundamental to safe and efficient vehicle operation. Under-inflated tires increase rolling resistance, lower fuel efficiency, wear out more quickly, and compromise handling and braking. Over-inflated tires reduce traction, making the vehicle less stable, especially in wet or slippery conditions. Punctures or failed valves can cause rapid or hidden pressure loss, increasing the risk of tire damage, and poor control, and even blowouts at speed.

tire-pressure monitoring system

Tire pressure naturally changes over time due to slow permeation (the gas moving thru the rubber), minor leaks, and seasonal temperature swings: cold weather lowers pressure, while hot weather or extended driving can temporarily increase it. Modern vehicles use a *tire-pressure monitoring system* (TPMS) to warn when a tire is under-inflated. Regular manual checks with a reliable gauge remain the best practice for preventative maintenance.

The TPMS is usually a small controller inside the body or chassis module. Loss of pressure raises a warning and may limit speed on some platforms. There are two kinds:

Direct systems transmit an absolute pressure (and often temperature) from each tire. Each sensor has an identifier that the controller "learns" to map to each a wheel. After a tire rotation, a sensor replacement, or a module swap, that map must be relearned or the display names the wrong corner. The sensors have a sealed primary cell battery in the valve stem or on the rim, which cannot be recharged. They last on the order of the tire's service life; a dead sensor is replaced.

Indirect systems do not use a sensor; instead they infer under-inflation by comparing rolling radius and wheel-speed behavior from the ABS sensors. They report a relative under-inflation warning, not an absolute pressure in kPa or psi, and they can miss a case where all tires lose pressure together.

Table 107: *The TPMS sense, decide, act and remember roles*

Role	Function
Sense	Per-wheel pressure and temperature (direct), or wheel speeds from ABS (indirect); sensor battery and RF reception; learn state and which identifier is assigned to which wheel.
Decide	Compare with the placard or a learned baseline. Warn on low pressure, rapid loss, high temperature, or a missing sensor. Separate a puncture from a slow leak by the rate of change.
Act	Light the TPMS warning and, on direct systems, show the corner and the value. A lamp that flashes and then stays on usually means a system fault (missing sensor, failed receiver, incomplete learn).
Remember	Learned sensor identifiers and corner positions, placard or baseline thresholds, and TPMS trouble codes.

98.8. DOMAIN CONTROLLER SOFTWARE ARCHITECTURE

AUTOSAR Adaptive is used on high-performance domain controllers (ADAS, central compute, some supervisory HPCs). Adaptive Applications sit on the runtime, communication is service-oriented (SOME/IP) and bound at runtime on a POSIX OS. The OS includes basic/extended prioritized tasks, priority ceiling for shared resources, and alarms for periodic work.

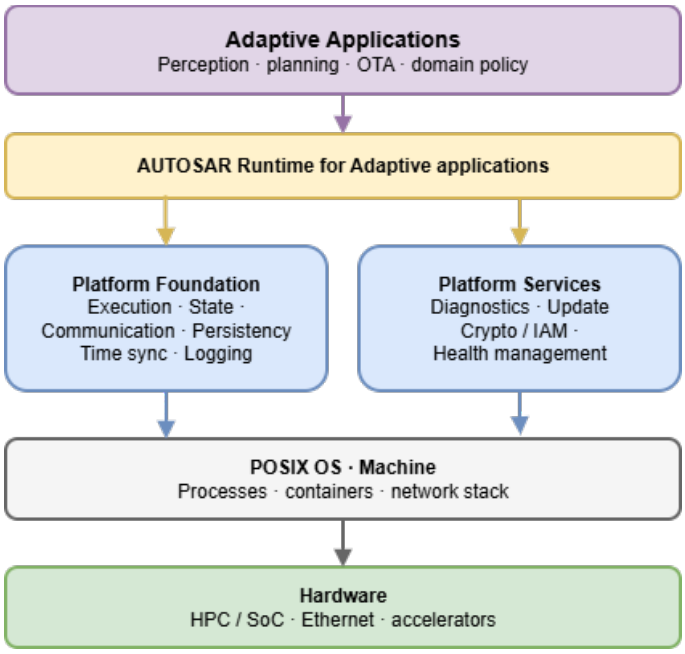


Figure 17: *AUTOSAR Adaptive layers*

Layer	Role
Application	Provides functions like perception, planning, diagnostics, and control.
Runtime	Manages application lifecycle, deployment, and communication (e.g., SOME/IP).
Platform services	Core services for diagnostics, logging, communication, and system health.
POSIX OS	Provides process scheduling, memory, files, and networking. Linux/QNX.
Hardware	CPU, memory, power, and interfaces supporting the software stack.

Table 108: *AUTOSAR Adaptive Layers*

Automobile controllers (e.g. engine, transmission, VCU, ABS/ESC, body, BMS, and similar ECUs) for engine-based vehicles are commonly built on the AUTOSAR Classic platform.

Many vehicles employ both AUTOSAR Adaptive and Classic. Classic owns the hard real-time loops; while Adaptive hosts perception, planning, and rich services. A gateway bridges Ethernet / SOME/IP to CAN.

98.9. REAL-TIME CONTROLLER SOFTWARE ARCHITECTURE

Automobile engine and transmission controllers are typically AUTOSAR Classic ECUs. The OS is based on OSEK/VDX (ISO 17356): priority-scheduled tasks, alarms for periodic work, and deterministic interrupt handling.

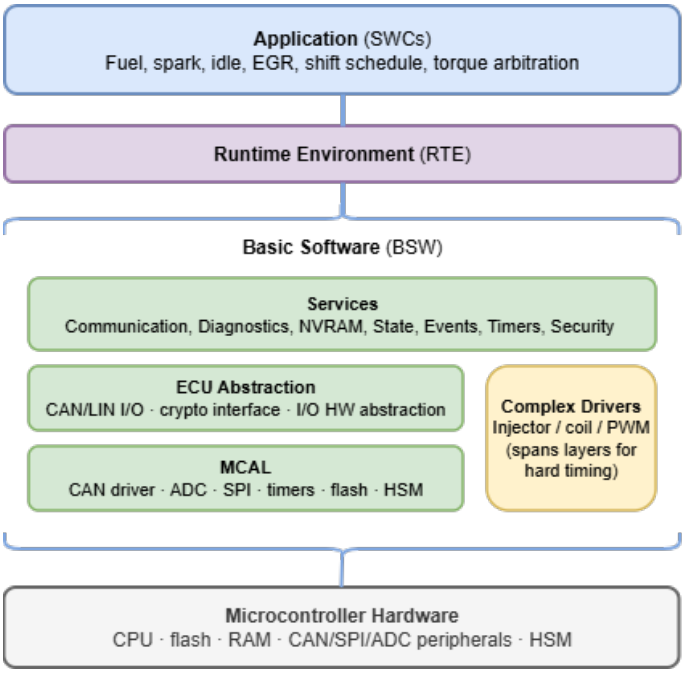


Figure 18: *AUTOSAR Classic layers in an ECU*

The application (fuel, spark, idle, EGR, shift schedule) runs as software components on a real-time environment. A basic software layer provides communication, diagnostics, NVRAM, and the OS; microcontroller drivers own the timers, ADC, and injector/coil outputs. Injection and ignition pulses are alarm or ISR-driven with hard deadlines, a late task is a misfire.

Safety-oriented power-train ECUs may add OS memory and timing protection. After treatment, transmission and hybrid motor controllers on the same vehicle usually share this architecture.

98.9.1 Application dataflow and parameter mapping

data mapping set

In AUTOSAR, application modules are connected to bus traffic, calibration, and operating modes through a *data mapping set*. This is a configuration regenerated on each build. If this mapping is missing or wrong in the build, modules compile but never exchange data.

The application on a controller is similar to a dataflow graph, or small component network:

- Each application module is a node;
- Each module declares *ports*, the typed input and output it expects. "Engine speed in rpm," might be an input, while "requested torque" might be an available output. A *data prototype* is the name and type of one such port slot on a node.
- The nodes are connected using *prototype mapping* of outputs to inputs so values flow.

The whole graph is run once every 10ms, cyclic processing with edges between modules. Some nodes also export values to the bus, memory, or scan tool (data-element and signal mapping); a *mode mapping* can enable or disable whole subgraphs.

data prototype mapping

Data prototype mapping is the internal wiring graph between components, such as connecting output ports from one module to input ports on another so data flows inside the ECU. For example, fuel control publishes torque request, and the transmission coordination reads it.

data elements and mapping

The named values are held in concrete variables called *data elements*, such as `CoolantTemp`, learned fuel-trim, last DTC snapshot. *Data element mapping* decides what happens to a value: where it is published, stored, or exposed for diagnostics. A data element may be mapped to one or more consumers:

- Mapping to an application module's input.
- The communication may broadcast on the in-vehicle network for other ECUs
- The non-volatile storage module may store in non-volatile memory across key cycles (adaptations, trouble code, freeze-frame)
- Diagnostics may expose the variables to scan tools

Example: coolant temperature might be mapped to each: the fuel-control loop uses it internally, it is broadcast to the instrument cluster, and it is readable as a diagnostic value.

parameter mapping

Modules read their parameters each cycle thru *parameter ports*. Each port maps to one of the following:

- A named variable
- A module output
- A setting
- A calibration (scalars, maps, and curves) and limits.
- A learned values (fuel-trim learn, throttle learn). These persist across key cycles, and are stored in non-volatile memory.

98.9.2 Mode mapping

mode mapping

Mode mapping coordinates when functions are allowed. It is a set of rules enable or disable subgraphs and outputs.

***Table 109:** Example mode mapping policies*

State	What mode mapping controls
Application mode	Normal run, limp, or manufacturing test.
Diagnostic session	Default session, programming (e.g. flash), or extended
ECU power	Sleep vs run vs shutdown
Network	Bus awake vs silent; which messages are transmitted.

mode manager

A *mode manager* applies if-then rules (action lists):

- In limp mode, disable boost control and cap torque;
- In programming session, disable normal functions and communication, accept flash commands until a proper exit or reflash.
- On bus wake, enable PDU group X.
- When EV Ready inhibited, disable propulsion.
- When fault is active, limp home, limit power until clear conditions and a drive cycle complete.

98.9.3 Communication

signal mapping

Values may leave the ECU when the data-element mapping passes them to the communication subsystem. Then *signal mapping* is applied, to which describes the over the wire encoding of the value on the bus. This includes the length, endian, raw integer vs physical value, scaling & offset (e.g. °C = raw × 0.5 − 40) used to map to logical units. It also includes a label, the update rate, and which unit carries the signal. If inconsistent mappings are used, two ECUs can expose the same label on the bus with different scaling or meaning.

98.9.4 Variants, supporting multiple vehicle configurations

variant

One ECU firmware image often supports many vehicle configurations. A *variant* is which configuration for this specific vehicle, the combination of hardware and software options must match. Examples:

- 2.0 L vs 2.5 L engine, manual vs automatic transmission
- US vs EU emissions calibration
- Electric vehicle pack size, axle count, tow package present or absent
- Optional ADAS or body features fitted at the factory

variant coding

The ECU image may contain several *calibration sets* (different maps, limits, and enable flags). *Variant coding* is the step that writes which variant applies to this vehicle. This is usually performed during flash, first boot, or with a scan tool. Parameter mapping then selects the right tables for that variant.

An incorrect variant coding means the application software will employ the wrong limits and maps. The ECU while runs, but behaves like the wrong car.

Note: Variant coding is separate from adaptations (learned trim updated by the ECU over drive cycles) and separate from feature enables (turning on optional equipment).

98.10. PASSENGER COMPARTMENT AND ENVIRONMENTAL CONTROLS

Environmental control systems provide comfort to passengers, and contribute to safety. Operator-present vehicles and work platforms used for extended periods need these to be serviceable:

Table 110: *Passenger environment control subsystem*

Subsystem	Role and Maintenance considerations
Cabin air filtration	Role: Removes dust, allergens, and particulates from incoming air to protect occupants and HVAC system components. Service: Regular filter inspection/replacement intervals, particularly for dusty or hazardous environments (construction, mining, agriculture). Filters should be replaceable without tools.
Cabin pressurization	Role: Sealed or semi-sealed cabins are used when operating in harsh or hazardous environments. (e.g., mining, agriculture, other specialty equipment); positive pressure inside the cabin is maintained to keep out dust, fumes, and other contaminants. Service: verify pressurization integrity and monitor pressure differential sensors
Defrost/defog	Role: Removes condensation and frost from windows to maintain clear visibility. Service: ensure window heating elements and airflow actuators are functional; verify

	sensor calibration after replacement
Heating, Ventilation, and Air Conditioning (HVAC) system	Role: Maintains cabin temperature, airflow, and comfort for occupants. Service: tool-less filter replacement, and modular, field-replaceable blower assemblies, with clear panel marking for each; diagnostics for temperature regulation, actuator faults, and refrigerant charge status
Operator interface	Role: Allows the operator to monitor and adjust environmental settings, view system status, and receive alerts or fault notifications. Service: physical and digital controls should provide clear fault indicators (e.g., service A/C, low refrigerant, filter blocked); error codes map to documented service steps.
Seat and steering wheel climate	Role: Provides comfort by heating or cooling the seat and steering wheel surfaces. Service: test heating/cooling elements and safety cutouts during scheduled service

Modern vehicles may store HVAC and comfort-related fault codes in the body or climate control module. Maintenance logs should capture firmware updates, sensor calibration, and notable operator-reported issues with environmental controls.

98.11. ELECTRICAL

98.11.1 Low-voltage (12 V / 24 V) architecture

Conventional vehicle electrical systems supply lighting, controls, infotainment, and ECU power. Serviceability depends on accessible fuse/relay boxes, labeled harness branches, corrosion-resistant connectors, and documented ground points. Weak batteries cause no-start and misleading fault codes; track rest voltage, cranking behavior, and replacement age.

98.11.2 High-voltage (EV and hybrid) architecture

High-voltage service adds isolation monitoring, contactor wear, charging-port damage, and thermal management faults. Service procedures must include verified de-energization, personal protective equipment (PPE), and OEM-specific re-enable steps. User-facing indicators should clearly distinguish “stop safely and call service” from “schedule maintenance.”

98.12. HYRDAULICS AND MECHANICAL SUBSYSTEMS

Many work vehicles rely on hydraulics for lift, steer, or implement functions. Serviceability features include sight glasses and dipsticks, replaceable filters and breathers, labeled bleed points, and pressure test ports.

Scissor lifts and boom lifts add structural and safety elements: limit switches, tilt sensors, descent rate controls, and emergency lowering.

99. MANAGEMENT INFORMATION AND DIAGNOSTICS

This section addresses the structure and function of onboard vehicle networks, health indicators, electrical and hydraulic system monitoring, techniques for diagnosing faults, and service processes.

99.1. HEALTH INDICATORS

Vehicle health indicators should be monitored at the whole-vehicle level, at major subsystems, and for key replaceable-units. These should track or estimate mechanical wear, electrical loading, hydraulic system status, and patterns of use. Components (or subsystems) typically report standard health indicators, environmental and exposure data (refer to chapter 8), as well as temperature and power measurements (see chapter 13).

Parameter	Description
Battery health (lead acid)	Rest voltage, cranking performance, age. **Units:** V, status
Battery health, state of charge	Pack capacity and charge state. **Units:** %
Distance, odometer	Accumulated travel. **Units:** km,miles
Operating hours	Total runtime since new or last major service. **Units:** hours
Recent service interval	Time or cycles since last inspection or fluid change. **Units:** days, hours
Rounds per charge / amp-hours per cycle	Utility cart range baseline (fleet carts). **Units:** count, Ah

Table 111: *Example of vehicle capacity metrics*

Parameter	Description
Charge session count, energy	Charging history and anomalies **Units:** count, kWh
Hydraulic pressure / temperature	System stress and overheating trends. **Units:** kPa, °C
Lift or load cycles	Platform raises, mast cycles, or load events. **Units:** count
Over-temperature events	Incidents exceeding safe thermal limits. **Units:** count

Table 112: *Example of vehicle load metrics*

Parameter	Description
ADAS calibration / availability	Aiming state; feature unavailable vs propulsion limp **Units:** status

Table 113: *Example of vehicle integrity metrics*

Brake wear	Pad/shoe wear estimates or service indicators. **Units:** %, status
HV isolation	Pack isolation and interlock health. **Units:** status
Interlock / safety trip count	Platform, seat, or guard faults. **Units:** count
logger health	Media free space, write errors, time sync, session armed/closed. **Units:** status, %
Suspension mode / height / fault	Active mode, ride height per corner, fail-soft or limited status. Units: mode, mm, status
Tire pressure	Per-wheel pressure, leak trends, and under-inflation warning state. **Units:** kPa, psi, warn/status
Trouble code count	Active or unaddressed diagnostic trouble codes. **Units:** count

For TPMS integrity metrics, treat low pressure and system fault as separate indicators.

99.2. HEALTH POLICY AND THRESHOLDS EXAMPLE

Use thresholds and policies so health indicators drive consistent service actions, such as warn, schedule service, or retire a device. *Warning thresholds* to schedule inspection or service when a developing issue is not yet severe enough to damage equipment or present safety issue:

- The trouble code count rises over multiple operating sessions or repeats after clear
- Hydraulic temperature or pressure trends drift from baseline under normal load
- Battery health falls below expected range for age and vehicle usage
- Utility cart range or charge time degrades versus fleet baseline for the same route count
- Lift cycle or hour counts approach scheduled inspection windows
- Repeated interlock trips or overload events in the same operating mode
- Suspension remains in fail-soft or limited mode across drive cycles, or ride height drifts from the calibrated baseline under normal load.
- Air-suspension compressor duty cycle or run-time rises well above fleet baseline (leak or dryer issue developing).
- Tire pressure falls below *placard pressure* (the recommended cold inflation pressure) when cold, or drops rapidly while driving. Note, the readings will typically rise as the tire warms in use.
- Tire pressure monitor system fault persists after cold pressures are set and learn/reset is complete.

- Logging or telemetry shows rising oil/coolant temp, damper asymmetry, or brake-pressure imbalance vs a baseline lap with no trouble code yet.
- Logger media nearly full, write errors, or repeated failed session downloads.

A fault is raised for serious issues – parameters or component health are such that normal operation could risk emissions violations, component damage, or safety. Restrict operation or require stopping in a safe position / state when continued use risks injury or major damage:

- Loss of critical safety interlock plausibility (tilt, overload, descent control)
- Confirmed high-voltage isolation fault or thermal runaway indication
- Hydraulic pressure loss with load elevated or unsupported structure
- Structural or restraint faults flagged by manufacturer diagnostic policy
- Active-suspension actuator or hydraulic fault that the OEM treats as a driveability or stability limit.
- Rapid pressure loss or multiple-wheel under-inflation where the OEM limits speed or requires a stop.

Use separate trigger and restore thresholds with hold-off periods to avoid oscillation between warn and fault states.

99.2.1 Service Reminders

Service reminders translate health indicators and usage counters into scheduled or prompted maintenance. These reduce unplanned downtime and help keep systems within design limits.

Consumable	Triggers
Electrical	12 V battery replacement; HV coolant or desiccant service per OEM (EV).
Filters	Reminders to service air, fuel, hydraulic, and cabin filters based on use and contamination exposure.
Fluids	Reminders to service engine oil, hydraulic fluid, coolant, brake fluid, and DEF where applicable.
Inspections and certifications	Annual or hour-based inspections for regulated lifts and fleet equipment.
Logging media / sensors	Logger cards or solid-state media at wear or error thresholds; suspension pots, brake/tire sensors, IMU at interval or after crash/impact.
Suspension (air / active)	Air-dryer service, compressor, leak-down checks; hydraulic fluid/filter on fully active circuits; dampers and air springs at wear or leak.
Tire pressure sensors	Service sealed in-wheel sensors at tire service life or when battery/RF fails; valve stems and seals when tires are dismounted
Wear Items	Tires, forks, chains, brushes, pads, and cutting edges

Table 114: *Preventative service notifications based on what to replace*

Some consumables can be inspected, with judgment passed. Few have automated measurement, or derived measurements before serious issues.

Reminders may be acknowledged or snoozed where safe; critical items may escalate to warnings or derate until service is performed, per calibration and regulation. Safety-critical items may lock out operation until service is recorded.

100. SERVICE TOOLS AND INTERFACES

During operation, physical repair is limited to what the operator or onboard systems can do immediately – isolating a fault, reducing load, engaging safing, or reaching a safe stop. At a shop, depot, or scheduled service visit, technicians can replace parts, run diagnostics, and restore function. Good design mixes fault tolerance with shop-based maintenance supported by onboard diagnostics, logging, and standardized interfaces.

100.1. FAULT & TROUBLE CODES

Light-duty platforms store standardized trouble codes accessible via the OBD-II connector. Codes may span power train, body, chassis, and network gateways.

Heavy-duty platforms report faults and diagnostic messages using J1939. The operating data and faults reporting are standardized across manufacturers using *parameter group numbers* (PGNs) and *suspect parameter numbers* (SPNs). Fleet tools aggregate active and historical faults.

Body controllers, lift controllers, BMS, and accessory ECUs may expose trouble codes. Service documentation should map these to subsystem, severity, and required action. Power-train code interpretation, emissions monitors, and engine-sensor troubleshooting are covered in chapter 21 *Vehicle Engines*.

100.2. SCAN TOOLS, PASS-THRU DEVICES, AND TEST EQUIPMENT

Service tooling must match the platform: OBD-II for light duty, J1939 adapters for fleets and heavy-duty, OEM security access and diagnostic suites for UDS/HV/Lift controllers and manufacturer-specific procedures for HV and safety ECUs, hydraulic pressure gauges and multi-meters for mechanical subsystems. Active and air suspension need OEM routines for ride-height calibration, plus leak-down and pressure checks on air or hydraulic circuits. TPMS work also needs a pressure gauge for cold placard checks and, for direct systems, an OEM or aftermarket tool that can trigger sensors and run the relearn routine.

Some example CAN and diagnostics tooling: Vector, Kvaser CANalyzer/CANoe, Peak-System.

100.3. SECURITY ACCESS AND DIAGNOSTIC SESSION

Modern vehicles require authenticated sessions for flash, clear, or reconfigure modules. Service organizations should govern credentials, audit tool use, and train technicians on HV and lift safety. (ISO/SAE 21434, UNECE R155/R156)

100.4. VEHICLE-SPECIFIC DIAGNOSTIC PROTOCOLS

Maintain run-books that map common symptoms to code sets, live parameters, and approved test sequences per model and attachment configuration.

100.5. EVENT LOGS, SNAPSHOTS, AND DATA RETENTION

Significant faults, interlock trips, derate events, charge sessions, and service actions should be logged with timestamps and operating context such as freeze-frame of key parameters at the time of fault. Higher tiers export service bundles for remote analysis. Retention policy should balance warranty support, fleet analytics, and privacy regulation.

Tier	Triggers
Freeze-frame / trouble code context	Captures a snapshot at time of fault. Used by shop to triage.
ECU internal buffers	Used by pre/post trigger captures.
Fleet / connected telematics	Captures (and reprots) data every second to few minutes. Used to dispatch, trend, remote prioritization.
High-speed logging	Captures data at hundreds of Hz to multi-kHz. Used to examine race/track dynamics, setup compare, plant faults without a trouble code.

Table 115: *Tier and logging*

High-performance and race programs add high speed logging and other motorsport loggers, channel-config tools, and telemetry receivers. After a session, download and archive the onboard file before the next run; a full card or failed pull is a tooling/process fault.

100.6. DIAGNOSTIC INTERFACE

Vehicle controller diagnostics are typically OEM-specific rather than a single regulatory parameter set. Diagnostic sessions that exercise HV contactors or propulsion commands require verified safe state, PPE, and OEM procedures.

101. CONFIGURATION MANAGEMENT

101.1. CALIBRATION AND PARAMETER SETS

Many subsystems require calibration after part replacement – steering angle, ride height, lift envelope, pressure limits, or BMS parameters. The calibration date and version should be stored in service logs. Calibration is part of the ECU parameter mapping set.

When a vehicle control unit is replaced or reprogrammed, it will need calibration and coding for axle count, pack size, drive modes, market and other vehicle factors. Wrong coding causes range, charge, or limp misbehavior.

101.1.1 Right-Height and Suspension Calibration

After arm, height-sensor, actuator, or module replacement:

1. Run the OEM ride-height calibration and a short road test in each mode (comfort, sport, load-level, access). Uncalibrated height looks like a sensor or linkage fault; a vehicle that sits crooked after module swap often lost it's learned or coded height offsets.

2. Confirm variant / option coding for air springs, adaptive dampers, or fully active hardware. All four corners wrong after a flash usually points at coding or the controller, not four failed plants.
3. On preview systems, confirm camera aiming after windshield or bumper work. A misaimed preview feed is a configuration/sensor alignment problem, not a damper failure.

101.1.2 TPMS learn and reset

With direct TPMS, after a tire rotation, sensor replacement, or TPMS/receiver module swap, run the OEM procedure that maps each sensor identifier to a wheel position. An empty or stale learn table makes every corner look failed, or shows pressure on the wrong corner.

With indirect TPMS, after setting cold pressures to the placard, run the OEM reset so the controller relearns the baseline rolling radius. Without that reset the warning often stays on with good tires.

Record learn/reset completion in the service log with the cold pressures used. Do not compare a hot reading to the placard and bleed pressure down; that leaves the tire low when it cools.

101.1.3 Race HSL channel and sensor calibration

After logger, harness, or sensor work:

1. Verify the channel list, engineering-unit scaling, and sample rates. Wrong configuration looks like a failed plant part in the pit display.
2. Calibrate race sensors (suspension pots, brake and tire-pressure sensors, steering angle, IMU) the same way wrong signal mapping ruins scan-tool data.
3. Confirm time sync among logger, ECU, GPS/timing beacon, and video before the next timed session.
4. Record channel-configuration revision with the session archive so later comparisons use the same mapping.

101.2. VARIANT CODING AND FEATURE ENABLES

variant

A *variant* is a descriptor for the vehicle configuration: engine size, transmission type, emissions market, pack size, axle count, tow/ADAS options, and similar factory differences. The ECU variant must match the vehicle. One firmware image may support many variants; *variant coding* tells each module which cal set and enable flags apply to this VIN, usually during programming, first boot, or an OEM coding session.

If the wrong variant coding is used, the module will run, but with another vehicle's maps and limits (wrong shifts, range, limp, monitors incomplete).

feature enable

Feature enables are closely related: optional equipment (attachments, telematics, and speed limits) may be enabled through configuration, when the hardware is present. Changes should be authenticated and logged.

101.3. ECU FLASHING AND SOFTWARE UPDATES

Flashing propulsion, body, lift, or BMS controllers requires compatible images, stable power, and cybersecurity-compliant sessions. OTA updates should support rollback or staged

application where failures would affect safety. Unauthorized flashing of the vehicle controller can leave the vehicle unable to enter Ready or to charge.

Flashing a classic AUTOSAR unit replaces generated firmware configuration plus application code and the full data mapping set (signal, data-element, parameter, mode, and prototype mappings). Adaptive updates are packaged applications and machine manifests; rollback and staging matter when the HPC owns drive policy.

Fleet and industrial vehicles often have stricter access control and cyber security requirements for reprogramming and diagnostic sessions (see ISO/SAE 21434).

102. OPERATIONAL TELEMETRY AND ANALYTICS

102.1. CONNECTED VEHICLES

Connected and "smart" vehicles add telematics, remote diagnostics, over-the-air (OTA) firmware updates, and manufacturer service portals. The vehicle reports health metrics centrally, may schedule service based on detected faults, and can receive software corrections without a shop visit for some classes of issue. This improves fleet and owner visibility but also ties long-term serviceability to vendor tooling, account access, and update policy.

102.2. FLEET MONITORING AND PREDICTIVE MAINTENANCE

Where tire pressure monitoring systems are fitted, fleet telemetry should distinguish low pressure from system fault, trends repeated warnings on the same VIN or trailer ID, and (on direct systems) capture corner pressure when the event fired. Heavy vehicles often expose the same pattern on J1939; trailer sensors are separate from the tractor module. A trailer warning with healthy tractor tires is the trailer's sensors or connection.

For active or air suspension, trend fail-soft / limited events, compressor or actuator run-time, and corner height when a fault is stored. Rising compressor duty across the fleet often precedes leak or dryer service; repeated height faults on one VIN after recent chassis work often precede a missed calibration rather than a parts campaign.

Fleet management identifies which vehicles are safe to dispatch, which need charge before use, and which require shop service without requiring technicians to read raw BMS data for every unit. The connected fleet aggregates hours, fuel or energy use, regeneration or after treatment events, geolocation, idle time, upcoming maintenance due dates and fault trends. Trend analysis can predict failures (battery degradation, hydraulic overheating, recurring trouble codes) before they cause downtime.

These are often integrated with depot maintenance systems.

For EV fleets, telemetry should support dispatch decisions: state of charge and estimated range at shift start, open HV faults, charge-station assignment, and SoH trends that trigger pack inspection or vehicle rotation. For utility-cart fleets, track rounds per charge, charge duration, and repeat motor or charger faults to identify carts due for battery replacement or bearing service.

102.3. HIGH-SPEED LOGGING AND RACE TELEMETRY

On high-performance and race vehicles, high-speed logging (HSL) and pit telemetry are the primary operational data path:

- Log more channels at higher rates than fleet telematics. This often includes suspension travel, brake pressures, IMU, tire and brake temperatures, driveline torques, and dense ECU internals where licensed.
- Stream a bandwidth-limited subset live; keep the full-rate recording on the car. When live data and the downloaded HSL disagree, trust the onboard file after checking radio path and channel config.
- Tag each session with setup, driver, track, and conditions so analytics compare like with like.
- Use high-speed logging to catch degradation before a trouble code: asymmetry, drift from a baseline lap, or thermal climb under the same duty.

103. FAULT FINDING AND TROUBLESHOOTING

Troubleshooting will employ systematic procedures, trouble codes, sensor data analysis, and guided tests to identify root causes.

103.1. SYMPTOM-BASED TRIAGE

Narrows the fault space before running deep tests: group the complaint, note when it appears, and tie it to likely subsystems.

1. Capture the complaint in the operator's words and note when it occurs (cold start, under load, after transport, at full elevation, etc.); whether it is constant, intermittent, or under specific conditions.
2. Note the operating context: recent service, attachments, terrain, load weight, ambient temperature, and whether multiple warnings appeared together.
3. Read stored and pending trouble codes first; note whether faults are current, intermittent, or history-only. For chassis, record active mode (suspension, TPMS learn state) before clearing anything.
4. Identify safety stops before further diagnosis (structural damage, uncontrolled motion, fuel or hydraulic leak under pressure, HV fault indicators).
5. Map symptoms to likely subsystems using the table below; escalate to subsystem-specific guided tests.
6. Reproduce or confirm the fault under controlled, safe conditions.
7. Isolate subsystem using codes, data, and divide-and-conquer: disable non-essential loads, swap known-good modules.
8. Repair or replace the failed unit.
9. Clear codes and reset maintenance counters per policy.
10. Run OEM self-tests functional checks, and a representative duty cycle.
11. Document actions for fleet, warranty, or regulatory inspection records.

Symptom	Possible causes
Charging failure	Charge port, onboard charger, cable, session logs, HV contactors Charger interlock, cord/connector, BMS fault, incorrect charger type, tripped

Table 116: *Bucketing symptoms with likely causes*

	breaker. If charge inhibited with no charger fault, check vehicle control unit.
Intermittent electrical	Connectors, grounds, CAN/LIN wiring, moisture ingress
Loss of power under load, limp mode	Power-train derate, HV thermal limits, fuel delivery, network fault, restricted intake/exhaust, clogged filter, overheating, limp due to sensor fault.
No start / long crank	Battery, starter, immobilizer, HV contactors, neutral / interlock, fuel delivery, etc. Vehicle control unit faults.
Overheating	Coolant, hydraulic fluid, radiator/airflow blockage, fan/pump fault, thermostat, water pump, head gasket signs—separate from pure "powertrain network" issues unless sensor-related.
Platform will not raise / slow lift	Hydraulics, interlocks, overload, tilt, low battery (electric lift).
Platform drift / descent fault	Cylinder seals, holding valves, descent controls, structural damage.
Short range / weak acceleration	Battery age or imbalance, charger fault, brake drag, motor/controller, tire pressure.
Harsh / low / crooked ride; suspension lamp	Active or air suspension: fail-soft mode, height calibration, leak, compressor, coding.
Ride height will not change / stuck in one mode	Mode inhibit, failed compressor or valve block, coding, height sensor, preview camera not aimed.
Compressor runs constantly or cycles often	Air leak, dryer saturated, wrong height target after missed calibration.
TPMS lamp (steady)	Under-inflation; confirm with a cold gauge before replacing parts.
TPMS lamp (flashes, then stays on)	System fault: missing sensor, dead sensor battery, incomplete learn, failed receiver.
Warning lamp / malfunction indicator	Power-train emissions (see Engines chapter), network, subsystem ECU.
Logging gaps, wrong scale, or desync	Channel configuration, sensor cal, logger media, timebase across logger/ECU/video.
Telemetry flat / missing	Check radio/path or bandwidth, it is not necessarily a failed sensor.

103.2. ACTIVE AND ADAPTIVE SUSPENSION

Triage for passive, adaptive, air, and active suspension issues:

1. Confirm active mode and the first suspension DTC before replacing springs or the chassis module. Fail-soft leaves the vehicle drivable with a harsher or fixed ride.

2. Pull live data: commanded vs actual height per corner, mode request vs mode active, compressor or actuator current, and height-sensor voltages. Compare wheels on a level surface.
3. Separate plant faults (leaking air spring, seized damper, weak compressor, low fluid) from mapping or coding faults (wrong variant, height offsets, camera preview not aimed).
4. One corner low with the others normal is usually that corner's hardware or its channel. All four wrong after a flash points at coding or the controller, not four failed plants.
5. After arm, height-sensor, actuator, or module replacement, complete OEM ride-height calibration and road-test each mode before condemning parts. A crooked stance after module swap often means lost height offsets.
6. Air systems: listen for leaks, run leak-down, check dryer and compressor duty. Fully active hydraulics: fluid level/condition, filter, and pressure at OEM test ports (same isolate-and-verify pattern as other vehicle hydraulics below).
7. Preview systems: after windshield, bumper, or grille work, confirm camera aiming before replacing dampers or the suspension ECU.
8. Do not treat "suspension malfunction" as an alignment-only complaint, and do not replace the module for a leak that the height sensors are correctly reporting.
9. After repair, clear codes per policy, verify mode changes on the HMI, confirm heights within OEM tolerance, and road-test comfort and firm modes over a representative surface.

103.3. TIRE PRESSURE MONITORING

Triage for tire-pressure monitoring issues:

1. Measure the cold pressures with a gauge and compare with the placard pressure.
2. Separate low tire warning (steady lamp, pressure below threshold) from system faults (lamp flashes then stays on, or a DTC for missing sensor / incomplete learn).
3. After tire rotation or sensor replacement on a direct system, complete relearn before condemning the receiver or chassis module. One corner wrong after a rotation is usually a position-mapping fault.
4. On an indirect system: Set the tire pressures to cold pressures, then run the OEM reset.
5. One corner missing after tire service is usually that sensor, its battery, or a wheel installed without a sensor.
6. Inspect for puncture, leaking valve, or bead leak once a low reading is confirmed.
7. On tractor-trailer combinations, isolate trailer sensors and the trailer link before replacing tractor TPMS module.

103.4. CABLE TESTS AND CONNECTOR TROUBLESHOOTING

Cable and connector faults are common root causes for intermittent engine and sensor faults, especially in high-heat, vibration, and moisture-exposed zones. Use the structured isolate-

and-verify cable troubleshooting work flow (Chapter 11) to troubleshoot a suspected cabling issue.

1. Narrow down to suspected harness branches using trouble codes, freeze-frame data, symptom timing, and subsystem schematics.
2. Inspect for chafing, melted insulation, rodent damage, pinched harnesses, and improper strain relief near engine movement points.
3. Check connector integrity: Inspect pin fit, bent/pushed-back terminals, corrosion, moisture ingress, broken locks, loose back-shells, loose terminals and seal damage; verify full seating and latch engagement on both ends.
4. Measure voltage drop under load (not static voltage alone) on supply and ground paths.
5. Provoke intermittent faults with controlled wiggle, vibration, or heat tests while watching live data and error counters to reproduce dropouts.
6. Run continuity, short-to-ground/short-to-power checks, and resistance checks against OEM limits; for communication lines, verify expected bus levels/termination.
7. Repair with approved repair methods: correct wire gauge, crimp type, seal, splice location and harness re-wrap/routing. Avoid temporary bypasses on safety circuits.
8. Verify after repair: clear codes per policy, run self-tests or drive cycle, and confirm stable readings over a representative duty cycle.

103.5. HYDRAULICS AND MECHANICAL SUBSYSTEMS

Many work vehicles rely on hydraulics for lift, steer, or implement functions. Serviceability features for work vehicle hydraulics include sight glasses and dipsticks, replaceable filters and breathers, labeled bleed points, and pressure test ports. Fully active suspension hydraulic circuits use the same checks (fluid, filter, pressure ports) after confirming the fault is plant-side rather than height coding or fail-soft mode.

Common fault patterns:

Symptom	Possible causes
Air suspension	Slow raise, one corner low, or compressor continuous run: leak, dryer, or valve block (not an alignment-only job)
Drift or creep	Cylinder seal wear, holding valve fault, or internal leakage.
Noise	Cavitation, aeration, or mechanical wear in pump or motor.
Overheating	low fluid level, cooler blockage, or continuous high duty beyond design.
Slow or jerky motion	Low fluid, aeration, worn pump, restricted filter, or valve sticking.

Table 117: *Hydraulics symptoms with likely causes*

Scissor lifts and boom lifts add structural and safety elements: limit switches, tilt sensors, descent rate controls, and emergency lowering. Troubleshooting should follow OEM sequences that verify ground vs platform control paths, interlock chains, and relief valve settings before component replacement.

Mechanical wear – tires, forks, pins, bushings, chains – is often diagnosed by inspection and measurement rather than trouble codes. Combine visual checklist results with hour/cycle counters and hydraulic trends for preventative maintenance.

103.6. VEHICLE CONTROL UNIT FAILURE MODES AND INDICATORS

Vehicle controller related faults often appear as: vehicle will not enter ready; charge always inhibited with no charger fault; torque derate without motor over-temperature; simultaneous communication trouble codes to multiple propulsion modules; or limp that clears only briefly after battery disconnect. External causes that look like VCU failure include:

- Weak 12 V / 24 V supply or poor ground causing resets
- Open HV interlock loop or isolation fault (BMS-owned, VCU-enforced)
- Contactor welded or stuck open
- Gateway offline so the VCU cannot see BMS or motor controllers
- Corroded or water-intruded VCU connector
- Incomplete OTA or interrupted flash leaving inconsistent software levels. A module that will not enter Run after a flash is often boot or configuration.
- "ADAS unavailable" with a healthy ready state is usually sensor blockage, calibration, or the ADAS ECU issue.

Logs should capture VCU resets, ready/charge inhibit reasons, contactor command vs feedback, bus-off counts, and thermal derate triggers so shop and fleet triage can separate supervisory policy from peer-module faults.

When diagnosing cascading faults, establish which module set the limit, which reported the first trouble code, and whether the network path between VCU and peers is intact.

VCU diagnostics are typically OEM-specific rather than a single regulatory PID set.

103.6.1 Replacement

When a vehicle controller is replaced or reprogrammed:

- Calibration and coding: Match software to vehicle variant (axle count, pack size, drive modes, market); wrong coding causes range, charge, or limp misbehavior.
- Identity and immobilizer: Pairing with keys, gateway, or security modules may be required before Ready is allowed.
- Learned and duty data: Some energy or thermal adaptations and fleet counters transfer only via OEM process; clearing codes does not restore coding.
- Peer alignment: After flash or swap, verify BMS, inverter, and charger software compatibility; mismatch versions are a common post-service no-charge or derate cause.
- Physical environment: Mounting, connectors, and sealing matter; water intrusion at the VCU connector mimics controller failure.

Treat VCU replacement as a last step after verifying LV power and grounds, HVIL and isolation, BMS and charger readiness, and network integrity to peer modules.

103.7. IN-OPERATION VS SHOP SERVICE

While operating, the vehicle focuses on safety and controlled degradation rather than physical repair. That is, it is limited to what the operator or onboard systems can do immediately, such as isolating a fault, reducing load, engaging safing, or reaching a safe stop:

- Detect fire, over-temperature, low fluid, structural overload, or critical sensor faults and warn the operator
- Shed non-essential load to preserve steering, braking, lighting, or descent controls
- Limit speed, torque, elevation, or hydraulic functions in derate or limp mode
- Suspension fail-soft (firm or passive dampers, parked air height) and TPMS warnings that limit speed, continue carefully to service; do not treat as propulsion ready failure
- Provide clear indication of fault and required next action (stop, lower platform, proceed to service)

Shop or depot service completes root-cause correction: read and clear codes, replace modules or consumables, bleed hydraulics, reflash firmware, calibrate sensors, and verify with road test, roller diagnostics, or load test as applicable.

103.8. HIGH-SPEED LOGGING AND RACE TELEMETRY

1. If the pit screen looks wrong, download the onboard log first. Live telemetry is a subset and can drop without a plant fault.
2. Confirm channel list, scaling, and sample rate after sensor, harness, or logger changes; a wrong configuration looks like a failed part.
3. Check logger media health, free space, and that the session actually armed and closed. A missing file is a logging fault, not proof the car was fine.
4. Verify time sync among logger, ECU, GPS/timing beacon, and video before comparing laps or overlaying channels.
5. Use log asymmetry and thermal climb vs a baseline lap to catch plant issues that never set a trouble-code; then confirm with conventional gauges or UDS where the platform supports them.
6. After repair, re-calibrate affected sensors, confirm channels, run a short validation log, and archive with setup metadata.

104. REFERENCES & RESOURCES

104.1. CAN

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Bosch, *CAN FD Specification Version 1.0*, 2012. https://www.bosch-semiconductors.com/products/ip-modules/can-protocols/can-fd/

ISO 11898-1:2024, *Road vehicles – Controller area network (CAN) – Part 1: Data link layer and physical coding sublayer,* https://www.iso.org/standard/86384.html

ISO 11898-2:2026, *Road vehicles – Controller area network (CAN) Part 2: High-speed physical medium attachment (PMA) sublayer,*
https://www.iso.org/standard/90697.html

104.2. DOIP

ISO 13400-2:2025 *Road vehicles – Diagnostic communication over Internet Protocol (DoIP) Part 2: Transport protocol and network layer services*
https://www.iso.org/standard/13400-2

104.3. EAST-ADL

Vehicle-level architecture description; Implementation maps to AUTOSAR

EAST-ADL Association, specifications (Vehicle / Analysis / Design / Implementation levels):
https://east-adl.info/Specification.html

EAST-ADL White Paper (alignment with AUTOSAR and ISO 26262):
https://www.maenad.eu/public/conceptpresentations/EAST-ADL_WhitePaper_M2.1.12.pdf

ObD-II

For serviceability, treat them as the same role: the primary power-train controller that owns engine-related DTCs, adaptive values, and emissions monitors.

J1939

- Scan-tool UDS is DCM. Freeze-frame and fuel-trim adaptations are NvM; interrupting shutdown (skipping write-all) can drop a just-finished relearn.
- A flash loads generated RTE/BSW/MCAL configuration plus the calibration SWCs. Wrong variant coding is often a configuration extract, not a bad ECU.

The ECU is the primary respondent for power-train diagnostics:

- OBD-II / EOBD (light duty): Standardized modes and PIDs for live data, DTCs, freeze-frame, and emissions readiness; accessed via the SAE J1962 connector.
- J1939 (heavy duty): Engine operating data and diagnostic messages (e.g., DM1/DM2) as SPN/FMI pairs for fleet tools.
- OEM-enhanced / UDS: Deeper routines: bi-directional actuator tests, coding, security-gated flashing, and module-specific service functions beyond the regulatory baseline.

104.4. FLEX RAY

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Wikipedia, FlexRay
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104.5. LIN

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https://www.iso.org/standard/85125.html

ISO 17987-2:2025 *Part 2: Transport protocol and network layer services*
https://www.iso.org/standard/85126.html

ISO 17987-3:2025 *Part 3: Protocol specification*
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ISO 17987-4:2025 *Part 4: Electrical physical layer (EPL) specification 12 V/24 V*
https://www.iso.org/standard/85128.html

ISO/TR 17987-5:2016 *Part 5: Application programmers interface (API)*
https://www.iso.org/standard/69116.html

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https://www.iso.org/standard/71044.html

Texas Instruments, *Introduction to LIN*
https://www.ti.com/lit/an/slla383b/slla383b.pdf

104.6. OSEK/VDK

AUTOSAR Classic is based on the OSEK/VDX (ISO 17356) standard, which defined the architecture for real-time operating systems, communications, and network management in automotive ECUs. It specifies requirements for task scheduling, alarms, communication protocols, network management, and standard interfaces for embedded controllers.

AUTOSAR Classic Platform: https://www.autosar.org/standards/classic-platform

AUTOSAR, *Layered Software Architecture* (EXP, Classic Platform), R22-11
https://www.autosar.org/fileadmin/standards/R22-11/CP/AUTOSAR_EXP_LayeredSoftwareArchitecture.pdf

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https://www.autosar.org/standards/adaptive-platform

ISO 17356-1:2005 *Road vehicles – Open interface for embedded automotive applications Part 1: General structure and terms, definitions and abbreviated terms*, 2005
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ISO 17356-4:2005 *Road vehicles – Open interface for embedded automotive applications Part 4: OSEK/VDX Communication (COM)* 2005
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ISO 17356-6:2006 *Road vehicles – Open interface for embedded automotive applications Part 6: OSEK/VDX Implementation Language (OIL)*, 2006
https://www.iso.org/standard/40080.html

104.7. UNIFIED DIAGNOSTIC SERVICES

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https://www.iso.org/standard/87962.html

ISO 14229-2:2021 *Road vehicles – Unified diagnostic services (UDS) Part 2: Session layer services*
https://www.iso.org/standard/77322.html

ISO 14229-3:2022 *Road vehicles – Unified diagnostic services (UDS) Part 3: Unified diagnostic services on CAN implementation (UDSonCAN)*
https://www.iso.org/standard/77323.html

ISO 14229-4:2012 *Road vehicles – Unified diagnostic services (UDS) Part 4: Unified diagnostic services on FlexRay implementation (UDSonFR)*
https://www.iso.org/standard/55285.html

ISO 14229-5:2022 *Road vehicles – Unified diagnostic services (UDS) Part 5: Unified diagnostic services on Internet Protocol implementation (UDSonIP)*
https://www.iso.org/standard/76882.html

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CHAPTER 21

Vehicle Engines

This chapter discusses the engine and transmission controllers, their sensors and actuators, as well as the in-vehicle networks used by the power train:

- Overview of features by product tier
- Engine management system and engine control unit
- Transmission Control Unit (TCU)
- Controller software (AUTOSAR / OSEK-VDX)
- Health indicators and service reminders
- Fault codes, troubleshooting, and verification

105. OVERVIEW

engine management system

An *engine management system*'s controls combustion to provide performance, compensate for degradation and changing environment, and to reduce vehicle-exhaust pollution. Most rely on a network of sensors and actuators governed by electronic controllers, constantly monitoring and adjusting parameters for performance, efficiency, and emissions.

With storage and communication, it's largely sealed packages. With engine and power train systems, there are significant harsh environment, many consumables, many things degrade, and each sensor and other item is often replaced/serviced.

Table 118: *Serviceability tiers for engine diagnostics & support*

Tier	Common features
Low-cost consumer	Minimal status indicators; limited codes; basic check power/connection, style troubleshooting. Little or no usage/health tracking
Premium consumer	OBD-II, Basic counters (hours, starts), limited logs; some freeze-frame. basic scan tool access; basic app support, standardized trouble codes; guided checks
Contractor / professional	Easier to replace units, e.g. air and oil filters. Longer life spans. Increased health tracking & diagnostics, such as OBD-II plus enhanced modes and J1939 for heavy-duty; Structured workflows, exportable service bundles, and service tools. Increased security.
Enterprise / industrial fleets	Expands contractor grade with remote diagnostics, predictive maintenance analytics. Centralized fleet-wide telemetry: operating hours, start/stop cycles, stress and duty monitoring, error rate trends; unified dashboards for analytics.

106. KEY COMPONENTS AND ARCHITECTURE

The engine and transmissions control units sit at the *controller* layer of the vehicle stack. They coordinate continuously: the engine controller publishes torque, RPM, and throttle; the

transmission controller schedules shifts and requests a brief torque cut so clutches can engage. The anti-lock braking system (ABS), electronic stability control (ESC), and ADAS may also request brake or intervention as well.

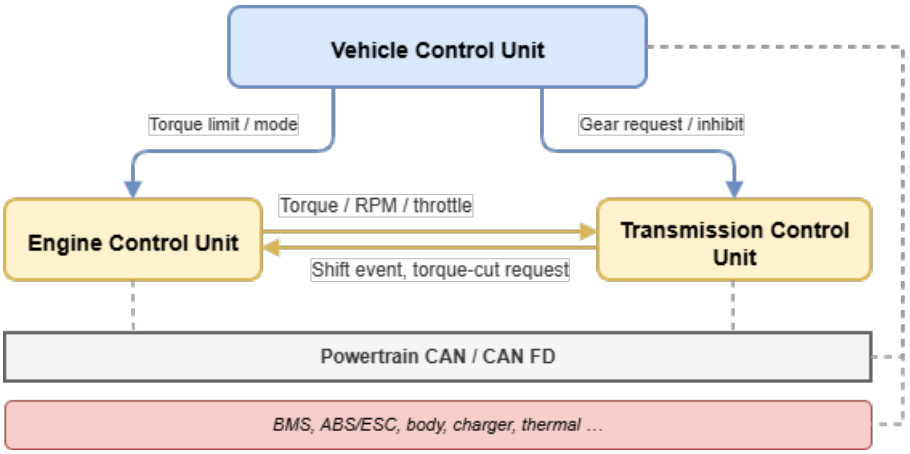


Figure 19: *The power train supervisor and controller layers*

Conventional internal combustion engine vehicles rarely have a propulsion supervisor. On electric and hybrid vehicles, a distinct vehicle or hybrid control unit is used to arbitrate engine torque, motor torque, battery limits, and enforces policies such as ready / limp / charge. Mild hybrids and some domain-controller platforms may also add a thin supervisory role.

106.1. ENGINE CONTROL UNIT

engine control unit

An *engine control unit* (ECU) monitors air & coolant temperatures, airflow, throttle position, crankshaft position, speed, and engine knock. In turn, it controls fuel injection, airflow, exhaust-gas recirculation, and spark-plug timing. The ECU has strict requirements so that it may not 'miss' sending the proper pulse or send it late.

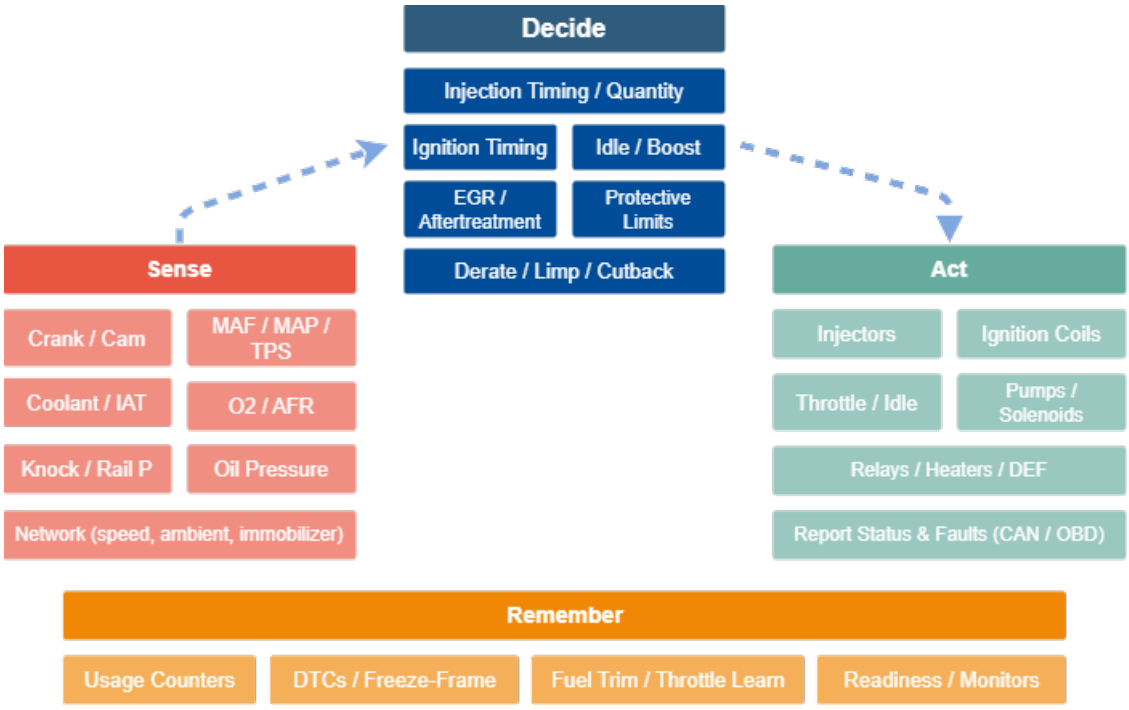


Figure 20: *The engine management integrates sensing and actions*

Role	Function
Sense	Sample crank/cam position, airflow or manifold pressure, coolant and intake temperatures, throttle position, oxygen/air-fuel ratio (AFR) sensors, knock, rail pressure, and related inputs; also receive network messages (e.g., vehicle speed, ambient conditions, immobilizer status, gear and torque requests from the TCU).
Decide	Compute injection timing and quantity, ignition timing (gasoline), idle and boost control, exhaust gas recirculation (EGR), after-treatment commands (diesel), and protective limits (derate, limp, over-speed/over-temp cutbacks). Honor TCU torque-reduction requests during shifts.
Act	Drive injectors, coils, throttle or idle actuators, pumps, solenoids, relays, and heaters; report status and faults on the diagnostic and powertrain buses. Report torque available, RPM, and faults so the TCU can schedule shifts.
Remember	Retain coordination information, freeze-frame, readiness/monitors, learned adaptations (fuel trim baselines, throttle learn, injector corrections), and usage counters across key cycles in non-volatile memory.

Table 119: *The engine controller's sense, decide, act and remember roles*

Note: controller names vary by manufacturer and region. *Engine control module* (ECM) might be used instead of ECU; or *power train control module* (PCM) when transmission functions are combined.

Class	Examples and notes
Actuators	Injectors, ignition coils, electronic throttle, EGR, turbo actuators, glow plugs, DEF dosing. Confirm power/ground and mechanical freedom before replacing the ECU.
Air path	Mass air flow (MAF), manifold absolute pressure (MAP), throttle position, ambient/intake air temp. Compare values vs expected load/RPM; vacuum and boost leaks mimic sensor faults
Combustion / emissions	Upstream and downstream O_2 or AFR sensors; NO_x (diesel/SCR). Heater circuits and exhaust leaks often cause "sensor" codes.
Fuel	Rail pressure, fuel temperature, injector feedback (where fitted). Pressure/volume tests before condemning injectors or the ECU
Network	CAN (and sometimes LIN) to transmission, ABS, body, instrument cluster, immobilizer. Communication DTCs may be wiring, termination, or another module offline
Position / speed	Crankshaft and camshaft sensors no-start or misfire if lost; check connectors and air gap.
Protection	Coolant temp, oil pressure (switch or sensor), knock. Over-temp and low-oil policies may derate or shut down.

Table 120: *Common ECU-connected elements*

ECUs greatly reduced needed for engine tune ups and extends component longevity and allowing the vehicle to perform well while its components deteriorate. One of the ECU functions is to translate pulses from the engine's timing gear. Several techniques are applied to accomplish this: it can fill in for missing teeth, or 'hide' extra teeth; it can cure knocking by

adding several phantom teeth after gap, and removing an equal number after; it can increase horsepower by reversing that process.

And ECUs adapt better to a range of environmental conditions. In advances models the timing tables are dynamically switched or adjusted based upon the crank trigger sensor, knock sensors and others, which help reflect the expected type of driving (accelerating, highway speeds etc.). These can be adjusted using aftermarket tools.

106.1.1 Diagnostic interface

The ECU stores fault history, exposes codes, live data, trouble codes and guided tests to scan tools, making fault-finding practical. Higher tiers deepen that access (enhanced OEM modes, UDS, authenticated flashing); lower tiers may expose only lamps and limited codes.

Diagnostic sessions often require key-on engine-off or specific security access. Unauthorized or interrupted flashing can brick the module or violate emissions/cyber-security rules; use approved tooling and procedures (see Configuration Management and Service Tools).

106.2. TRANSMISSION CONTROL UNIT

transmission control unit

The *transmission control unit* (TCU) manages a vehicle's automatic transmission, delivering the requested torque smoothly while avoiding over-revving, engine lugging, or transmission overheating. It controls gear selection, clutch engagement, and hydraulic or pressure control. It often communicates with stability or traction control systems to coordinate torque management and prevent wheel slip under adverse conditions. They are used in automatic, *dual-clutch transmission* (DCT), automated-manual, and many *continuously variable transmissions* (CVT).

The engine and transmissions controllers coordinate on every shift. The engine controller provides the transmission controller with the engine speed (RPM), throttle position, engine temperature, and current torque output. The transmission controller, in turn, controls shift and pressure solenoids, clutch valves, and shift actuators. It reports gear selection, transmission fluid temperature, and upcoming shift events or torque-cut requests to other vehicle systems. Without a timely torque cut, shifts are harsh or aborted; without a valid torque/RPM stream, the TCU may hold gear or limp.

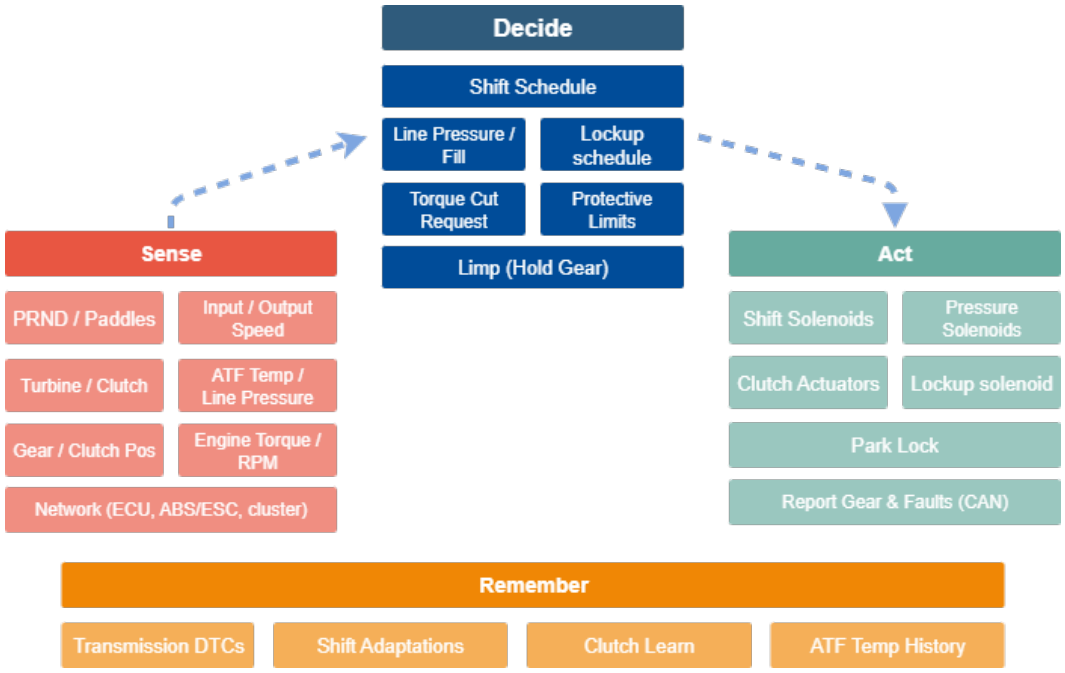


Figure 21: *The transmission control integrates sensing and actions*

Role	Function
Sense	Range selector (park-reverse-neutral-drive (PRND) or equivalent), input and output shaft speeds, turbine or clutch speed where fitted, transmission fluid temperature and line pressure, clutch or gear-position sensors; also engine torque, RPM, and throttle or pedal from the ECU, and wheel-speed or brake status from ABS/ESC.
Decide	Shift schedule (when to upshift/downshift), line pressure and clutch fill, lockup schedule, and protective limits (over-temp, over-speed, limp in a single gear). It requests engine torque reduction during shifts so clutches can engage without shock.
Act	Drive shift and pressure solenoids (or mechatronic valves), clutch actuators, lockup solenoid, and park-lock where electronic; report gear, ratio, and faults.
Remember	Shift and clutch adaptations, fluid-temperature history, and transmission trouble codes.

Table 121: *The transmission controller's sense, decide, act and remember roles*

Note: controller naming varies by manufacturer and region. *Transmission control module* (TCM) is common; on some platforms engine and transmission share a *power-train control module* (PCM). The TCU may be built into the mechatronics unit on the gearbox.

Class	Examples and notes
Actuators	Shift/pressure solenoids, TCC lockup, clutch valves. Confirm mechanical freedom, fluid condition, and power/ground before replacing the TCU.
Driver / range	PRND, paddles, sport/tow mode. Implausible range vs actual gear; check inhibitor/neutral-safety before condemning the TCU.
Engine interface	Torque available/requested, RPM, throttle. No-shift or limp can be an engine

Table 122: *Common TCU-connected elements*

	torque-limit or lost CAN, not a failed gearbox.
Fluid / pressure	ATF temperature, line pressure. Over-temp derate and delayed shifts; low fluid or a failed pump mimics solenoid or TCU faults.
Network	CAN to ECU, ABS/ESC, cluster. Communication DTCs may be wiring, termination, or the engine ECU offline.
Position	Gear, clutch, park lock. DCT clutch-position faults often need adaptation after repair, not a new TCU.
Speeds	Input, output, turbine/clutch speed. Ratio errors and harsh or missing shifts; compare to engine RPM and wheel speed.

107. MANAGEMENT INFORMATION AND DIAGNOSTICS

107.1. HEALTH INDICATORS

Engine and power-train health has several related views. The same symptoms (derate, rough idle, MIL, limp) can come from any view. Read them together rather than in isolation:

- Combustion and AFR control: Is closed-loop fueling and ignition healthy?
- Air path and load sensing: Do MAF, MAP, and throttle agree with expected load/RPM?
- Thermal and lubrication: Are coolant and oil within safe envelopes?
- After treatment and emissions: Are catalyst, DPF, SCR/DEF, and monitors meeting design intent?
- Sensor vs root cause: Is the signal bad, or is the plant (leak, injector, restriction) driving it?
- Controller and network: Is the ECU/TCU or the power train bus the fault?
- Adaptations and readiness: Have learns completed; are OBD monitors ready?

107.2. ENGINE AND POWERTRAIN METRICS

Health Measure	Description
Coolant temperature	Temperature of engine coolant. **Units:** °C or °F
Engine hours	Total runtime since installation. **Units:** hours
Fuel use	Measured total or rate of consumption **Units:** liters, gallons, L/hr, gal/hr
Intake air temperature	Measured at throttle or intake. **Units:** °C or °F

Table 123: *Example of engine health metrics*

Misfire count	Number of detected misfire events. **Units:** count
Oil pressure	Lubrication oil pressure during normal operation. **Units:** kPa, psi
Over-temperature events	Incidents where temp exceeded safe range. **Units:** count
Recent oil/filter change	Time or cycles since last service. **Units:** hours/cycles/days
Regeneration cycles	Diesel particulate filter regeneration. **Units:** cycles (count)
Sensor drift detected	Evidence of sensor zero or gain drift. **Units:** status/Δ value
Throttle learn status	Whether adaptive values re-learned. **Units:** OK/Needs Service
Total start cycles	Number of engine start events. **Units:** count
Trouble code count	Number of unaddressed diagnostic trouble codes. **Units:** count

Each oxygen (O_2) sensor (there are multiple) may have metrics like:

Health Measure	Description
Sensor bias / offset	Mean offset of O_2 sensor voltage compared to expected baseline under steady-state conditions. **Units:** V (or normalized ratio)
Sensor contamination / aging indicator	Evidence of slow switching, reduced amplitude, or persistent bias attributable to contamination/aging. **Units:** status / counts
Sensor heater health	Heater control performance (heater current/voltage stability, time-to-light-off, and heater-related trouble codes). **Units:** status / Δ values
Sensor response time	Time to reach target voltage after a commanded air-fuel ratio (AFR) change (slow response indicates aging or contamination. **Units:** seconds
Switching frequency	Rate of lean/rich voltage switching during closed-loop operation (or switching cycles per unit time).

Table 124: *Example oxygen (O_2) sensor health indicators*

	Units: Hz or cycles/min

These indicators complement broader health indicators and allow for trend monitoring. The devices have the common health indicators and environmental exposure tracking (see chapter 8), temperature and power measurements (see chapter 13.).

107.3. ABSTRACT HEALTH INDICATORS

Abstracted health indicators fuse several metrics, counters, rates, and measurements to form a stable signal. They normalize units and intervals, trend over time (slopes, baselines, percentiles), and combine related measures so one noisy counter does not dominate.

Table 125: *Abstract engine / power-train integrity indicators*

Indicator	Patterns
Adaptations / readiness incomplete	Throttle learn, injector corrections, or similar adaptations not complete after repair; OBD readiness/monitors incomplete across drive cycles without a matching hard fault.
After-treatment unhealthy	DPF regen frequency high or regen fails to complete; rising differential pressure; SCR/DEF level, crystallization, dosing, or NO_x conversion outside baseline; catalyst efficiency codes with corroborating downstream O_2/AFR.
Combustion / AFR control unhealthy	Fuel trims swinging abnormally; misfire rate above baseline (esp. clustered by mode); upstream O_2/AFR slow, stuck, or biased while closed-loop is active; catalyst damage risk from sustained rich/lean or misfire.
Controller / network unhealthy	ECU or TCU resets, watchdog events, supply-voltage excursions, or bus-off; absurd values on many channels at once; lost communication to TCU/ABS with a healthy local sensor loop; actuator commands fail with good wiring and power.
Sensor vs root-cause ambiguity	Sensor bias, aging, or heater fault codes alongside plant can create AFR-like symptoms (fuel pressure, injector leak, vacuum leak, MAF/MAP error, exhaust leak, after-treatment effect). Isolate the plant before condemning the sensor or ECU.
Transmission coordination unhealthy	Harsh or aborted shifts, limp in gear, or ratio errors while engine torque/RPM stream or torque-cut handshake with the TCU is missing or late (see TCU section); distinguish gearbox hydraulics from lost CAN or ECU torque limit.

Table 126: *Abstract engine / power-train load indicators*

Indicator	Patterns
Air-path / load sensing degraded	MAF, MAP, and throttle disagree with expected load/RPM; vacuum or boost leaks mimic sensor faults; intake restriction inferred from high vacuum or low mass flow at given RPM.
Thermal / lubrication at risk	Coolant sustained above expected range; recurrent over-temperature events; oil pressure below minimum for calibrated duration; time near thermal limit with fan command/tach mismatch.

107.4. POLICIES AND THRESHOLDS

Use thresholds and policies so health indicators drive consistent service actions, such as warn, schedule service, or retire a device:

Issue a *warning* for problems not yet severe enough to impact engine safety or emissions compliance; and to schedule service:

- If O_2 switching frequency drops below baseline for a sustained interval (e.g., repeated slow switching events).
- If O_2 response time exceeds an allowed maximum or if heater time-to-light-off drifts beyond tolerance.
- If coolant temperature is sustained above an expected range for a defined interval (e.g., recurrent over-temperature events or overheating in normal load).
- If oil pressure is below a minimum threshold for more than a calibrated duration, or if oil-pressure events correlate with wear/sensor drift trends.
- If misfire count exceeds a baseline rate or if misfire events cluster by operating mode (idle, acceleration, high-load).
- If sensor drift indicators persist beyond tolerance (zero/gain drift) or when throttle learn status indicates repeated adaptive failures.
- If injector corrections repeatedly fail; monitors remain incomplete.
- In diesel systems, if regeneration cycles occur unusually frequently (suggesting exhaust backpressure/fuel-quality issues), or when regeneration fails to complete within expected limits.
- If trouble code count increases rapidly or remains non-zero across multiple drive cycles.

Issue a *service notification* when derate, limp, or emissions readiness is at risk but the vehicle can still operate under policy (e.g., after-treatment nearing forced derate, adaptations incomplete after a major repair, or TCU–ECU coordination unhealthy with intermittent harsh shifts).

A fault is raised for serious issues such as parameters or component health have deteriorated to the where normal operation could risk emissions violations, component damage, or safety:

- Disable closed-loop features when uncorrectable O_2 sensor faults occur or when bias prevents meeting stoichiometric control limits.
- Adjust engine operating mode (e.g., disable aggressive timing/air-fuel adjustments) when misfire events exceed critical thresholds that risk catalyst/engine damage.
- Enforce thermal or low-oil shutdown / severe derate when envelopes are exceeded.
- Treat sustained controller/network failure that prevents safe torque or shift coordination as a power-train fault.

107.4.1 Service Reminders

Service reminders translate health indicators and usage counters into scheduled or prompted maintenance. These reduce unplanned downtime and help keep emissions-related systems within design limits.

Consumable	Triggers
Belts and pumps	Timing belt or chain service windows; coolant pump or accessory drive service if tied to hours or predictive signals.
Cooling system	Coolant exchange or inspection interval; Reminders after repeated over-temperature events.

Table 127: *Preventative service notifications based on what to replace*

Emissions and after-treatment (diesel)	DPF ash service, differential-pressure checks, or regeneration policy when regen frequency or failure trends exceed normal.
Fuel and air path	Air filter replacement or inspection; fuel filter replacement (especially on diesels or where contamination is likely).
Oil and filters	Remind when Engine hours; distance; and/or calendar time since last oil change exceeds policy; include oil filter and sometimes air filter intervals.
Spark / ignition	Remind per OEM component intervals (hours, distance, or wear indicators).
Timing belts or chains	Use service window

Some consumables can be inspected, with judgement passed. Few have automated measurement, or derived measurements before serious issues.

Reminders may be snoozed or acknowledged; critical items may escalate to warnings or derate until service is performed, per calibration and regulation.

108. FAULT FINDING AND TROUBLESHOOTING

Troubleshooting employs systematic procedures, trouble codes, sensor data analysis, and guided tests to identify root causes.

108.1. SYMPTOM-BASED TRIAGE

Symptom-based triage narrows the fault space before running deep tests: group the complaint, note when it appears, and tie it to likely subsystems (air/fuel/spark or compression, emissions, cooling, lubrication, electrical, or network/diagnostic access).

1. Capture the complaint: exact wording (e.g., “hard start,” “stall at idle,” “limp mode under load,” “MIL on”); whether it is constant, intermittent, or cold/hot only.
2. Note the operating context: key cycle, ambient temperature, fuel quality, recent service, towing or high-load use, and whether multiple warnings appeared together.
3. Map to possible causes, likely subsystems (see table below)
4. Safety: identify symptoms that require stopping before doing any further diagnosis.
5. If triage points to a subsystem, move to guided diagnostics and service tests for that subsystem; if trouble codes are absent, use logs and repeatable conditions to reproduce.

Symptom	Possible causes
No start / long crank	Battery, starter, fuel delivery, crank/cam sensors, immobilizer, compression (rare first pass).
Rough idle / stall	vacuum leaks, idle air, fouled plugs/injectors, mass airflow sensor (MAF) / manifold absolute pressure (MAP), exhaust gas recirculation (EGR) stuck open, fuel quality.
Loss of power under load	fuel pressure/volume, turbo/boost leaks, restricted intake/exhaust, clogged filter, overheating, limp due to sensor fault.

Table 128: *Bucketing symptoms with likely causes*

Overheating	coolant level, thermostat, fan, water pump, head gasket signs–separate from pure "powertrain network" issues unless sensor-related.
Emissions / malfunction indicator	catalyst efficiency, O_2 sensors, EVAP, EGR–often paired with fuel trim trends.

Read the indicators together. Combustion complaints often share fuel-trim, misfire, and O_2/AFR signatures with air-path leaks or after-treatment restriction. Controller/network faults can flood trouble codes that look like sensor failures. Confirm which module owns the trouble code and whether freeze-frame is from the engine ECU before replacing parts.

108.2. CLOGGED FILTER TROUBLESHOOTING

A clogged filter (air intake, fuel, oil, cabin, or – on diesels – exhaust aftertreatment) can show up as reduced power, poor fuel economy, rough idle, limp mode, trouble codes, or abnormal sensor readings. e.g., high intake vacuum or *mass air flow* (MAF) / *manifold absolute pressure* (MAP) vs expected, fuel rail pressure shortfall, elevated pressure differential across a filter or *diesel particulate filter* (DPF).

1. Note the symptoms, and operating condition (idle, load, cold vs hot); read stored trouble codes and freeze-frame; check service reminders and filter life counters if present.
2. Map to possible causes / buckets
3. Replace or clean
4. After replacement or cleaning, clear non-persistent codes, run a drive cycle or self-test to verify, and confirm metrics return to baseline (no recurring faults).

Item, things to check

Service Item	What to check
Diesel particulate filter (DPF) exhaust after treatment (diesel)	Frequent or failed regen, high differential pressure, and back pressure related trouble codes can indicate ash overload or plugging– follow OEM regen and cleaning/replacement procedures.
Fuel filter	Check for low rail pressure, long crank, fuel-related trouble codes; verify pressure/volume tests where specified; replace on schedule or when contamination is suspected.
Intake air filter	Inspect for restriction, debris, collapsed media, or wrong part; compare or mass air flow (MAF) / manifold absolute pressure (MAP) vs expected for load/RPM; replace per interval or when restriction exceeds policy
Oil filter	Usually diagnosed via oil pressure trends and related trouble codes; confirm correct filter and torque; address if pressure is low and filter change is overdue or filter is wrong/collapsed.

Table 129: *Preventative service notifications replace based on what*

108.3. OTHER CONSUMABLES TROUBLESHOOTING

Other wear items and fluids behave like consumables: they age with hours, heat cycles, and duty. Triage by symptom, interval, and codes; confirm correct part grade and specification (viscosity, coolant type, plug heat range, belt type). After service, reset maintenance counters where applicable, clear codes per policy, and confirm stable temps, fuel trim, and no recurrence over a representative drive cycle.

Table 130: *Consumable*

Service Item	Description, symptoms, what to check
Belts (accessory / timing)	Squeal, chirp, or thrown belt points to tensioner, alignment, or wear; timing belt/chain service is interval or noise driven. Follow OEM guidance; do not ignore coolant pump on same service if applicable.
Coolant	Low coolant, wrong mix, or exhausted inhibitor can cause overheating, boil-over, or heater problems; verify level when cold, cap and hose condition, thermostat behavior, and pressure test if head gasket is suspected.
DEF / AdBlue (diesel SCR)	Low level, crystallization, wrong fluid, or sensor faults cause derate; refill with correct spec, clean spilled DEF, and follow OEM for tank/heater faults.
Engine oil	Low oil level or wrong viscosity can cause noise, pressure warnings, or cam/lifter issues; check level, leaks, and service history; look for fuel dilution (diesel/gdi) or coolant contamination; replace with OEM specified oil and interval.
Fuel quality	Water, wrong octane/CN, or contamination can cause knock, hard start, or filter plugging; drain/replace fuel if indicated; document for fleet or field return analysis.
PCV / crankcase ventilation	Stuck PCV or excessive blow-by can affect idle and oil consumption; inspect hoses and valve per application.
Spark plugs and ignition wear (gasoline)	Misfires, rough idle, or coil overload often trace to worn or wrong plugs, gap erosion, or oil fouling from valve seals; replace as a set where specified; verify torque and connector integrity.

108.4. OXYGEN SENSOR TROUBLESHOOTING

Oxygen (O_2) or air-fuel (A/F) sensor faults commonly appear as fuel-trim issues (lean/rich correction), poor catalyst control, elevated emissions, or intermittent closed-loop failures.

1. Start with the complaint and context: Note symptoms (malfunction indicator, stalling, rough idle, reduced power), whether it happens hot/cold, and which bank/sensor (upstream vs downstream, Bank 1 vs Bank 2). Record relevant freeze-frame data (RPM, load, coolant temp, fuel trim, catalyst temp where available).
2. Read and interpret codes: Identify whether the engine controller (ECU) reports heater circuit faults, slow response, stuck voltage, bias/offset problems, or catalyst efficiency related codes (downstream sensor use). Confirm pending vs stored codes to separate intermittent from persistent issues.
3. Correlate with fuel trim and closed-loop status: If upstream O_2 switching is absent/slow but fuel trims swing abnormally, focus on heater/power, sensor contamination, exhaust leaks, or an AFR control issue feeding the sensor.

4. Check heater and power/ground (most common preventable cause): Verify O_2 heater operation (time-to-light-off), check fuses/relays, sensor connector integrity, and ground paths. A weak heater can cause “slow response” codes even when the sensor element is otherwise healthy.
5. Inspect wiring and exhaust integrity: Look for damaged insulation, pin fit issues, corrosion, loose terminals, and routing near heat sources. Check for exhaust leaks upstream of the sensor (even small leaks can bias readings toward lean). Verify that sensor threads are correct and that installations do not introduce contamination.
6. Assess sensor dynamic behavior: Compare expected switching activity/response time with observed data (e.g., slow response indicates aging/contamination; stuck voltage indicates failure or persistent bias). If you have both bank sensors, compare banks to isolate whether the fault follows a sensor vs an exhaust/AFR condition.
7. Test for root cause beyond the sensor: Validate upstream causes that create abnormal AFR: fuel pressure/volume issues, injector leakage, intake vacuum leaks, MAF/MAP errors, exhaust restriction/backpressure, or regenerator/aftertreatment effects.
8. Replace only after isolation and verification: When replacing an O_2 sensor, follow OEM procedure (correct sensor type, torque/spec, and anti-seize guidance). After replacement, clear codes per policy and confirm restored switching/response, stable fuel trim, and no recurrence over a drive cycle.

108.5. CONTROLLER PROBLEMS

A single symptom can originate in the sensor, wiring, actuator, mechanical path, or the engine controller itself. Isolate external causes before replacing the controller.

108.5.1 Engine Controller Troubleshooting

Engine controller related faults show up as:

- Absurd sensor values on all channels indicate an ECU failure. Absurd values on one channel are usually sensor, wiring, or MCAL/ADC.
- Lost communication with the TCU or ABS with a healthy local loop is network or communication configuration, not the combustion strategy.
- Failed actuator commands with good wiring, or recurring trouble codes that clear only briefly after flash/relearn.

Common external causes that look like ECU failure include:

- Weak battery or poor ground causing resets and spurious codes
- Water intrusion or corrosion at the ECU connector
- Shorted actuator or sensor pulling a supply rail down
- Harness damage near heat or moving parts
- Failed immobilizer or missing security response

Log and telematics (where available) should capture ECU resets, watchdog events, supply-voltage excursions, and bus-off counts to distinguish controller faults from intermittent wiring.

108.5.2 Mapping and configuration problems

If any link in the ECU signaling chain is wrong live data looks implausible even when the sensor and wiring are fine. This can include wrong signal packing (scale/PDU), wrong variant coding, or a bad flash.

If the data identifier points to the wrong element, the scan tool may show a believable number while the control loop uses a different (or stale) value. That looks like a sensor fault but is an incorrect mapping.

A missing data mapping set is a common post-flash "everything looks dead" symptom with no wiring fault.

108.5.3 Cascading problems across controllers

The engine controller shares the power train domain with transmission, after treatment, and hybrid/EV motor controllers where fitted. A fault in one module can cascade (e.g., transmission limp after engine torque-limit messages). When diagnosing, confirm which module owns the trouble code, whether freeze-frame is from the engine ECU, and whether network or gateway faults are masking the root cause.

108.5.4 Engine Control Unit Replacement

When an ECU is replaced or reprogrammed, serviceability depends on restoring identity and learned state:

- Calibration and coding: Match software/calibration to engine variant, emissions market, and options; incorrect coding presents drivability and readiness failures.
- Immobilizer / anti-theft pairing: Many platforms require pairing the ECU with keys and the immobilizer or body module; without it, the engine will not run.
- Adaptive values: Throttle learn, idle adaptation, injector corrections, and similar tables may need explicit relearn or a defined drive cycle after repair.
- Counters and history: Lifetime and emissions-related counters may be protected or transferred per OEM policy; clearing codes does not reset all learned or regulatory data.
- Physical mounting: Controllers sit in harsh thermal and vibration environments; connectors, seals, and grounding must be correct after swap.

108.5.5 Transmission Control Unit Replacement

A single "will not shift" or "stuck in gear" complaint can originate in fluid and hydraulics, a solenoid, speed sensors, the engine torque interface, or the TCU. Confirm which module owns the trouble code and whether the TCU commanded a shift the hardware did not complete.

When a transmission controller is replaced or reprogrammed, match calibration to gearbox variant and engine, restore or relearn clutch and shift adaptations, and pair with the immobilizer or ECU as required. Treat TCU replacement as a last step after verifying supply and grounds, fluid level and condition, speed sensors, solenoids, and network integrity to the engine ECU.

CHAPTER 22

Residential Home & Facility Management

This chapter is a think piece on describing facilities, serviceability and emergency-response practices for smart homes, and commercial buildings..

- Home residential energy systems, and automation
- Facility operations management and emergency readiness

109. OVERVIEW

Residential systems combine many of the same disciplines as commercial facilities – HVAC, electrical distribution, life safety, and access control – but at smaller scale, with lower formal maintenance structure, and with more consumer-oriented automation platforms.

Industrial buildings have long relied upon robust, proven protocols. They have higher cost tolerance and larger budgets.

The residential market is highly cost-sensitive, accepting a lower reliability and quality in exchange for price and ease of installation. It avoids new or extensive wiring, using solutions novel like power line communication (e.g., X10, Wi-Fi extenders) or low-cost wireless technologies. There are fewer established standards in this space, which both limits consistency and encourages rapid innovation (in certain areas). Vendors and service providers in the residential sector are often subject to frequent change and turnover.

As standards like Matter become more widely adopted, residential products are offering capabilities that were previously limited to commercial systems, with increasing reliability and vendor-neutral interoperability.

Designed for ease of installation and operation, typically serving one residence.

110. RESIDENTIAL HOMES

Residential home automation depends on sensors, controllers, and cloud or local automation to detect problems early and keep comfort and safety functions running. The elements include:

- Thermostats and HVAC controls that schedule heating and cooling, respond to occupancy, and report fault or offline states.
- Heating plant equipment – boilers, furnaces, heat exchangers, and air-conditioning units. These are controlled by the thermostat and other HVAC controls.

- Life-safety devices such as smoke and carbon-monoxide detectors with low-battery reporting, and (in advanced models) periodic self-test and network monitoring.
- Telecommunications infrastructure, including internet connectivity, networking for data, voice, and video. See chapter 15, *Communication*.

Home automation – smart homes – may include:

- Environmental sensing for temperature, humidity, occupancy or presence, air quality, and light levels. Door and window sensors. Rain sensors for sprinklers.
- Home energy systems including rooftop solar, battery storage, transfer switches, and off-grid or backup-power configurations.
- Home automation hubs and controllers (for example, Home Assistant) that integrate devices, run automations, and expose status for troubleshooting.
- Lighting and audio video automation (for example, WLED or scene-based control) used for convenience, energy savings, and occupancy-driven behavior.
- Audio and video systems with integrated automation, supporting multi-room sound, smart speakers, home theater, and streaming devices.
- Security systems covering intrusion detection, surveillance cameras, smart locks, and alarm integration. Window and door sensors.
- Lawn and garden irrigation monitor precipitation or soil moisture to control irrigation, and prevent overwatering.
- Autonomous robots may mow the yard on a schedule (with weather awareness) using boundary wires or mapping, sensors for obstacles, and auto-docking for charging.
- Autonomous floor cleaning

110.1. INTEGRATION HUB

integration hub

Bridges are used to make the sensors, switches, lighting, and appliances using wireless device networks such as Zigbee, Z-Wave, Thread, and BLE accessible on the local network. These often perform basic on/off, dimming, and motion rules. At a minimum, controllable with a phone "app."

Richer logic – multi-room scenes, calendar awareness, integration with HVAC or security – is provided by another service or hub. This integrates the sensors, actuators, thermostats, lighting controllers, and internet services, and provides the automations, dashboards, and notification paths the owner relies on for day-to-day operation and troubleshooting. A hub provides:

- A unified entity model so devices from different vendors appear as consistent states. This allows automation logic, visualization, and alerting to treat devices as peers.
- Automations and scripts that encode response policy locally – occupancy-driven lighting, humidity warnings, or shedding non-essential loads when on battery backup
- Centralized visibility of the current status, low-battery reports, maintenance alerts, unavailable entities (e.g., notification if a device is unresponsive or stuck and last-seen timestamps).

- Activity history, trending, logging and fault codes for post-incident review and for distinguishing transient glitches from sustained faults.
- Bridging between subsystems (for example, HVAC, lighting, security, energy storage), so that automation policies can reference whole-system state – enabling, for example, shutting off nonessential lighting during peak demand or battery operation.

For convenience and fast on boarding, many vendors provide cloud hosted services. Software can also be run on a local PC. Unlike the cloud, it can continue to carry out policies and respond to commands during an internet service outage. This should not be the only path for essential local behavior. Design so that loss of the hub leaves lights manually operable and, where policy requires, motion-activated paths still function on the controller tier.

110.1.1 Readiness for Integration

Home automation is a system integration problem. This coordinates devices and data from multiple domains and vendors into a cohesive network with unified control and visibility. Many smart home products are designed to operate independently, rather than with a clear plan to be integrated into a unified system.

- The high-end market devices suited for integration have traditionally been expensive and catered to a narrow audience, often relying on proprietary controls & touch panels.
- The premium consumer devices are often standalone products managed through a dedicated mobile app, with potential automation dashboard and cloud hosted services. Often limited in terms of cross-device coordination and deeper automation. Integration with a central smart hub may be possible, but manufacturer support is often low, and requires community efforts.
- Low-end units might have a phone application, or bespoke remote control.

110.2. AUDIO VIDEO

Audio and video (AV) systems include televisions, multi-room sound systems, smart speakers, home theater receivers, projectors, and streaming devices. These may be connected to the automation system for coordinated actions such as synchronized playback of music across zones, automated volume and source selection based on time of day or occupancy, and scene-based control (for example, dimming lights and closing shades for movie mode).

110.2.1 Media Servers

Media servers provide centralized storage and management of music, movies, and other digital media. They enable streaming and sharing of content across multiple devices within the home, support metadata organization, and often offer remote or mobile access.

Examples: Plex, iTunes, and Jellyfin.

110.2.2 Network Video Recorders (NVRs)

Network Video Recorders are dedicated appliances designed to store and manage security camera footage on-site. By keeping recordings local, NVRs avoid ongoing cloud subscription fees and deliver quick, low-latency playback both onsite and remotely for authorized users.

110.3. CATEGORIES OF SENSORS

Residential sensors fall into two broad groups: those that infer human presence or activity in a space, and those that measure ambient environmental conditions. Both feed automations.

PRESENCE AND OCCUPANCY sensors support convenience automations and, indirectly, fault detection (for example, heating a room that should be occupied but is not warming as expected).

Type of sensor	Description
Human presence detectors	Often millimeter-wave or similar radars that detect fine motion (breathing, small movements) and can distinguish occupancy from simple passage. Useful where lights should stay on while someone is still but not moving much.
Motion sensor Passive infrared (PIR) or similar	Detects movement in a cone of view. A practical stand-in for room occupancy when precise headcount is not required.
Room occupancy	May combine motion, door events, CO_2 rise, or seat/pressure sensors to approximate whether a room is in use and sometimes how many people are present; accuracy varies by sensor mix and room layout.
Proximity or distance	Used in targeted lighting, displays, or safety interlocks; less common in whole-home automation but relevant where fixtures should respond to position in a zone.

Table 131: *Presence and occupancy sensors*

ENVIRONMENTAL sensors measure ambient conditions in the volume around the sensor. Placement matters: a sensor on an exterior wall, in direct sun, or above a heat source will misrepresent the space occupants experience.

Type of sensor	Description
Temperature	Comfort control, freeze-risk warnings, and HVAC fault hints (for example, set point not met for an extended period).
Humidity	Mold risk, HVAC performance, and coordination with dehumidification or ventilation.
Air pressure	Barometric trends; occasionally used with weather-aware automations rather than room-level comfort.
Air quality	Volatile organic compounds (VOC), particulate, or CO_2 proxies; useful for ventilation triggers and for alerting when levels stay outside expected bands.
Ambient light	Daylight and shade control; can reduce false triggers when coordinating with occupancy-based lighting.
Sound	Noise monitoring for security or nuisance detection; privacy and false-positive rate should be considered in policy design.

Table 132: *Environmental sensors*

Note: treat sensor offline state, stale readings, and battery level as first-class health indicators – not only the measured value. A humidity sensor that has not reported in twelve hours is a different problem than a humidity reading of 65%.

110.4. LIGHT CONTROL

Residential lighting automation typically provides remote control of lighting and scene recall. *Scene-based control* (sometimes described as theatre or mood modes) stores preset settings for the light fixtures, dimming, and color temperature for activities like watching a movie or reading. For example, dimmed warm light for film viewing, brighter cool light for reading, or per-user preferences. These recall them with one action rather than adjusting each fixture individually.

Lighting automation often includes sensors – such as motion detectors and door contacts – that can trigger lights to turn on upon entry, maintain illumination while occupancy is detected, and turn off once the space is empty. This approach can be applied to both indoor and outdoor lighting and enhances both convenience and energy efficiency.

- Motion and door sensors can turn lights on when someone enters, hold them on while activity continues, and turn them off after a timeout when the space is empty.
- Time-of-day and ambient light can select different scenes automatically so artificial light supplements daylight only when needed.
- Manual override should remain available at the wall switch or equivalent so occupants are not trapped behind automation during a hub or network fault.

The same applies to outdoor lighting: path and security lights triggered by motion, with scenes for entertaining versus overnight security levels. Outdoor gear adds serviceability concerns – weather exposure, corrosion, insect intrusion in enclosures, and higher replacement churn. Prefer fixtures and controllers with clear offline indication and replaceable modules where possible.

110.5. HEATING, VENTILATION, AND AIR CONDITIONING (HVAC)

Modern homes have reliable, efficient heating, cooling, and air/water circulation.

Table 133: *HVAC systems*

System	Description
Central heating, cooling systems	Variations: Forced-air furnaces/AC, heat pumps, and air handlers. Managed via thermostats and (optional) room sensors. Zoning allows separate temperature settings per area, reacting to occupancy, schedules, or even window/door status.
Radiators & in-floor heating	Variations: hydronic or electric. May use individually actuated valves or relays for precise, supporting schedules, feedback from floor/room temperature sensors, and disabling heating in unused zones.
Fans and ventilation	Variations: Exhaust and fresh-air fans, energy recovery ventilators (ERVs/HRVs), and make-up air systems Controlled via air quality (CO_2, VOC, humidity), occupancy, or scheduled cycles. Central or room fans can supplement cooling or distribute heat.
Boilers Sensors	Monitor water pressure, temperature, burner cycles, and error codes. Fault detection algorithms can alert for low water pressure, failure to ignite, continuous burning, scaling, or other health/maintenance indicators.
Hot water	Variations: tank and on-demand (tankless)

	Temperature, flow, and energy sensors help detect leaks, excessive cycling, or failing heating elements. Home automation can schedule pre-heating for peak usage or react to user presence.
Hot water recirculation	Pumps circulate hot water on demand or per schedule, minimizing wait times at taps/showerheads and reducing water waste, with occupancy-triggered or button-press activation options.
Leak and freeze detection	Water and pipe temperature sensors in mechanical areas provide early warning of leaks or freezing risk, with automations for valve shut-off or heat activation.

110.6. HOME ENERGY SYSTEM

home energy management system

Home energy management systems (HEMS), or home energy management platforms, automate power management within residential homes, handling source selection, load shedding, health monitoring, and event logging. See also chapter 13, *Power Management*.

***Table 134:** Home energy management systems functions*

Function	Description
Monitoring	Gathers data electricity generation, storage, and use from solar panels, batteries, utility meters, smart breakers, heavy loads (EV chargers, heat pumps, water heaters), and connected appliances. Typically per-circuit or per-device granularity.
User interface	Largely automated, akin to a thermostat. Presents dashboards, analytics, and health/service notifications, so occupants and technicians can track energy flows, spot faults, and manually override as needed.
Forecasting	Using weather, solar output predictions, tariff schedules, and usage history to plan charging/discharging and load shifting.
Automation	Scheduling loads (washing machines, EV charging) for off-peak or high-solar periods, shed non-essential loads when needed; controlling battery/inverter set-points for optimal use or grid support.
Integration	Linking to the smart home systems, demand-response signals from utilities, or distributed energy resource (DER) aggregators. Inverter to track stored energy, and generation.
Equipment	Advanced systems include solar panels (often roof top), inverters, battery storage, and transfer switches.

An energy manager may be an embedded function of a smart inverter or whole-home energy controller.

110.6.1 Power monitoring

Whole-home or circuit-level monitors include smart panels, data from the inverter and utility meter, and current transformer clamps on mains and branch circuit. A *current transformer clamp* (CT clamp) is a device that "clips" around a conductor and measures the electric current flowing through; it does not make direct electrical contact or disconnect the circuit. These monitors show current flows in & out, and per-circuit consumption in near real time. This allows:

- Detecting abnormal draw (stuck relay, failing compressor, phantom load) before it trips a breaker or inflates a bill

- Verifying that load-shedding automations actually reduced demand during a battery or generator event
- Correlating HVAC, water heater, or EV behavior with total load for troubleshooting
- Recording event history: grid loss, transfer to backup, generator start, return to utility

110.6.2 Buying and storing power off-peak

Home energy management may shift seamlessly between grid, battery, and generator sources according to policy, status, and pricing. Energy can bought during cheap periods, delaying activities (such as washing dishes, laundry, heat storage), or storing power in compatible batteries. The stored power can be used during expensive peak times. Some grid operators offer variable-rate or time-of-use (TOU) pricing, where electricity costs less during certain hours (e.g. overnight or off-peak). Homes with compatible batteries can buy and store energy during these cheap periods, then use stored power during expensive peak times, reducing bills.

The automations and policies for time-of-use planning include:

- Charging batteries only when grid prices are lowest, reserving capacity to absorb solar surplus later.
- Scheduling tasks like EV charging, water heating, or laundry to run during off-peak windows.
- Exporting surplus during peak tariff hours if net metering or grid-supporting programs are available.

In some areas, utility programs or energy markets give homeowners the option to sell stored or generated power to the grid. This often occurs during peak demand periods, when the payout rate is higher, allowing homeowners to benefit financially.

110.6.3 Battery storage and inverters

Home energy storage usually involves three main parts: inverters, batteries, and switches that control where the power comes from.

An *inverter* takes DC (direct current) from solar panels or power stored in batteries and converts it into AC (alternating current) for household loads or export to the grid. There are several setup configurations:

- Grid-tied only works when connected to the main utility grid. It doesn't provide backup during outage unless paired with storage batteries and the logic to switch over.
- Hybrid or off-grid systems can run when the grid is down, by using stored battery power.

For keeping things working smoothly, it’s important to know if the inverter is

- Connected to the grid, or running separately (“islanded”),
- Limited on how much power it can supply
- Showing any fault and warning codes
- Safely communicating with other devices: communication link status, and last successful telemetry

During a grid outage, anti-islanding requirements mean grid-tied inverters automatically shut down (for safety-reasons). If the system is designed for backup, it must coordinate inverter

behavior with transfer switches and generator interlocks to disconnect from the grid. Note: power-transfer logic should not depend on cloud connectivity (although it may employ it).

Home *batteries* store extra solar energy, provide backup during outages, and can reduce peak import from the grid. Monitor state of charge (SOC), charge and discharge power, temperature, and state of health (if possible).

Define a minimum charge reserve for emergencies, based on expected outage duration. When SOC falls below a reserve threshold, shed non-essential loads (entertainment, hot tub or pool pumps, discretionary HVAC) to preserve power for important uses, such as refrigeration, medical equipment, lighting, and communications.

Battery chemistry and warranty terms often limit depth of discharge and cycle count. And extreme temperatures can shorten their life. Track charge cycles (how often they're charged) and temperature excursions. This information can support warranty claims and replacement planning.

110.6.4 Solar

Rooftop solar panels work with inverters, batteries, and transfer switches. The vendors often provided homeowners and contractors access to real-time data on solar production, battery status, and grid connectivity.

Example vendors: SolarEdge, Enphase, Tesla Powerwall, or Modbus-connected hybrid inverters.

EXPECTED VARIATION. Photovoltaic output follows predictable daily and seasonal patterns: production ramps after sunrise, peaks near local solar noon, and falls to zero after sunset. Cloud cover, shading, and panel soiling add day-to-day variation. Automations and policy should treat these rhythms as normal rather than as faults – for example, scheduling high-load tasks (EV charging, pool pumps, laundry) when historical production or forecast sun is adequate. It should avoid alerts that fire every evening when output drops to zero. Seasonal change matters too: a winter afternoon may deliver a fraction of summer peak even on a clear day. Trend production over weeks to distinguish gradual degradation (dirty panels, failed string, inverter derating) from expected seasonal decline.

CLEANING. Most rooftops do not need frequent cleaning except in especially dirty or dry areas. Dust, pollen, and bird droppings can reduce panel output, especially in dry climates. Water is used periodically to clean panel surfaces. Use scheduled reminders to clean, and to prompt users about seasonal tilt changes, and trigger washing based on local conditions (e.g. after a dust storm, or when output drops unusually).

SEASONAL ADJUSTMENT OF SOLAR PANEL ORIENTATION. The orientation or tilt angle of rooftop panels should be adjusted as the seasons change. Although many home systems are fixed (not adjustable), some have adjustable racks. Automated tracking systems are rare in residential settings due to cost and complexity.

In the northern hemisphere, steeper panel angles capture more low winter sun, while a shallower tilt performs better during summer months when the sun is higher in the sky. Panels can be repositioned twice a year , in spring & fall (or quarterly), using close to the location's latitude in winter and around 15° less in summer.

110.6.5 Off-grid operation

When physical grid connection is lost or intentionally disconnected (e.g., for resilience, during blackouts, or by design in remote settings), the home must strictly manage generation (solar, generator), storage (batteries), and loads. Critical automations:

- Prioritize loads, and temporarily switching off or throttling non-essential loads when grid power is lost or battery is low.
- Prioritizing refrigeration, communications, medical equipment, and lighting.
- Notifying users when manual intervention is needed (e.g., fuel for generator, reducing consumption, or reconnecting grid service).

Real-time monitoring of battery state, generation, and forecast consumption is vital. Off-grid systems typically require more granular control than grid-tied homes, with fail-safes to avoid deep battery discharge or overload.

110.7. FLOOR CARE

Robot vacuums and mops provide hands-off floor cleaning. Most robots can be programmed to start cleaning at specific times or days, enabling unattended, routine operation without user intervention. Cleaning routines can react to occupancy, events, or even energy pricing

Higher-end models allow real-time monitoring, custom commands, and deep integration into complex automation routines.

110.8. LAWN AND GARDEN CARE

110.8.1 Irrigation

Lawn and garden irrigation has often been automated with timers to initiate watering, and sensors to monitor precipitation and control irrigation systems, and prevent overwatering.

Sprinkler sensors may measure soil moisture levels or detect whether sprinkler heads are functioning properly (e.g., via flow or pressure sensors). Rain sensors typically detect rainfall and can signal the irrigation controller to skip or pause watering cycles when sufficient rain is detected, conserving water and reducing costs.

Both typically are directly connected to irrigation controllers, functioning independently of the home's main automation or monitoring system. Advanced setups may integrate this data for smarter irrigation, leak detection, or water usage tracking. The home automation may provide weather-aware scheduling.

110.8.2 Lawn mowing

Autonomous lawn mowing robots handle regular grass cutting for the yard, reducing manual effort for homeowners. These define their mowing area using boundary wires or smart mapping. Sensors to detect obstacles or pets, and programmable schedules for unattended operation. They return to a charging dock automatically when low on battery. More advanced systems may allow weather-aware scheduling, or multiple zone management.

These are standalone products, and at a minimum can be monitored remotely through a mobile app or automation dashboard. They often can integrate with the home automation system, providing the mower's status, activity log, and fault alerts (such as being stuck or requiring maintenance)

110.9. POLICY

For policy and response design, warn on rising temperature or humidity outside expected bands, shed non-essential loads when on battery backup, isolate a failing circuit or zone, and escalate when a life-safety or HVAC-critical device reports fault or goes offline.

Only local-only automations can continue when internet service fails. Policies should be designed accordingly. Heating limits and power-transfer logic should not depend on cloud connectivity.

110.10. SERVICEABILITY

A homeowner or technician can inspect and repair most home equipment directly. Consumables, such as air and water filters should be accessible to regular replacement or cleaning. Typically these are serviced based on a schedule, but may also be enhanced by tracking usage. Sensors may be available to detect clogs, trigger fan speed changes and/or send alerts for required maintenance.

Home automation systems can assist. They can provide reminders and alerts arrive through phone apps, dashboards, or device LEDs. They should provide diagnostic information – such as device online/offline status, connectivity, firmware version, and error/fault reporting – and support troubleshooting and ongoing maintenance. Ideal they're able to apply device firmware updates.

Note: even with automation systems, maintenance remains informal: there isn't a CMMS, formal work-order queue, or dedicated operations staff.

The equipment will need replacement. The equipment is built to a cost, and lower durability. The vendors in this space are volatile; sensors, hubs, and appliances may outlive their cloud support or firmware update path.

110.11. COMMUNITY

Home automation has hobby groups and community form around it. Being largely a hobbyist, cost-sensitive market with unreliable equipment, has increased the need for community, and the uses of work arounds. Only narrow business areas have had the tech specialists to support customers.

111. FACILITIES MANAGEMENT

Facilities management combines building systems, maintenance workflows, and emergency procedures into a coordinated operating model. This integrates:

- *Building Automation Systems* (BAS) to monitor and control HVAC, electrical distribution, and plumbing infrastructure. In response to gas leaks or fires it automatically regulates HVAC, pressurizes stairwells for smoke control, and triggers alarms.
- Life-safety systems such as fire detection, sprinkler networks, fire suppressants (e.g., FM-200 in data centers), emergency lighting, and smoke evacuation systems.
- Alarm and Notification: team manage the full range of emergency situations from fire, flood & smoke

- Access control and security systems, including badging, surveillance and automated lockdown systems, integrated with local police and fire dispatch.
- *Computerized Maintenance Management Systems* (CMMS) track facility assets, process work orders, and schedule preventive maintenance to prevent structural and mechanical failures. Example: FMX.
- *Building Information Modeling* (BIM) and as-built documentation represent a building's structure, piping, and electrical lines. Primarily used for maintenance, it can be used by emergency response teams to locate shutoff valves and structural supports quickly.

111.1. BUILDING AUTOMATION SYSTEM

Building Automation Systems (BAS) orchestrate large building core infrastructure: heating, ventilation, air conditioning (HVAC), lighting, security, energy management, and fire safety systems. These systems can range from simple programmable thermostats or standalone alarm panels, up to advanced, building-wide *supervisory control and data acquisition* (SCADA) systems.

111.1.1 Architecture: Supervisors, Control and Protocols

Building management systems coordinate operations across extensive, multi-zone facilities, connecting a wide range of equipment from many vendors. *Supervisory Control and Data Acquisition* (SCADA) systems are advanced industrial automation systems that may be used for comprehensive monitoring and control of facility infrastructure, esp. utilities, large buildings, manufacturing, and critical environments.

The system monitors indoor climate, equipment status, occupancy, and energy usage, providing both automated and user-directed control. The supervisor, a high level centralized control layer, continuously collects real-time data from distributed sensors and controllers, such as *Programmable Logic Controllers* (PLCs) and remote terminal units. It may also provide remote diagnostics, cloud-based monitoring, automatic event logging, compliance tracking & enforcement, and machine learning to enhance operation.

Table 135: *Building automation system elements*

System Element	Description
Sensing	Temperature, humidity, CO_2, smoke, light levels, and occupancy. These sensors provide real-time environmental data to automation controllers.
Controllers and Actuators	Specialized control units execute automation logic, sending commands to actuators (such as HVAC dampers, relays, valves, or lighting circuits) to adjust device operation according to schedules or dynamic policy. These are proprietary, durable and environmentally sealed.
Scheduling and Zoning	Building spaces can be divided into zones, each managed according to time-of-day, occupancy, or priority. For example, office lighting and cooling may ramp down automatically after working hours or server room cooling may be escalated based on load.
Energy Optimization	Historic trends, weather forecasts, and occupancy predictions used to improve energy use, reduce operating costs and environmental impact. Demand-response and time-of-use pricing can be leveraged to shift loads automatically.
Fault Detection	Identifies abnormal patterns, failed equipment, or out-of-range conditions, and can

and Diagnostics	direct alarms or operator intervention before minor issues become major failures.

111.1.2 Monitoring and Reporting

Building automation systems support centralized facility-wide oversight, to visualize conditions, issue manual or automated commands. This displays status, alarms, and trends, with automated responses and allows operators to issue commands and the capability for emergency overrides. This high level supervision is used to integrated safety protocols, automating complex operations, and supporting both immediate response and long-term performance.

111.1.3 Alarming, Notifications, and Overrides

A key function is the ability to detect abnormal, unsafe, or critical conditions, and to trigger automated responses or dispatch alerts to operators and first responders.

Alarms are centrally configured to trigger needed actions, such as local actions (flashing lights, audible sirens), automated building responses (closing valves, unlocking or locking doors, isolating HVAC dampers), and operator responses. An alarm event can be generated when a monitored value (e.g., temperature, smoke, water leak, unauthorized entry) crosses a threshold, deviates from set points, or when other conditions are met. These alarms are prioritized (e.g., critical, warning, informational) to ensure the most urgent situations receive immediate attention. See chapter 10, *Error and Service Notifications*.

111.1.4 Communication infrastructure

The communication between controllers, sensors, and actuators is often organized along multiple network buses, providing isolation and robustness in larger or complex facilities:

- The *primary bus* (aka trunk or backbone) connects major control panels, supervisory systems, and key subsystems. This bus serves as the main data pathway for central coordination and management.
- *Secondary buses* (or field bus) connect thermostats, lighting controllers, HVAC actuators and other devices within a particular zone or floor.

By segmenting communication in this way, systems gain improved reliability: if one secondary bus experiences a fault, only that local segment is affected, while the primary bus remains operational.

The communication protocols provide robust support for real-time automation, prioritization, and emergency response in building systems. These protocols include BACnet, KNX, or LonWorks. Each uses different data models and signaling methods.

BACnet

Building Automation and Control Network (BACnet) uses standardized "virtual object" types, such as analog inputs, binary outputs, schedules, and calendars, to represent device functions within the automation system. Each has a defined set of properties that describe the device's real-time state, its operational parameters, and the available control options. There are 16 message priority levels, enabling urgent commands (e.g., fire or safety alerts) to override routine communication. For example, a smoke detector transmits an explicit signal to indicate a smoke condition.

KNX

KNX uses a shared bus or via IP networks to communicate with lights, blinds, HVAC, security systems, and meters in commercial buildings. Each device stores its configuration and operates as part of a distributed, decentralized system. KNX supports programmable

logic for automation tasks, prioritization of messages, and local/manual overrides. It is primarily used in Europe.

LonWorks

LonWorks is based on a network of variables – one for each device input and output. Devices interact by updating these variables, and system logic is driven by bindings that trigger actions when variable states change, supporting both one-to-one and one-to-many relationships. The protocol gives precedence to the most recent command received on any variable. LonWorks reserves separate communication channels for critical signals (such as emergency overrides), to ensure prompt action. For instance, a fire alarm typically transmits a periodic "all clear" or "fire" signal; if the periodic signal is lost, the system defaults to an alarm state. LonWorks is legacy, with new installations preferring BACnet.

Other protocols may be employed for niche and proprietary uses, such as modbus to connect to meters.

111.2. POLICY

Example automation policies:

- Automatic ramp-down of HVAC and lighting in unoccupied areas.
- Locking of exterior doors and activation of alarms on schedule, or when security systems detect a threat.
- Humidity regulation in museums or archives to preserve artifacts.
- Smoke detection automatically closing dampers and shutting down air handling units to prevent fire spread.

111.3. SERVICE

Commercial, industrial, and institutional building facilities can deploy trained teams to inspect, isolate, and repair affected areas. They employ equipment allowing personnel to detect, analyze, and handle situations, with goals of:

- Protecting people
- Limiting equipment damage
- Restoring normal operation in a controlled way

111.4. HOSPITALS

Hospitals include additional resilience capabilities:

- Emergency power architecture that automatically restore powers to critical care units using backup generation.
- Layered communications using hardwired and satellite communication arrays.
- Isolation-capable HVAC zoning, with negative-pressure airflow zones and filtered airflow pathways isolate and contain airborne pathogens or chemical spills.
- Secure storage tanks for medical-grade oxygen, vacuum pressure, and potable water, for extended utility disruption.

- Mass Casualty Triage Spaces – special spaces (e.g. external inflatable field tents) to process large influxes of patients while keeping the main emergency room from getting overwhelmed.
- Security procedures with lockdown zones to control access and protect staff and patients during violent events.

Emergency and damage control depends on whether human repair crews can access the system during the incident.

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Part VI

Appendices

- References and Resources. This provides further reading and referenced documents.
- Glossary. The abbreviations, acronyms, initialisms, terms and phrases used, and gloss of their meaning.
- After Market Support. How to keep servicing and repairing when the original company has gone away?
- 2001. A visual of the idealized data center
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[This page is intentionally left blank for purposes of double-sided printing]

References & Resources

Note: most references appear in the margins, significant references will appear at the end of their respective chapter.

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113.2. MONITORING AND ALERTING

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An introduction to *Google's Four Golden Signals* of what to monitor for system health.

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Appendix A

Abbreviations, Acronyms, Glossary

Table 136: Common acronyms, abbreviations, and initialisms

Abbreviation / Acronym	Phrase
ADAS	advanced driver assistance systems
AFR	air-fuel ratio
AOC	active optical cable
API	application program interface
BER	bit error rate
BIOS	basic input-output system
BLE	Bluetooth low energy
BMS	battery management system
CAN	controller area network
CAN FD	Flexible Data-rate
CAPA	corrective and preventive actions
CMMS	computerized maintenance management system
CRC	cyclic redundancy check
CRL	certificate revocation list
CSRK	connection signature resolving keys
DCIM	data center infrastructure management
DEK	data-encryption keys
DER	distributed energy resource
DFT	design for test
DHR	device history file
DMR	device master record
DoD	depth of discharge
DoD	denial of diagnostic
DPU	data processing unit

DTC	diagnostic trouble code
DVM	digital volt meter
ECC	error correction code
ECO	engineering change order
ECU	engine control unit
EDS	energy dispersive X-ray spectroscopy
EMS	engine management system
EEPROM	electrical-erasable programmable read-only memory
eMMC	embedded multimedia card
EOBD	European on-board diagnostics
ESD	electrostatic discharge
FFT	fast Fourier transform
FMEA	failure mode effects analysis
FRU	field replaceable unit
FTA	fault tree analysis
FTIR	Fourier-transform infrared spectroscopy
HEMS	home energy management system
IC	integrated circuit
IFU	instructions for use
IPMI	intelligent platform management interface
IPSec	internet protocol security
IRK	identify-resolving key
JEDEC	joint electron device engineering council
JSON	javascript object notation
LIN	local interconnect network
LiPo	lithium-polymer
LTK	long-term key
MIB	management information base
MIL	malfunction indicator lamp
NCSI	network channel side interface
NFC	near-field communication
NIC	network interface card
NVRAM	non-volatile RAM
OBD-II	on-board diagnostics (2nd generation)
OCSP	online certificate status protocol
OTA	over the air update
OTDR	Optical time domain reflectometry
PER	packet error rate

PGN	parameter group number
PHI	protected health information
PID	proportional-integral-derivative
PID	parameter identifier
PII	personally identifiable information
PKI	public key infrastructure
PLC	programmable logic controller
PMBus	power management bus
PPE	personal protective equipment
PSU	power supply unit
RAM	random access memory; aka data memory
RF	radio frequency
RMA	return merchandise authorization
ROM	read only memory
RPA	return product analysis
RTC	real-time clock
RUL	remaining useful life
SATA	serial AT attachment (ATA)
SBOM	software bill of materials
SCADA	supervisory control and data acquisition
SDI	serial digital interface
SFP	small form factor pluggable
SMBus	system management bus
SOC	state of charge
SoH	state of health
SOM	system on module
SOP	standard operating procedure
SPN	suspect parameter number
SSD	solid state disc
TCU	transmission control unit
TDR	time domain reflectometry
TLS	transport layer security
TPM	Trusted platform module
UDI	unique device identifier
UDS	unified diagnostic services
UFS	universal flash storage
UPS	uninterruptible power supply
USB	universal serial bus

VCU	vehicle control unit
VPN	virtual private network

Table 137: *Glossary of common terms and phrases*

Phrase	Description
abstract machine	A theoretical, a theoretical, high-level model of a computer system and its capabilities. *See also virtual machine*
accessory	An item – e.g., case, cable, mount, or adapter – that works with a main product to add function, protection, or convenience.
advanced driver assistance systems (ADAS)	A system in vehicles that use sensors, cameras, and software to enhance driver safety and automate certain driving functions.
adapter	A circuit board driven by software. See also *driver*.
anomaly detection	Identification of behavior that deviates from the normal pattern, indicating potential faults, issues, or emerging failures.
anonymous page	A memory page allocated for process data (not backed by a file) whose contents exist only in RAM or paging store.
anti-rollback protections	Security features to prevent a system (or device) from reverting to an older version of health or counter values (akin to preventing odometer rollback) or software. Older versions are potentially less secure or non-compliant.
audit log	A secure, chronological record of events, actions, or changes within a system; often used to verify compliance and trace activity for review or investigation.
authentication	A process of verifying that a device is manufactured by an authorized party.
bad block management	Identifies and marks unusable blocks in flash memory to prevent data corruption.
baseboard management controller (BMC)	A dedicated management controller, independent of the host processor, and main power state; allows remote system health monitoring, event logging, power control, and recovery often via *intelligent platform management interface (IPMI).*
baseline	A reference level for a metric.
battery authentication	A process to verify the authenticity of the battery to ensure it meets the required specifications and standards.
battery connected	The battery was previously not connected and then reconnected.
battery disconnected	The battery is physically removed from chassis by the user, or the firmware cannot detect the presence of a battery.
battery life	See *battery service life*
battery management	The monitoring and control of a battery's charging, discharging, temperature, and protection to prolong its service life and improve performance.
battery service life	Duration for which a battery can operate before needing a recharge or replacement.
battery protection	Mechanisms and protocols to safeguard the battery from conditions that could cause damage, such as overcharging, deep discharging, or short circuits.
battery temperature not OK	The battery temperature is outside the charging range (for example, 0°C to 45°C) or discharging range (for example, 0°C to 60°C).
battery temperature OK	The battery temperature is within the acceptable ranges for charging and discharging.
battery under voltage	A condition of the battery, where its voltage is below a hardware-defined threshold.

block management	The process of encoding data to make it unreadable without the proper decryption key, ensuring data security and confidentiality.
body domain	Comfort and convenience electronics outside the power-train: doors, locks, windows, mirrors, seats, exterior lighting, wipers, keyless entry.
building automation system (BAS)	Integrated networks of hardware and software that monitor and control building facilities such as HVAC, lighting, security, and energy systems to improve comfort, safety, and efficiency.
building information management (BIM)	Creating, managing, and sharing information about a building throughout its lifecycle to support planning, design, construction, and operation.
calibration fixtures	Specialized tools or devices used to set, adjust, or verify the accuracy of equipment during calibration procedures.
captive fastener	A screw, bolt, or other fastener designed to be permanently attached to an assembly so it cannot be fully removed or lost when loosened. See also *fastener.*
captive screw	See *captive fastener*
change point detection	The automatic identification of significant shifts or jumps in monitored variables, indicating a possible onset of a fault or a new operating condition.
chassis intrusion	A feature that detects opening of a chassis. Unauthorized opening or tampering might be logged for security reasons.
cockpit domain	Driver and passenger interface: instrument cluster, head-up display, center screen, infotainment, HMI. Displays status and takes driver input.
computerized maintenance management system	Tools to support maintenance operations – managing assets, preventive schedules, work orders, spare parts, technician assignments, and compliance records.
condition-based maintenance	Maintenance actions triggered by real-time monitoring of equipment condition or performance, rather than by a preset schedule.
constant charge current	The steady current supplied to the battery during the main phase of the charging process until the battery reaches its charge target voltage.
consumables	Items that are gradually depleted or fouled during normal operation and replaced periodically based on usage or condition. For example: filters, tubing, cartridges. See also *maintenance items*, *single-use item*.
control authority	The ability of a control system to directly affect and maintain the desired process variable (e.g. temperature, pressure, flow) over its full required range, even in the presence of disturbances or system limitations.
control block	A data structure holding all relevant information about a managed item, such as a process, task, file, communication channel, hardware or other resource; or an item doing work or work to be performed.
controller	A device or software component that monitors and directs system or subsystem operation by processing inputs and issuing commands to achieve desired outcomes.
controller area network (CAN)	A robust vehicle bus standard designed to allow microcontrollers and devices to communicate with each other within a vehicle without a host computer. Widely used for in-vehicle networking due to its reliability and real-time capabilities.
crash reporting	Automatic capture and transmission of diagnostic information when a failure (such as a crash) occurs.
critical field replaceable units	Field-replaceable units whose failure or misconfiguration can directly impact safety or compliance; to address these risks they may require authenticity checks, calibration, and post-replacement verification.
current transformer clamp	A device used to measure electric current flowing through a wire without needing to make direct electrical contact or disconnect the circuit.
customer care	Services to support the user when encountering questions and problems.

data center infrastructure management	Tools to monitor and manage data center physical infrastructure – power, cooling, racks, cabling, space/capacity, and environmental sensors – with dashboards, alerts, and capacity planning to keep uptime and efficiency high. See also *computerized maintenance management system.*
data processing unit (DPU)	A specialized processor or adapter designed to offload networking, security, or storage tasks from the main CPU.
data transfer	The exchange of data between a device and external sources via a USB connection.
diagnostic	A "process by which hardware malfunctions may be detected" [IEEE 2000]
diagnostic trouble code (DTC)	Codes stored by the ECU that indicate specific faults in vehicle systems, used for diagnostics and troubleshooting.
disassembly	"process whereby a product is taken apart in such a way that it could subsequently be reassembled and made operational." [EN45554:2020]
disassembly depth	"the number of steps required to remove a part from a product, without damaging the product." [JRC128672] *See also step*
distributed energy resource (DER)	A small-scale unit of power generation, storage, or load management – such as rooftop solar panels, home batteries, or controllable loads – connected to the electric grid or operating locally to provide energy services.
domain	A functional grouping of related vehicle electronics by what they do (power-train, chassis, ADAS, body, cockpit), not by wiring location. See also *zone*.
driver	Software items that bridge with hardware, such as an adapter, to allow the software and hardware devices to function together correctly. See also *adapter.*
eAxle	*electronic axle*. An integrated drive module combining an electric motor, power electronics, and a gearbox (optional), to directly drive a vehicle's wheels.
engine control unit (ECU)	Manages an engine's operation, such as fuel injection, ignition timing, and emissions control.
engine controller	Overseeing the operation and performance of the engine by monitoring sensors and actuating controls to ensure optimal function.
engine management system	An integrated system of sensors, actuators, and control algorithms that monitors and regulates key engine functions (such as fuel injection, ignition timing, and emissions control) for performance, efficiency, and compliance.
entitlement	An entitlement is a family of features or resources that the program or owner is allowed to use.
environmental histograms	Graphical representation displaying the frequency or distribution of environmental conditions (e.g. temperature or humidity) experienced by a (sub)system over time.
error correction code (ECC)	A method of detecting and correcting errors in data to ensure data integrity.
exportable service bundle	A packaged set of diagnostic and operational data (logs, health metrics, fault codes, configuration snapshots, and related metadata) that can be exported from a device and shared with support, field service, or return-product analysis without requiring physical access to the unit.
fastener	"A hardware device that mechanically or magnetically connects or fixes two or more objects, parts or pieces. A fastener is generally non-permanent, i.e., it can be easily removed or disassembled without damaging the objects, parts or pieces connected or fixed together (e.g., screws or clips). Welds and some glues are, in contrast, permanent fixings." [Source: French Score, 2020]. See also *captive fastener.*
fault	Any deviation from the normal or expected operation of a unit or system.

fault finding	The process of identifying and diagnosing issues or malfunctions.
feature flags aka feature toggle	A setting that enables and disables features.
field replaceable unit (FRU)	A modular assembly designed for quick swap by users or a service representative.
field service playbook	A comprehensive guide or manual outlining practices, procedures, and troubleshooting steps for performing service and maintenance in the field.
fixed guard	A "guard affixed in such a manner (for example, by screws, nuts, and welding) that it can only be opened or removed by the use of tools or by destruction of the means by which the guard is affixed" [ISO 12100:2010, 3.27.1] See also *guard, moveable guard*
flash storage	A type of non-volatile memory that can be electronically erased and reprogrammed.
frame	See *page frame*.
freeze-frame data	A snapshot of key parameters and measurements captured at the moment a trouble code is set.
fuel gauge	A circuit or algorithm in a battery management system that estimates the remaining charge, and usable capacity of a battery.
full reset	A process that restores a (sub)system to its original factory settings, erasing all user data and configurations. See also *reset*
functional testing	Testing functions of the product to identify where it fails.
garbage collection	A process internal to SSDs to reclaim blocks by moving valid data and erasing old blocks,
guard	A "physical barrier, designed as part of the machine, to provide protection" [ISO 12100:2010, 3.27] See also *fixed guard, moveable guard*
guided diagnostics	**A** structured troubleshooting process where a service tool or procedure leads a person through step-by-step checks based on symptoms, codes, and measured results.
guided replacement workflow	A step-by-step, often interactive, process designed to assist users in correctly replacing a component or device.
histogram	Counts of values within given range bands.
home energy management System (HEMS)	A system that monitors, and controls home energy generation, storage, and consumption within a home, often integrating solar panels, batteries, utility meters, and smart appliances.
hot swap	The ability to dynamically replacing an software module with a new one, without service interruption.
installable module	A module that can be added to an operating system without core system relinking, typically providing additional functionality such as device drivers, file systems, or utilities. Not necessarily loadable.
instructions for use (IFU)	A document or manual that provides detailed information on the proper handling, operation, and safety of a medical device or product.
intelligent platform management interface	A standardized interface used for remote monitoring and management computer systems, using a dedicated management controller. *See baseboard management controller*.
interlock	A mechanism or logical step to ensure proper setup, and that the equipment (or a subsystem) can operate safely. [IEC 60204-1:2016, 3.1.37]
inventory ID	A unique identity for a physical medium (barcode, RFID/Tag, etched label, or

	similar).
labelling	Information provided on or with a product, including instructions, warnings, identification, and regulatory details, to ensure proper use and compliance.
Li-ion	Lithium-ion
link flapping	When a network or interconnect link that rapidly alternates between connected (up) and disconnected (down) states, causing instability and interruptions.
loadable module	An *installable module* that can be dynamically loaded into or removed from an operating system at runtime.
low state of charge	Battery raw state of charge is too low to support intended device operation.
maintenance items	Parts with finite service life that are replaced on schedule or condition, often tracked with health indicators (e.g., RPM, state of health). For example, fans, batteries. See also *consumables*.
management information	Data and insights collected and organized to support decision-making, monitoring, and control within an organization or system.
management information base (MIB)	A structured database or collection of information used for managing the components and operations of a network or system.
manager	An individual or system responsible for overseeing, organizing, and directing operations, resources, or personnel to achieve defined objectives.
media server	A device or software that stores, organizes, and streams digital media (e.g. music, videos, and photos) on a home network.
memory pressure	A condition in which available physical memory is low or heavily utilized.
mode	A mode is a categorization of how the system operates: its behaviors and functions. It is often a user-selected setting (e.g., automatic mode, manual mode, power-save mode).
monitoring	The ability to identify if a system malfunctioning; typically uses observability to track measures, and counts, applying thresholds to classify a subsystem as working, impaired, faulty, etc.
movable guard	A “guard which can be opened without the use of tools” [ISO 12100:2010, 3.27.2] See also *fixed guard, guard*
network video recorder (NVR)	A device that records, stores, and manages video from IP-based security cameras over a network, providing centralized surveillance footage storage and access.
on-board diagnostics (OBD-II)	A standardized automotive system that monitors, records, and reports vehicle health, performance, and emissions-related data via a diagnostic port.
observability	The ability to understand what’s happening inside a system, to administer, improve it (e.g. debug) or justify its behaviour.
operational profile	A quantitative characterization of system use and tasks performed by each element or subsystem often starts with user perspective and progressively narrows down.
operations center	A centralized facility/team to coordinate multiple teams.
over current	Power output exceeds a hardware-defined threshold for a specified duration.
page frame	A fixed-size block of physical memory used by a virtual memory system to hold a page.
parameter identifier (PID)	Standardized codes used in OBD-II/EOBD protocols to request data from a vehicle's *electronic control unit* (ECU), such as engine speed or temperature.
part identifier	Any label, number, or code used to distinguish and track a particular part among others; typically used to ensure precise identification within inventory, manufacturing, or procurement systems.

personalization	Tailoring the system to the user's settings and preferences.
planned maintenance	Scheduled service activities performed at regular intervals to prevent equipment failure and ensure optimal performance. A type of *preventative maintenance.*
policy	"Statements, rules or assertions that specify the correct or expected behavior of an entity." [GFD-I.044]
power delivery	The management and negotiation of power from an external USB or power supply to the device. This involves ensuring that the correct amount of power is delivered to the device safely and efficiently.
power management	The process of controlling the power consumption and usage of the system, including managing sleep and hibernation states.
power management bus (PMBus)	An I^2C-based communication protocol used for monitoring and controlling power supply devices in an electronic system.
power supervisor	A component (or microcontroller feature) to detect and stop the microcontroller if the power supply to voltage is too low (brownout). It serves as a risk control on faults in power supply. Also called *brownout detect* or *power monitor*.
power supply unit (PSU)	A device that converts mains AC power into regulated DC voltage to safely power the components of electronic equipment, such as computers and medical devices.
power train	All the vehicle parts that create power and send it to the wheels, such as the engine or motor, transmission, and driveshafts. In electric or hybrid vehicles, this also includes electric motors, inverters, and batteries.
pre-charge current	The initial current supplied to the battery at the beginning of the charging process, often used to safely charge a deeply discharged battery.
preventative maintenance	Routine service activities performed to prevent equipment failure and extend the lifespan of a product or system.
priming	Preparing a device, especially one involving fluid pathways, by filling it with fluid to remove air or contaminants to ensure proper and safe operation.
printhead	A component of a plotter, printer, or 3D printer (such as FDM material extruders) that deposits ink, toner, melted plastic filament, or other substances onto a surface or into layers to create text, images, or 3D objects.
programmable logic controller (PLC)	A ruggedized industrial computer used to automate monitoring, control, and sequencing of machinery and processes.
proportional set size (PSS)	The amount of shared and private memory apportioned to a process the size of shared pages divided by all processes using them.
protected health information (PHI)	Individually identifiable health information that relates to a person's health, care, or payment and is protected by law.
personally identifiable information (PII)	Any data that can identify a specific individual, directly or when combined with other data (e.g., name, address, email, device IDs, biometrics), governed by general privacy laws rather than HIPAA.
power supply unit (PSU)	A device that converts mains AC power into regulated DC voltage to safely power the components of electronic equipment, such as computers and medical devices.
priming	Preparing a device, especially one involving fluid pathways, by filling it with fluid to remove air or contaminants to ensure proper and safe operation.
protocol analyzer	specialized tool to detect communication errors, timing violations, and unexpected message patterns, as well as capture, decode, and display the data exchanged.
quick-start guide	A concise guide that provides essential instructions to help users set up and begin using a product rapidly.
recovery mode	A separate, independent operating system that some products can boot into for purposes of downloading software to replace a damaged partition.

refurbishment	The process of restoring a used or worn product, device, or component to a like-new condition through cleaning, repair, and replacement of necessary parts.
repair	The process of restoring a malfunctioning or damaged product or system to its proper working condition.
return product analysis	The structured, documented investigation of devices or components that are sent back from the field.
rolling log	A continuously updated record that automatically removes the oldest entries as new ones are added, maintaining a fixed size or time span.
safe hold	The safe state for a space craft, usually to protect the equipment.
safe state	A state that the equipment may be placed into where the relevant risks have been addressed to an acceptable risk index.
safing routine	A sequence of autonomous actions a spacecraft takes to protect itself when it detects a major problem. In munitions, safing mechanisms prevent accidental detonation.
scheduled maintenance	Maintenance tasks that are planned and carried out at predetermined intervals, regardless of equipment condition.
scrubbing	Scanning storage media, performing integrity checks (e.g. reads and checks CRC) of structures/memory, recovery and updating health information.
secure element	A tamper-resistant hardware component designed to securely store sensitive data (e.g. cryptographic keys, biometric data, and credentials) and perform cryptographic operations.
self-protect	Behavior safeguards equipment and users when operating limits are approached or exceeded.
serviceability	The ease and efficiency with which a product or system can be maintained, repaired, or serviced to ensure optimal performance and longevity.
service-only	Components that are replaceable only by qualified technicians following documented procedures due to safety, calibration, or access constraints.
service procedures	Standardized steps or instructions for performing maintenance, repairs, or other support activities on a product or system.
service training	Structured programs or sessions designed to teach users or technicians the knowledge and skills necessary to operate, maintain, or service the product.
service windows	Designated time periods during which maintenance, upgrades, or repairs can be performed on a (sub)system with minimal disruption to operations.
single-use items	Components intended for one-time use or a single patient interaction that must be discarded after use and are not refurbished or reused.
storage management	The processes and policies to ensure data is available, secure and cost-effective; it allocates resources, protection, manages organization, and monitors the conditions.
supervisor	An individual or system that oversees the activities, safety, and performance of others to ensure tasks are carried out correctly and efficiently.
supervisory control and data acquisition (SCADA)	A type of industrial automation system for remotely monitoring, controlling, and collecting data from facility equipment and infrastructure.
system management bus (SMBus)	An I^2C-based communication protocol for low-speed system management communications between devices within a single PCBA or chassis.
spare-parts governance	The policies and processes for managing, controlling, and tracking spare parts to ensure availability, compliance, and effective usage.
step	“A step consists of an operation that finishes with the removal of a part (or bundle), and/or with a change of tool.” [JRC128672]
storage management	The processes and policies to ensure data is available, secure and cost-effective; it

	allocates resources, protection, manages organization, and monitors the conditions.
thermal throttling	Reducing performance (e.g. clock speed) and power consumption.
tier	A distinct level used to classify resources, services, or data by their characteristics or importance.
tiering	The process of organizing resources, or services at different levels (tiers) based on criteria like performance, or cost.
toolhead	The part of a machine that holds and manipulates the tool performing work on the *workpiece* such as cutting, printing, or engraving. See also *printhead*
transmission control unit (TCU)	An electronic module that manages and controls the operation of a vehicle's automatic transmission.
trending	The process of observing a metric over time to determine its direction and rate of change.
triggered maintenance	Maintenance actions initiated in response to specific events, conditions, or indicators signaling that service is needed.
unique device identifier (UDI)	A distinct code assigned to a device that enables its identification, tracking, and traceability throughout its distribution and use.
universal flash storage	A storage specification for use in smartphones, tablets, and other portable electronics. Faster, more robust and uses less power than eMMC.
usable capacity	A measure of the energy storage capability of the battery that is actually available for use.
usage tracking	The monitoring and recording of how often and in what manner a product, system, or service is utilized.
USB power applied	USB cable plugged in with a voltage of 4.75V to 5.2V. No support for power supply with higher voltage.
user-replaceable	Parts intended to be safely swapped by end users without special tools or training, guided by the instructions and without opening sealed enclosures.
user-service kit	A set of tools, parts, or materials supplied to end users to enable them to perform routine maintenance or repairs on a product or device.
vehicle control unit	An electronic module to manage specific vehicle subsystems or functions such as power train, braking, or steering.
wear leveling	A technique used to evenly distribute write and erase cycles across the memory blocks of flash memory, extending its lifespan.
witness mark	A mark or indicator applied to a part during assembly or service to show a component was properly installed, or to indicate tampering or movement later.
workpiece	A raw material or part that is being machined, processed, or assembled into a finished product.

APPENDIX B

After Market Support

How to keep servicing & repairing a companion robot? What to do when the company has gone away?

- Sourcing parts
- The range of features and customer services available in products for serviceability
- Dividing up the features and services up by different product tiers

3rd parties offering to repair and create service information

114. SOURCING PARTS

Tables of parts and where to get parts. This should prioritize consumables, parts that wear out or tend to break.

Parts are easier when they are standard; or when the volume of the original product was sufficient that others made excess parts or made will-fit/work alight parts.

Often parts that started as standard but had a modification: perhaps shortened or lengthened cable to fit, or alternate connector. Some cables are easier to modify, others (such as flex or complex cables) are not.

Finding batteries

Harder to source or synthesize unusual screws and other fasteners.

Scavenge scrap parts

115. SYNTHESIZING PARTS

Aftermarket parts, Creating will-fit parts

- Research reported information, diagrams from FCC (etc) reports, guides/manuals, patents, articles, blog posts, game assets, marketing, etc. Thus will require reconciling information that is contradictory, inaccurate, low resolution, etc.
- Reverse engineering, Scan form replacements

Make 3D prints by scanning parts to make will fit

Others can measure and create diagrams and alternate designs.

115.1. SCANNING PARTS

Handheld and portable 3D scanners have become practical tools for aftermarket repair. A worn clip, broken shell tab, or missing cosmetic cover can be scanned, cleaned up in mesh software, and printed or cast without access to original CAD. The workflow is:

1. Scan the part (or a mirror-image counterpart from a donor unit). Scanner choice depends on part size, surface finish, required accuracy, and budget. Shiny, black, or transparent surfaces often need scanning spray or adhesive markers; some newer hybrid scanners reduce or eliminate that prep.
2. Export STL/OBJ/PLY. Some mesh-to-CAD conversion tools include Geomagic, Fusion 360, Blender, etc.
3. Repair the mesh,
4. Optionally convert to solid CAD, then
5. Print or machine a will-fit replacement.

115.1.1 Entry / hobby tier

These products are suitable for small plastic brackets, clips, covers, and interior trim on companion robots and consumer products. Accuracy is often 0.05–0.1 mm, which is adequate for many will-fit mechanical parts when tolerances are not critical.

Creality Ferret Pro (wireless IR structured light, ~$330–$400). Compact, phone/PC workflow via CrealityScan. Good first scanner for small parts; 0.1 mm rated accuracy.

Revopoint POP 3 Plus / POP 4 (~$600–$900). Infrared VCSEL and/or blue-laser hybrid; higher accuracy than entry IR-only units; popular in maker and small-shop reverse engineering.

115.1.2 Mid-range tier

Hybrid laser + structured-light scanners with wireless operation, onboard displays, and marker-free modes on some surfaces.

Creality Pika (~$600–$700). Pocket form factor with 3-inch touchscreen. Dual-engine hybrid:

- 7-line parallel blue laser: up to 0.03 mm accuracy, marker-free on many objects; working distance ~160–400 mm.
- NIR structured light: up to 0.1 mm accuracy, eye-safe, 24-bit color, marker-free geometry/texture alignment; removable IR filter for outdoor use (≤80,000 lux).

Software: CrealityScan with AI re-texturing and body completion; exports STL, OBJ, PLY, ASC; Open SDK via Orbbec. Scan volume from ~10 mm cubes (laser) to 2 m objects.

Creality RaptorX (~$1,200+). Wireless hybrid blue-laser and NIR scanner; 34 cross blue laser lines, one-shot 3D imaging, scans black/metal without spray in many cases; larger volume and faster point rate than Pika; integrates with Geomagic Control X, Design X, and Verisurf for inspection workflows.

115.1.3 Professional tier

For shops doing repeated reverse engineering, first-article inspection, or parts with tight tolerances.

Revopoint MetroX Pro (~$1,200). Blue laser metrology tier.

Revopoint MetroY Pro (~$2,000+). ~0.02 mm blue-laser accuracy; higher scan speed.

EinScan Rigil (~$5,000). Tri-mode standalone (crossed blue laser, parallel laser, IR VCSEL); onboard computer and 6.4-inch touchscreen; up to ~0.04 mm accuracy; captures black, shiny,

and metallic surfaces without spray; common in automotive aftermarket and heritage digitization.

Revopoint Trackit (~$5,000). Marker-free optical tracking for large objects (10 mm–6 m); useful for scanning whole robot bases, furniture-scale fixtures, or vehicle-scale assemblies.

115.1.4 Industrial / metrology tier

ISO-traceable accuracy for aerospace, automotive quality control, and certified inspection – not usually needed for simple will-fit parts, but relevant when replacements must meet GD&T or regulatory documentation.

Shining3D FreeScan UE Pro2, Creaform HandySCAN BLACK Series, Artec Leo, ZEISS T-SCAN hawk 2. Pair with Geomagic, PolyWorks, or ZEISS INSPECT for certified pipelines.

115.1.5 Practical notes for aftermarket service:

0.1 mm accuracy is often sufficient for clips, covers, and cosmetic shells; aim for 0.03–0.05 mm when mating surfaces, bearing seats, or gear interfaces matter.

Scan both the broken part and a mating surface or donor assembly to verify fit before printing final parts.

Document scan settings, alignment mode (markers vs marker-free), and export scale in repair notes so others can reproduce the part.

Combine scanning with caliper measurement of critical dimensions; mesh models drift on thin walls and deep recesses.

For companion-robot parts, prioritize scanners with good performance on matte plastics and mixed-size objects in the 20–300 mm range; Pika and Ferret Pro cover much of that range at lower cost than industrial units.

115.2. 3D PRINTING PARTS

The norms for what can help a customer have changed as well. Some areas that have:

1. File to support 3D printing of parts and accessories (such as STL files, CAD files, or just solid diagrams).
2. Some open-source release

115.3. CASTING PARTS

Clear screens, lenses, covers are hard— notoriously, but advances and skills such as from automobile restoration have help.

116. CREATING DOCUMENTATION

Troubleshooting manual

- How to check for software problems
- How to update software

Repair manuals. E.g. with Vector

- How to replace battery

- How to clean wheels/sensors
- Replace treads

Ideally these would include diagram of assemblies, and where the parts fit in, such as using exploded diagrams. This is

There are often wiki, but these are rarely cohesive as an authored work: scattershot, incomplete,

Reduced customer care, services, community support. I'm not sure it was a great business

Greatly reduced documentation writing. They are not great for customers.

There are forums, but these have an ephemeral nature. Somehow the information needs to be collated and made concise and useful.

Places like iFixit do some commercial work, and support volunteers creating more, with style guidance on how to present information. This is good, but often for most mainstream.

APPENDIX C

2001

Act III of *2001 A Space Odyssey* from Kubrick presents the Odyssey and HAL 9000 as an idealized version of an automated professionalism, down to treating people as fallible, replaceable parts. Infamously, an idealized IBM data center.

The office space was hyper clean: The workers have no personal effects; the personal messages are received coldly, without reaction.

117. THE AUTOMATION

HAL 9000 was deliberated extrapolated from real IBM work and Apollo-era materials; the self-monitoring, fault prediction, and diagnostics were grounded in real mainframe concepts:

- Long-lived autonomous system
- Capable of self-assessment, diagnostics and degradation detection
- Redundant computing
- Telemetry / remote monitoring

Diagnostics

- Self-test for mission-critical systems
- Predictive maintenance models: Statistical failure forecasting, Mean Time Between Failures (MTBF) analysis
- Predictive failure indicators, with confidence levels, degrees of certainty, error classes
- Schedule preemptive replacement

It showed testing the board, signal probes and display of schematics and board layout.

118. RESOURCES AND REFERENCES

Jerome Agel, *The Making of 2001: A Space Odyssey*, 1970

Discusses Kubrick's consultations with IBM engineers. Shows how these features were based in mainframe concepts.

APPENDIX E

Vehicle Facility management

This appendix is a think piece on describing facility-oriented serviceability and emergency-response practices in maritime, aircraft, and spacecrafts.

- Maritime vessels operations, and damage control systems
- Aircraft maintenance processes and emergency operations
- Spacecraft emergency operations

119. OVERVIEW

Emergency and damage control depends on whether human repair crews can access the system during the incident. Building facilities and ships can deploy trained teams to inspect, isolate, and repair affected areas. Aircraft and spacecraft cannot and must rely more heavily on onboard automation, redundancy, and procedural recovery while in operation. They all employ equipment allowing personnel to detect, analyze, and handle situations, with the common goals of:

- Protecting people
- Preserving mission or service continuity
- Limiting equipment damage
- Restoring normal operation in a controlled way

120. MARITIME DAMAGE CONTROL

damage control system

Maritime damage control is the shipboard discipline for detecting hazards, containing spread, and restoring survivability after incidents such as fire, flooding, structural damage, or contamination. The process emphasizes rapid detection, clear command authority, and checklists that support distributed teams under time pressure. A *damage control system* (DCS) "is an information-retrieval and equipment-control system that allows ship personnel to detect, analyze, and handle various types of situations which are hazardous for shipboard safety." The elements include:

- Compartmentalization and systematic closure controls to contain the spread of hazards.
- Coordination and management of fire main systems (seawater supply for fire hydrants) and fire suppression equipment
- Flooding detection, dewatering operations, and stability assessment
- Sensors track smoke, temperature, flooding, and system health

- Incident playbooks and decision aids that define immediate and follow-on actions

120.1. MILITARY VESSELS

Military vessels employ *battle damage control systems*, which enable teams to address a broader range of emergencies not just fire, flooding, and smoke, but radiation, chemical hazards, and potential unexploded ordnance. These systems are typically fully integrated with the ship's overall *Integrated Platform Management System* (IPMS).

The primary role is incident management. Establish cooling and smoke boundaries, initiating automated smoke extraction, relocating combustible materials, and determine evacuation or response routes based on the current ship closure state. The system can even predict the likely path of a blast from unexploded ordnance to help minimize further damage.

Emergency checklists – called *compartment cards* or *kill cards* – guide sailors and damage control operators through the rapid mitigation, isolation, and recovery of combat or accident-related damage. Automated systems direct response actions – often remotely – while continuously monitoring the status of relevant sensors and systems. They are dynamic, triggering requests for specific actions and enforce timeouts on uncompleted tasks. They may provide direct links to operational interfaces and on-line procedural guidance.

Note: There is another, dog tag like item called an "kill card". Personnel carry "small laminated [cards that] contain essential information about the person carrying it, including blood type, and [an identifier] made up of the first initial of his Company, the first initial of his last name, and the last four digits of his Social Security number." They are to quickly identify casualties.

121. AIRCRAFT

Aircraft cannot use physical repair crews, or physical damage control parties to perform *patching and repairing* mid-flight. They rely on crew procedures, automated systems, structural redundancy (like backup flight controls), and onboard suppression to manage emergencies.

121.1. STRUCTURAL DAMAGE TOLERANT DESIGN

Aircraft designs reduce single-point failure risk through "damage-tolerant" structures, redundancy, inspection intervals.

- Fuselages and wings are designed with overlapping layers and crack-arresting features, so that a localized crack or tear does not compromise the entire wing or depressurize the cabin.
- Redundant Hydraulic & Electrical Systems: Vital systems are deeply segregated. Military jets often have two or three separate hydraulic systems running in different paths, so a single projectile cannot disable all of them at once.
- Self-Sealing Fuel Tanks: Lined with materials that expand or react when punctured, preventing jet fuel from leaking and catching fire. (Military)
- Armor and Ballistic Protection: Critical components (such as the cockpit and engine) are shielded by ceramic or composite armor plates to protect pilots and prevent catastrophic engine failure. (Military)

121.2. GROUND-MAINTENANCE DAMAGE PREVENTION

Aircraft damage prevention systems (ADPS) safeguard planes both in the air and on the ground. They combine strict debris management programs, automated ground-support hardware, and in-flight collision avoidance technology, to prevent catastrophic structural or engine failures.

A large portion of aircraft damage occurs during ground operations rather than flight. Ramp damage from baggage loaders, passenger stairs, and maintenance vehicles is a leading cause of aircraft repair costs. Improved ground handling can prevent these:

controlled approach

- *Ground support equipment* (GSE) coordinates the approach to an aircraft using proximity sensing (radar, ultrasonic sensors), and soft contact bumpers. When a vehicle approaches the aircraft, it slows automatically, and the bumpers stop and reverse the machine before it can impact the fuselage.
- Systems like ADPS constantly monitor aircraft movements (e.g., weight changes from fueling/loading) and adjusts connected passenger stairs or belt loaders.
- Interlocks prevent engine cowlings from being improperly latched or left open, avoiding aerodynamic failure or damage during takeoff.

Foreign Object Debris (FOD) is to any loose object that can damage aircraft components, particularly jet engines. Prevention:

- Airfield maintenance sweeps, vacuums, and uses magnetic sweeps on runways. It also involves physical nets and wind barriers to stop airborne debris from hitting aircraft.
- Automated optical and radar scanning systems are used on airport runways and ramps to spot stray debris or hardware before aircraft run over them.
- Maintenance areas mandate strict inventory logging (such as shadow-boards) to ensure no wrenches or hardware are left inside aircraft panels or on the tarmac.

121.3. IN-FLIGHT COLLISION AVOIDANCE

- Weather Radar & Wind Shear Detection scans for hazardous meteorological conditions, warning pilots to adjust routes and avoid extreme turbulence or downdrafts.
- *Traffic Collision Avoidance System* (TCAS) scans the surrounding airspace for other aircraft using transponder signals. If a collision risk exists, it issues alerts and suggests coordinated maneuvers (e.g., climb or descend) to both aircraft.
- *Ground Proximity Warning System* (GPWS) prevents *Controlled Flight Into Terrain* (CFIT) by alerting pilots if their aircraft is flying too close to the ground or an obstacle.
- *Automatic Ground Collision Avoidance System* (Auto-GCAS) is an autonomous safety where if the pilot becomes incapacitated or the plane is plummeting out of control, the system automatically takes over to pull the aircraft up and prevent a crash.

121.4. DAMAGE CONTROL

Aircraft focus on damage control such as *isolating failures* and keeping the aircraft airworthy using the following core systems:

- Checklists and crew resource management procedures to stabilize, isolate, and divert.

- Fire detection & suppression in engine nacelles, cargo compartments, and auxiliary power units (APUs). They often feature built-in sensors and fire extinguishers (such as Halon) that discharge automatically or manually by the crew to starve flames of oxygen.
- If a generator fails or wiring is damaged, the aircraft's electrical system automatically drops power to non-essential systems (like in-flight entertainment) to ensure the cockpit displays, navigation, and critical flight controls stay operational. *load shedding*
- Redundant Hydraulic & Electrical Systems: Vital systems are deeply segregated. If one hydraulic line is severed, backup systems (e.g., secondary pumps, Ram Air Turbines) deploy automatically to maintain control of the landing gear and flight surfaces.
- Redundant flight-control paths. Electronic networks route control inputs rather than fixed mechanical cables. If a control cable or hydraulic line is damaged, the commands are automatically routed to functioning control surfaces. (Fly-By-Wire)
- Adaptive flight controls automatically recalculate how to use the remaining intact surfaces to keep the aircraft flying if a flight control surface is damaged. They protect the flight envelope, actively preventing the pilot from making maneuvers that would overstress or break an already structurally compromised airframe. *See Automatic Ground Collision Avoidance system above*

122. SPACECRAFT

Like aircraft, spacecraft cannot use physical repair crews, relying instead on autonomous fault management and conservative design margins.

122.1. OPERATOR CONTROLS

A touch screen may be used for control involving data entry, navigation maps, or secondary system monitoring. But a toggle switch is likely used in high-stress or emergency environments, such as time-critical (needed in seconds), safety-critical (fire, oxygen), or must be operated during high-vibration or high-acceleration phases.

Physical switches are preferred in high-stress or emergency environments for several reasons: *embodiment*

- Tangible switches provide tactile feedback. Operators can locate and operate switches by feel alone, allowing them to keep their eyes on other instruments.
- Touch-screens are notoriously difficult to use when shaking violently or undergoing force; a tangible switch provides a solid object to find and register (clicks) into place.
- Mechanical switches are easily operated while wearing bulky, pressurized spacesuit gloves.
- The position of a toggle (up vs. down) provides direct visual confirmation.

122.2. STRUCTURAL DAMAGE TOLERANT DESIGN

Spacecraft designs reduce single-point failure risk through “damage-tolerant” structures, with conservative design margins and redundancy,

Rigorous engineering standards prevent hardware from failing due to cracks, flaws, or structural fatigue over a mission's lifespan. Engineers make structures “damage tolerant” by selecting materials and utilizing non-destructive testing before launch.

Micrometeoroid and Orbital Debris (MMOD) shielding includes active physical systems designed to break up or vaporize debris before it breaches pressurized habitable areas, such as the Whipple shield used on the International Space Station.

Redundant avionics and segregated power and control subsystems for graceful degradation.

122.3. MAINTENANCE AND DAMAGE PREVENTION

Risk-informed operations planning, and for long-duration missions, telemetry review, anomaly trending, and procedural discipline are central to maintaining mission availability.

122.4. FAULT MANAGEMENT, AND SAFING ROUTING

Fault management and protection through highly automated engineering disciplines. The onboard computers (like the *Command and Data Subsystem*) constantly monitor thousands of telemetry points. It monitors for anomalies like:

- System Failures, sudden hardware malfunctions, such as a failing gyroscope or computer; pointing away from the sun.
- Environmental Hazards, e.g. a burst of solar radiation or a cosmic ray striking sensitive electronics.
- Command Loss: If the spacecraft hasn't heard from Earth for a set period, it may assume something is wrong and enter safe mode to re-establish contact.
- Power or Thermal Issues: Batteries getting too low or the spacecraft becoming too hot/cold.

If a failure, anomaly, or orbital debris impact is detected, this system automatically executes preplanned recovery responses to isolate the problem, switch to backup hardware, or put the spacecraft into a safe, low-power state using a safing routine.

122.4.1 Safing Routines

A *safing routine* is a sequence of autonomous actions a spacecraft takes to protect itself when it detects a major problem. These routines transition the vehicle to a power-positive, thermally safe, communications-safe state – they disable high-voltage power supplies, closes valves, or turns off instruments to protect sensitive hardware when an anomaly is detected. In munitions, safing mechanisms prevent accidental detonation. They are "emergency fallback plans" designed to keep the vehicle stable and healthy until human controllers on Earth can figure out what went wrong. The follow these priorities:

- Stop Non-Essential Tasks: immediately cancels scientific observations and shuts down instruments to save power.
- Point to the Sun: The spacecraft reorients itself so its solar panels get maximum light and its thermal shields are properly aligned.
- Establish Contact: It points its antenna toward Earth and begins broadcasting a simple "low-gain" signal that is easier for ground stations to pick up.
- Wait for Help: It enters a quiescent or benign state – minimizing activity and staying in a known, stable configuration while awaiting new instructions.

122.4.2 Real-World Examples

The Hubble Space Telescope has entered safe mode several times over its life, often due to issues with its gyroscopes.

The MAVEN Mars Orbiter recently used its safing routine to swap to a backup computer after an orientation failure.

The Mars Reconnaissance Orbiter (MRO) encountered multiple safing events in 2009, requiring months of careful recovery by engineers.

The GOES-19 when into a safe hold on 2026 July 15.

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